Praxia Update

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Delayed Surgical Treatment of Pediatric and Adolescent Medial Meniscus Posterior Root Tears Is Associated With Increased Odds of Medial Tibiofemoral Compartment Cartilage Injury: A Multicenter Study.

Moran J, Amaral JZ, Ortiz E, Farhat AG, Jones RH, Groff KD, Tollefson LV, LaPrade CM et al.

BACKGROUND: Medial meniscus posterior root tears (MMPRTs) have been associated with rapidly progressive cartilage degeneration when left untreated in adults. However, their clinical presentation, tear morphology, and associated cartilage pathology remain poorly characterized in pediatric and adolescent patients.

PURPOSE: To identify risk factors for medial tibiofemoral compartment cartilage injury in pediatric and adolescent patients undergoing MMPRT repair, and secondarily to characterize MMPRT morphologies and skeletal maturity-associated injury patterns.

STUDY DESIGN: Case series; Level of evidence, 4.

METHODS: Patients <19 years of age who underwent a transosseous MMPRT repair between 2015 and 2025 across 5 institutions were included. Operative records were reviewed to classify MMPRT morphology using the LaPrade classification and to document concomitant ligamentous procedures. The presence, location, and severity of arthroscopically identified medial tibiofemoral compartment cartilage injuries involving the medial femoral condyle (MFC) and/or medial tibial plateau were graded using the International Cartilage Regeneration & Joint Preservation Society (ICRS) classification. Patients with and without medial compartment cartilage injury were compared, and multivariable logistic regression was used to identify risk factors associated with its presence at the time of surgery […]

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Effect of Medialized Versus Anatomic Repair of Medial Meniscus Posterior Root Tears on Joint Contact Mechanics: A Cadaveric Model of Simulated Gait.

Uppstrom TJ, Chastain KL, Brusalis CM, Steineman B, Gomoll AH, Maher SA, Strickland SM

BACKGROUND: The biomechanical consequences of the repair of medial meniscus posterior root (MMPR) tear-whether anatomic or nonanatomic-during simulated gait have not been quantified.

HYPOTHESIS: Medialized root repair would result in increased peak contact stress and proportion of compartment load acting through the meniscus and decreased contact area compared with anatomic root repair throughout simulated gait.

STUDY DESIGN: Controlled laboratory study.

METHODS: Six human cadaveric knees were mounted on a robotic test system programmed to apply dynamic forces, moments, and flexion angles to mimic level walking. Contact area, peak contact stress, and proportion of compartment force acting through the meniscus were measured throughout the stance phase of the simulated gait cycles for the following nonrandomized conditions: (1) native meniscus-"intact" condition, (2) root repair at the anatomic location, and (3) root repair 10 mm medially. The difference between the anatomic and medialized repair conditions was calculated for each metric, and then the mean and 95% CI were computed.

RESULTS: No difference was found in contact area, peak contact stress, or proportion of compartment force through the meniscus between the medialized and anatomic repair conditions. During the early stance phase of simulated gait (10%-22% of gait cycle), medialized repair of MMPR tear resulted in s […]

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Association Between Preoperative Joint Line Convergence Angle and Clinical Outcomes After Isolated Medial Meniscus Posterior Root Tear Repair.

Ackermann J, Moews LD, Morgan JT, Song CB, Chahla J

BACKGROUND: While coronal malalignment influences outcomes after medial meniscus posterior root tear (MMPRT) repair, alignment parameters such as the hip-knee-ankle angle (HKAA) may not fully capture the biomechanical setting of the medial compartment. The joint line convergence angle (JLCA) reflects the intra-articular environment but has not been well studied as a prognostic factor after MMPRT repair.

PURPOSE/HYPOTHESIS: The purpose of this study was to evaluate the association between the preoperative JLCA and patient-reported outcomes (PROs) after MMPRT repair and to determine whether the JLCA provides prognostic information independent of the HKAA. It was hypothesized that a higher preoperative JLCA would be associated with inferior postoperative PROs and that the JLCA would demonstrate a stronger association with outcomes than the HKAA.

STUDY DESIGN: Case series; Level of evidence, 4.

METHODS: A retrospective review was performed of patients undergoing isolated MMPRT repair by a single surgeon between 2017 and 2025. The preoperative and postoperative JLCA were measured on weightbearing knee radiographs. In a subset of patients with full-length radiographs, the HKAA was also measured, with positive values indicating varus alignment and negative values indicating valgus alignment. PROs included the International Knee Documentation Committee (IKDC) and Knee injury and Ost […]

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Biomechanical Comparison of 4 Radial Meniscal Repair Techniques Including a Novel Dual Tie-Grip Configuration.

Bachmaier S, Krych AJ, Bedi A, Müller PE, Smith PA

BACKGROUND: Complex repairs of radial meniscal tears have shown increased fixation strength compared to conventional sutures. A simplified 2-suture hybrid and a novel dual tie-grip repair with interconnected vertical mattress sutures spanning the tear were tested and compared to all-inside double-horizontal and tie-grip repairs.

HYPOTHESIS: The dual tie-grip repair would increase the initial compression load across the tear and show the least cyclic displacement and highest failure strength.

STUDY DESIGN: Controlled laboratory study.

METHODS: A total of 40 porcine medial menisci were assigned to 4 repair groups: double-horizontal, hybrid, tie-grip, and dual tie-grip (10 per group). Results from tie-grip repairs were used to assess the effect of vertical mattress suture interconnection. After suture placement according to the described technique and fixation, the initial compressive load, stiffness, and relief displacement were measured. The repaired specimens underwent cyclic loading between 5 and 30 N over 1000 cycles (0.75 Hz), while cyclic stiffness and displacement were measured. Ultimate stiffness and load-to-failure were analyzed at 3.15 mm/s.

RESULTS: Analysis of variance revealed that dual tie-grip and hybrid repairs showed higher initial compressive load and relief displacement than the other techniques, with dual tie-grip repair reaching the highest values for bot […]

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Increased Posterior Tibial Slope Correlates With Greater Odds of Reoperation and Failure After Meniscal Allograft Transplantation.

Bi AS, Lemme NJ, Mufti Y, Sachs JP, Franzia CH, Yanke AB, Cole BJ

BACKGROUND: Higher posterior tibial slope (PTS) is associated with a greater rate of failure in meniscus root tears and anterior cruciate ligament reconstruction.

PURPOSE: To analyze the correlation of PTS with patient-reported outcomes (PROs) and rates of reoperation and failure in meniscal allograft transplantation (MAT).

STUDY DESIGN: Cohort study (prognostic); Level of evidence, 3.

METHODS: A retrospective review of a prospectively maintained database was performed to assess outcomes after MAT in patients between 2003 and 2021 with minimum 2-year follow-up. PTS was measured on lateral knee radiographs, with medial PTS (MPTS) and lateral PTS (LPTS) measured on magnetic resonance imaging (MRI). PROs were collected preoperatively and at minimum 2-year follow-up. MAT failure was defined as revision MAT or conversion to arthroplasty. Multivariable regression was used to correlate PTS with PROs and rates of reoperation and failure. Failure and reoperation were further analyzed separately in medial and lateral MATs for MPTS and LPTS.

RESULTS: In total, 175 knees (174 patients) met inclusion criteria with a mean ± SD age of 27.4 ± 9.1 years and follow-up of 8.3 ± 3.8 years. By radiograph, the mean PTS was 8.8°± 3.2°; by MRI, the mean MPTS and LPTS were 5.1°± 2.7° and 5.8°± 3.3°, respectively. MRI measurements significantly underestimated radiographic measurements by 3.3°± 2.9°. […]

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Oral Contraceptive Use and Improved Graft Maturation After Anterior Cruciate Ligament Reconstruction in Female Patients: An MRI-Based Study.

Masferrer-Pino A, Martínez-Peñas J, Ormazabal I, Pizza N, Perelli S, Monllau JC

BACKGROUND: Female athletes have a greater risk of anterior cruciate ligament (ACL) injuries, a disparity partly attributed to hormonal influences on ligament biomechanics and collagen metabolism. Oral contraceptives (OCs) may stabilize hormonal fluctuations and have been associated with a reduced risk of ACL injury; however, their potential effect on biological healing after ACL reconstruction (ACLR) remains unclear.

PURPOSE: To investigate whether the use of OCs affects graft maturation, assessed by the signal-to-noise quotient (SNQ) of postoperative magnetic resonance imaging (MRI), in female patients who have undergone ACLR with hamstring tendon autograft and associated lateral extra-articular tenodesis (LET).

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: This retrospective comparative cohort study included a series of female patients with an active menstrual cycle, aged 16 to 45 years, who underwent anatomic ACLR with hamstring tendon autograft and LET between January 2022 and December 2023. Patients were stratified based on OC use at the time of surgery and postoperatively. Thus, the study included patients who underwent ACLR+LET with and without OC treatment. The 2 groups were comparable based on all the criteria analyzed. Follow-up MRI (3.0 T) was performed at a mean of 7 months postoperatively to analyze graft maturity using circular region of interest […]

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Contralateral ACL Rupture as an Indicator of Intrinsic Risk for Graft Failure After ACL Reconstruction: A Cohort Study of 7718 Patients From the SANTI Study Group.

Santamaria F, Culebras Almeida LA, Carrozzo A, Campos JP, Saithna A, Vieira TD, Sonnery-Cottet B

BACKGROUND: Graft failure after anterior cruciate ligament (ACL) reconstruction (ACLR) may be influenced by both extrinsic exposure and intrinsic patient susceptibility. A contralateral ACL rupture could reflect an inherent predisposition to ligament injury.

PURPOSE: To determine whether a history of bilateral native ACL ruptures increases the risk of graft failure after primary ACLR.

STUDY DESIGN: Cohort study; Level of evidence, 2.

METHODS: A total of 7718 consecutive patients who underwent primary ACLR by a single surgeon (2003-2022) were included. Graft failure, defined as a clinically and magnetic resonance imaging-confirmed rupture, was compared between patients with bilateral versus unilateral native ACL ruptures. Multivariable logistic regression identified independent predictors, including age, sex, Tegner score, sport type, and a history of bilateral rupture.

RESULTS: The cohort included 6327 patients with isolated native ACL rupture and 1391 patients with bilateral native ACL rupture. The overall graft failure rate was 6% (461-7718), with a mean follow-up of 135.9 months. Patients with bilateral ACL ruptures had a higher failure rate (9.3%) than those with unilateral injuries (5.2%) ($P < .0001$). Independent predictors included younger age (odds ratio [OR], 0.92 [95% CI, 0.90-0.93]; $P < .0001$), male sex (OR, 1.46 [95% CI, 1.10-1.96]; $P = .0089$), higher Tegner scor […]

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Progressive Cartilage Degeneration After Anterior Cruciate Ligament Reconstruction: Longitudinal Evidence From the Swedish Knee Ligament Registry.

Agostinone P, Romandini I, Rotzius P, Zaffagnini S, Forssblad M, Sandon A

BACKGROUND: Chondral lesions and subsequent knee osteoarthritis often occur after anterior cruciate ligament (ACL) injuries, and although ACL reconstruction (ACLR) is common, cartilage degeneration remains more prevalent than in the general population, with inconsistent evidence on whether surgery offers a protective effect.

PURPOSE/HYPOTHESIS: This study aimed to evaluate the progression of cartilage damage after ACLR using second-look arthroscopy in patients who underwent both primary and revision procedures and to identify risk factors for cartilage degeneration. It was hypothesized that chondral lesions would progress between primary and revision surgery, primarily influenced by characteristics of the initial injury and primary surgery.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: A retrospective cohort study was conducted using prospectively collected data from the Swedish Knee Ligament Registry. All patients who underwent both primary and revision ACLR between January 1, 2005, and December 31, 2023, were included. Patients with multiligament injuries or missing key data were excluded. The prevalence, location, and severity of cartilage lesions assessed intraoperatively at both primary and revision ACLR were reported and compared. Secondarily, potential risk factors for cartilage damage at the time of revision surgery were evaluated using multivariate logi […]

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Cutting Technique of Soccer Players After ACLR: On-Field Matched Control Study.

Di Paolo S, Mendicino M, Viotto M, Aparo F, Ambrosini V, Grassi A, Zaffagnini S

BACKGROUND: Anterior cruciate ligament (ACL) ruptures in soccer most commonly occur during cutting maneuvers with noncontact injury mechanisms. Biomechanical assessment of the cutting technique has become a critical component of return-to-sport (RTS) evaluation after ACL injury, particularly in young football players at elevated risk of reinjury. However, the cutting technique in players with ACL reconstruction (ACLR) has never been examined under sport-specific conditions.

PURPOSE/HYPOTHESIS: This study aimed to evaluate the cutting technique in pediatric soccer players who had ACLR after RTS clearance against matched healthy controls during on-field soccer (or football)-specific (FS) movements. It was hypothesized that players with ACLR would exhibit risk factors for ACL injury and biomechanical differences relative to healthy counterparts.

STUDY DESIGN: Case-control study; Level of evidence, 3.

METHODS: A total of 61 young soccer players-21 with ACLR (age, 16.7 ± 1.5 years) and 40 healthy matched controls-performed planned and unplanned FS cutting tasks on a regular soccer pitch. Kinematics of the lower limbs, pelvis, and trunk were collected using 8 wearable inertial sensors. Cut-stance phase kinematics were extracted as peak values, range of motion (ROM), initial contact values, and peak knee flexion. A linear mixed-effects model tested the effects of injury ($\beta 1$, ACLR v […]

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The Anterior Cruciate Ligament Injury Severity Scale (ACLISS) as a Predictor of Short-Term Reoperation and Functional Outcomes After ACL Reconstruction.

Graham GD, Haffner M, Testa EJ, White A, Lifsey C, Gachigi K, Padley J, Mueller J et al.

BACKGROUND: The Anterior Cruciate Ligament Injury Severity Scale (ACLISS) was developed to classify the magnitude of damage to knee structures beyond the anterior cruciate ligament (ACL) (meniscus, cartilage, collateral ligaments, etc) at the time of ACL rupture. However, its validity in predicting clinical outcomes after ACL reconstruction (ACLR) has never been assessed.

PURPOSE: To determine whether ACLISS correlates with reoperation and patient-reported functional outcomes after ACLR.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: The records of all patients who underwent primary ACLR at a single institution between 2019 and 2022 with minimum follow-up of 2 years were reviewed. Patients were excluded if they had concomitant collateral ligament or posterior cruciate ligament repair/reconstruction or prior ipsilateral ACLR. ACLISS scores (0-12) and grades (grade 1: scores 0-3; grade 2: scores 4-7; grade 3: scores 8-12) were determined using preoperative magnetic resonance imaging and intraoperative arthroscopic findings based on the original published technique. The primary outcome was reoperation after ACLR. Secondary outcomes included International Knee Documentation Committee (IKDC) subjective scores and Marx activity scores. Bivariable and multivariable logistic regression analyses were performed to identify predictors of reoperation. Cox proportional hazard […]

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Comparison of Multiligament Reconstruction in Individuals With BMI >30 and <30 kg/m2.

Tagliero AJ, Smith JR, Iyer S, Yanamala S, Johnson T, Krych AJ, Levy BA

BACKGROUND: Despite the growing clinical relevance of obesity in the context of multiligament knee injuries, there remains a paucity of data evaluating patient-reported outcomes in this population.

PURPOSE: To compare clinical and functional outcomes, including postoperative patient-reported outcome measures (PROMs) and complication/revision rates, of multiligament knee reconstruction (MLKR) in individuals with a body mass index (BMI) >30 kg/m2 versus <30 kg/m2 at a minimum 2-year follow-up.

STUDY DESIGN: Case series; Level of evidence, 4.

METHODS: A retrospective review was performed to identify all patients who underwent MLKR between 2001 and 2022 at a single institution. Patients with a BMI >30 kg/m2 were propensity-matched to patients with a BMI <30 kg/m2 on a 1:1 basis by age, sex, laterality, and Schenck classification (knee dislocation). Medical records were reviewed for patient characteristics, time to surgery, year of surgery, number and type of ligament interventions, graft choice, single versus staged reconstruction, and concomitant injuries. Postoperative clinical examination findings and PROMs for each cohort, including visual analog scale score, Tegner activity scale score, Lysholm score, and International Knee Documentation Committee (IKDC) subjective score, were analyzed and compared. Postoperative complication and revision rates were evaluated and compared. […]

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The Effect of Minimalist Versus Motion Control Shoes on Patellofemoral Joint Forces in Adolescents With Patellofemoral Pain During Running: A Randomized Crossover Study.

Kayll SA, Hinman RS, Bryant AL, Bennell KL, Rowe PL, Chi PW, Starkey S, McManus F et al.

BACKGROUND: High joint forces can contribute to the pathogenesis of patellofemoral pain in adolescents. Minimalist shoes may reduce patellofemoral joint forces during running compared with commonly worn motion-control shoes.

HYPOTHESIS: Patellofemoral joint, lateral patella, and quadriceps forces will be lower, while gastrocnemius forces will be higher when adolescents run in minimalist shoes compared with motion control shoes.

STUDY DESIGN: Cross-sectional study; Level of evidence, 2.

METHODS: This within-session 2-period randomized crossover study was conducted from March 2023 to September 2024. A total of 51 physically active adolescents aged between 12 and 19 years (mean age, 16.9 ± 2 years) (23 female adolescents [45%]) with patellofemoral pain were recruited from the general population. Kinematic, kinetic, and electromyography (EMG) data were collected during overground running in minimalist and motion-control shoes, with testing randomized within the same session. Patellofemoral joint force was measured during the stance phase of running using an EMG-informed neuromusculoskeletal model. The primary outcome was the resultant patellofemoral joint force (N). Secondary outcomes included lateral patellar force (defined as the peak force acting on the patella in the frontal plane) and quadriceps and gastrocnemius muscle forces (N). All outcomes were analyzed at peak using p […]

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Triamcinolone Acetonide Is Not Harmful to Healthy and Osteoarthritic Human Knee Cartilage.

Porter A, Nguyen J, Peng Y, DiStefano S, Asres N, Snyder-Mackler L, Axe M, Lu XL

BACKGROUND: Intra-articular triamcinolone acetonide (TA) injections are commonly used to manage joint synovitis despite reports that they accelerate cartilage degeneration. Previous studies found that TA minimally affected bovine cartilage, but effects in human cartilage remain unclear.

PURPOSE: To determine how TA in physiologically relevant concentrations influences chondrocyte viability and matrix metabolism in healthy and osteoarthritic (OA) human cartilage.

STUDY DESIGN: Controlled laboratory study.

METHODS: Human cartilage was harvested from cadaver donor knees (mean age, 38 years; 4 male) or total knee replacement remnants (mean age, 66 years; 9 female, 7 male). Samples were exposed to TA, 200 µM (0.087 mg/mL) or 1 nM, to evaluate chondrocyte viability and gene expression. A high-resolution click chemistry assay quantified glycosaminoglycan (GAG) and collagen synthesis and tracked GAG loss. Two dosing regimens were tested: a continuous 14-day exposure (multiple-injection simulation) and a 2-day exposure followed by a 14-day recovery (single-injection simulation).

RESULTS: TA did not reduce chondrocyte viability and did not increase baseline GAG loss from the cartilage samples. In healthy cartilage, TA downregulated MMP13 (mean ± SD; 1.0 ± 0.8 vs 0.4 ± 0.2; P = .04) and COL2A1 (1.0 ± 0.3 vs 0.5 ± 0.2; P = .003). In OA cartilage, 200 µM TA downregulated MMP13 (1.0 ± 0. […]

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Rat Model of Gluteus Medius Tendon Tear: Biomechanical and Histological Effects of Autologous Fascia Lata Augmentation.

Aydemir S, Akin E, Havitcioglu C, Ibrahim MM, Karabag UA, Karabay U, Husemoglu RB, Hapa O

BACKGROUND: Primary repair of full-thickness gluteus medius tears is challenging, particularly in retracted tears with poor tissue quality. Augmentation with autologous fascia lata may enhance repair strength and biologic healing.

PURPOSE: To establish a gluteus medius tear detachment model in rats and to evaluate whether acute repair augmented with autologous fascia lata results in higher failure load and improved tendon-bone interface organization as compared with primary repair.

STUDY DESIGN: Controlled laboratory study.

METHODS: Forty adult female Sprague-Dawley rats were randomly assigned to 4 groups: intact control (group C), unrepaired tear (group 1), primary repair (group 2), and augmented repair with autologous fascia lata (group 3) (n = 6 for biomechanical testing, n = 4 for histological analysis per group). In groups 1 to 3, the gluteus medius tendon was detached from the greater trochanter. Group 1 underwent no repair. In groups 2 and 3, repair was performed acutely during the index procedure immediately after tendon detachment, using intraosseous tunnels and 5-0 Prolene sutures; in group 3, an autologous fascia lata graft was incorporated as an onlay augmentation. At 4 weeks, animals were euthanized for biomechanical and histological evaluation.

RESULTS: All tendons failed at the tendon-bone interface. Failure load was lower in group 1 (mean ± SD, 15 ± 6 N; P = […]

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Short-Term Outcomes Following Periportal Versus Interportal Capsulotomy With Closure in Nondysplastic Women Undergoing Hip Arthroscopy for Femoroacetabular Impingement.

Johnson BT, Metz AK, Puzzitiello RN, Hunter CDR, Singh A, Featherall J, Aoki SK

BACKGROUND: The periportal capsulotomy for hip arthroscopy minimizes iliofemoral ligament violation compared to T- and interportal capsulotomies, theoretically reducing the risk of postoperative instability secondary to capsular insufficiency.

PURPOSE/HYPOTHESIS: The purpose of this study was to compare patient-reported outcomes (PROs) and revision rates between women without dysplasia who underwent an interportal versus a periportal capsulotomy approach for the treatment of femoroacetabular impingement syndrome (FAIS). It was hypothesized that women without dysplasia would achieve better outcomes and a lower revision rate through a periportal approach.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: Consecutive female patients with normal acetabular coverage (lateral center-edge angle >25° and ≤40°) who underwent primary hip arthroscopy with complete capsular closure by a single surgeon for the treatment of FAIS between February 2020 and November 2022 were prospectively added to a database and their records retrospectively reviewed. An interportal capsulotomy was utilized in all patients before June 2021, and periportal capsulotomies were utilized thereafter. Femoral osteoplasty was performed in all patients, and a labral repair was performed when indicated. PROs were collected preoperatively and at a 2-year minimum follow-up, including pain scores at rest and ac […]

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The Modified Budin View: A Reliable and Accessible Screening Tool for Femoral Version Deformities in Joint-Preserving Hip Surgery.

Heimann AF, Carrel J, Schmaranzer F, Stetzelberger VM, Schwab JM, Hanauer M, Marti D, Laurencon J et al.

BACKGROUND: Femoral version deformities are key in the diagnostic evaluation of young patients with hip pain. While computed tomography (CT) and magnetic resonance imaging (MRI) are commonly used for femoral version assessment, their limitations include a lack of standardized reference values, high costs, and accessibility constraints. An accessible and reproducible alternative is needed, particularly in the context of joint-preserving hip surgery.

PURPOSE: To evaluate the reliability and reproducibility of femoral version measurements on the modified Budin view, to compare these measurements with 4 established cross-sectional imaging methods (CT or MRI), and to determine the diagnostic accuracy of the modified Budin view in identifying abnormal femoral version.

STUDY DESIGN: Case series; Level of evidence, 4.

METHODS: The authors retrospectively analyzed the records of 93 patients (107 hips; mean age, 31 ± 9 years [range, 16-55 years]; 43 female hips [40%]) evaluated for joint-preserving hip surgery at a specialized tertiary center. Femoral version was measured using the modified Budin view and compared to 4 cross-sectional imaging methods (CT and MRI) using Bland-Altman analysis and diagnostic accuracy metrics.

RESULTS: The mean femoral version measured on the modified Budin view was 13°± 7° (95% CI, 12°-14°). The technique demonstrated excellent intra- and interobserver […]

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Femoral Version Measurements Vary Significantly Between Commonly Used Methods: Implications for Defining Diagnostic Thresholds.

Vorimore C, Smit K, Rakhra K, Tien-Yueh Schneider M, Beaulé P, Speirs A, Grammatopoulos G

BACKGROUND: An accurate measurement of femoral version is essential for diagnosing version abnormalities, determining the need for a surgical intervention, and establishing the extent of correction required. However, in the literature, numerous measurement techniques have been described.

PURPOSE: To (1) evaluate the differences among 4 different 2- and 3-dimensional (2D and 3D, respectively) techniques for measuring femoral version, (2) assess the effect of femoral version on differences between measurement techniques, and (3) examine the accuracy of an automated method in determining femoral version using commonly employed techniques.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: A total of 100 patients who underwent hip preservation surgery at a single academic center were analyzed (measured and segmented). Pelvic computed tomography was used to measure femoral version using 4 established methods: 2 were 2D axial techniques (Murphy and Reikeras), and 2 were 3D reconstructions (Sugano and Lee). Overall, 4 assessors performed the axial measurements. Measurements were performed relative to the posterior condylar axis (PCA) and transepicondylar axis (TEA). Discrepancies in femoral version between different techniques and interobserver correlations were determined. Linear regression equations to convert femoral version values between methods were established. Using […]

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Clinical Utility of the Moving Valgus Stress Test and Milking Maneuver for Medial Ulnar Collateral Ligament Injuries of the Elbow.

Sciascia AD, Bowman EN, Camp C, Chalmers P, Smith MV, Wolf BR, Griffith T, Erickson B et al.

BACKGROUND: Several special tests for detecting medial ulnar collateral ligament (MUCL) injuries of the elbow have been described, but data on the clinical utility of these tests are limited.

PURPOSE: To determine the clinical utility of selected elbow special tests to detect MUCL injuries in an athletic population.

STUDY DESIGN: Cross-sectional study; Level of evidence, 3.

METHODS: A total of 84 consecutive patients with sports-related elbow pain underwent a standardized clinical examination. Overall, 6 special tests (moving valgus stress test, static valgus stress test, modified valgus stress test, milking maneuver, pronation flexion resistance test, and temple/cheek press test) were performed. Sensitivity, specificity, accuracy, positive/negative predictive value, and positive/negative likelihood ratio (LR+ and LR-, respectively) were calculated for individual tests and combinations of tests to detect MUCL injuries. Higher values for the LR+ and lower values for the LR- represent stronger clinical utility.

RESULTS: For detecting MUCL injuries, the moving valgus stress test resulted in the highest sensitivity (98%) and accuracy (93%) with the lowest LR- (2%). The milking maneuver had the highest LR+ (5.1) with the second highest sensitivity (77%) and accuracy (79%). The pronation flexion resistance test and the temple/cheek press test had the lowest LR+ (1.5 and 1.8, resp […]

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Surgical Management of SLAP Lesions in Baseball Throwers: Posterosuperior and Posterior Labral Debridement With or Without Anterosuperior Repair Yields High Return to Play Rates.

Kin Y, Sugaya H, Onishi K, Takahashi N, Tokai M, Ueki H, Komatsu T, Inoue T et al.

BACKGROUND: Reported rates of return to the preinjury level of play (RTPL) after superior labrum anterior and posterior (SLAP) repair in overhead athletes are relatively low (22%-64%). Current fixation techniques that rigidly fix the posterosuperior labrum may lead to overconstraint and persistent pain.

PURPOSE/HYPOTHESIS: The purpose of this study was to retrospectively evaluate the clinical outcomes and RTPL rates of our surgical strategy in competitive baseball throwers with SLAP lesions. This strategy includes debridement of the posterosuperior labrum without fixation, with selective anterosuperior fixation when anterior extension and instability of the labral detachment are present. The authors hypothesized that posterosuperior and posterior debridement with or without anterosuperior repair would yield high return to play rates in baseball throwers with SLAP lesions.

STUDY DESIGN: Case series; Level of evidence, 4.

METHODS: A total of 80 shoulders in 80 baseball players (mean age, 20 years [range, 15-38 years]) underwent arthroscopic surgery for SLAP lesions (Type II or Type VIII with posterior extension). The cohort included 16 professional, 31 collegiate, 23 high school, and 10 recreational players. Based on intraoperative findings, isolated debridement of the posterosuperior and/or posterior labrum was performed for stable lesions, defined as posterosuperior detachme […]

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Incomplete Restoration of Coracoclavicular Ligament Thickness After Surgical Treatment of Rockwood Type 5 Acromioclavicular Dislocations: A Contralateral-Controlled MRI Study With Functional Outcomes.

Koluman AC, Tingir M, Ziroglu N, Ercin E, Moroder P

BACKGROUND: Surgical fixation of Rockwood type 5 acromioclavicular (AC) dislocations can restore coracoclavicular (CC) distance radiographically, but whether the CC ligaments regain their native morphology is unknown.

PURPOSE: To evaluate postoperative CC ligament thickness on magnetic resonance imaging (MRI) compared with the contralateral side and to assess its relationship with functional outcomes.

STUDY DESIGN: Cohort study; Level of evidence, 2.

METHODS: A total of 46 patients (28 hook plate, 18 EndoButton) with ≥24 months of follow-up received bilateral radiographs and sagittal oblique MRI sequences. CC distance and conoid and trapezoid ligament thickness were measured, with contralateral shoulders serving as individualized controls. Functional outcomes included the visual analog scale, Constant, University of California Los Angeles, and Quick Disabilities of the Arm, Shoulder and Hand scores.

RESULTS: Surgical shoulders showed increased CC distance and reduced trapezoid and total CC thickness, whereas conoid thickness was comparable to that of the contralateral side. Neither fixation method achieved complete restoration of CC ligament thickness. Functional scores were favorable across all patients, with no clinically relevant differences between fixation techniques. The overall rate of complications was similar between groups; however, complication patterns differed, […]

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The Long-term Radiographic Fate of the Chronically ACL-Deficient Knee: A Systematic Review and Meta-analysis of Matched Cohort Studies.

Shah A, Gumarathas K, Marks P, Marx RG, Kiss A, Wasserstein D

BACKGROUND: Determining the long-term risk of arthritis in patients with anterior cruciate ligament (ACL) injury treated nonoperatively versus those who undergo ACL reconstruction (ACLR) remains an important and unanswered question for patients and surgeons.

PURPOSE: (1) To define the cumulative arthritis rate and severity after nonsurgical management of ACL injury-the chronically ACL-deficient (ACLD) knee; (2) to compare rates and severity of arthritis in patients who have ACLD knee with similar patients who underwent ACLR; and (3) to identify clinically relevant risk factors for arthritis.

STUDY DESIGN: Systematic review; Level of evidence, 3.

METHODS: Three databases (Medline, Embase, PubMed) were searched for primary studies examining radiographic outcomes in patients with chronic ACL deficiency (>12 months of ACL deficiency). Studies with a matched ACLR control group were included. Quality assessment was performed with the MINORS (Methodological Index for Nonrandomized Studies) tool. Arthritis prevalence over time was plotted and modeled to best-fit using the Akaike information criterion. Data were extracted for meta-analysis for the primary outcome of osteoarthritis. The cumulative odds ratio of prognostic factors was calculated where appropriate.

RESULTS: Nineteen full-text studies met inclusion criteria (11 matched cohort studies comparing ACLD and ACLR) including 1 […]

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The Effects of Adherence to the FIFA 11+ Program on Injury Risk in Soccer Players: A Systematic Review and Meta-analysis.

Zarei M, Aslani M, Yaghoubitajani Z, Sheikhhoseini R

BACKGROUND: The FIFA 11+ warm-up program for injury prevention was developed in soccer given its high risk of injuries. However, the overall effect on injury rates remains uncertain because of the existence of variability in how players adhere to this program.

PURPOSE: To assess the effects of adherence to the FIFA 11+ program on injury risk in soccer players.

STUDY DESIGN: Systematic review and meta-analysis; Level of evidence, 1.

METHOD: The authors systematically searched randomized controlled trials in the databases, including Web of Science, PubMed, and Scopus, for publications in English from 2008 onward following the PRISMA guidelines. Data from 5 randomized controlled trials were included in this meta-analysis, involving 2025 participants. A random effects model was used to calculate the risk ratios and the I2 statistic for heterogeneity.

RESULTS: The risk ratios revealed a significant reduction in injury risk for soccer players who adhered to the FIFA 11+ program (95% CI, 0.12-0.22; P = .001). There was substantial heterogeneity (I2 = 88.98%), suggesting noteworthy variability in the outcomes of the included studies. Additionally, no significant predictors of effect size were identified in meta-regression analysis, implying that unmeasured factors may play a role in the program's effectiveness.

CONCLUSION: Findings revealed that adhering to the FIFA 11+ program ef […]

PubMed · DOI

Management of Osteochondral Lesions of the Tibial Plateau: A Systematic Review of Operative Techniques and Reported Outcomes.

Goli PS, Touhey DC, Brophy RH, Matava MJ, Smith MV, Knapik DM

BACKGROUND: Management of tibial plateau osteochondral (TP-OC) lesions remains challenging due to limited reported treatment options and outcomes.

PURPOSE: To systematically review the literature to better understand current operative indications, techniques, and outcomes for TP-OC lesions.

STUDY DESIGN: Systematic review; Level of evidence, 4.

METHODS: Studies reporting on the treatment of TP-OC lesions from inception to August 2025 were identified using the Cochrane Library, EMBASE, and PubMed databases. Inclusion criteria included studies reporting on patients undergoing operative management with injury cause, lesion size and location, reported technique (repair vs restoration), and postoperative outcomes, including complications and patient-reported outcome measures (PROMs).

RESULTS: A total of 24 studies, consisting of 581 patients (mean age, 42.2 years; range, 12-77 years), were identified. Weighted mean follow-up was 106.0 months (range, 3-408 months). Traumatic injury was reported in 82.6% (n = 256/310) of patients. Lesions were most commonly located on the lateral tibial plateau (65.8%; n = 340/517), with a weighted mean lesion area of 2.29 cm2 (range, 0.80-6.00 cm2). Operative treatments included osteochondral allograft transplant (OCAT) (80.2%; n = 466/581), microfracture (6.7%; n = 39/581), osteochondral autograft transfer (OAT) (6.4%; n = 37/581), autologous ch […]

PubMed · DOI

Outcomes of Simultaneous Bilateral, Staged Bilateral, and Unilateral Hip Arthroscopy for Femoroacetabular Impingement Syndrome: A Systematic Review and Meta-analysis.

Thamrongskulsiri N, Morgan JT, Casanova F, Vega TF, Mirahmadi A, Chahla J

BACKGROUND: Femoroacetabular impingement (FAI) is a common cause of hip pain, often treated with arthroscopy. The optimal approach among unilateral, staged bilateral, and simultaneous bilateral procedures remains unclear due to limited comparative data.

PURPOSE: To compare clinical outcomes and complication rates among simultaneous bilateral, staged bilateral, and unilateral hip arthroscopy in patients with FAI syndrome.

STUDY DESIGN: Systematic review and Meta-analysis; Level of evidence, 3.

METHODS: A systematic review and meta-analysis were conducted in accordance with PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines. The PubMed, Scopus, and Ovid MEDLINE databases were searched through April 2025. Eligible comparative studies included patients with FAI undergoing hip arthroscopy via unilateral, simultaneous bilateral, or staged bilateral approaches and reported outcomes such as patient-reported scores, complications, and revision or conversion to total hip arthroplasty (THA).

RESULTS: Nine studies with a total of 4040 hips were included. All surgical approaches showed significant improvements in pain and functional outcome scores. There were no statistically significant differences in postoperative visual analog scale score, Modified Harris Hip Score, Non-Arthritic Hip Score, International Hip Outcome Tool-12 score, rates of revision […]

PubMed · DOI

Platelet-Rich Plasma in the Treatment of Musculoskeletal Disease in 2025 and Beyond.

Rothrauff BB, Featherall JT, Uppstrom TJ, Gohring G, Nelson AL, Evans TA, Millett PJ, Provencher MT et al.

Platelet-rich plasma (PRP) is a blood-based orthobiologic used to treat a myriad of musculoskeletal conditions. While in vitro and preclinical studies on PRP have been promising, clinical results have been mixed. The heterogeneity in clinical benefits is attributable to both the complexity and variability of PRP as a biologic as well as the diversity of targeted tissues and ailments. Many variables have been proposed to affect PRP's bioactivity and clinical effects, with differing levels of evidence demonstrated for each variable. These variables can be broadly categorized as biological, technical, and abnormality-specific factors. Additionally, insufficient characterization of PRP in clinical studies has been a major limitation in both determining the efficacy of PRP for a given clinical condition and understanding the basic biology of PRP. This review highlights the current landscape of PRP as a treatment of musculoskeletal conditions, including both the regulatory environment and clinical applications, and considers the influence of numerous factors affecting PRP's bioactivity and clinical effects. Emerging technologies that may further enhance the utility of PRP as an orthobiologic are also discussed. Rigorous basic, translational, and clinical research remains fundamental to realize the promise of PRP treatment for musculoskeletal disease.

PubMed · DOI

The Long-term Radiographic Fate of the Chronically ACL-Deficient Knee: Response.

Shah A, Gumarathas K, Marks P, Marx RG, Kiss A, Wasserstein D

PubMed · DOI

The Long-term Radiographic Fate of the Chronically ACL-Deficient Knee: Letter to the Editor.

Bolaños Bermúdez IA, Acosta NG, Panqueva Giraldo LN, Rivero Rapalino OM

PubMed · DOI

Metformin Reduces the Incidence of Shoulder Stiffness After Arthroscopic RC Repair: Letter to the Editor.

Qin K, Zhao L, Wang Y

PubMed · DOI

Metformin Reduces the Incidence of Shoulder Stiffness After Arthroscopic RC Repair: Response.

Zhu Z, Chen H

PubMed · DOI

What Is the "Perfect" Lateral Radiograph? Effects of Beam Directionality and Condylar Alignment on the Perceived Location of the Medial Patellofemoral Ligament: Letter to the Editor.

Kose O

PubMed · DOI

What Is the "Perfect" Lateral Radiograph? Effects of Beam Directionality and Condylar Alignment on the Perceived Location of the Medial Patellofemoral Ligament: Response.

Aoki SK, Johnson BT, Khalil AZ, Featherall J, Metz AK, Ernat JJ

PubMed · DOI

Corrigendum to "The Rectus Femoris Tendon as a Graft for Combined ACL and Anterolateral Reconstruction: A Biomechanical Evaluation".

PubMed · DOI

BJJ

The Bone & Joint Journal

Volume 108-B, Issue 7 · July 2026 · 17 new articles

Interesting data, but further corroborative studies are often needed.

Haddad FS

PubMed · DOI

Missing not at random : how much missing patient-reported outcome measure data is too much?

Williamson TR, Clement ND, Yapp LZ, Duckworth AD, Haddad FS, Scott CEH

PubMed · DOI

From surgical backlog to opioid burden : the case for opioid stewardship in arthroplasty pathways.

Jamal N, Clement ND, Duckworth AD, Collinson CV

The prolonged waiting times for hip and knee arthroplasty have resulted in numerous negative effects for patients being reported, with none more important than increasing opioid reliance due to chronic musculoskeletal pain. Opioid use prior to surgery is a factor associated with prolonged use postoperatively, and while opioids remain a valuable resource in the management of acute surgical pain, ongoing use carries substantial clinical, psychological, and societal risks. Recent Medicines and Healthcare products Regulatory Agency (MHRA) guidance and the 2024 UK consensus statement have resulted in a shift for perioperative pain management, moving away from routine modified release opioid use towards individualized care. To address these issues and changes, interventions need to be focused both preoperatively and postoperatively using improved risk stratification, patient education, and community engagement. This annotation discusses these issues and emphasizes the importance of opioid stewardship as a unifying framework to embed principles across the arthroplasty pathway, with the aims of reducing harm, improving outcomes, and supporting safer sustainable care.

PubMed · DOI

The management of distal humeral fractures in older patients.

Yoong AWH, Super JT, Tan GJS, Watts AC, Ollivere BJ, Duckworth AD

The number of fractures of the distal humerus in older adults is rising due to an ageing population and an increasing incidence of fragility fractures. The management is based on the patient's preference, the pattern of the fracture, the quality of the bone and soft-tissue, the patient's comorbidities and potential for rehabilitation, as well as balancing the risk of complications such as infection, nonunion, instability, or loosening. Nonoperative treatment remains appropriate for a few low-demand or medically unfit patients. Open reduction and internal fixation (ORIF) or arthroplasty, either hemiarthroplasty (HA) of the elbow or total elbow arthroplasty (TEA), provide good outcomes when appropriately selected. An increase in the use of arthroplasty for trauma is shown in registry data, although the indications and survival remain controversial with few robust comparative and long-term data. Similar short- to mid-term functional outcomes between ORIF and TEA have been reported in recent comparative studies, with a trend towards an increased reoperation rate after ORIF, but the evidence is derived largely from small retrospective observational studies. HA of the elbow may be a viable alternative for active older patients, particularly when stable fixation is not achievable, although instability and conversion to TEA may be required. The aim of this review was to synthesize the […]

PubMed · DOI

The importance of axial rotation of the lower limb : a systematic review of measurement methods.

Favier B, Acker A, Miles J, de Cesar Netto C, Burssens A, Goldberg A

AIMS: Assessment of lower limb alignment is a cornerstone of orthopaedics. Few studies look at rotational alignment in the axial plane as measured by femoral version (FV) and tibial torsion (TT), both of which have implications for hip, knee, and ankle pathology. This review provides an overview of the axial rotation and evaluates CT-based measurement methods for FV and TT to identify the most reliable and reproducible techniques for use in clinical practice.

METHODS: A systematic PRISMA-guided review assessed original CT-based methods, examining inter- and intraobserver reliability (intraclass correlation coefficient (ICC)), frequency of use, and validation.

RESULTS: Seven FV and nine TT CT-based techniques were identified. FV had a weighted mean of 17.8° anteversion (-9° to 60°). TT had a weighted mean of 30.8° (2° to 82°). ICCs ranged from good to excellent. The Murphy method (FV) and Goutallier method (TT) had the highest reliability and clinical utility.

CONCLUSION: Lower limb axial rotational profile plays an important role in the management of hip, knee, and ankle arthroplasty surgery as well as many other orthopaedic pathologies. The Murphy and Goutallier methods should be adopted as standard for measuring FV and TT. Their high reproducibility and validation make them ideal for consistent clinical and research use.

PubMed · DOI

Primary total hip arthroplasty after prior osteosynthesis versus osteotomy : distinct risk profiles and the modifying effect of femoral fixation.

Wagener N, Wu Y, Perka C, Hardt S

AIMS: To compare outcomes after primary total hip arthroplasty (THA) following prior ipsilateral osteosynthesis with prior ipsilateral osteotomy, and to assess whether femoral fixation modifies risk.

METHODS: Using the German Arthroplasty Registry (EPRD; 2012 to 2024), two 1:1 matched analyses were performed: prior osteosynthesis versus no prior surgery (8,643 pairs) and prior osteotomy versus no prior surgery (5,569 pairs), matched on age, sex, BMI category, and Elixhauser score. Kaplan-Meier and Cox regression were used, censoring at death, with adjustment for femoral fixation (cemented, cementless, hybrid, reverse-hybrid).

RESULTS: After prior osteosynthesis, hazards were higher for revision (HR 6.93 (95% CI 6.07 to 7.92), periprosthetic femoral fracture (PFF; HR 17.22 (95% CI 12.28 to 24.15)), and infection (HR 3.87 (95% CI 3.05 to 4.91)) compared with controls (all $p < 0.001$); cemented and hybrid fixation had lower revision and PFF hazards than cementless fixation. After prior osteotomy, revision (HR 2.30 (95% CI 1.89 to 2.80)) and dislocation (HR 9.43 (95% CI 4.55 to 19.56)) hazards were higher; PFF and infection were not significant. Reverse-hybrid fixation was associated with higher revision (HR 2.01 (95% CI 1.27 to 3.19)) and infection (HR 3.71 (95% CI 1.71 to 8.06)) hazards than cementless fixation.

CONCLUSION: Prior osteosynthesis denotes a high-risk, conversion-t […]

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Coronal alignment parameters of the knee predict osteoarthritis development : a Coronal Plane Alignment of the Knee classification-based analysis using the Multicenter Osteoarthritis Study data.

Kim JS, MacDessi SJ, Lee DW, Ro DH

AIMS: Mechanical varus or valgus alignment measures have been demonstrated to be closely related to knee osteoarthritis (KOA) development, but few attempts have been made to correlate the Coronal Plane Alignment of the Knee (CPAK) classification and disease development. This study hypothesized that certain CPAK types or deviations in CPAK parameters might enhance the risk of KOA development.

METHODS: KOA was defined as a Kellgren-Lawrence grade of 2 or higher. Data from 2,541 knees without KOA at baseline from the Multicenter Osteoarthritis Study (MOST) over five years of follow-up were examined. Differences in baseline CPAK types and parameters were analyzed between knees that developed KOA and those that did not. The association of CPAK type, arithmetic hip-knee-ankle angle (aHKA), and joint line obliquity (JLO) with the risk of disease development and joint line convergence angle (JLCA) changes were assessed.

RESULTS: Knees that developed KOA (480 knees; 18.9%) had significantly lower JLO (p = 0.022) and aHKA (p = 0.008) values than those that did not (2,061 knees; 81.1%). JLO was significantly associated with KOA development (odds ratio 1.048 (95% CI 1.001 to 1.098)). For knees with JLO values under the cut-off value of 171.6°, the risk of KOA development was 1.36 times higher than for those with values over the cut-off value (95% CI 1.053 to 1.770). aHKA showed a signifi […]

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Robotic arm-assisted bi-unicompartmental knee arthroplasty compared with conventional total knee arthroplasty : five-year clinical outcomes of a randomized controlled trial.

Foxcroft WD, Doonan J, MacLean AD, Jones BG, Rowe P, Blyth MJG

AIMS: The aim of this study was to compare the outcomes between constitutionally aligned, robotic arm-assisted bi-unicompartmental knee arthroplasty (RA-bi-UKA) and conventional, mechanically aligned total knee arthroplasty (TKA) at five years' follow-up. The results at one-year follow-up have previously been reported.

METHODS: This was a prospective, randomized, single-centre study comparing the clinical outcomes of RA-bi-UKA with conventional TKA in patients with medial and lateral osteoarthritis (OA) of the knee. A total of 80 patients were initially randomized to receive either a cemented, fixed-bearing RA-bi-UKA or a cemented, posterior-stabilized, mechanically aligned TKA using conventional instrumentation over a three-year recruitment period. The data which were collected included patient-reported outcome measures (PROMs) and complications. The patients were reviewed at three months, one, two, and five years.

RESULTS: After randomization, 80 patients were enrolled in the study (42 in the RA-bi-UKA group and 38 in the TKA group). At five-year follow-up, there were no significant differences between the PROMs including the Oxford Knee Score, Forgotten Joint Score, EuroQol five-dimension three-level questionnaire (EQ-5D-3L), University of California, Los Angeles (UCLA), New Knee Society Score, and Pain or Stiffness visual analogue scale scores, in the two groups. There we […]

PubMed · DOI

Lower septic revision rates following cementless compared with cemented robotic-assisted total knee arthroplasty using a single implant design : a five-year survivorship analysis.

Burgio C, Salmons HI, Zepeda KE, Quevedo Gonzalez FJ, Mayman DJ, Jerabek SA, Debbi EM, Vigdorchik JM

AIMS: The choice of fixation for total knee arthroplasty (TKA) remains controversial. Cementless designs incorporating highly porous titanium surfaces may achieve durable biological fixation and mitigate cement-related complications. However, little has been written comparing the methods of fixation for robotic-assisted (RA) TKA. The aim of this study was to compare the implant survival and complication rates between cemented and cementless RA-TKA using a single robotic system and uniform implant design.

METHODS: This retrospective review involved 2,647 consecutive primary RA-TKAs undertaken by 27 surgeons at a high-volume academic centre, between 2016 and 2024, using a single CT-based robotic platform and implant design (Triathlon). Patients aged > 18 years with primary osteoarthritis and at least one year of follow-up were included. The method of fixation was determined by the operating surgeon. This resulted in 1,289 cemented and 1,358 cementless RA-TKAs. The mean follow-up was 3.5 years (SD 2.1). Outcomes included Kaplan-Meier survival free from revision (overall, septic, and aseptic), all-cause further surgery, and 90-day readmissions and nonoperative complications. Multivariable Cox regression analysis adjusted for age, BMI, sex, race, ethnicity, and operating time was performed to identify independent predictors.

RESULTS: The five-year revision-free survival was 98.0% […]

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Single-use surgical gowns are associated with an increased rate of periprosthetic joint infection : analysis of 9,239 primary hip and knee arthroplasties from a single centre.

Firth AM, Sian R, Nightingale J, van Duren BH, Higgins M, Manktelow ARJ, Bloch BV

AIMS: Periprosthetic joint infection (PJI) remains a serious complication of arthroplasty, associated with increased morbidity, mortality, and healthcare costs. While patient-related and procedural risk factors are well established, the effect of surgical gown type on infection risk is unclear. Single-use surgical gowns are often used due to their perceived superior sterility in the absence of clear evidence. The purpose of this study was to determine the effect of single-use compared with reusable gown use on the incidence of PJI in the 12 months following primary hip and knee arthroplasty.

METHODS: Between January 2015 and September 2023, 9,239 consecutive primary elective hip and knee arthroplasties were performed at our institution. Data were obtained from a prospectively maintained local registry aligned with the UK Health Security Agency Surgical Site Infection database and National Joint Registry, and combined with surgeon gown preference data. A retrospective logistic regression was undertaken to evaluate the association between gown type and PJI in the 12 months after surgery, adjusting for age, sex, BMI, diabetes, and chronic obstructive pulmonary disease (COPD).

RESULTS: A total of 76 infections (0.82%) were identified, comprising 49/5,024 knees (0.98%) and 27/4,215 hips (0.64%). Infection occurred more frequently in the single-use gown group (n = 4,314; 1.0%) than […]

PubMed · DOI

Comparison of the modified Broström-Gould procedure for chronic lateral ankle instability in adolescents with and without an os subfibulare : a propensity score-matched study.

Xiong S, Li Y, Liu Z, Zhao F, Jiang D, Hu Y, Jiao C, Xie X et al.

AIMS: An os subfibulare (OS) is frequently seen in adolescents with chronic lateral instability of the ankle, potentially impairing participation in sport and the quality of life. The aim of this study was to compare the clinical outcomes of the modified Broström-Gould procedure in adolescent patients with chronic lateral instability of the ankle with and without an OS.

METHODS: We retrospectively reviewed adolescent patients who underwent a modified Broström-Gould procedure for chronic instability of the ankle between August 2019 and July 2023. Propensity score matching (1:1) was undertaken based on age, sex, side, BMI, the duration of symptoms, pre-injury Tegner score, and joint hypermobility, yielding 82 matched cases (41 OS and 41 non-OS). Preoperative imaging parameters and intraoperative findings were recorded. Primary outcomes included patient-reported outcome measures (PROMs): a visual analogue scale (VAS) for pain, the Karlsson score, the Tegner score, and the Foot and Ankle Outcome Score (FAOS). Secondary outcomes included the incidence of re-sprain, the time to return to sport, satisfaction, and complications. Logistic regression analysis identified risk factors for re-sprain.

RESULTS: The mean follow-up was 47.8 months (SD 11.2) in the OS group and 47.9 months (SD 9.6) in the non-OS group (p = 0.949). Both groups had significant improvements in all PROMs, with no […]

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Effect of prior rotator cuff repair technique and timing on clinical outcomes of reverse shoulder arthroplasty.

Holt KE, Hao KA, Bindi VE, Kaiser AHA, Hones KM, Wright JO, Wright TW, Struk AM et al.

AIMS: Reverse shoulder arthroplasty (RSA) after prior rotator cuff repair (RCR) is commonly performed for recurrence of pain and arthritic progression; however, the influence of prior RCR timing and technique on outcomes of RSA has not been adequately explored. Our primary aim was to evaluate the relationship between the postoperative American Shoulder and Elbow Surgeons (ASES) score, whether prior RCR was performed open or arthroscopic, and the time interval between RSA and prior RCR. Secondary outcomes were forward elevation (FE) and external rotation (ER).

METHODS: A prospectively collected single-institution shoulder arthroplasty database was queried to identify patients undergoing primary RSA after prior RCR between 2004 and 2022 for non-tumour and non-fracture indications with minimum two-year follow-up. We evaluated the relationship between the postoperative ASES score, FE, and ER stratified by prior RCR approach (open vs arthroscopic) and time between RSA and RCR. Multivariable analysis was used to adjust for potential confounders.

RESULTS: We included 104 RSAs (99 patients) with a mean 8.8 years (SD 7.3) after RCR and a mean age of 68 years (SD 8.3). Prior RCR was performed arthroscopically in 40 RSAs (38%). Mean follow-up was 4.8 years (SD

2.8). On multivariable analysis, arthroscopic RCR was independently associated with a 13-point greater postoperative ASES score […]

PubMed · DOI

Time-course changes in bone mineral density after anatomical and reverse total shoulder arthroplasty : a dual-energy X-ray absorptiometry-based study.

Nakazawa K, Manaka T, Ito Y, Hirakawa Y, Nishiura R, Oi N, Terai H

AIMS: Bone resorption around cementless short humeral stems is increasingly recognized after shoulder arthroplasty, but quantitative longitudinal data are lacking. We measured bone mineral density (BMD) around the humeral component after anatomical total shoulder arthroplasty (aTSA) and reverse total shoulder arthroplasty (RSA) using dual-energy X-ray absorptiometry (DEXA) and evaluated the time-course changes over two postoperative years. Additionally, we investigated the association between osteoporosis and clinical outcomes.

METHODS: The subjects were patients who underwent aTSA or RSA, and for whom DEXA evaluation was available at three weeks, six months, one year, and two years postoperatively. The area around the humeral component was divided into five zones, and the BMD of each zone was measured. To evaluate clinical outcomes, we investigated their association with osteoporosis and measured the postoperative American Shoulder and Elbow Surgeons (ASES) score using the minimal clinically important difference (MCID) as the reference. Overall, 94 RSAs and 43 aTSAs were observed.

RESULTS: In RSA, BMD decreased significantly over time in zones 3 and 5 (both $p < 0.05$). In aTSA, BMD decreased in zones 2, 3, and 4 from three weeks to six months, with little sustained decrease thereafter. Additionally, in RSA, a decrease in zone 3 BMD was significantly associated with osteoporos […]

PubMed · DOI

Preoperative alignment and risk of proximal junctional failure : a framework for upper instrumented vertebra selection in adult spinal deformity.

Hills J, Lenke LG, Smith JS, Shaffrey CI, Lafage V, Lafage R, Bess S, Kelly MP et al.

AIMS: Proximal junctional kyphosis (PJK) remains a major complication after surgery for adult spinal deformity (ASD). While postoperative alignment is a recognized modifiable risk factor, objective methods for selecting the upper instrumented vertebra (UIV), a key modifiable factor, are lacking. We aimed to determine whether preoperative sagittal alignment, specifically cervicothoracic alignment, predicts the risk of PJK, and whether this risk can be mitigated by UIV selection, focusing on factors available at the time of surgical planning.

METHODS: From a multicentre, prospective ASD registry, we identified patients who had undergone fusion to the sacrum or pelvis and had an upper (T1-T5) or lower thoracic (T9-L1) UIV, with a two-year or more radiological follow-up, excluding those with a previous fusion over more than four levels. The primary outcome was PJK within two years. Multivariable logistic regression modelled the risk of PJK by UIV region, preoperative C2-T9 pelvic angle (PA), age, sex, and pelvic incidence, testing for interaction between UIV region and C2-T9 PA. Adjusted absolute risk reduction (ARR) and number needed to be exposed (NNEB) were calculated. Multivariable linear regression estimated two-year patient-reported outcome measures, adjusting for baseline scores, age, UIV, and PJK.

RESULTS: A total of 627 patients across 20 centres were included (median ag […]

PubMed · DOI

Hospital costs attributable to further fracture, reoperation, and death during the first year following hip fracture in England : a nationwide cohort study.

Baji P, Gregson CL, Patel R, Judge A, Chesser T, Griffin XL, Ben-Shlomo Y, Johansen A et al.

AIMS: Within one year after hip fracture in older adults living in England, this study aimed to estimate the inpatient hospital costs associated with subsequent complications, including further fractures, reoperations, and death.

METHODS: The study linked national datasets to identify all patients aged ≥ 60 years who underwent an index hip fracture operation in all hospitals in England between 1 April 2016 and 31 March 2019, with one-year follow-up. Patients were categorized into eight mutually exclusive groups based on combinations of further fracture (after 30 days from index hip fracture, at any skeletal site, excluding facial, digit, and skull fractures), reoperation, and death in the year after hip fracture. Inpatient costs were valued using 2019/20 NHS reference costs. Marginal cost differences over patients with no complications were estimated using generalized linear models.

RESULTS: Among 164,691 patients (70.8% female (n = 116,534); mean age 82.7 years (SD 8.7)), across 159 hospitals, the mean one-year inpatient cost was £14,581 (SD 8,583). Further fracture (£24,306 (SD 11,635); n = 11,226) and reoperation (£23,923 (SD 13,537); n = 7,522) were each associated with approximately twice the cost of patients without complications (£12,859 (SD 7,097); n = 107,638). Additional costs, adjusted for prefracture costs, fracture type, and patient case-mix variables, were highe […]

PubMed · DOI

Incidence of medical and surgical complications, and subsequent mortality, after hip fracture surgery : a nationwide cohort study from 2010 to 2021.

Haugen KG, Hansen CM, Jensen SS, Pedersen AB

AIMS: Hip fracture patients are at high risk of complications affecting their treatment, recovery, and survival. Previous studies report considerable variation in complication rates, and comprehensive data are limited. We examined the cumulative incidences of 20 medical and surgical complications after hip fracture surgery and their association with mortality.

METHODS: Using Danish nationwide registries, we identified 72,693 hip fracture patients aged ≥ 65 years who underwent surgery (2010 to 2021). Primary outcome was 30-day cumulative incidence of hospital-treated registered complications. The analysis was conducted overall and stratified by age, sex, and comorbidity. Secondary outcome was mortality within 31 to 60 days after surgery in patients with and without each specific complication.

RESULTS: The cumulative incidence of any medical complication was 23% (n = 14,404), while the incidence of any surgical complication was 4% (n = 2,708). The most frequent medical complications were pneumonia (6%; n = 4,364), urinary tract infection (5%; n = 3,943), gastrointestinal complication (3%; n = 1,893), delirium (2%; n = 1,454), and dehydration (2%; n = 1,301), and the most common surgical complications were major reoperations (2%; n = 1,655) and prosthesis dislocation (2%; n = 1,053). The cumulative incidence of any medical complication was higher in patients aged > 85 years comp […]

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Erratum.

Haddad FS

PubMed · DOI

CORR Clinical Orthopaedics & Related Research

Volume 484, Issue 7 · July 2026 · 40 new articles

Editorial: Something Simple, Easy, and Useful for Call-taking Orthopaedic Surgeons.

Leopold SS

PubMed · DOI

Editor's Spotlight/Take 5: Cigarette Smoking Was Not Associated With Lower Odds of Radiographic Fusion After Combined TLIF and Posterolateral Lumbar (270°) Arthrodesis: A CT-based Retrospective Cohort Evaluation.

Leopold SS

PubMed · DOI

Cigarette Smoking Was Not Associated With Lower Odds of Radiographic Fusion After Combined TLIF and Posterolateral Lumbar (270°) Arthrodesis: A CT-based Retrospective

Cohort Evaluation.

Ohana N, Benharroch D, Abu-Gameh A, Koch JE, Baruch Y, Schleifer D, Engel I

BACKGROUND: Smoking is widely regarded as a risk factor for impaired spinal fusion, particularly in posterolateral fusion, and prior studies have generally relied on plain radiographs for assessment. However, data evaluating the association of smoking on transforaminal lumbar interbody fusion (TLIF) with adjunctive posterolateral arthrodesis using modern CT-based evaluation remain limited.

QUESTIONS/PURPOSES: (1) Is cigarette smoking associated with a reduced likelihood of achieving radiographic spinal fusion, as assessed by thin-slice CT, after TLIF with adjunctive posterolateral arthrodesis? (2) Does the quality of TLIF (interbody) and posterolateral (intertransverse) fusion differ between patients who smoked and those who did not? (3) Does smoking intensity, measured in pack-years, correlate with reduced fusion quality or increased odds of cage subsidence? (4) Are patient-related factors such as gender, age, and BMI associated with differences in fusion quality or cage subsidence?

METHODS: Between 2013 and 2017, one surgeon at a single private hospital performed 360 TLIF procedures with adjunctive posterolateral fusion for degenerative lumbar disease. During this period, surgery was offered to patients who smoked after individual assessment of overall surgical risk, including medical stability and ability to adhere to follow-up recommendations. Of these 360 patients, 18% ([…]

PubMed · DOI

Residency Diary: Percival Ignatius Penguin.

Friedman LGM

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The Forward Movement: Amplifying Black Voices on Race and Orthopaedics-Medical Literacy is a Fallacy.

Owusu-Akyaw K

PubMed · DOI

Beyond the Bone: Value Over Volume-Redefining Success in Orthopaedic Care.

Kim TK

PubMed · DOI

On Patient Safety: Worsening Healthcare Costs Threaten Americans' Safety.

Rickert J

PubMed · DOI

Art in Science: Perspectives Through Fire and Sand.

Green SA

PubMed · DOI

Classifications in Brief: The Rotator Cuff Healing Index.

Frederickson M, Menendez ME

PubMed · DOI

How Is Lumbar Fusion Associated With Compensatory Hip Motion After THA?

Tokuyasu H, Tsushima E, Takemoto M, Vergari C, Kanoe H, Kim Y

BACKGROUND: Although patients with THA may increase their use of hip flexion to compensate for loss of spinal motion after lumbar fusion, no studies have directly examined this relationship. Furthermore, the association between the number of fused levels on postoperative compensatory hip motion is not clear, making it difficult to anticipate these changes prior to lumbar fusion.

QUESTIONS/PURPOSES: (1) Is lumbar fusion after THA associated with resting spinopelvic alignment or compensatory changes at the hip? (2) Is the number of fused spinal segments associated with an increase in hip compensatory motion? (3) Can this increase in hip compensatory motion be anticipated prior to lumbar spine fusion?

METHODS: Between January 2017 and December 2023, we retrospectively identified 109 patients with lumbar fusion after THA. The minimum follow-up time after lumbar fusion was set at 6 months to account for implant stabilization and early dislocation events. Of these patients, 59% (64) were excluded because of incomplete questionnaires (17% [19 of 109]), neuromuscular disease (3% [3 of 109]), unavailable or obscured functional lateral radiographs before and/or after lumbar fusion (36% [39 of 109]), or lumbar fusion performed within 6 months after THA (3% [3 of 109]), leaving 41% (45 of 109) of patients for analysis. Among the included patients, 82% (37 of 45) were female, with a mean […]

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CORR Insights®: How Is Lumbar Fusion Associated With Compensatory Hip Motion After THA?

Kurmis AP

PubMed · DOI

Acute or Delayed TKA for Tibial Plateau Fracture? An Observational Study From the Swedish Arthroplasty Register.

Olerud F, Garland A, Dahl AW, Hailer NP, Wolf O

BACKGROUND: Acute TKA has been proposed as an alternative to open reduction and internal fixation for complex tibial plateau fractures in patients who are older and who have compromised bone quality. The alternative is a delayed TKA after primary fracture management with an unfavorable outcome. However, the long-term outcomes and risk for reoperation after acute TKA compared with delayed TKA for fracture sequelae remain unclear.

QUESTIONS/PURPOSES: When comparing acute TKA (< 3 months after injury) for tibial plateau fracture with delayed TKA for fracture sequelae, we asked: (1) Do the risks of reoperation or revision for any cause differ? (2) Do the risks of reoperation or revision for infection differ? (3) Do the risks of reoperation or revision for loosening differ?

METHODS: Data for all TKAs performed between 2014 and 2023 with the indication of acute tibial plateau fracture (n = 152) or fracture sequelae (n = 950) were extracted from the Swedish Arthroplasty Register. Patients who underwent TKA for acute tibial plateau fractures were older (73 versus 66 years), more often women (78% [118 of 152] versus 57% [539 of 950]), had lower BMI (26.7 versus 27.9 kg/m^2), and received constrained or hinged implants more frequently (59% [89 of 152] versus 33% [311 of 950]). Reoperations were identified in the Swedish Arthroplasty Register through subsequent procedures on the index […]

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CORR Insights®: Acute or Delayed TKA for Tibial Plateau Fracture? An Observational Study From the Swedish Arthroplasty Register.

Cornell CN

PubMed · DOI

A Novel Classification System for Fifth Metatarsal Base Fractures Is Reliable and Clinically Validated.

Marcus RE, Rascoe AS, Pasternack J, Guyton GP

BACKGROUND: One goal of classifying fractures at the base of the fifth metatarsal is to reliably identify which fracture patterns have an increased risk of delayed union and nonunion and therefore merit consideration for operative fixation. The most common classification system of proximal fifth metatarsal fractures is the three-part Lawrence and Botte scheme, which has been shown to have poor interobserver and intraobserver reliability and can be ambiguous in terms of clinical treatment and prognosis. In particular, the assignment of the middle classification (Type II) has unclear implications for treatment and prognosis.

QUESTIONS/PURPOSES: We developed a two-part classification system based on prior anatomic and biomechanical studies that we performed, and we asked: (1) What is the interobserver and intraobserver reliability of the system? (2) What association does it have with the risk of delayed union or nonunion?

METHODS: This novel classification system was based on previous work on the effect of the peroneus brevis insertion on fracture stability. The fractures were divided into two categories, avulsion or indirect, based on their location with regard to the peroneus brevis footprint and with regard to the mechanism of injury. Fractures were considered avulsion if they occurred as the result of a direct avulsive force as ascertained by whether they were located within […]

PubMed · DOI

CORR Insights®: A Novel Classification System for Fifth Metatarsal Base Fractures Is Reliable and Clinically Validated.

Wilson NA

PubMed · DOI

Is Arm Dominance Associated With Clinically Meaningful Differences in Outcomes After Shoulder Arthroplasty?

Lazaridou A, Giannakis P, Schneller T, Hofer CK, Poeran J, Sideris A, Memtsoudis SG, Marx RG et al.

BACKGROUND: Limited available evidence seems to suggest that the increased use of the dominant (versus nondominant) limb may allow for earlier return to function and better ROM in the dominant limb at 12-month follow-up after anatomic or reverse total shoulder arthroplasty (TSA). Nevertheless, whether the earlier achievement of physical therapy milestones is associated with a clinically meaningful difference in patient-reported outcome measures (PROMs) is yet to be determined.

QUESTIONS/PURPOSES: (1) What are the 12-month minimum clinically important difference (MCID) and patient acceptable symptom state (PASS) thresholds for the Shoulder Pain and Disability Index (SPADI), QuickDASH, numeric rating scale (NRS) for pain, and Constant-Murley score? (2) Is there a difference between the dominant- and nondominant-side TSAs in terms of the proportions of patients achieving an MCID or PASS at 12-month follow-up?

METHODS: This retrospective, comparative study analyzed data from a longitudinally maintained shoulder arthroplasty registry at a specialized orthopaedic institution. Patients were eligible for inclusion if they underwent primary anatomic or reverse TSA from 2006 to 2024 for cuff tear arthropathy or primary osteoarthritis and had 12-month follow-up for at least one PROM. We collected relevant baseline patient-related and procedure-related characteristics. The main associati […]

PubMed · DOI

CORR Insights®: Is Arm Dominance Associated With Clinically Meaningful Differences in Outcomes After Shoulder Arthroplasty?

Accousti KJ

PubMed · DOI

Is Higher Surgeon Volume Associated With Lower Complication and Revision Risk After Distal Radius Fracture Surgery? A Population-based Cohort Study of 13,389 Patients.

Persitz J, Zagorski B, Baltzer H, Calzavara A, Wolfstadt J, Avisar E, Zuo KJ, Chan A et al.

BACKGROUND: Surgeon case volume has been linked with outcomes across many orthopaedic procedures, but its influence on distal radius fracture fixation remains uncertain.

QUESTIONS/PURPOSES: (1) For distal radius fracture surgery, at what surgeon annual case volume does the risk of complications plateau? (2) For distal radius fracture surgery, at what surgeon annual case volume does the risk of revision surgery plateau?

METHODS: A retrospective, population-based study was performed using administrative health databases in Ontario, Canada, accessed through the Institute for Clinical Evaluative Sciences, an independent, nonprofit research institute that houses linkable, individual-level health administrative data for Ontario's publicly funded healthcare system. Between 2010 and 2020, a total of 27,945 adult patients (≥ 18 years of age) underwent surgical fixation for acute isolated distal radius fracture. After applying prespecified inclusion and exclusion criteria, including exclusion of patients with open fractures, polytrauma, compartment syndrome, neurovascular injury, emergent presentations, incomplete administrative records, or prior distal radius surgery, a final cohort of 13,389 patients (48% of the initial cohort) was included (71% [9533] females; mean ± SD age 56 ± 15 years). Surgeon annual case volume, defined as the number of distal radius fracture fixations performe […]

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CORR Insights®: Is Higher Surgeon Volume Associated With Lower Complication and Revision Risk After Distal Radius Fracture Surgery? A Population-based Cohort Study of 13,389 Patients.

O'Shaughnessy MA

PubMed · DOI

Can We Develop a Web-based Nomogram That Could Predict the Risk of Local Recurrence After Treatment for Giant Cell Tumor of Bone of the Extremities by Curettage?

Masunaga T, Tsukamoto S, Kurakami H, Mavrogenis AF, Honoki K, Fujii H, Donati DM, Errani C et al.

BACKGROUND: For giant cell tumor of bone (GCTB) in the extremities, joint-preserving surgery (curettage) is often favored over en bloc resection to maintain postoperative function, although it carries a notable risk of local recurrence. We decided to develop a nomogram to predict the risk of local recurrence after curettage of GCTB of the extremities so physicians can shorten the interval between radiographs, add CT or MRI, and even monitor tartrate-resistant acid phosphatase 5b levels during surveillance for patients at particularly high risk of local recurrence.

QUESTIONS/PURPOSES: Can we develop a web-based nomogram that could predict the risk of local recurrence after treatment for GCTB of the extremities by curettage?

METHODS: A total of 535 patients were histologically diagnosed with GCTB of the extremities and underwent surgery at the authors' institutions between 1980 and 2022. Of these, 38% (204) underwent en bloc resection, five underwent amputation, and one patient with an unknown Campanacci stage was excluded. A retrospective analysis was performed on the remaining 325 patients who underwent curettage. The median age was 30 years, and 67% (216) of the patients had a tumor of the distal femur or proximal tibia. The incidence of Campanacci Stage III tumors was 21%, and pathologic fractures were observed in 6% (19) of the patients at presentation. Nomograms were deve […]

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CORR Insights®: Can We Develop a Web-based Nomogram That Could Predict the Risk of Local Recurrence After Treatment for Giant Cell Tumor of Bone of the Extremities by Curettage?

Smolle MA

PubMed · DOI

What Are the Patient-reported Outcomes After Elective Transtibial Amputation? A Comparison of CRPS, Neuropathic Pain, and Other Conditions That Lead to Late Amputation After Limb Salvage.

Lansford JL, Cantor AG, Oplinger SL, Hoyt BW, Bozzay AB, Potter BK

BACKGROUND: In the United States, approximately 2.3 million people live with limb loss, of whom about 91% have undergone lower extremity amputation. Despite these numbers, relatively few studies have evaluated outcomes after elective transtibial amputation. In particular, we do not know whether the outcomes after amputation for intractable neurogenic pain such as complex regional pain syndrome (CRPS) are comparable to outcomes after late amputations (6 weeks or more after limb salvage) performed for other reasons such as recurrent infection, soft tissue problems, recalcitrant nonunion, poor function, and chronic nonneurogenic pain.

QUESTIONS/PURPOSES: (1) Did patients who underwent late amputation for CRPS or neuropathic pain have inferior Patient-Reported Outcomes Measurement Information System (PROMIS) scores at follow-up compared with patients who had nonneurogenic elective amputation? (2) Did patients who underwent late amputation for CRPS or neuropathic pain have more pain, use more pain medication, or wear prosthetics less often than patients who have undergone nonneurogenic elective amputation? (3) Were patients who underwent late amputation for CRPS or neuropathic pain more likely to be revised to a higher level of amputation or express decision regret than patients undergoing nonneurogenic elective amputation?

METHODS: This retrospective comparative study examined 70 […]

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CORR Insights®: What Are the Patient-reported Outcomes After Elective Transtibial Amputation? A Comparison of CRPS, Neuropathic Pain, and Other Conditions That Lead to Late Amputation After Limb Salvage.

Villamin CAC

PubMed · DOI

What Is the Effect of Robot Reduction in Displaced Pelvic Fractures? A Multicenter Randomized Clinical Trial.

Zhao C, Ge Y, Cao Q, Yang M, Gao F, Xiao H, Liu G, Sun D et al.

BACKGROUND: Displaced pelvic fractures present real surgical challenges because of complex three-dimensional deformity patterns and proximity to vital structures, with conventional manual reduction techniques limited by accuracy constraints and radiation exposure. Although robotic assistance shows promise in preclinical studies, its clinical effectiveness remains unproven in randomized clinical trials (RCTs).

QUESTIONS/PURPOSES: (1) Does robotic closed reduction improve reduction quality compared with manual closed reduction in displaced pelvic fractures? (2) Can robotic closed reduction reduce intraoperative radiation exposure while maintaining functional outcomes?

METHODS: In this multicenter RCT conducted at six tertiary trauma centers in China involving 10 senior orthopaedic traumatologists, 92 adult patients with acute closed, displaced pelvic fractures (Tile Type B or C) were randomized 1:1 to robotic closed reduction (n = 46) or manual closed reduction (n = 46) groups. At 12 weeks, loss to follow-up for patient-reported outcomes was 9% (4 of 46) in the robotic group and 4% (2 of 46) in the manual group; the remainder were handled in a prespecified per-protocol analysis. In the robot group, reduction was planned using CT-based three-dimensional reconstruction with contralateral pelvic symmetry as the target and executed by a robotic arm with adjunct elastic traction and […]

PubMed · DOI

CORR Insights®: What Is the Effect of Robot Reduction in Displaced Pelvic Fractures? A Multicenter Randomized Clinical Trial.

Mittwede PN

PubMed · DOI

Single Epidural Analgesia With Opioid-free IV-PCA Reduces Opioid Consumption in Lumbar Spine Surgery: A Randomized, Multicenter Trial.

Ham DW, Kang H, Won YS, Lee J, Park SM, Kim YB, Chang DG, Song KS

BACKGROUND: Postoperative pain management after lumbar spine surgery often involves modest to high doses of opioids, which can contribute to the risk of dependence. Epidural analgesia has emerged as a promising opioid-sparing alternative, but its efficacy compared with conventional opioid-based intravenous patient-controlled analgesia (IV-PCA) in a randomized trial has not been established in these patients.

QUESTIONS/PURPOSES: We asked whether an opioid-sparing protocol, compared with conventional opioid-based IV-PCA, (1) provides superior pain control, (2) reduces opioid consumption, and (3) lowers the frequency of opioid-related adverse events.

METHODS: In this multicenter, randomized, double-blind, parallel-group trial, we enrolled 98 patients undergoing single-level lumbar fusion or decompression. Eligible participants were adults age 20 to 80 years diagnosed with lumbar spinal stenosis or spondylolisthesis. Exclusion criteria included history of prior lumbar surgery, coagulation disorders, or opioid dependence. Patients were randomized to either the epidural opioid-free IV-PCA group (intraoperative single-shot epidural ropivacaine injection with opioid-free IV-PCA; epidural group) or the conventional opioid-based IV-PCA group (fentanyl-based IV-PCA; control group). In the epidural opioid-free group, fentanyl was administered solely as a rescue analgesic for breakthrough […]

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CORR Insights®: Single Epidural Analgesia With Opioid-free IV-PCA Reduces Opioid Consumption in Lumbar Spine Surgery: A Randomized, Multicenter Trial.

Reitman CA

PubMed · DOI

Expert Performance on a Hip Wire Navigation Simulator Provides a Basis for Establishing Benchmarks to Define Skill Proficiency.

Long S, Thomas GW, Anderson DD, Downes JM, Wagstrom EA, Karam M

BACKGROUND: Simulator-based training addresses a pressing need to move surgical skill acquisition from the operating room to the laboratory, reducing patient risk associated with traditional trial-and-error training methods. The Core Requisites of Orthopedic Wire Navigation Skills (CROWNS) curriculum employs a hybrid reality simulator to teach surgeons the foundational skill of placing a guidewire in bone via fluoroscopic imaging. Simulator-based training has already been shown to improve subsequent operating room performance, but objective benchmarks to assess skill proficiency are lacking.

QUESTIONS/PURPOSES: The objective of this study was to measure and compare expert and novice performance on a simulated hip wire navigation task as a basis to define candidate proficiency training benchmarks.

METHODS: This study was designed as an experimental assessment of a surgical simulation platform to differentiate between skill levels during a proctored examination. In all, 113 surgeons participated, including 68 orthopaedic residents (novices) and 45 experts (28 Orthopaedic Trauma Association fellows and 17 practicing orthopaedic surgeons). To achieve the primary objective of measuring and comparing expert and novice performance, all participants performed a simulated hip wire navigation task replicating guide wire placement for intertrochanteric fracture fixation. Experts complet […]

PubMed · DOI

CORR Insights®: Expert Performance on a Hip Wire Navigation Simulator Provides a Basis for Establishing Benchmarks to Define Skill Proficiency.

DeFrancesco CJ

PubMed · DOI

Letter to the Editor: Editorial: AAOS Orthobiologics Registry-Sometimes, More is Less.

Jevsevar DS, Kelly JD, McCarty EC, Jacobs JJ, Huddleston JI, Glassman SD, Matzkin EG, Parks ML et al.

PubMed · DOI

Reply to the Letter to the Editor: Editorial: AAOS Orthobiologics Registry-Sometimes, More is Less.

Gebhardt MC, Gioe TJ, Manner PA, Porcher R, Rimnac CM, Wongworawat MD, Leopold SS

PubMed · DOI

Letter to the Editor: How Can the Use of Patient-Reported Outcomes Data by Orthopaedic Surgeons Be Improved? A National Qualitative Study.

Ayati Firoozabadi A

PubMed · DOI

Reply to the Letter to the Editor: How Can the Use of Patient-Reported Outcomes Data by Orthopaedic Surgeons Be Improved? A National Qualitative Study.

Heath EL, Harris IA, Ackerman IN

PubMed · DOI

Letter to the Editor: What Factors and Patient-reported Outcome Measures Are Associated With Stress Fracture After Periacetabular Osteotomy?

Cui M, Gao F, Dong X

PubMed · DOI

Reply to the Letter to the Editor: What Factors and Patient-reported Outcome Measures Are Associated With Stress Fracture After Periacetabular Osteotomy?

Jochl OM, Trotzky ZA, Beltrame G, Muffly BT, Sink EL

PubMed · DOI

Letter to the Editor: How Does Anterior Vertebral Body Tethering Compare To Posterior Spinal Fusion for Thoracic Idiopathic Scoliosis? A Nonrandomized Clinical Trial.

Abudayeh A, Fishchenko I

PubMed · DOI

Letter to the Editor: Your Best Life: Aging with Grace-Keys to Success in the Autumn of Life.

Flanagan JP, Monsivais JJ

PubMed · DOI

Reply to the Letter to the Editor: Your Best Life: Aging with Grace-Keys to Success in the Autumn of Life.

Kelly JD

PubMed · DOI

Letter to the Editor: High Risk of Venous Thromboembolism With Aspirin Prophylaxis After THA for High-riding Developmental Dysplasia of the Hip: A Retrospective, Comparative Study.

Qu S, Ding J, Song J, Jiang H

PubMed · DOI

Reply to the Letter to the Editor: High Risk of Venous Thromboembolism With Aspirin Prophylaxis After THA for High-riding Developmental Dysplasia of the Hip: A Retrospective, Comparative Study.

Gharanizadeh K, Abolghasemian M

PubMed · DOI

JBJS

JBJS (American)

Volume 108, Issue 13 · July 2026 · 16 new articles

Give the Cefazolin, Even in Patients with a Cephalosporin Allergy: Commentary on an article by Josef E. Jolissaint, MD, et al.: " Cefazolin and the R1 Side Chain. Why Patients with a Cephalosporin Allergy Can Be Safely Given Cefazolin While Undergoing Joint Arthroplasty ".

Masri BA

PubMed · DOI

Cefazolin and the R1 Side Chain: Why Patients with a Cephalosporin Allergy Can Be Safely Given Cefazolin While Undergoing Joint Arthroplasty.

Jolissaint JE, Mallett KE, Thomson AL, Carli AV, Austin MS

BACKGROUND: Cefazolin, a first-generation cephalosporin, is the standard antibiotic for perioperative prophylaxis in patients undergoing hip or knee arthroplasty. Research has demonstrated significantly higher periprosthetic joint infection (PJI) rates when non-cefazolin antibiotics are used for prophylaxis. Notably, cefazolin contains an R1 side chain that has not shown cross-reactivity with other cephalosporins. However, in patients with a reported cephalosporin allergy, there is often uncertainty about the optimal antibiotic choice. This study aimed to determine the safety of perioperative cefazolin in patients with a documented cephalosporin allergy undergoing joint arthroplasty.

METHODS: We reviewed the records of 1,268 patients who had a documented cephalosporin allergy and underwent total hip or knee arthroplasty at a high-volume academic center from 2016 to 2024. We compared patients who received cefazolin despite a cephalosporin allergy (n = 482) and patients who received an alternative antibiotic prophylaxis (n = 786). The primary outcome was the incidence of immunoglobulin E (IgE)-mediated allergic reactions or "severe" Type-IV delayed hypersensitivity reactions with end organ dysfunction within 72 hours postoperatively. The secondary outcomes included 90-day rates of complications including PJI, Clostridioides difficile infections, adverse events related to the ant […]

PubMed · DOI

The Value of Specific Special Questions on Case Lists Submitted for ABOS Certification: Commentary on an article by Gregory P. Guyton, MD, et al.: " Insights into Evolving Trends and Controversies in Orthopaedic Surgery from the ABOS Case Collection System. Data from Current Procedural Terminology-Specific Special Questions, 2022 to 2024 ".

Meller MM

PubMed · DOI

Treatment Alternatives for Displaced Closed Humeral Shaft Fractures: Practical Implications for Shared Decision-Making from a Randomized Pragmatic Trial: Commentary on an article by Cyrill Suter, MD, et al.: " Cost-Effectiveness of Surgery Versus Functional Bracing for Humeral Shaft Fractures in Adults. A Prespecified Economic Evaluation of the Finnish Shaft of the Humerus (FISH) Trial ".

Band PA, Zuckerman JD

PubMed · DOI

Putting the Problem of Dislocation Following Total Hip Arthroplasty into Perspective by Recalling the Past: Commentary on an article by Amanda D. Klaassen, MSc, et al.: " Minimal Movement Restrictions Do Not Increase Hip Dislocations Following Total Hip Arthroplasty. A Before-and-After Study of 10,357 Patients ".

Robinson RP

PubMed · DOI

Antimicrobial Resistance in Periprosthetic Joint Infection: Use Correct Definitions for Better Clinical Decisions: Commentary on an article by Ruikang Guo, MBChB, et al.: " Microbial Resistance Patterns in Periprosthetic Joint Infection of the Knee. A 24-Year Longitudinal Study ".

Baldy Dos Reis F, Salles MJ

PubMed · DOI

Moving from Bedside to Bench, Increased Critical Shoulder Angle Impairs Tendon-to-Bone Healing in Rotator Cuff Repair in Rats: Commentary on an article by Yi Long, MD, et al.: " Increased Critical Shoulder Angle Impairs Tendon-Bone Healing in a Rat Model of Chronic Rotator Cuff Tears ".

Chin PC

PubMed · DOI

What's New in Orthopaedic Trauma.

Padubidri A, Patterson BM

PubMed · DOI

"First, Do No Harm": Revisiting the Hippocratic Tradition.

Malizos KN

PubMed · DOI

From Hip and Knee to Shoulder: Is the Obesity Paradox Becoming a Surgical Reality?

Kunutsor SK

PubMed · DOI

Interbody Cages: Surface Technologies in Spinal Implants.

Nassar JE, Ammar LA, Toavs TL, Kim J, Knebel A, Daher M, Shaffrey CI, Daniels AH

➢ Implant surface characteristics play a critical role in promoting osseointegration and long-term spinal fusion success. ➢ Titanium and polyetheretherketone (PEEK) are the most commonly utilized materials in interbody cages, each with distinct advantages and limitations. ➢ Surface modifications such as roughening, porosity, and hydroxyapatite coatings enhance osseointegration and early fusion outcomes. ➢ Emerging materials, including silicon nitride and porous tantalum, demonstrate favorable biological and mechanical properties but require further clinical validation. ➢ No single implant material or surface technology has shown consistent clinical superiority, highlighting the need for ongoing research and evidence-based selection.

PubMed · DOI

Insights into Evolving Trends and Controversies in Orthopaedic Surgery from the ABOS Case Collection System: Data from Current Procedural Terminology-Specific Special Questions, 2022 to 2024.

Guyton GP, Harrast JJ, Martin DF, Armstrong AD, Azar FM, Garvin KL, Jeray KJ, McComis GP et al.

➢ The American Board of Orthopaedic Surgery (ABOS) collects sequential Current Procedural Terminology (CPT)-coded surgical case lists both from Candidates for initial certification and from Diplomates participating in Maintenance of Certification (MOC). Collectively, these case lists represent a unique and extensive sampling of orthopaedic surgical practices across the United States. ➢ Because revisions or additions to the CPT system often take many years to reflect changes in surgical practice, in 2022 the ABOS began adding special questions for recorded cases in which the CPT codes alone do not identify potentially new or controversial surgical choices. ➢

The responses are used primarily for the ABOS internal case selection process for oral examination, but the ABOS recognizes that these previously unreported data are also of broad interest to the orthopaedic community. ➢ The data from the ABOS CPT-specific special questions from 2022 to 2024 were collected and analyzed. Responses from both early-career Candidates for initial certification and mid-career MOC Diplomates are compared in this report. They provide insight into both established orthopaedic practices across the United States and evolving trends in surgical choices and education across multiple subspecialties.

PubMed · DOI

Cost-Effectiveness of Surgery Versus Functional Bracing for Humeral Shaft Fractures in Adults: A Prespecified Economic Evaluation of the Finnish Shaft of the Humerus (FISH) Trial.

Suter C, Ibounig T, Reito A, Mattila H, Sumrein BO, Launonen AP, Paavola M, Järvinen TLN et al.

BACKGROUND: Humeral shaft fractures commonly affect working-age adults and can lead to prolonged work absence and substantial economic burden. Although surgical fixation and functional bracing offer comparable functional outcomes, their relative cost-effectiveness remains unclear.

METHODS: We conducted a prespecified economic evaluation alongside a multicenter, superiority, randomized clinical trial at 2 Finnish university hospitals between 2012 and 2018. Eighty-two adults (mean age, 48.9 years; 38 women) with displaced, closed humeral shaft fractures were randomly assigned to surgical fixation (n = 38) or functional bracing (n = 44) and followed for 2 years. The primary outcome was the incremental net monetary benefit (INMB) based on quality-adjusted life years (QALYs) measured with the 15-dimensional (15D) instrument, analyzed from both societal and health-care perspectives.

RESULTS: From a societal perspective, surgical treatment was both more effective and less costly than bracing. The mean total cost per patient was €23,680 for surgery and €30,389 for bracing, yielding an INMB of €9,423 (95% confidence interval [CI], €4,139 to €14,609). Cost-effectiveness acceptability curves showed that surgery was highly likely to be cost-effective across all willingness-to-pay thresholds up to €120,000 per QALY. The cumulative QALYs from 6 weeks to 2 years post-injury were 1.776 (95% […]

PubMed · DOI

Minimal Movement Restrictions Do Not Increase Hip Dislocations Following Total Hip Arthroplasty: A Before-and-After Study of 10,357 Patients.

Klaassen AD, Willigenburg NW, Musters JWQ, Jager J, Poolman RW, the Santeon Hip Osteoarthritis Group

BACKGROUND: Hip dislocation is the most common reason for early revision after total hip arthroplasty (THA). To prevent dislocation, patients are often prescribed movement restrictions following THA; however, these restrictions can be uncomfortable and limiting for patients and may delay a return to daily activities. The present study aimed to investigate whether a protocol with reduced movement restrictions following THA would lead to an increased incidence of early hip dislocations.

METHODS: In this uncontrolled, multicenter, before-and-after study, 2 groups were retrospectively compared. Patients who underwent a THA between January 1, 2015, and March 31, 2020, were included. Patients in the first group (n = 7,666) were prescribed strict movement restrictions following THA, and patients in the second group (n = 2,691) were prescribed minimal movement restrictions. Patient and prosthesis characteristics were collected, as well as data regarding hip dislocations within 90 days postoperatively as the primary outcome.

RESULTS: The incidence of early hip dislocations was not significantly different between movement groups, with 112 (1.46%) of 7,661 hips in the restricted group and 52 (1.93%) of 2,691 hips in the minimally restricted group experiencing a dislocation (p = 0.093).

CONCLUSIONS: The use of a protocol with minimal movement restrictions following THA did not increase […]

PubMed · DOI

Microbial Resistance Patterns in Periprosthetic Joint Infection of the Knee: A 24-Year Longitudinal Study.

Guo R, Tay ML, Kim K, Zhu M, Young SW

BACKGROUND: Understanding the causative microorganisms and initiating appropriate empirical antibiotics early are important in the management of knee periprosthetic joint infections (PJIs). The aim of this study was to identify trends in PJI microorganisms and antibiotic resistance profiles over 24 years to guide empirical antibiotic selection.

METHODS: This study included 487 first-episode PJIs identified between 2000 and 2023 following primary total knee arthroplasty (TKA) at 3 large tertiary hospitals. PJIs were classified using the Tsukayama classification, which is based on the timing from the primary TKA and the source of infection. Multivariable logistic regression was used to analyze risk factors for polymicrobial and resistant infections.

RESULTS: A total of 487 PJI cases with 608 culture specimens were identified. The mean patient age (and standard deviation) was 70 ± 11 years, with 65% male patients and 35% female patients. All ethnicity data were self-reported. Of the patients in this study, 57% were New Zealand European, 14% were other European, 14% were Pacific Islander, 10% were New Zealand M■ori, and 6% were Asian. The most common pathogen for PJIs was Staphylococcus aureus . The proportion of resistant cases (19% to 24%) was consistent across the 24-year period. A prosthesis in situ for <1 year was found to be the most important risk factor for polymicrobial […]

PubMed · DOI

Increased Critical Shoulder Angle Impairs Tendon-Bone Healing in a Rat Model of Chronic Rotator Cuff Tears.

Long Y, Su Y, Li X, Chen Y, Wang Z, Deng X, Yao Z, Yang R

BACKGROUND: The role of an elevated critical shoulder angle (CSA) in rotator cuff healing following rotator cuff repair (RCR) remains a subject of clinical controversy. The present study aimed to investigate the effect of increased CSA on tendon-bone interface healing following RCR.

METHODS: A bilateral chronic rotator cuff tear model was established in 48 Sprague-Dawley rats. Acromion lateralization (Acr) surgery was performed unilaterally to increase CSA. After 4 weeks, bilateral RCR was performed. Micro-computed tomography was utilized to measure CSA. Tendon-bone interface healing was assessed at 3, 6, and 9 weeks post-RCR with use of magnetic resonance imaging (MRI), biomechanical testing, gait analysis, and histological evaluation.

RESULTS: The mean CSA in the Acr group was significantly greater than that in the RCR-only group (37.2° ± 2.6° versus 29.7° ± 3.1°; $p < 0.001$). At 6 and 9 weeks postoperatively, the Acr group demonstrated significantly poorer outcomes on MRI (i.e., higher signal-to-noise quotient), biomechanical strength (i.e., lower ultimate failure load and stiffness), and gait parameters compared with the RCR-only group ($p < 0.05$). Histological analysis revealed inferior tendon-bone interface integration in the Acr group ($p < 0.01$), including reduced fibrocartilage formation, disorganized collagen fibers, and a lower collagen I/III ratio. Immunohistochemist […]

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JA The Journal of Arthroplasty

Volume 41, Issue 7S1 · July 2026 · 22 new articles

Proceedings of The Knee Society 2025 Members Meeting.

Cushner FD

PubMed · DOI

Evaluating the Optimal Timing Between Staged Bilateral Total Knee Arthroplasties for Improved Clinical Outcomes.

Khury F, Padon B, Trudeau MT, Meftah M, Macaulay W, Schwarzkopf R

BACKGROUND: Simultaneous bilateral total knee arthroplasty (BTKA) is avoided due to higher perioperative risk, favoring staged procedures. This study evaluated how the interval between staged TKAs affects patient-reported outcome measures (PROMs) and compared complications between the first and second procedures.

METHODS: We retrospectively reviewed patients undergoing primary, elective, staged, BTKAs at a high-volume academic center between 2011 and 2024. Intra- and perioperative data and complications were compared between the surgeries. Patients were stratified by interval: less than three months ("extremely short waiters"), three to six months, six to nine months, nine to 12 months, one to two years, two to five years, and greater than five years. The PROMs were compared across these groups. A total of 4,210 patients underwent 8,420 staged, BTKAs at a mean 21.4-month interval. The most common intervals were six to nine months (27.4%) and nine to 12 months (21.1%).

RESULTS: Patients were more likely to return to the emergency department (22 versus 16, P = 0.030) and be readmitted (48 versus 26, P = 0.113) after the second surgery, primarily due to infection. Revision rates did not differ. "Extremely short waiters" had the greatest improvement in Patient-Reported Outcomes Measurements Information System Pain Intensity and interference scores nine months after the second TKA […]

PubMed · DOI

Impact of Implant Size Variation on Surgical and Clinical Outcomes in Staged, Bilateral Total Knee Arthroplasty.

Khury F, Maheu AR, Sarfraz A, Novikov D, Schwarzkopf R, Lajam CM

BACKGROUND: This study evaluated differences in surgical and clinical outcomes among patients who have identical versus different implant sizes in sequential total knee arthroplasty (TKA) surgeries.

METHODS: We retrospectively reviewed patients who underwent primary, elective, staged, bilateral, same-surgeon, same-prosthesis TKA between 2011 and 2024 at a large academic health system. Patients were grouped by femoral and tibial implant size consistency: same femoral and tibial (SS), different femoral, same tibial, different tibial, same femoral, and different femoral and tibial (DD).

RESULTS: A total of 4,536 TKAs were performed in 2,268 patients. The SS had the shortest length of stay compared to DD (51.8 versus 58.2 hours, P < 0.001). The majority had the same femoral (75.6%) and tibial (76.6%) sizes in both knees, whereas polyethylene thickness varied. Undergoing contralateral surgery within one year was associated with receiving the same implant sizes (P < 0.001). The DD were more common in manual surgery, and the SS were more common using navigation assistance (P < 0.001). Different assistance modalities between surgeries increased different femoral, same tibial and DD, whereas the same navigation assistance increased SS (P < 0.001). Complication and revision rates were not significantly different between the groups. All groups showed improvement in their Knee injury and […]

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Association Between Adoption of Robotic Total Knee Arthroplasty in Canada and Major Surgical Complications.

Pincus D, Ekhtiari S, Lex JR, Schemitsch E, Ruangsomboon P, Paterson JM, Ravi B

BACKGROUND: Robotic-assisted total knee arthroplasty (rTKA) is being adopted rapidly, but data on its real-world safety compared with conventional TKA (cTKA) outside specialized centers are limited. The purpose of this study was to evaluate the association between rTKA and major surgical complications in a large, population-based cohort.

METHODS: We conducted a propensity score-matched, population-based cohort study of all adults undergoing primary rTKA versus cTKA for osteoarthritis in Ontario, Canada, from April 1, 2019, to October 31, 2023, with follow-up through October 31, 2024. Use of robotic assistance versus conventional instrumentation during primary TKA was identified for each case. During the study period, 74,359 TKAs were performed across 62 hospitals by 345 surgeons: 1,613 rTKAs and 72,746 cTKAs. The primary outcome was major surgical complications within one year, defined as a composite of revision arthroplasty, deep infection requiring surgery, or fracture requiring surgery. Outcomes were compared between matched groups using Cox proportional hazards regressions to estimate hazard ratios and absolute risk differences.

RESULTS: Robotic adoption increased from 1.7% of cases in 2019 to 5.8% in 2023 (P < 0.001). Compared to matched cTKA patients, those undergoing rTKA had a significantly higher rate of major complications within one year (2.0 versus 1.0%; absolute […]

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Dynamic Alignment of the Knee (DyAK) Classification: Advancing Knee Evaluation in Robotic Arthroplasty.

Andriollo L, Koutserimpas C, Gregori P, Vermue H, Hirschmann MT, Lustig S

BACKGROUND: Total knee arthroplasty (TKA) has moved toward a more personalized approach, but achieving the ideal coronal alignment is still challenging, as most knees naturally change alignment throughout the range of motion. This study introduced the dynamic alignment of the knee (DyAK) classification, based on robotic hip-knee-ankle measurements in extension (rHKA-E) and flexion (rHKA-90F), and evaluated the clinical impact of changes in DyAK type.

METHODS: This study analyzed 350 patients who underwent robotic TKA using functional alignment principles. The DyAK classification matrix included nine knee phenotypes, with neutral defined as -3 (varus) to 3° (valgus) in rHKA-E and rHKA-90F, respectively. Initial (precut) and final (postimplantation) DyAK types were compared to assess variations in dynamic coronal alignment and their clinical impact, Knee Society Score knee and function, Forgotten Joint Score-12, and Kujala Anterior Knee Pain Scale.

RESULTS: Regarding the initial DyAK, type D1 was the most prevalent (58.3%), followed by type D2 (15.1%) and type D4 (9.2%). After implant placement, in the final DyAK, the distribution shifted, with type D5 becoming the most common (46.0%), while type D1 decreased to 26.9% and type D4 increased to 13.1%. Overall, 56.9% of patients experienced a change in DyAK type following TKA. Among them, 32.9% showed a modification in either rHKA […]

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Underperformance of Machine Learning Models Predicting Readmission and Prolonged Length of Stay Following Total Knee Arthroplasty Among Underrepresented Populations: Fairness Analysis and Mitigation Strategies.

Shimizu MR, Raza MM, Xiao P, Li Z, Freeman IA, Kwon YM

BACKGROUND: While machine learning (ML) models demonstrate high predictive accuracy, recent studies reveal that ML models underperform for smaller subcohorts such as racial and ethnic minorities, suggesting inherent ML biases that may exacerbate health disparities. To assume a "one-size-fits-all" approach perpetuates inequities in decision-making for underrepresented groups. This study assessed validated ML model "fairness" for total knee arthroplasty (TKA) outcomes and explored bias mitigation strategies to enhance equitable prediction performance.

METHODS: There were four ML models that were developed and validated to predict readmission and prolonged lengths of stay (pLOSs) following TKA. Bias assessment was performed using protected attributes (age, sex, race, and ethnicity), and various fairness metrics were evaluated. There were three bias mitigation strategies incorporated and trialed for each algorithm, and the most effective strategy for a given protected attribute was integrated with the original model and reassessed based on the fairness metrics.

RESULTS: The random forest model had the best predictive performance (ReadmissionAUC = 0.98; pLOSAUC = 0.92). Inferior predictive equality for readmission was demonstrated in women (0.47 versus men), non-White (0.75 versus White), and Hispanic or LatinoX (0.53 versus not Hispanic/LatinoX) subcohorts. Significant difference […]

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Primary Total Knee Arthroplasty Using a Functional Alignment Concept With Kinematic Versus Inverse Kinematic Technique: Does it Make a Difference?

Henningsen JD, Ware WT, Whitaker JE, Clark GW, Lee SJ, Collopy DM, Bhimani RB, Smith LS et al.

BACKGROUND: There is an increasing trend toward a functional alignment in total knee arthroplasty (TKA). Functional alignment can be performed using a kinematic technique, FA(k), or an inverse kinematic approach, FA(ik). The purpose of this study was to compare the results of kinematic versus inverse kinematic techniques during primary TKA using a functional alignment concept.

METHODS: A retrospective review was conducted of patients undergoing robotic-assisted TKA with 1-year follow-up, excluding patients who had a body mass index greater than 45 or valgus deformity by coronal plane alignment of the knee classification. The cohorts included 166 FA(k) and 146 FA(ik) cases with similar demographics, coronal plane alignment of the knee, and comparable preoperative tibial and femoral angles. Intraoperative data included coronal alignment of the femoral and tibial components and final hip-knee-ankle

(HKA). Postoperative outcomes included Forgotten Joint Score (FJS), visual analog scale pain, and Knee Injury and Osteoarthritis Outcome Score for Joint Replacement.

RESULTS: The FA(ik) group had less varus on the final tibial component (3.0 versus 4.5°, $P < 0.00005$) and less varus at final HKA compared to the FA(k) technique (0.92 versus 2.6°, $P < 0.00005$). There was no difference in postoperative Knee Injury and Osteoarthritis Outcome Score for Joint Replacement or, visual analog sc […]

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The Impact of Femoral Component Rotation on Patellar Position and Functional Outcomes in Total Knee Arthroplasty.

Leipman JH, Su L, Bash HD, Hohmann AL, Sellig MT, Dipane MV, Boghozian O, Stavrakis AI et al.

BACKGROUND: Patellar malalignment may occur after total knee arthroplasty (TKA) and can lead to pain and dysfunction. The purpose of this study was to examine the relationship between femoral component rotation and patellar position in TKA.

METHODS: This was a retrospective, multicenter study of patients who underwent conventional or robotic-assisted TKA. In conventional cases, femoral rotation was set to 3° of external rotation relative to the posterior condylar axis; in robotic cases, femoral component rotation was planned to create balanced flexion gaps and defined relative to the posterior condylar axis. Preoperative and postoperative radiographs were analyzed to determine patellar translation, tilt, and coronal limb alignment. Knee injury and Osteoarthritis Outcome Score-Joint Replacement, range of motion (ROM), and postoperative forgotten joint scores (FJS) were collected. Cases were analyzed in incremental groups stratified by femoral component rotation, ranging from greater than 6° of external rotation to greater than 0 to 3° of internal rotation.

RESULTS: Postoperative patellar translation and tilt were significantly different across all femoral rotation groups ($P = 0.002$, $P = 0.005$). Compared to all other femoral rotation groups, lateral patellar translation was reduced when femoral components were internally rotated between 0 and 3°. Conversely, those with femoral […]

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Cementless Total Knee Arthroplasty Is Associated With Improved Outcomes in Patients Who Have a Body Mass Index of 35 or More.

Misch M, Katanbaf R, Smitterberg C, Mont MA, Nace J, Delanois RE

BACKGROUND: Indications for cementless fixation in total knee arthroplasty (TKA) remain unclear despite increasing use. Previous findings have suggested that obese patients have favorable outcomes with cementless TKA, but evidence is sparse. This paper compared the following outcomes in patients who have a body mass index (BMI) of 35 or more: (1) mechanical complications, including periprosthetic fracture and aseptic loosening; (2) infectious complications, including periprosthetic joint infection (PJI) and surgical site infection; 3) venous thromboembolism; and 4) aseptic revision through five years postoperatively.

METHODS: A national all-payer database identified patients who had a BMI of 35 or more undergoing primary TKA for osteoarthritis from 2010 to 2021 ($n = 91,221$). Patients who had a history of trauma, malignancy, or rheumatoid arthritis were excluded. International Classification of Diseases-10th edition codes determined fixation type. Propensity score matching for age, Charlson Comorbidity Index, sex, alcohol abuse, tobacco use, and diabetes yielded cementless ($n = 5,416$) and cemented ($n = 18,525$) cohorts. Chi-square analyses were completed for respective complications at 90 days, one year, two years, and five years postoperatively. Odds ratios (ORs) were calculated using 95% confidence intervals (CIs).

RESULTS: Cementless fixation was associated with decreased ra […]

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Stable Fixation in Cementless Total Knee Arthroplasty Even for Low Local Bone Mineral Density.

Quevedo Gonzalez FJ, Teeter MG, Sculco PK, Zepeda KE, Yared TM, Burgio C, Kolli H, Lipman JD et al.

BACKGROUND: Cementless fixation is increasingly popular for total knee arthroplasty (TKA). Prior research suggested that volumetric bone mineral density (vBMD) measured in preoperative computerized tomography (CT) scans could be useful to identify suitable candidates for cementless TKA with sufficient bone strength to avoid aseptic loosening. However, the clinically relevant thresholds of vBMD for cementless knees have not been defined. As a step toward defining such thresholds, we sought to relate the preoperative vBMD to the migration of tibial baseplates after TKA as a marker of aseptic loosening.

METHODS: We prospectively enrolled 15 patients undergoing unilateral primary TKA with cementless tibial baseplates and cruciate-retaining inserts. Patients received a preoperative CT scan as standard of care, including a BMD reference phantom, and postoperative CT scans the day of surgery, at 6 weeks postoperatively, and at 6 months postoperatively. We calculated the implant motion relative to the day of surgery scan and related it to the vBMD in the four mm immediately under the baseplate.

RESULTS: The maximum total point motion (MTPM) ranged from 0.1 to 0.9 mm at 6 weeks and between 0.2 and 1.1 mm at 6 months. The largest motion occurred vertically and, on average, was consistent with the posterior tilt of the baseplate. The preoperative vBMD was highest under the posterior-med […]

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Third-Generation Flexible Metaphyseal Cones in Revision Total Knee Arthroplasty: Minimum Two-Year Follow-up.

Meding LK, Meneghini RM, Buller LT, Deckard ER, Meding JB

BACKGROUND: Porous metal cones are commonly utilized in revision total knee arthroplasty (rTKA) to manage severe bone loss. Flexible metaphyseal cones (FMCs) allow for macrodeformation, more uniform stress distribution, and diminished peak stresses to the compromised cortical bone, potentially reducing fracture risk during impaction. This study aimed to evaluate the early outcomes of FMCs used to treat severe bone loss in rTKA.

METHODS: There were 241 FMCs inserted during 200 rTKAs. A combination of femoral and tibial FMCs was used in 41 knees. The most common reasons for revision were flexion instability (90 cases), aseptic loosening with or without osteolysis (55 cases), and reimplantation postinfection (27 cases). The mean age at operation was 68 years (range, 37 to 82). The Anderson Orthopaedic Research Institute bone loss classification was used. Serial radiographs were reviewed for evidence of loosening or osseointegration, and survivorship was determined. The mean follow-up was 3 years (range, 2 to 4.5).

RESULTS: At final follow-up, no radiolucent lines were identified around the FMCs. Survivorship was 98% for any revision, 99% for aseptic revision, and 100% for aseptic loosening. Excluding infection, all remaining FMCs (240 cones) demonstrated evidence of stable fixation. There were two intraoperative fractures (1%) that occurred (one femur and one tibia), both in kne […]

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Influence of Conscious Sedation Versus Spinal Anesthesia on Range of Motion and Pain Scores Following Manipulation Under Anesthesia After Total Knee Arthroplasty.

Kutzer KM, Ruderman LV, Jing C, Wu KA, Ryan SP, Wellman SS, Seyler TM

BACKGROUND: Manipulation under anesthesia (MUA) is a common intervention for postoperative stiffness following total knee arthroplasty (TKA). While monitored anesthesia care (MAC) is most frequently used, short-acting spinal anesthesia (SA) is increasing in popularity. There is limited research comparing outcomes of these anesthetics, specifically functional and patient-reported outcomes. This study aimed to compare the impact of MAC versus SA on postoperative range of motion (ROM) and Patient-Reported Outcome Measure Information System (PROMIS) scores following MUA.

METHODS: This retrospective cohort study included 197 patients who underwent MUA after TKA. Patients were grouped by anesthesia type: MAC ($n = 138$) and SA ($n = 59$). Knee ROM measurements, PROMIS scores, and medication requirements, including opioids, muscle relaxants, and corticosteroid use, were collected. These measures were compared using Kruskal-Wallis tests.

RESULTS: Patients receiving MAC had significantly greater improvements in knee ROM (change in ROM = 30.8° ± 19.2) at two months following MUA compared with those receiving SA (change in ROM = 20.0° ± 20.1, $P = 0.0021$) after MUA. There was greater improvement in both flexion (change in flexion = 27.8° ± 18.3 for MAC versus 18.4° ± 18.4 for SA, $P = 0.0037$) and extension (change in extension = -3.0° ± 4.7 for MAC versus -1.6° ±

5.0 for SA, P = 0.013). Addit […]

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Obesity Severity and Stiffness After Total Knee Arthroplasty Revisited: A Contemporary Analysis of Patients Requiring Manipulation Under Anesthesia.

Weinblatt AI, Harris AB, Cushner FD, Pervizaj A, Westrich GH, Lan R, Chalmers BP, Della Valle AG et al.

BACKGROUND: Obesity has historically been associated with worse outcomes following total knee arthroplasty (TKA), and concerns regarding postoperative stiffness and poor response to manipulation under anesthesia (MUA) have contributed to body mass index-based restrictions on surgical eligibility. Contemporary optimization and rehabilitation protocols may have mitigated these risks. This study evaluated the association between obesity severity and postoperative stiffness requiring MUA and post-MUA range of motion outcomes in a large modern obese cohort.

METHODS: We retrospectively reviewed primary elective TKAs performed in patients who had a body mass index ≥ 30 at a single high-volume tertiary center between 2016 and 2022. Patients were categorized by World Health Organization obesity class. The primary outcome was stiffness requiring MUA. The secondary outcomes included repeat MUA, subsequent procedures for stiffness, maximum flexion, and post-MUA success, defined as achievement of ≥ 90 degrees of flexion without revision. Comparisons across obesity classes used Chi-square and nonparametric testing.

RESULTS: Among 12,237 TKAs, MUA was performed in 396 cases (3.2%) and did not differ across obesity classes (P = 0.132). Pairwise analysis showed a lower MUA rate in class III versus class II obesity (P = 0.018). Among 239 patients included in post-MUA analyses, overall success […]

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The Impact of Anticoagulant Choice on Rates of Manipulation Under Anesthesia, Hematoma Formation, and Venous Thromboembolic Events.

Crowley B, Hollimon M, Joaquin T, Lawrence J, Levine BR

BACKGROUND: Anticoagulation after total knee arthroplasty (TKA) remains a balance between clot prevention and postoperative bleeding. This study aimed to analyze anticoagulant choice for deep vein thrombosis prophylaxis on rates of stiffness requiring manipulation under anesthesia (MUA) and hematoma formation after TKA.

METHODS: A large national database was queried for all primary TKAs. We included patients who filled a prescription for an anticoagulant within three days before or five days after surgery. We excluded those who filled prescriptions for multiple or no anticoagulants. Logistic regression was used to compare potent anticoagulants (warfarin, Xa inhibitors, and low molecular weight heparin (LMWH)) to our control group receiving aspirin. The primary outcomes of interest were MUA or hematoma formation within 90 days postoperatively. After applying the exclusion criteria, 462,869 patients were available to review. Of those, 148,159 (32.05%) received aspirin, followed by factor Xa inhibitors (123,909 [26.80%]), warfarin (96,338 [20.84%]), and LMWH (84,962 [18.38%]).

RESULTS: Using logistic regression analyses, patients receiving factor Xa inhibitors (odds ratio [OR] 1.38 [95% confidence interval (CI) 1.32 to 1.43]), indirect factor Xa inhibitors (OR 1.33, [95% CI 1.19 to 1.48]), warfarin (OR 1.29 [95% CI 1.23 to 1.35]), and LMWH (OR 1.26, [95% CI 1.20 to 1.32]) had hi […]

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Tibial Post Fracture in Posterior-Stabilized Total Knee Arthroplasty: Is Fatigue Alone the Culprit?

Garvin KL, Zitsch B, Weisenburger J, Hartman CW, Xiong W, Haider H

BACKGROUND: Tibial post fracture in posterior-stabilized total knee arthroplasty (PS TKA) is a rare but serious complication resulting from insufficient post strength and polyethylene wear. Although highly cross-linked polyethylene (HXLPE) reduces wear compared to ultra-high-molecular weight polyethylene (UHMWPE), its mechanical strength is not improved. This study reports a series of tibial post fractures and evaluates whether fatigue alone accounts for such failures.

METHODS: There were six patients (seven knees) who underwent revision for tibial post fractures reviewed retrospectively for demographic, implant, and surgical factors. Complementary cantilever fatigue tests were conducted using PS TKA tibial inserts fabricated from UHMWPE and HXLPE. Reversing antero-posterior (AP) loads were applied via surrogate femoral components at 0° and 75° flexion, without axial compression. There were two post designs tested: a chamfered post and a nonchamfered post with a larger proximal cross-section. Cyclic loading at two Hz began at ± 209 N for 10,000 cycles, with stepwise increase every 10,000 cycles. Additional tests at physiological and supraphysiological loads continued for up to 10 million cycles (Mc), simulating decades of in vivo use.

RESULTS: Fractures occurred a mean of 12.8 years postoperatively (range, 4.5 to 17.8). There were five fractures that had UHMWPE and two that h […]

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Achieving Anatomic Kinematics in a Noncruciate Total Knee Arthroplasty: Preclinical Evaluation Using a Crouching Machine.

Parody N, Warren S, Hennessy D, Rozell JC, Bosco J, Walker PS

BACKGROUND: Studies on patients have shown that normal anatomic motion is often not achieved with current total knee arthroplasty (TKA) designs. The purpose of this study was to determine whether anatomic motion could be restored using a design where intercondylar guiding surfaces were positioned between the medial and lateral femoral condyles.

METHODS: A crouching machine was constructed, which included simulations of the collateral ligaments, the quadriceps mechanism, the hamstrings, and the gastrocnemius. Medial pivot, medially congruent, ultracongruent, and replica intercondylar TKAs were designed and 3-dimensionally printed. The femoral-tibial kinematics were measured by determining lateral and medial contacts, as the knee underwent flexion and extension from 12 to 130°.

RESULTS: The replica intercondylar design showed an almost constant position of the medial contact, with a progressive posterior displacement laterally during flexion. In contrast, the other designs showed nearly parallel motion with minimal variation in contact location.

CONCLUSIONS: It was concluded that intercondylar guiding surfaces could produce anatomical motion in a TKA where the cruciate ligaments were resected.

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A Magnetic Resonance Imaging Study Comparing Varus Arthritic to Normal Knees Indicates that the Medial Collateral Ligament Does Not Contract in up to 15° of Varus Deformity, Whereas the Lateral Collateral Ligament May Elongate.

Mullaji AB, Nadal RR

BACKGROUND: The question of whether the superficial medial collateral ligament (MCL) undergoes contracture in the presence of osteoarthritic changes remains unresolved. Addressing this fundamental scientific inquiry is crucial for determining the surgical technique of correcting varus deformity and evaluating the need for deploying alternative alignment philosophies.

METHODS: Magnetic resonance (MR) images of 100 knees with established osteoarthritis were compared to those of 100 normal control knees. We measured the width of the transepicondylar axis (TEA), lengths of the MCL and lateral collateral ligaments (LCL) along their tortuosity, the straight line distance between their attachments (MCL1 line and LCL1 line), the hip-knee-ankle angle, and the width of medial femoral and tibial osteophytes.

RESULTS: The MCL/TEA ratios did not differ significantly with deformity. The mean MCL length was reduced by five mm in knees greater than 15° varus ($P < 0.001$). The LCL/TEA ratios were greater in knees with 5 to 10° ($P = 0.001$) and 10 to 15° varus ($P = 0.007$). The MCL difference (MCL1 line minus MCL length) significantly correlated with femoral ($P = 0.009$) and tibial osteophyte width ($P = 0.044$).

CONCLUSIONS: In varus arthritic knees, the MCL does not shorten but is tented over medial osteophytes. The LCL may elongate by up to two mm with increasing deformity. At extreme deformity […]

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Synovial Fluid Electrochemical Properties Are Altered After Total Knee Arthroplasty: A Necropsy Retrieval Study.

Bond B, Kurtz MA, Kurtz SM, Aslani S, Holland CT, Mihalko WM

BACKGROUND: Primary total knee arthroplasty (TKA) relies on cobalt chrome alloy (CoCrMo) implants to replace the degraded joint and restore patient function. However, retrieval analysis documents wear processes on femoral components, associated with metal accumulation in the periprosthetic tissue. It remains unclear how TKA affects synovial fluid and whether arthroplasty promotes oxidation within the joint capsule. To address these gaps, we asked, does TKA alter the synovial fluid's physical and electrochemical properties?

METHODS: Synovial fluid was aspirated from 18 pairs of deidentified necropsy TKA and native contralateral knee specimens. The aspirated volume and pH were recorded. Then synovial fluid was transferred into an electrochemical cell consisting of a wrought CoCrMo working electrode, an Ag/AgCl reference electrode, and a platinum counter electrode. To assess electrochemical properties, open circuit potential, impedance spectra, and linear polarization data were acquired. After data collection, impedance spectra were circuit fitted, and the inverse of the oxide polarization resistance (Rp-1) was calculated to measure the instantaneous corrosion rate.

RESULTS: In necropsy TKA synovial fluid samples, we documented increased synovial fluid volumes (6 ± 2.77 versus 1.75 ± 1.41 mL, $P < 0.001$) and higher instantaneous corrosion rates (1.70 x 10-7 ± 1.03 x 10-7 versus 8 […]

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A Systematic Review of Registry and Large Database Studies Comparing Contemporary Cementless and Cemented Total Knee Arthroplasty.

Lombardi AV, Rainey JP, Gililland JM, Mont MA

BACKGROUND: Interest is growing in durable, cementless fixation in total knee arthroplasty (TKA). This study aimed to systematically review registry and large database studies, evaluating their quality of evidence and assessing implant survivorship. Additionally, we compared these findings to a systematic review of randomized controlled trials.

METHODS: A search of PubMed and Embase was conducted to identify eligible registry and large database studies on contemporary cementless TKA survivorship published between January 1, 2010, and December 31, 2024. Overlapping studies from the same registry were excluded. Modified Coleman methodology (MCM) scores were used to assess methodological quality, with a score of 85 to 100 considered excellent, 70 to 84 considered good, 55 to 69 considered fair, and below 55 considered poor. In addition, level of evidence assessments were made.

RESULTS: Among eight included registry studies, the mean MCM score was 35.5 (range, 29 to 39), reflecting poor quality, and all were level III evidence. Differences in revision rates between cementless and cemented TKA were typically less than 1%. Cementless prostheses comprised less than 5% of total implants evaluated for most studies, with one study evaluating 1,056,492 cemented versus 26,768 cementless TKAs. For the two large database studies, the mean MCM score was low at 36.0 (range, 35 to 37), and bo […]

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Spontaneous Osteonecrosis of the Knee Is Associated With Increased Five-Year Implant Complications Compared to Osteoarthritis Patients Undergoing Primary Total Knee Arthroplasty.

Gordon AM, Nian PP, Smitterberg CW, Mont MA

BACKGROUND: Spontaneous osteonecrosis of the knee (SPONK) is a potential, although uncommon, indication for total knee arthroplasty (TKA). Unlike secondary osteonecrosis, SPONK typically affects a single femoral condyle in patients over 50 years of age and is sometimes treated with TKA once joint-preserving strategies fail. However, implant survivorship of TKA for SPONK versus osteoarthritis (OA) remains under-characterized. This study aimed to compare outcomes, including (a) mechanical complications (aseptic loosening, periprosthetic fracture), (b) periprosthetic joint infection, and (c) all-cause revision, between patients undergoing TKA for SPONK and OA at two and five years.

METHODS: We retrospectively queried a nationwide database to identify 43,680 patients who underwent TKA from 2010 to 2021, including 7,282 patients who had SPONK matched by age, sex, and comorbidities to 36,398 patients who had OA. Implant-related complications were assessed at two and five years. Multivariable logistic regressions computed the odds ratios (ORs) and 95% confidence intervals with a significance threshold of $P < 0.001$.

RESULTS: Among mechanical complications, patients undergoing TKA for SPONK had a significantly higher risk of aseptic loosening at both two (OR: 1.52, $P < 0.001$) and five years (OR: 1.63, $P < 0.001$) compared to OA. Similarly, elevated risk of periprosthetic fracture was o […]

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Outcomes of Simultaneous Versus Staged Hardware Removal and Total Knee Arthroplasty.

Khury F, Fong C, Ruff G, Sarfraz A, Aggarwal VK, Schwarzkopf R, Rozell JC

BACKGROUND: This study compares clinical and functional outcomes between simultaneous hardware removal during total knee arthroplasty (TKA) and staged TKA after prior hardware removal.

METHODS: We retrospectively reviewed 155 patients who had prior knee hardware and underwent elective primary TKA between 2012 and 2024 at an urban academic institution. Patients were categorized into "simultaneous" removal during TKA (n = 127) or "staged" TKA after removal (n = 28), and stratified by hardware type (minor/moderate/major).

RESULTS: Simultaneous procedures involved significantly less "major hardware," single incisions, and tibial stem extensions than staged procedures (32.3 versus 78.6%, $P < 0.001$; 81.9 versus 100%, $P = 0.007$; and 0.8 versus 10.7%, $P = 0.019$, respectively). Hardware, particularly the major type, was more often retained or partially retained in the simultaneous group (48.0 versus 21.4%, $P = 0.008$). Reoperation, revision, and infection rates did not significantly differ based on timing or hardware location. Simultaneous patients had smaller 3-month Patient-Reported Outcomes Measurement Information System (PROMIS) Pain Intensity and Interference score reductions (-1.6 versus -9.9, $P = 0.006$ and +0.4 versus - 7.2, $P = 0.007$, respectively), but greater 2-year Knee Injury and Osteoarthritis Outcome Score for Joint Replacement improvements (+25.0 versus - 1.1, $P = 0.006$) […]

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The Presence of Osteonecrosis in Patients Who Have Systemic Lupus Erythematosus Undergoing Total Knee Arthroplasty Is Not Associated With Worse Outcomes.

Kim BI, Deeb AT, Mintz DN, Goodman SM, Rodriguez JA, Figgie MP, Blevins JL

BACKGROUND: Systemic lupus erythematosus (SLE) patients seeking total knee arthroplasty (TKA) typically present with secondary osteonecrosis of the knee. Whether osteonecrosis negatively affects TKA outcomes in this high-risk population has not previously been studied. This study aimed to compare TKA survivorship, clinical outcomes, and postoperative complication rates between SLE patients who did and did not have osteonecrosis.

METHODS: A retrospective analysis of SLE patients undergoing TKA between 2015 and 2023 at a large academic institution was conducted. The cohort was stratified by the presence of osteonecrosis on preoperative X-ray and magnetic resonance imaging when available. Imaging diagnosis was validated by an experienced musculoskeletal radiologist. Patient characteristics, clinical outcomes, and patient-reported outcome measures were compared between groups using univariable and multivariable analyses. We identified 179 knees in 147 patients who had lupus undergoing TKA. There were twelve percent (n = 21 of 179) of the knees that had osteonecrosis. The mean follow-up was 4.8 years (range, 1.0 to 9.1).

RESULTS: Osteonecrosis was associated with younger age ($P < 0.01$) and lower body mass index ($P = 0.04$). Osteonecrosis was not a risk factor for postoperative readmission or revision, but younger age was associated with increased risk for both ($P = 0.02$ and $P < 0.0$ […]

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Fractures of the coronoid process: state of the art.

Marinelli A, Riva M, Coliva F, Minerba M, Carbone G, Guerra E

Coronoid fractures are rarely isolated and are much more frequently associated with other osseous or ligamentous structures injuries. On the basis of the coronoid fracture patterns, described by the O'Driscoll classification, it is possible to recognize three main patterns of injury that differ on traumatic mechanism and on associated lesions: posterolateral rotatory instability, posteromedial rotatory instability, and axial load injuries. The management of coronoid fractures is challenging and varies according to characteristics of the fracture, associated lesions, and amount of elbow instability. In general, operative treatment is indicated in every case the fracture is at least 50% of the whole coronoid, whether the sublime tubercle is involved, and whether the ulno-humeral joint is not perfectly reduced. In conclusion, the correct management of the coronoid, especially in the setting of complex elbow instability, represents a predictive factor for patient outcomes and functional results. The stability of the elbow, rather than the size of the coronoid fragment, is the main parameter for surgical indication, aimed to fix the coronoid and/or repair the associated lesions.

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Total hip arthroplasty in Italy: an observational, population-based study on surgical volume growth from 2001 to 2023 and forecasts until 2050 with six different statistical models.

Ciminello E, Cuccu A, Romanini E, Venosa M, Cazzato G, Tucci G, Boniforti F, Carpanese L et al.

BACKGROUND: The number of total hip arthroplasty (THA) procedures has been steadily increasing worldwide, driven by aging population, improvements in surgical techniques and implant design. This study aimed to analyze the temporal trends of elective THA in Italy since 2001-2023 and forecast THA volumes up to 2050 to provide insights for healthcare planning.

MATERIALS AND METHODS: International Classification of Diseases, 9th Revision, Clinical Modification (ICD9-CM) coding system was used to extract records of interest (elective THA) from the Italian National Hospital Discharge Record database. Six statistical models were applied to forecast future THA volumes: logistic regression; Poisson regression; logarithmic regression; inverse/power regression; Poisson log-normal regression; and hierarchical Poisson regression with temporal effects (HPTE). Model performances were assessed by using error metrics and internal validation on the basis of a rolling-origin approach. An out-of-sample validation was conducted to ensure a robust assessment of forecasting reliability. THA volume forecasts were provided with 95% prediction intervals.

RESULTS: A total of 1,318,400 records for primary elective THAs performed in Italy since 2001-2023 were analyzed. The number of THAs increased by approximately 80%, rising from 68.270 in 2001 to 122.777 in 2023. Among the tested models, HPTE generally […]

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Clinical relevance of patient-reported outcome measures in the surgical management of focal chondral defects of the knee: a systematic review.

Migliorini F, Maffulli N, Memminger MK, Hofmann UK

INTRODUCTION: To evaluate clinical outcome following surgical management of focal chondral defects in the knee, patient-reported outcome measures (PROMs) are used. To give these measures meaning, parameters such as the minimal clinically important difference (MCID), patient-acceptable symptom state (PASS), minimally detectable change (MDC), clinically important difference (CID) and substantial clinical benefit (SCB) have been introduced. This systematic review investigated the MCID, SCB, CID, PASS and MDC of the most commonly used PROMs for assessing patients following surgical repair of focal chondral defects of the knee.

METHODS: This systematic review was conducted in accordance with the 2020 Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines and the recommendations of the Cochrane

Handbook for Systematic Reviews of Interventions. All clinical studies investigating tools to assess the clinical relevance of PROMs in the surgical repair of focal chondral defects of the knee were reviewed. In April 2025, the following databases were accessed: PubMed, Web of Science and Embase. The PROMs of interest included: the International Knee Documentation Committee (IKDC) questionnaire, the Knee Injury and Osteoarthritis Outcome Score (KOOS) and its related subscales activities of daily living (ADL), pain, quality of life (QoL), sports and recreational […]

PubMed · DOI

Simultaneous versus staged bilateral total hip arthroplasty: a meta-analysis.

Soler F, Hernández J, Lamo-Espinosa JM, Benlloch M, Mariscal G

BACKGROUND: There is debate regarding the optimal timing of bilateral total hip arthroplasty (THA), with simultaneous or staged approaches considered. Cost-effectiveness is an important factor that influences resource allocation. The main aim of this study was to assess the cost implications of bilateral simultaneous (sim-THA) versus staged (st-THA) total hip arthroplasty. The secondary objective was to evaluate the efficacy and safety of sim-THA compared those with of st-THA.

METHODS: A systematic review and meta-analysis were conducted according to the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) guidelines. Studies comparing simultaneous and staged bilateral THA in terms of cost, complications, length of hospital stay, and patient outcomes were eligible. Odds ratios (ORs), mean differences (MD), and standard mean differences (SMD) with 95% confidence intervals (CIs) were calculated. The risk of bias was assessed using the Methodological Index for Non-randomised Studies (MINORS). Meta-analyses were performed using Review Manager version 5.4 (Cochrane, Oxford, United Kingdom).

RESULTS: A total of 20 observational studies including 13,984 patients were included. Sim-THA was associated with significantly lower total costs (SMD -0.54, 95% CI -0.92 to -0.16; p = 0.005) and shorter hospital stay (MD -2.90, 95% CI -4.38 to -1.42; p = 0.0001) than tho […]

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Long-term outcomes of the modified Dunn procedure in moderate and severe slipped capital femoral epiphysis: a prospective case series with 7-year follow-up.

Fahmy M, Abdelazeem AH, Shawky MA

INTRODUCTION: Slipped capital femoral epiphysis (SCFE) is the most common hip disorder in adolescents and may lead to femoroacetabular impingement, early osteoarthritis, and long-term functional disability if inadequately treated. While in situ pinning remains the standard treatment for mild slips, it fails to correct the deformity in moderate and severe cases, potentially predisposing to degenerative changes. The modified Dunn procedure (MDP) was developed to restore proximal femoral anatomy through surgical hip dislocation while preserving vascular supply. The aim of the study is to evaluate the long-term radiological and functional outcomes of the MDP in patients with moderate (14 cases) and severe (10 cases) SCFE, and to assess the incidence of avascular necrosis (AVN), osteoarthritis, and other complications.

METHODS: A prospective case series was conducted between August 2015 and January 2019 at a single tertiary institution. A total of 24 hips with moderate-to-severe SCFE and open physis were treated using the MDP via surgical hip dislocation. Mild and acute-only slips were excluded. MDP was used as a primary procedure, performed early in severe slips and in selected moderate slips after clinical assessment. Patients were followed clinically and radiologically for a mean duration of 84 ± 2.6 months (range 80-88 months). Functional outcomes were assessed using the Harris […]

PubMed · DOI

The presence of a positive blood culture should be regarded a risk factor for DAIR failure in cases of haematogenous total knee arthroplasty infection.

Pedemonte-Parramón G, Reynaga E, Molinos S, de Los Ríos JD, Vivero A, García-Oltra E, López-Pérez V, Paredes R et al.

PURPOSE: Late acute haematogenous total knee arthroplasty (LAH TKA) infections represent approximately 20% of all prosthetic infections. Debridement, antibiotics, and implant retention (DAIR) is a widely accepted treatment; however, it carries a significant risk of failure. The study aims to identify risk factors for DAIR failure in LAH TKA infections, focusing on blood cultures (BC) as a potential contributor.

METHODS: A retrospective cohort study was conducted on 37 cases of LAH TKA infections from 2015 to 2023. Patients were divided into two groups: 20 in the success group (SG) and 17 in the failure group (FG). The study analyzes various factors, including demographics, comorbidities, serum and joint fluid biomarkers, blood and intraoperative cultures, prior infection or antibiotic use, and surgical and post-surgical variables.

RESULTS: Positive BC were more frequent in the FG compared with the SG (P = 0.03), and were also more common in patients with a history of prior infection (P = 0.03). Logistic regression identified positive BC as the only significant predictor of DAIR failure (odds ratio (OR) 12, 95% confidence interval (CI) 1.1-18, P = 0.04), even after adjusting for other variables. Positive intraoperative cultures were more frequent in the FG, particularly in those with a prior infection history (P = 0.08), even if they were receiving antibiotics (P = 0.05).

CON […]

PubMed · DOI

Iatrogenic femoral neck fractures or separation of the proximal femoral epiphysis during closed reduction of irreducible femoral head fracture-dislocations in children: a review of 12 cases.

Lu Y, Xue C, Canavese F, De Salvo S, Yao H, Huang D, Lin R, Zheng J et al.

OBJECTIVES: This study aimed to analyze the radiologic characteristics of irreducible pediatric femoral head fracture-dislocations to prevent potential iatrogenic femoral neck fractures (FNF) or separation of the proximal femoral epiphysis (SPFE).

METHODS: This is a retrospective review of patients who were skeletally immature and diagnosed with traumatic hip dislocations combined with femoral head fractures. The collected data included patient demographics, fracture classification, fragment ratio, combined injuries, urgent reductions, treatment strategies, complications, and final outcomes.

RESULTS: We treated 12 patients with femoral head fractures and dislocations; 11 out of 12 patients underwent urgent closed reduction (91.7%). Six of the patients failed closed reduction and experienced FNF (n = 2, 33.3%) or SPFE (n = 4, 66.7%). Five patients presented with avascular necrosis of the femoral head after open reduction with internal fixation via a surgical hip dislocation approach, and two patients required further surgical treatment. Analysis of radiographs and computed tomography (CT) scans of irreducible femoral head fracture-dislocations revealed that the fractured femoral head was perched on the sharp angle of the posterior wall of the acetabulum, with a fragment ratio of 18-27%. After recognizing the irreducibility, one case with a fragment ratio of 26% underwent immed […]

PubMed · DOI

Minimal clinically important difference (MCID), patient-acceptable state (PASS), minimally detectable change (MDC), and substantial clinical benefit (SCB) in patients who have undergone arthroscopic surgery for femoroacetabular impingement: a systematic review.

Migliorini F, Maffulli N, Schäfer L, Jeyaraman M, Betsch M, Mahmoud M

INTRODUCTION: Hip arthroscopy has been established as a successful management for femoroacetabular impingement (FAI) syndrome. Patient-reported outcome measures (PROMs) are frequently used to assess treatment results in patients. However, clinically important outcome values have recently been calculated to better understand and interpret PROMs and define successful treatment. This systematic review aimed to investigate the minimal clinically important difference (MCID), patient-acceptable state (PASS), substantial clinical benefit (SCB), and minimally detectable change (MIC) of the most commonly used PROMs for assessing patients who have undergone arthroscopic surgery for FAI.

MATERIAL AND METHODS: This study was conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses: the 2020 PRISMA statement. In October 2025, the following databases were accessed: PubMed, Web of Science, and Embase. All clinical studies investigating tools to assess the clinical relevance of PROMs used to evaluate patients who have undergone arthroscopic surgery for FAI were accessed. Only studies that assessed the minimal MCID, PASS, SCB, and/or MIC were considered.

RESULTS: The methodological quality of the 21 included studies was assessed, and the overall risk of bias was found to be low-to-moderate. The main characteristics of the included studies were analyzed and […]

PubMed · DOI

Efficacy and safety of MIPO versus ORIF in distal tibial fractures: a systematic review and meta-analysis.

Hassaan AM, Hassan AA, Abdelkhalik WS, Saad AM, Hosny AS, Elfeky BAEH, Mandour I, Ragheb MG et al.

BACKGROUND: Distal tibial fractures are challenging to manage owing to limited soft-tissue coverage and compromised blood supply. Minimally invasive plate osteosynthesis (MIPO) and open reduction and internal fixation (ORIF) are among the widely used surgical options, but the optimal approach remains unclear. MIPO preserves periosteal blood flow with smaller incisions, while ORIF offers direct visualisation but increases soft-tissue damage. With growing comparative evidence, an updated evaluation of these techniques is necessary.

AIM: To compare the efficacy and safety of MIPO versus ORIF in adult patients with distal tibial fractures.

METHODS: Following PRISMA guidelines and the Cochrane Handbook, we searched in PubMed, Embase, Cochrane, Scopus, and Web of Science till November 2025. Randomised trials and cohort studies comparing MIPO and ORIF were included. Data extraction was performed independently by two reviewers. Risk of bias was assessed using RoB 2 for RCTs and the Newcastle-Ottawa Scale for observational studies. Random-effects models were used to estimate pooled mean differences and odds ratios, and heterogeneity was assessed using I2. Sensitivity and subgroup analyses were conducted by study design.

RESULTS: Nine studies involving 530 patients met our inclusion criteria. No significant differences were found between MIPO and ORIF in AOFAS scores, union time, oper […]

PubMed · DOI

Vascularized versus nonvascularized free fibular grafts in reconstruction of post-traumatic critical long bone defects, a comparative study.

Fouaad AA, Massoud AH, Shaban M, Abdel-Naby M, Abulsoud MI

BACKGROUND: Reconstruction of critical-sized long bone defects is a complex orthopedic challenge. Free fibular grafts, vascularized (FVFG) and nonvascularized (NVFG), are established reconstructive options, but comparative clinical outcomes remain uncertain.

PURPOSE: To compare clinical and radiological outcomes of FVFG and NVFG in patients with post-traumatic critical bone defects more than 10 cm.

METHODS: A randomized controlled trial was conducted with 50 patients assigned equally to FVFG or NVFG groups. The primary outcome was time to union, while secondary outcomes included graft hypertrophy, functional scores (DASH and LEFS), complication rates, and donor site morbidity.

RESULTS: The mean time to union was 5.78 months in the FVFG group and 6.17 months in the NVFG group, showing no statistically significant difference ($p = 0.447$). Rates of graft hypertrophy, functional recovery, and complications were comparable between the groups.

CONCLUSIONS: Both FVFG and NVFG provide effective reconstruction for critical bone defects, with nearly similar healing times and functional outcomes. NVFG may represent a less complex alternative in selected cases.

PubMed · DOI

Five-year outcomes of medial open-wedge high tibial osteotomy with lateral supplemental lag screw in patients with obesity.

Tian YH, Lee KH, Huang CW, Lin SY, Yang JC

BACKGROUND: Obesity contributes to the accelerated progression of knee osteoarthritis. Medial open-wedge high tibial osteotomy (MOWHTO) is a joint-preserving surgical intervention for unicompartmental knee osteoarthritis; however, its efficacy in patients with obesity remains controversial. This study aimed to evaluate the

5-year clinical and radiographic outcomes of MOWHTO with lateral supplemental lag screw fixation in patients with obesity.

MATERIALS AND METHODS: This retrospective cohort study included patients with obesity who underwent MOWHTO between 2017 and 2020, with a minimum follow-up of 5 years. All procedures were performed by a single surgeon using three-dimensional printed, patient-specific instrumentation, locking plates, and a lateral supplemental lag screw. Clinical outcomes were assessed using the visual analog scale (VAS) and Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC). Radiographic parameters, including the weight-bearing line percentage (WBL%) and medial proximal tibial angle (MPTA), were evaluated.

RESULTS: Significant improvements in VAS and WOMAC scores were observed at all postoperative time points ($p < 0.001$) and were accompanied by improved radiographic alignment, with WBL% shifting toward the Fujisawa point and increased MPTA values. At 5 years, mild regression in alignment was noted; however, the overall correction was […]

PubMed · DOI

Outcomes of surgical fixation of osteochondral fracture: a systematic review.

Pang AS, Oviya R, Tan JJ, Lim KSA, Hui JHP, Tan SHS

BACKGROUND: Dealing with knee osteochondral defects presents a substantial clinical challenge due to the variable nature of repair outcomes and intricate biomechanical complexities inherent to the knee joint. Numerous surgical alternatives exist for addressing knee osteochondral lesions, yet there is limited evidence available for comparing their respective outcomes.

METHODS: According to the Preferred Reporting Items for Systematic Review and Meta-analyses (PRISMA) guidelines, a systematic literature search was conducted to identify publications from inception to September 2023 on PubMed, Cochrane, and Medline academic search engines with the MeSH terms "osteochondral" AND ("patella" OR "patellar" OR "patellofemoral" OR "femoral condyle lesions" OR "trochlear lesions"). Inclusion criteria were clinical human studies, English language, subjects who underwent surgical management (fixation, reconstruction, or debridement) for knee osteochondral lesions with reported treatment outcomes, and a minimum sample size of ten patients. The methodological index for non-randomized studies (MINORS) was used to evaluate the non-randomized studies' methodological quality. Main outcome measurements Pediatric International Knee Documentation Committee (Pedi IKDC) scale, Tegner-Lysholm Scoring Scale, Tegner Activity Score, Visual Analog Scale, Knee Society Score (KSS) Score, Kujala Score, and F […]

PubMed · DOI

Do modular tapered long stems differ in stability and function? A prospective comparison of two stems in complex femoral revision THA.

Fahmy M, Shawky MA

BACKGROUND: Revision total hip arthroplasty (rTHA) in the setting of substantial proximal femoral bone loss is technically challenging. Modular tapered fluted stems provide predictable diaphyseal fixation while allowing independent adjustment of version, offset, and limb length. Among these, two commonly used systems-modular tapered fluted stem Type A (Revitan™) and Type B (LIMA Modulus™)-have limited direct comparative evidence. This study aimed to prospectively compare radiographic stem subsidence (primary outcome), as well as functional outcomes, complications, survivorship, and secondary outcomes, between Type A and Type B modular long stems in femoral rTHA.

METHODS: In this single-center randomized prospective study, 110 patients undergoing femoral revision rTHA were randomly assigned to receive either Type A ($n = 55$) or Type B ($n = 55$) stems. All procedures were performed by experienced revision surgeons under standardized perioperative and rehabilitation protocols. Radiographs were analyzed for stem subsidence, osseointegration, and limb-length restoration. Functional outcomes were assessed using the Harris Hip Score (HHS), Oxford Hip Score (OHS), and European Quality of Life Visual Analogue Scale (EQ-VAS) at baseline and final follow-up (mean 61.4 months). Complications and stem survivorship were recorded prospectively. Statistical analysis included paired and unpaired […]

PubMed · DOI

Clinical comparison of single posterolateral plate with medial-cannulated-screw fixation and double-plate fixation for extra-articular distal humerus fractures.

Jung HS, Chu MS, Lee JS

BACKGROUND: Double-plate fixation is the gold standard for extra-articular distal humerus fractures, but it carries a substantial risk of postoperative ulnar neuropathy. Fixation using a single posterolateral plate with a medial cannulated screw may reduce ulnar neuropathy while maintaining fracture stability. This study aimed to compare the clinical outcomes of single-plate-with-medial-screw fixation versus double-plate fixation for extra-articular distal humerus fractures.

MATERIALS AND METHODS: Fifty-six patients who underwent surgery for extra-articular distal humerus fractures (Arbeitsgemeinschaft für Osteosynthesefragen/Orthopaedic Trauma Association [AO/OTA] classification A2 or A3) between January 2018 and August 2024 were divided into a double-plate group and a single-plate-with-medial-screw group. We conducted a retrospective, nonrandomized comparative study. The double-plate fixation was used in 30 patients from January 2018 to October 2021, while the single-plate fixation with a medial screw was used in 26 patients from November 2021 to August 2024. All surgeries were performed using a posterior paratricipital approach. Bony union, radiographic healing, and loss of reduction were evaluated. Postoperative pain scores (visual analog scale at 2 days after the operation), operative time (minutes), elbow range of motion, elbow function (Mayo Elbow Performance Score [MEP […]

PubMed · DOI

A study on the association between tibial plateau fractures and intra-articular soft-tissue injuries under valgus injury mechanisms.

Duan S, Wang S, Zhu T, Li Q, Zhang M

BACKGROUND: While recent investigations have focused on injury mechanism classifications of tibial plateau fractures (TPFs), the association between valgus TPFs and concomitant soft-tissue damage involving menisci and ligaments remains insufficiently elucidated. This study aimed to characterize intra-articular soft-tissue injuries associated with various valgus TPFs and assess the predictive value of lateral plateau depression (LPD) and widening (LPW). Additionally, the analysis extended to other injury mechanisms.

MATERIALS AND METHODS: This study included adult patients with acute tibial plateau fractures who had complete imaging data, excluding patients with open fractures and multiple fractures throughout the body. Imaging examinations were used to assess the fracture mechanism and intra-articular soft-tissue damage.

RESULTS: The study retrospectively analyzed the clinical data of 138 patients with valgus injury TPFs in our hospital. The study compared the incidence of TPFs with intra-articular soft-tissue damage under different injury mechanisms. The result demonstrated that the incidence of valgus hyperextension TPFs combined with medial collateral ligament injuries was relatively high (61.5%) compared with TPFs with other injury mechanism. Multivariable logistic regression and smooth curve fitting revealed significant dose-response relationships of LPD (odds ratio [OR] […]

PubMed · DOI

The coronoid as the key fragment of trans-ulnar fracture-dislocations of the elbow: Insights from a retrospective cohort comparison using the coronoid-centric Mayo classification system.

Henssler L, Kerschbaum M, Zanklmaier M, Wieczorek LA, Alt V, Klute L

BACKGROUND: Trans-ulnar fracture-dislocations of the elbow are rare injuries with complex fracture patterns and variable outcomes. Traditional classification systems offer limited prognostic value. A recently introduced coronoid-centric Mayo classification distinguishes injury subtypes based on coronoid attachment and identifies trans-ulnar basal coronoid (TUBC) fractures as a particularly challenging entity. This study evaluated outcomes across Mayo fracture types and explored factors associated with inferior results in TUBC injuries.

MATERIALS AND METHODS: In this retrospective cohort study, surgically treated trans-ulnar elbow fracture-dislocations managed at a level I trauma center between 2010 and 2022 were identified and classified according to the Mayo system. Demographic data, injury characteristics, surgical management, radiographic outcomes, and complications were recorded. Functional outcomes were assessed after a minimum follow-up of

12 months using the Mayo Elbow Performance Score (MEPS); Oxford Elbow Score (OES); Quick Disabilities of Arm, Shoulder and Hand Questionnaire (QuickDASH); European Quality of Life Five-Dimension, Five-Level Version (EQ-5D-5L); and range-of-motion measurements. Radiographs were analyzed for union, instability, heterotopic ossification, and post-traumatic osteoarthritis (OA).

RESULTS: A total of 52 patients were included (14 trans-olecr […]

PubMed · DOI

Feasibility study of a robot-assisted endoscopic technique for plate osteosyntheses of the acetabulum and anterior pelvic ring.

Hinz N, Althoff G, Thomaschewski M, Bonke LA, Eberenz A, Münch M, Behrends L, Männel G et al.

BACKGROUND: Minimally invasive stabilization of acetabular and pelvic ring fractures using endoscopic techniques has become increasingly important. The logical advancement of conventional endoscopic techniques is a robot-assisted approach, which benefits from the advantages of robot-assistance systems (e.g., more degrees of freedom for instruments and improved visualization). The aim was therefore to investigate the feasibility of a robot-assisted endoscopic technique for plate osteosyntheses of the acetabulum and anterior pelvic ring.

MATERIALS AND METHODS: In this two-part feasibility study, four different plate osteosyntheses were performed endoscopically using the Hugo™ robot-assisted surgery system, first on ten synthetic pelvic models and then on ten human cadavers. During robot-assisted dissection for the preperitoneal approach, identification of ten relevant anatomical landmarks was also assessed. In both parts, the success, number of drilling errors, and time for each plate were described as learning curves and analyzed using linear regression.

RESULTS: The infrapectineal plate could be successfully performed in 100% of synthetic models, the posterior column plate in 100%, the suprapectineal plate in 90%, and the superior pubic ramus plate in 80%. Learning curves could be observed for the number of drilling errors per plate (e.g., 0.67 to 0, $p = 0.009$) and the time r […]

PubMed · DOI

Favorable subjective clinical outcomes after revision hip arthroscopy for femoroacetabular impingement syndrome despite a high rate of capsular defects and increased capsular thickness.

Cao Z, Gao G, Lin W, Zhu Y, Zhou X, Wang J, Xu Y

BACKGROUND: The capsular healing status and capsular thickness changes in patients with femoroacetabular impingement syndrome (FAIS) following revision hip arthroscopy are poorly documented, and their relationship with subjective clinical outcomes remains unclear. The purpose of this work is to evaluate the incidence of capsular defects and changes in capsular thickness using magnetic resonance imaging (MRI) following hip arthroscopy in patients with FAIS, and to compare the subjective clinical outcomes between patients with and without capsular healing after revision hip arthroscopy.

PATIENTS AND METHODS: Consecutive patients with FAIS who underwent revision hip arthroscopy between 2013 and 2023 were included. Patients were categorized into two groups on the basis of capsular healing status after revision hip arthroscopy. Patient-reported outcomes (PROs) were collected preoperatively and at minimum 2-year follow-up, including the modified Harris Hip Score (mHHS), International Hip Outcome Tool-12 (iHOT-12), Hip Outcome Score-Activity of Daily Living Scale (HOS-ADL), Hip Outcome Score-Sport Specific Subscale (HOS-SSS), and visual analog scale (VAS). Achievements in minimal clinically important difference (MCID) and patient acceptable symptom state (PASS), patient satisfaction, and re-revision rates were evaluated and compared between groups. Capsular thickness was assessed by […]

PubMed · DOI

From bench to bedside: a novel suture-augmented prosthesis for greater trochanteric fixation in osteoporotic hip arthroplasty.

Yang M, Yang J, Wang X, Wu Z, Bian S, Sheng R, Liao F, Qin K et al.

BACKGROUND: Stable fixation of the greater trochanter during hip arthroplasty for unstable osteoporotic intertrochanteric fractures remains a significant challenge. Conventional techniques depend on securing the bone-implant interface, which is particularly compromised in osteoporotic bone. Here, we propose a novel conceptual approach that establishes a direct mechanical bridge from the abductor tendon to the prosthesis, thereby reducing reliance on the fragile bone-implant interface.

METHODS: In this two-stage translational study, we progressed from biomechanical validation to clinical application. First, using a decalcified caprine femoral model simulating osteoporosis, three fixation constructs were compared: locking plate (LP), suture-augmented locking plate (LPSA), and Kirschner-wire tension band (KWTB). Ultimate load and construct stability were evaluated. Biomechanical testing confirmed the principle of suture-mediated load sharing and highlighted the intrinsic weakness of screw fixation in osteoporotic bone. Guided by these results, we designed a novel femoral prosthesis that eliminates screw fixation, employs sutures as the primary load-bearing element, and incorporates integrated suture anchor tunnels. This prosthesis was then assessed in a retrospective series of 15 consecutive elderly patients with osteoporotic intertrochanteric fractures. Clinical outcomes were ev […]

PubMed · DOI

Updating radiographic parameters for the healthy growing hip: are current parameters still valid?

Hütter K, Smolle MA, Butter S, Schroedter R, Tschauner S, Kraus T

BACKGROUND: This study aimed to evaluate whether the acetabular angle (AA) by Tönnis and lateral center-edge angle (LCEA) by Wiberg-key radiographic parameters in pediatric hip assessment-have changed in a contemporary pediatric population compared with historical reference values, considering trends in earlier skeletal maturation and body composition.

MATERIALS AND METHODS: This retrospective diagnostic accuracy study assessed changes in AA and center-edge angle (CE) angles in a contemporary cohort. A total of 1774 anteroposterior pelvic radiographs (3548 hips) from patients aged 0-18 years without hip pathology were analyzed. Radiographs were obtained between 2006 and 2018 and measured using the Supervisely digital platform. Standardized anatomical landmarks were applied. Data were stratified by age and sex and compared with historical values using two-sample t-tests ($p < 0.05$). To minimize measurement bias, two independent observers conducted all assessments using standardized digital protocols.

RESULTS: A total of 1774 patients aged 0-18 years (mean age, 8.30 ± 5.20 years; 666 females, 1108 males) were evaluated. AA values did not show statistically significant differences in 11 of 16 age groups compared with historical data ($p > 0.05$). In contrast, LCEAs were significantly higher in the contemporary cohort, especially in the 5-8, 9-12, and 12-16 age groups (all $p < 0.01$) […]

PubMed · DOI

In-hospital deep vein thrombosis after tibial plateau fractures: incidence, laterality/anatomy, and risk factors in a multicenter retrospective cohort of 3366 patients.

Yang S, Li G, Li Y, Long Y, Zhang J, Liu L, Zhao Z, Sun C et al.

BACKGROUND: We aimed to determine the cumulative in-hospital incidence, anatomic distribution, and predictors of duplex ultrasonography-detected lower-extremity deep vein thrombosis (DVT) in patients hospitalized with tibial plateau fractures under routine thromboprophylaxis and protocolized surveillance.

METHODS: We retrospectively analyzed consecutive adults (≥ 18 years) admitted to two level-I trauma centers (November 2014-January 2024). Bilateral duplex ultrasonography was performed on admission/preoperatively (as soon as feasible after arrival and prior to surgery) and repeated serially during hospitalization (approximately every 5-7 days and when clinically suspected). The primary outcome was cumulative in-hospital DVT from admission to discharge; isolated intermuscular calf-muscle vein thrombosis (soleal/gastrocnemius) was recorded descriptively but excluded from the primary endpoint. Multiple injuries (polytrauma) were defined as ≥ 1 additional acute traumatic lesion beyond the index tibial plateau fracture documented on admission. Pulmonary embolism (PE) was assessed only when clinically suspected and confirmed by computed tomography pulmonary angiography (CTPA). Multivariable logistic regression identified independent predictors; receiver operating characteristic (ROC) analysis evaluated discrimination for key continuous predictors.

RESULTS: Among 3366 patients, 675 […]

PubMed · DOI

Retained articulating spacers (1.5 stage) in chronic knee periprosthetic joint infections: an analysis of associate factors and outcomes in a single center cohort of patients.

Balato G, Ascione T, Festa E, De Mauro D, Di Gennaro D, Rosa D, Mariconda M

BACKGROUND: Two-stage revision remains the gold standard for chronic periprosthetic joint infection (PJI), but it can significantly impact patients' quality of life and functional recovery. The 1.5-stage revision, involving indefinite spacer retention, has emerged as a potential alternative in selected high-risk patients. Our questions were: (1) What are the early functional outcomes, including their infection-free survival? (2) Which pre- and perioperative factors are associated with spacer retention? and (3) What is the mechanical survivorship of retained spacers?

MATERIALS AND METHODS: This was a retrospective observational study including 108 consecutive patients with chronic PJI between 2016 and 2020 who were followed prospectively for clinical outcomes. Functional status and quality of life were assessed using the EuroQol-5-Dimension-5-Level (EQ-5D-5L), Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC), and Knee Society Score (KSS) before spacer placement and prior to scheduled reimplantation. Infection-free status was defined as the absence of clinical, radiographic, and laboratory signs of infection recurrence over a 24-month follow-up. Univariate and multivariate logistic regression were used to identify factors associated with spacer retention.

RESULTS: All patients initially underwent articulating spacer implantation as part of a planned two-st […]

PubMed · DOI

Moderate leg length discrepancy: a long-term risk factor for hip osteoarthritis?

Aprato A, Donis A, Via RG, Scandurra A, Tuè F, Audisio A, Fusini F, Massè A

BACKGROUND: Leg length discrepancy (LLD) has been implicated as a biomechanical factor contributing to hip osteoarthritis (OA), yet the extent of its influence remains unclear. This study examines the correlation between moderate LLD (≥ 10 mm) and hip OA progression, focusing on asymmetrical OA distribution in patients over 65 years old.

MATERIALS AND METHODS: A retrospective analysis was conducted from a database of 1672 full-length standing X-rays. Patients under 65 years, with deformities, unilateral limb issues, or prosthetics, were excluded; therefore, the study group was composed of 220 patients. Tibial and femoral lengths were measured bilaterally, and hip OA severity was assessed using the Tönnis classification. Statistical analyses included Pearson's Chi-squared test and linear regression to explore correlations between LLD and OA distribution.

RESULTS: Among the sample, 18% showed an LLD ≥ 10 mm. A significant correlation was found between LLD and the asymmetrical distribution of hip OA ($p = 0.002$), with higher OA severity observed in the hypometric limb. Linear regression analysis suggested that each millimeter of LLD corresponded to a 0.74-unit change in OA severity difference between hips.

CONCLUSIONS: This study highlights a significant association between moderate LLD and contralateral hip OA in the elderly, emphasizing the biomechanical impact of asymmetrical […]

PubMed · DOI

Authorship, titles and open access as drivers of citation performance in orthopaedics: a scientometric analysis.

Migliorini F, Vaishya R, Rivera F, Eschweiler J, Kobbe P, Betsch M, Oliva F, Maffulli N

BACKGROUND: Bibliometric analyses are increasingly used to explore how scientific knowledge is created, disseminated, and perceived. In orthopaedics, research output has expanded rapidly over the past decade, yet the factors determining whether an article achieves wide visibility and scholarly impact remain poorly understood. Beyond the inherent quality of a study, elements such as authorship patterns, title construction, and open access (OA) availability may play an essential role in shaping citation performance. However, evidence in this field is still limited and sometimes contradictory, highlighting the need for large-scale, field-specific analyses.

METHODS: Orthopaedic publications from 2010 to 2020 were identified in Scopus using the keyword 'orthopaedic'. After duplicate removal, 97,806 unique articles were included with complete data on authorship, titles, citation counts, study design, and OA status. Citation rates were normalised per year since publication. Associations between bibliographic features and citation performance were assessed using multiple linear regression, while differences across title styles and study designs were evaluated with comparative statistical testing. Exploratory modelling was performed to identify combinations of authorship and title characteristics linked to the highest predicted citation rates.

RESULTS: Larger author teams were associa […]

PubMed · DOI

Interpreting nonsignificant findings: Reassessing the safety of self-locking standalone cages in octogenarian patients undergoing ACDF.

Abudayeh A, Fishchenko I

PubMed · DOI

A novel, simple, and affordable technique for arthroscopic soft-tissue tenodesis of the long head of the biceps tendon.

Fossati C, Ioppoli A, Ravaglia R, Vaja F, Menon A, Randelli PS

BACKGROUND: The purpose of this study was to describe a novel, simple, and implant-free arthroscopic soft-tissue tenodesis technique of the long head of the biceps tendon (LHBT) to the rotator cuff using an absorbable monofilament suture and to evaluate its clinical and functional outcomes at a minimum 1-year follow-up.

METHODS: A retrospective case series of 23 patients (mean age 58.0 ± 7.8 years) who underwent arthroscopic rotator cuff repair with concomitant LHBT soft-tissue tenodesis between June 2021 and June 2023 was analyzed. Functional outcomes were assessed using Constant-Murley Score (CMS), American Shoulder and Elbow Surgeons (ASES) score, Single Assessment Numeric Evaluation (SANE), visual analog scale (VAS) for pain, and the Long Head of Biceps (LHB) score. Supination strength was measured with a handheld dynamometer and compared with the contralateral side. Incidence of Popeye deformity and tenderness over the bicipital groove were recorded.

RESULTS: Objective Popeye deformity was observed in 13% of patients, with subjective concern reported by only 4.3%. Supination strength and LHB scores were similar to the contralateral side (means of 104.65 versus 104.64 N; LHB 93.7 versus 94.6 points). The mean CMS and ASES scores were 90.3 ± 12.4 and 89.6 ± 15.9 points, respectively. The SANE score averaged 87.4 ± 20.9, and the VAS for pain was low (1.65 ± 2.59 cm).

CONCL […]

PubMed · DOI

ORIF of three- and four-part proximal humeral fractures: a retrospective study of long-term clinical and complication outcomes of two-plate fixation methods.

Conteduca J, Gasbarra E, Rausa I, Casto A, Valentino L, Solarino G, Meccariello L, Rollo G

BACKGROUND: The aim of this study was to compare clinical outcomes and complication rates during long-term follow-up (≥ 6 years) after treatment for three- and four-part proximal humeral fractures using two different types of locking plates.

MATERIALS AND METHODS: A total of 113 patients with three- and four-part proximal humeral fractures who underwent surgery between September 2012 and January 2019 were enrolled retrospectively. Data for 49 patients [9 males, 40 females; mean age [standard deviation] (SD) 68.9 (5.8) years} treated with a PGR (intrauma) plate (group A) and 51 patients [10 males, 41 females; mean age (SD): 66.0 (7.5) years] treated with a PHILOS humeral plate (PHP, group B) were available at the last follow-up. The mean follow-up periods in groups A and B were 10.8 and 10.2 years, respectively, with a minimum follow-up of 6 years. At the final follow-up evaluation, functional outcomes were assessed using the Oxford Shoulder Scale (OSS), Simple Shoulder Test (SST), and Disabilities of the Arm, Shoulder and Hand (DASH) questionnaires. X-ray evaluation was also performed, and complications were recorded.

RESULTS: The mean OSS score in the PGR and PHP groups at follow-up was 42.9 and 39.1, respectively ($p > 0.05$). The SST score (PGR: 7.5 ± 2.0; PHP: 7.3 ± 4.0) and DASH score (PGR: 22.1 ± 5.5; PHP: 20.5 ± 4.0) were similar in both groups ($p > 0.05$). In our serie […]

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Incidence rates and treatment of the transcervical fracture of the neck of femur in Italy: is total hip arthroplasty an increasingly preferred approach? A population study on trends between 2001 and 2023 based on 1,120,770 hospital discharge records.

Ciminello E, Romanini E, Venosa M, Cazzato G, Tucci G, Boniforti F, Carpanese L, Cuccu A et al.

INTRODUCTION: Transcervical femoral neck fractures (TFNFs) are among the most devastating fragility fractures in the elderly. TFNF are associated with excess 1-year mortality rates ranging from 15% to 30%. Treatments include conservative methods, internal fixation, and arthroplasty (partial or total hip arthroplasty). This study aims to analyze the changes in incidences of TFNF in the Italian population between 2001 and 2023 and the evolution of the choices of treatment.

MATERIALS AND METHODS: Using hospital discharge record (HDR) data from 2001 to 2023, records with ICD9-CM codes for femoral neck fractures (820.0 and 820.1) among diagnoses were selected and categorized into four treatment groups: total arthroplasty, partial arthroplasty, fixation, and conservative. Time series were analyzed with stratification by sex and age.

RESULTS: The extracted data included 1,120,724 records of TFNFs, with 871,161 cases treated surgically (total or partial arthroplasty or internal fixation) and 249,563 treated conservatively; the average patient age was 79.1 years, with a higher proportion of women (72.8%). Partial hip arthroplasty was the preferred treatment overall. For younger patients, in the age classes < 45 and 45-54 years, fixation was the most chosen treatment. Over time, the use of the conservative treatment decreased from 27.5% in 2001 to 14.6% of cases in 2023. The use of par […]

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Does latissimus dorsi tendon transfer provide a normal scapular rhythm and external rotation strength in posterosuperior massive irreparable rotator cuff tears? A kinematic analysis in a retrospective cohort.

Calvo C, Fiumana G, Brigo A, Ruggiero ED, Bonfatti R, Donà A, Micheloni GM, Giorgini A et al.

BACKGROUND: Posterosuperior massive irreparable rotator cuff tears (PMIRT) are rare and disabling conditions. When conservative treatment fails, latissimus dorsi tendon transfer (LDTT) is a viable surgical option for symptom relief in carefully selected patients. However, its effectiveness in restoring glenohumeral function and its influence on scapulothoracic rhythm remain subjects of ongoing debate.

PURPOSE: The purpose of this study was to evaluate the clinical outcomes of LDTT and assess its impact on scapulothoracic rhythm using kinematic and electromyographic analysis.

MATERIAL AND METHODS: A total of 18 patients with PMIRT underwent LDTT. Functional scores consisted of Constant-Murley score (CMS), America Shoulder and Elbow Surgeons (ASES) score, and Quick Disabilities of the Arm, Shoulder, and Hand (QuickDASH) score. Electromyography (EMG) activity of the latissimus dorsi muscle was evaluated in conjunction with three-dimensional (3D) kinematic tracking system. Differences in external rotation strength were compared with and without active adduction.

RESULTS: The mean age was 55.9 years (range 40-69 years), with a mean follow-up of 20.4 months (range 6-39 months). At final follow-up, the mean CMS was 68.2 (95% CI 64.9-71.5), ASES was 76.9 (95% CI 66.1-87.6), and QuickDASH was 17.6 (95% CI 9.5-25.6). A significant difference in external rotation strength was observed […]

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Is there a need for exploration in pulseless supracondylar fractures of the humerus? A systematic review and individual patient data meta-analysis.

Haoyang C, Chen JG, Lai H, Lim AKS, Tan SHS, Hui JHP

BACKGROUND: Different approaches have been proposed to treat patients with pulseless supracondylar humeral fractures (SHF). We aim to analyze the current outcomes of patients who have undergone different treatment options for pulseless SHF, namely watchful waiting versus urgent surgical exploration.

METHODS: Electronic databases including PubMed, Embase, and Cochrane Library were explored. Our study included patients from all age groups but only included English-language articles. Single case reports and case studies with adequate description of patient population, injury, and outcomes were included. An individual patient data meta-analysis was done to evaluate key outcomes of surgical exploration versus no surgical exploration in pulseless patients with SHF.

RESULTS: Overall, the data for 1070 individual patients from a total of 48 studies were included. Patients with pulseless SHF with open fractures have a higher probability of requiring vascular intervention ($p < 0.001$) as they are more likely to have disrupted arteries ($p = 0.050$) and more likely to require vascular repairs ($p < 0.001$) than patients with closed fractures. Similarly, patients with pulseless SHF with pucker sign ($p = 0.003$) and ecchymosis ($p = 0.002$) were more likely to undergo surgical exploration. However, the neurological status had no relation to them undergoing surgical exploration ($p = 0.382$), and al […]

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Increasing complexity and plateauing outcomes in AO type C distal humerus fractures: a 50-year institutional case series.

Hönck K, Wenzelburger P, Eibinger N, Sadoghi P, Puchwein P

BACKGROUND: Distal humerus fractures in adults are rare but complex injuries, particularly AO type C3 fractures, often involving comminution, instability, and osteoporotic bone. Demographic trends indicate rising incidence, especially among elderly patients, with increasing functional demands challenging modern osteosynthesis. Open reduction and internal fixation (ORIF) with bicolumnar plating remains the preferred reconstructive approach, although complications remain substantial. This study aimed to evaluate trends in fracture complexity, patient demographics, and functional outcomes over five decades, focusing on the most recent cohort (2009-2018, series E) and comparing with historical cohorts (series A-D, 1969-2008).

METHODS: This retrospective analysis included five consecutive 10-year institutional cohorts (series A: $n = 43$; B: $n = 29$; C: $n = 47$; D: $n = 58$) of 233 patients with 235 supradiacondylar distal humerus fractures (AO type C1-3) treated with ORIF. Functional outcome parameters (Jupiter and Cassebaum Scores) and demographic data were available for all cohorts; additional parameters (range of motion (ROM), QuickDASH Score) were available for series D and E, Mayo Elbow Performance Score (MEPS), Short Form (SF)-36, and strength were assessed specifically in series E ($n = 56$; follow-up $n = 21$). Cross-cohort comparisons of categorical variables used chi-square tests; […]

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Magnetic resonance imaging-based fat infiltration grading improves reparability prediction in massive rotator cuff tears.

Hu Q, Zhang G, Wang D, Tung TH, Ding J, Zhou X

PURPOSE: This study aims to investigate the extent of fat infiltration (FI) in the rotator cuff (RC) muscles in more medial slices from the Y-shaped view (Y-view) on shoulder magnetic resonance imaging (MRI) and assess whether these slices provide a more accurate prediction of the repairability of massive RC tears (massive RCTs).

METHODS: The retrospective study included 57 patients with massive RCTs who successfully underwent arthroscopic surgery between 1 January 2023 and 31 December 2023. Patients were categorized into two groups: the irreparable group and the reparable group. All patients underwent shoulder MRI covering the region from the acromion to the medial border of the scapula in oblique coronal and oblique sagittal orientations. The FI stage of the RC was assessed across three different views to determine which view most effectively predicts the repairability of patients with massive RCTs.

RESULTS: The FI stage of the infraspinatus (IS) muscle across three distinct views demonstrated a significant correlation with the repairability of massive RCTs. The FI stage was more severe in the irreparable group. Receiver operating characteristic (ROC) analysis revealed that the area under the curve for the 1/2 scapula view (0.737) and the most severe view (0.745) exceeded that of the traditional Y-view (0.644). Paired-sample ROC curve analysis revealed a significant differe […]

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Mid-term clinical and functional outcomes after reverse shoulder arthroplasty with latissimus dorsi transfer.

Colombini AG, Rab P, Macken AA, Soares MN, Kimmeyer M, Shirinskiy IJ, Popescu IA, Lafosse L et al.

BACKGROUND: Although reverse total shoulder arthroplasty (rTSA) with concomitant latissimus dorsi transfer (LDT) has been shown to effectively treat external rotation (ER) deficits, there are limited data regarding its outcomes with modern implants and its impact on activities of daily living (ADLs) requiring ER. The purpose of this study was to assess the mid-term clinical and radiographic outcomes of rTSA with concomitant isolated LDT in patients with an ER lag sign and posterior rotator cuff deficiency.

METHODS: This retrospective cohort study with prospective follow-up included consecutive patients who underwent rTSA with concomitant isolated LDT between 2010 and 2022 with a minimum follow-up of 2 years. Primary outcomes included the resolution of ER lag sign, active ER, and the Activities of Daily Living and External Rotation (ADLER) score. Secondary outcomes included the Auto-Constant Score (CS), Subjective Shoulder Value (SSV), activities of daily living requiring internal rotation (ADLIR) score, visual analog scale (VAS) for pain, and radiographic analysis of standardized radiographs.

RESULTS: In total, 32 procedures in 32 patients were identified. Of these, 22 procedures in 22 patients (68% female, 72.9 ± 8.4 years at surgery) were available for follow-up at 4.8 ± 2.2 years postoperatively (response rate 73%). The ER lag sign resolved in 95.5% of patients, the active […]

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Partial versus full revision in cementless single-stage exchange for hip periprosthetic joint infection: a multicenter comparative study with 10-year survivorship analysis.

Mu W, Jarusriwanna A, Xu B, Brown SA, Zhang X, Dawes A, Parvizi J, Cao L

BACKGROUND: Periprosthetic joint infection (PJI) is one of the most challenging complications following total hip arthroplasty (THA). Traditional management typically involves full revision to ensure comprehensive infection eradication. However, for patients with well-fixed implants, partial revision in a single-stage cementless approach may provide a viable alternative, potentially preserving bone stock and reducing operative time. The relative efficacy of this approach compared with full revision remains to be fully explored. This multicenter study aims to determine whether partial revision in cementless single-stage exchange surgery offers comparable infection control outcomes to full revision for select patients with well-fixed implants.

METHODS: We conducted a retrospective, multicenter cohort study involving 226 patients who underwent cementless single-stage exchange hip arthroplasty for PJI between 1 October 2013 and 31 July 2022. Patients were divided into partial revision (n = 61) and full revision (n = 165) groups. The primary outcome was treatment success, defined as the absence of clinical symptoms and signs of infection at a minimum follow-up of 2 years.

RESULTS: The success rates were 77.0% for the partial revision group and 80.0% for the full revision group, with no significant difference (p = 0.629). Both groups showed comparable 10-year survival rates for ove […]

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Journal of Orthopaedic Trauma

Volume 40, Issue 7 · July 2026 · 15 new articles

Biomimetic Hematoma Promotes Superior Bone Regeneration With Ultra-low-Dose rhBMP-2 in a Goat Large-Defect Model.

Glatt V, Aguilar L, Agarwal A, Woloszyk A

OBJECTIVE: The aim of this study was to evaluate whether an ultra-low dose of recombinant human bone morphogenetic protein-2 (rhBMP-2) delivered within a biomimetic hematoma (BH) scaffold could regenerate a large bone defect in goats more efficiently than a high dose of rhBMP-2 delivered on an absorbable collagen sponge (ACS). The BH is an autologous scaffold, created from whole blood and defined concentrations of coagulants, and is designed to closely mimic the structural and biological properties of a naturally healing fracture hematoma. It initiates the body's intrinsic healing cascade, thereby recapitulating the sequential phases of bone repair.

METHODS: A 2.5-cm defect was created in goat tibias and treated with either a BH scaffold containing ultra-low doses of rhBMP-2 (210 and 42 µg) or ACS with 2.1 mg of rhBMP-2. Empty defects and contralateral tibias served as controls. Bone regeneration was evaluated biweekly using radiographs and further assessed with micro-CT and histology at 8 weeks.

RESULTS: A total of 12 specimens, three per group, were used for radiographic, micro-CT, and histological evaluations. By 8 weeks, radiographic scores showed complete regeneration in the BH+210 µg (5.0 ± 0.0) and BH+42 µg (4.7 ± 0.2) groups. The ACS+2.1 mg group scored 4.3 ± 0.4 with some unmineralized regions while the ED group showed limited healing (2.6 ± 0.2; P < 0.001 vs. all ot […]

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Breaking Down the Evidence: A Multicenter Analysis of Venous Thromboembolism Among Trauma Patients with Lower Extremity Fractures.

Knowlton LM, Guorgui JG, Wang S, Arnow K, Knudson MM, CLOTT Study Group

OBJECTIVES: To determine whether chemoprophylaxis initiation within 24 hours reduces venous thromboembolism (VTE) risk among trauma patients with lower extremity long bone fractures.

DESIGN: This was a retrospective cohort study.

SETTING: Seventeen Level I trauma centers as a part of the Consortium of Leaders in the Study of Traumatic Thromboembolism study group.

PATIENT SELECTION CRITERIA: Patients aged 18-40 years with a diagnosis of lower extremity long bone fracture between January 1, 2018, and December 31, 2020, were included.

OUTCOME MEASURES AND COMPARISONS: Primary outcome was VTE during admission. Patients were compared based on VTE chemoprophylaxis initiation within 24 hours (early prophylaxis, E-PROPH) versus not (late or no prophylaxis, L-PROPH) using inverse probability weighted Cox survival analysis.

RESULTS: In total, 120 (5.3%) among 2264 patients with lower extremity fractures developed VTE. 57.5% received E-PROPH and 42.5% received L-PROPH. The E-PROPH group had fewer patients with an injury severity score ≥16 (30.3% vs. 52.4%, P < 0.001) and a lower proportion of associated head injury (8.1% vs. 26.7%, P < 0.001). VTE incidence was significantly higher in the L-PROPH group than in the E-PROPH group (8.6% vs. 2.8%, P < 0.001). In the adjusted model, E-PROPH was independently associated with nearly a half reduction in VTE incidence (hazard ratio: 0.54, 95% […]

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Response to "Competing-Risks Re-estimation of Five-Year Second Hip Fracture Risk: Accounting for Death as a Competing Event".

Konda SR

PubMed · DOI

Competing-Risks Re-estimation of Five-Year Second Hip Fracture Risk: Accounting for Death as a Competing Event.

Shaikh FA, Curran T, Nemeth ZH

PubMed · DOI

Letter to the Editor Regarding "Proton Pump Inhibitors Are Associated With Increased Risk of Site-Specific Nonunion After Open Reduction Internal Fixation".

Ayas O, Acar A

PubMed · DOI

Response to the Letter to the Editor Regarding "Proton Pump Inhibitors Are Associated With Increased Risk of Site-specific Nonunion After Open Reduction Internal Fixation".

Romoff M, Kim MS, Scolaro JA

PubMed · DOI

The Effect of Preoperative Tranexamic Acid on Blood Transfusions in Geriatric Hip Fracture Surgery: A Randomized Controlled Trial.

Sullivan M, Perea LL, Bradburn E, Guglielmo K, Bresz K, Horst M, Martin B, Heinle C et al.

OBJECTIVES: To evaluate whether preoperative intravenous tranexamic acid (TXA) reduces postoperative blood transfusion rates in geriatric patients undergoing operative treatment for hip fracture.

DESIGN: Prospective, double-blinded, randomized controlled trial terminated early after interim futility analysis.

SETTING: Single-center Level I trauma center.

PATIENT SELECTION CRITERIA: Patients aged ≥65 years who underwent operative treatment for femoral neck (AO/OTA 31-B), intertrochanteric region (AO/OTA 31-A), or subtrochanteric (AO/OTA 32-A/B/C) fractures between June 2019 and June 2022 were included. Procedures included arthroplasty (hemiarthroplasty or total hip arthroplasty) and internal fixation (ORIF) with intramedullary nailing or sliding hip screw fixation. Exclusion criteria were recent thromboembolic events, cancer, hypercoagulable disorders, or TXA allergy.

OUTCOME MEASURES AND COMPARISONS: The primary outcome was postoperative transfusion. Secondary outcomes included hospital length of stay, 30-day readmission, and 90-day complications and mortality. Statistical comparisons were performed using odds ratios, chi-squared, and t-tests, with significance thresholds adjusted by O'Brien-Fleming criteria.

RESULTS: A total of 283 patients were analyzed {TXA: 146 [mean age 84 (range 65-100), 71.9% female; placebo: 137 [mean age 83.1 (range 65-100), 77.2% female]}. Baseli […]

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Perioperative Use of Tranexamic Acid Is Associated With Reduced Transfusion Rates in Femoral Shaft Fractures: A Propensity-Matched Cohort Study.

Kim D, Shen V, Norris Z, Ranson R, Sterling R, Villa J

OBJECTIVES: To evaluate the association between perioperative tranexamic acid (TXA) use and transfusion rates, perioperative blood loss, and 90-day postoperative complications in patients undergoing intramedullary nailing of femoral shaft fractures.

DESIGN: Retrospective cohort study.

SETTING: Multicenter academic and community hospitals participating in the TriNetX research network.

PATIENT SELECTION CRITERIA: Adult patients (≥18 years) who underwent open intramedullary nailing of OTA/AO 32 (femoral diaphyseal) fractures between 2010 and 2025 were included. Patients with a history of malignant neoplasm, deep vein thrombosis (DVT), and pulmonary embolism (PE) were excluded. Patients were required to have ≥90 days of postoperative follow-up. The TXA group included patients who received perioperative TXA, while the comparison group did not.

OUTCOME MEASURES AND COMPARISONS: Primary outcomes included the incidence of nonautologous red blood cell transfusion and 90-day postoperative complications. Minor and major complications analyzed were mortality, DVT, PE, myocardial infarction, stroke, and renal failure. Propensity-score matching was performed to balance demographics and comorbidities. Chi-squared tests were used to compare outcomes.

RESULTS: A total of 7644 patients were included in the study. The mean age was 57.0 ± 23.2 in the TXA group (n = 3,822, 54.1% women) and 57. […]

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Long Distally Unlocked Hip Nails Versus Short Distally Locked Nails for Peritrochanteric Fractures: Operative Characteristics and Peri-implant Fracture Risks.

Ayoub M, Tallapaneni J, Robles E, Vue Y, Lindvall E

OBJECTIVES: To compare the primary outcome of surgical time for long, distally unlocked, intramedullary nails (LUIMNs) versus short, distally locked, intramedullary nails (SLIMNs) and to compare secondary outcomes of blood loss, peri-implant fractures, reoperation events, and mortality between these 2 groups.

DESIGN: Retrospective cohort study.

SETTING: One Level I Trauma Center and one community hospital.

PATIENT SELECTION CRITERIA: Patients older than 18 years treated with SLIMNs or LUIMNs for peritrochanteric hip fractures, OTA/AO 31A1, 31A2, 31A3, were included.

OUTCOME MEASURES AND COMPARISONS: The primary outcome was surgical time for LUIMNs versus SLIMNs. Secondary outcomes were estimated blood loss, peri-implant fracture rates, reoperation for nonunion or shortening, all-cause reoperation, and mortality within 2 weeks and 3 months of surgery for LUIMNs versus SLIMNs.

RESULTS: There were 602 patients in the LUIMN group (68.6% female, 31.4% male), with an average age of 77.3 (range, 29-108) years. There were 142 patients in the SLIMN group (63.4% female, 36.6% male) with an average age of 75.3 (range 33-101) years. Average OR time for LUIMNs was 46.9 minutes compared with 53.7 minutes for SLIMNs ($P < 0.001$). Peri-implant fracture rates were 0.6% for LUIMNs and 4.1% for SLIMNs ($P = 0.012$). There was no difference in estimated blood loss between LUIMNs (103.8 ± 55.1) […]

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Comparing the Efficacy Between ChatGPT 5, Grok 3, and Claude 4.5 Sonnet in Analyzing Orthopedic Trauma-Related Imaging.

Holmstrom JA, Braithwaite CL, Alhankawi AR, Moore ML, Patel KA, Miller BH

OBJECTIVES: To evaluate and compare the ability of three popular open-source artificial intelligence platforms to diagnose common trauma-related fractures using radiologic imaging.

DESIGN: Retrospective diagnostic performance comparison study.

SETTING: Publicly accessible online radiologic imaging databases.

PATIENT SELECTION CRITERIA: Five common orthopedic trauma fractures were assessed: ankle, tibial plateau, intertrochanteric, femoral neck, and humerus. Radiographs and computed tomography (CT) images were collected. Images were randomly selected from confirmed diagnoses on Radiopaedia.org .

OUTCOME MEASURES AND COMPARISONS: ChatGPT 5, Grok 3, and Claude 4.5 Sonnet were queried with each image. Diagnostic accuracy, sensitivity, specificity, positive and negative predictive values, and performance by modality (X-ray vs. CT) were assessed. The reference standard was the expert-verified diagnosis provided by Radiopaedia.org , limited to cases labeled with a "diagnosis certain" tag.

RESULTS: Each model was provided with 30 radiographs and 20 CT images whenever possible. ChatGPT 5, Grok 3, and Claude 4.5 Sonnet accurately diagnosed diseased images in 26.8%, 18.8%, and 22.4% of cases, respectively. By fracture type, ChatGPT 5 demonstrated the highest correct classification rates for ankle (10%), femoral neck (38%), humerus (40%), and tibial plateau (44%) fractures. Grok 3 dem […]

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Preoperative Quadriceps Muscle Quality Predicts Early Ambulation after Hip Fracture Surgery.

Achebe C, Cureton R, Rothberg D, Higgins T, Marchand L, Haller J

OBJECTIVES: To investigate whether preoperative muscle characteristics measured on CT scans could predict early ambulation capacity after hip fracture surgery.

DESIGN: Retrospective cohort study.

SETTING: Single level I academic trauma center.

PATIENT SELECTION CRITERIA: Patients aged ≥55 years who underwent surgical fixation for hip fractures (OTA/AO types 31A, 31B) between January 2015 and December 2024 were included. Inclusion required preoperative CT imaging and documented ambulation assessments at postoperative day 1 (POD1) and discharge. Patients with neuromuscular disorders affecting muscle mass were excluded.

OUTCOME MEASURES AND COMPARISONS: The primary outcome was distance walked (feet) during physical therapy at POD1 and discharge. Preoperative CT scans were opportunistically analyzed for muscle quality (Hounsfield units) and size (cross-sectional area) in psoas, gluteal, and quadriceps muscles. Associations were analyzed using linear mixed-effects models adjusted for body mass index, Charlson Comorbidity Index, preinjury wheelchair use, time to surgery, and fixation method.

RESULTS: A total of 140 patients (mean age 75 ± 8 years, 59% female) contributed 210 ambulation observations. Ambulation improved from 0 (IQR 0-20) feet at POD1 to 20 (IQR 0-86) feet at discharge ($P < 0.001$). In the fully adjusted mixed-effects model, quadriceps quality was the only muscle […]

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Reliability of Hip Fracture Classification and Treatment Planning on Smartphones Versus Picture Archiving and Communications Systems.

Schultz MJ, Kohring A, Bridges TN, DiCiurcio WT, Smith EB, Pulido S, Deirmengian GK

OBJECTIVES: To compare the reliability of classifying and recommending treatment for femoral neck and intertrochanteric fractures when reviewing radiographs on smartphones versus Picture Archiving and Communications Systems (PACS).

DESIGN: Retrospective cross-sectional radiographic review.

SETTING: Private academic medical center.

PATIENT SELECTION CRITERIA: Consecutive patients ≥18 years with isolated unilateral femoral neck or intertrochanteric fractures (AO/OTA 31A or 31B) and complete radiographs (anteroposterior pelvis, anteroposterior and lateral hip, and anteroposterior and lateral femur) who presented from January 1 to December 31, 2021, were included.

OUTCOME MEASURES AND COMPARISONS: Three orthopaedic surgeons (1 orthopaedic trauma attending and 2 adult reconstruction attendings) independently evaluated radiographs on 2 separate occasions, first on their smartphone and then on PACS using their personal laptop. After reviewing patient images, a survey assessing fracture characteristics and treatment plans was completed. Smartphone images were accessed through an embedded URL link within the survey, providing zooming functionality comparable with that of standard text message. The primary outcome was intraobserver agreement (smartphone vs. PACS) for fracture classification and treatment recommendations (general and specific) using Cohen kappa (k). Additional outcome […]

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Similar Outcomes in Proximal Humerus Fractures Treated With Conventus Intramedullary Cage Device and Open Reduction Internal Fixation.

Kurkowski SC, Taleghani ER, McMillan P, Solomon EG, Keller JE, Gerak SK, Wyrick JD, Grawe BM

OBJECTIVES: To compare clinical outcomes between open reduction and internal fixation (ORIF) with or without Conventus intramedullary cage (CC) for proximal humerus fractures.

DESIGN: Retrospective cohort study.

SETTING: Level 1 trauma center.

PATIENT SELECTION CRITERIA: Included were patients who sustained OTA/AO 11B and 11C fractures treated with ORIF with or without CC with minimum 1 year follow-up.

OUTCOME MEASURES AND COMPARISONS: Primary outcomes included complication and reoperation rates. Secondary outcomes were American Shoulder and Elbow Surgeons and Short Form 12 (SF-12) scores. Outcomes were compared between fixation strategies.

RESULTS: Fifty-six patients were included, 33 ORIF and 23 CC, with mean follow-up of 5.0 (range 1.1-11.4) and 5.3 (range 2.3-7.6) years, respectively. The ORIF and CC groups were 27.3% and 21.7% men, respectively ($P = 0.759$). The ORIF group had a higher mean age (68.2 ± 5.3 vs. 64.2 ± 8.7 years, $P = 0.037$). There was a higher proportion of OTA/AO 11C fractures in the CC cohort (65.2% vs. 33%, $P = 0.029$). CC group had a higher unplanned reoperation rate (18.2% vs. 34.8% for CC, $P = 0.158$) and complication rate (24.2% vs. 39.1% for CC, $P = 0.233$) that did not reach statistical significance within this available study population. Unplanned reoperations in the CC cohort included revision fixation (3) and conversion to hemiarthroplasty (2) […]

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Short-Term Outcomes of Supination External Rotation IV Bimalleolar and Deltoid Variant Fractures With No Deltoid Repair.

Malempati M, Danford NC, Chien BY, Greisberg JK

OBJECTIVES: To evaluate whether radiographic and clinical outcomes differ between supination-external rotation (SER) type IV bimalleolar (BM) fractures and deltoid ligament injuries treated without deltoid repair. A secondary aim was to compare outcomes between universal tubular (TB) and fibula-specific (FS) plates for fibular fixation.

DESIGN: Retrospective cohort study.

SETTING: Academic medical center.

PATIENT SELECTION CRITERIA: Skeletally mature patients with SER-IV (AO/OTA 44-B2) ankle fractures (either BM or deltoid ligament injury without medial malleolar fracture) treated between November 2011 and July 2024. Exclusions: pathologic/pilon fractures, congenital deformity, nonunion repair, posterior malleolar fixation, or <3 months of follow-up. Secondary comparison included rotational ankle fractures (SER, pronation external rotation) stabilized with TB or FS plates.

OUTCOME MEASURES AND COMPARISONS: Radiographic outcomes included medial clear space (MCS), superior clear space (SCS), tibiofibular clear space (TFCS), talocrural angle, and Takakura score. Clinical outcomes included complication rates, secondary surgeries, infections, and conversions to arthroplasty. Comparisons were made between intraoperative and final postoperative radiographic measurements.

RESULTS: A total of 132 patients with SER-IV fracture were included in the primary study arm, with 89 in the B […]

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"Throwing the Flag": Patient Behavior Reporting Affects Outcomes After Orthopaedic Trauma Surgery.

Mercer NP, Egol AJ, Jacobi S, Padon B, Lashgari A, Egol KA

OBJECTIVES: To evaluate the association between electronic health record-based behavioral flag designation and postoperative outcomes in orthopaedic trauma patients undergoing surgical fixation for acute fractures.

DESIGN: Retrospective cohort study with 1:1 propensity score matching.

SETTING: Level I trauma center.

PATIENTS SELECTION CRITERIA: Adult orthopaedic trauma patients who underwent operative fixation for an acute fracture and received a long-term behavioral flag issued for documented disruptive, threatening, or violent behavior toward health care staff after institutional review either before surgery or within 1 year postoperatively were included. Those with pathologic fractures and those with inadequate follow-up were excluded. Each flagged patient was matched to an unflagged control based on age, sex, body mass index, smoking status, comorbidity burden, and fracture type.

OUTCOME MEASURES AND COMPARISONS: Primary outcomes included 1-year rates of major postoperative complications (eg, fracture-related infection, nonunion, painful hardware) and reoperation. Subgroup analyses examined outcomes by timing of flag assignment.

RESULTS: A total of 116 patients were included (58 flagged patients, 58 unflagged). Flagged patients had a mean age of 53.4 ± 18.1 years (range, 19-74) and were 55.2% male; unflagged controls had a mean age of 49.0 ± 13.6 years (range, 21-71) a […]

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Archives of Orthopaedic & Trauma Surgery

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Impact of severe varus deformity on the outcome of mobile-bearing unicompartmental knee arthroplasty: a comparison of patients with varus deformity of less than and more than 15 degrees using matched-pair analysis.

Mullaji AB, Gupta AK, George GK, Kazi R

INTRODUCTION: The outcome of patients undergoing UKA with larger preoperative deformity (> 15º varus) and lesser deformity is sparsely compared in the literature. This matched-pair retrospective study attempts to fill this research gap by comparing the effect of preoperative and postoperative residual varus on outcome.

METHODS: Patients undergoing UKA with preoperative varus deformity > 15º (HKA > 15º group) were selected and matched for age and gender with those with < 15º of preoperative varus deformity (HKA < 15º group). There were 72 age and gender-matched pairs knees. HKA was measured and Forgotten Joint Score, OKS, VAS and WOMAC scores were obtained at a minimum follow up of 5 years. Postoperative residual varus was classified into 3 groups based on postoperative HKA: varus < 3º, varus 3 to 7º and varus > 7º.

RESULTS: Mean correction of deformity was significantly more ($p < 0.001$) in the HKA > 15º group (10.5º) when compared to HKA < 15º group (4.5º). There was no significant difference in outcome measures OKS ($p = 0.21$), FJS ($p = 0.35$) and VAS ($p = 0.6$) between patients in the two groups. There was no significant correlation between the preoperative and postoperative HKA of the patient with postoperative OKS, FJS and VAS in the two groups. There was no statistically significant difference between FJS ($p = 0.73$) of the three postoperative varus groups. Excellent to good […]

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A multimodal imaging approach to predict carpal tunnel syndrome after distal radius fractures.

Önüt M, Alkol N, Önalo■lu Y, Özer M, Apayd■n E, Talmaç MA

OBJECTIVE: Distal radius fractures are commonly associated with the development of carpal tunnel syndrome (CTS), yet current evidence primarily focuses on radiographic alignment parameters. The role of carpal tunnel morphology, particularly total carpal tunnel cross-sectional area (CSA) measured on computed tomography (CT), remains insufficiently investigated. This study aimed to evaluate the relationship between CSA, volar tilt, and the development of CTS in conservatively managed distal radius fractures, and to assess the predictive performance of these parameters when used in combination.

METHODS: This retrospective case-control study included 150 patients, comprising 50 patients who developed CTS and 100 controls without CTS. CSA was measured on CT scans obtained at initial presentation, while volar tilt was assessed using standard radiographs. Additional variables, including the presence of ulna fracture and fracture type, were analyzed. Group comparisons were performed using nonparametric tests. Multivariable logistic regression analysis was conducted to identify independent predictors of CTS. Diagnostic performance was evaluated using receiver operating characteristic (ROC) curve analysis.

RESULTS: Patients who developed CTS had significantly lower CSA and higher volar tilt values (both $p < 0.001$). The presence of an ulna fracture was inversely associated with CTS (p = […]

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Unicompartmental knee arthroplasty and total knee arthroplasty in patients with 15 or more degrees of varus deformity: age- and gender-matched comparative study of radiographic and clinical outcomes.

Mullaji A, Chopperla S, George GK, Kazi R

INTRODUCTION: To date, few studies have directly compared matched cohorts of patients with severe varus deformity undergoing UKA versus TKA. The current study seeks to address this gap. The hypothesis was that, in patients with deformity of > 15°, postoperative PROMs following UKA would be inferior to those achieved with TKA.

METHODS: Records of patients operated for UKA and TKA from January 2016 to December 2021 were evaluated, and the data for patients with deformity of ≥ 15° were selected. The UKA patients were then matched for age and gender with TKA patients. 52 matched pairs of patients were obtained. The HKA, mLDFA, MPTA were measured. OKS, SF12, and FJS were collected. The magnitude of change in the mLDFA and MPTA was assessed using the Estimated Marginal Means.

RESULTS: The mean follow-up times for UKA and TKA were 3.7 years and 5.5 years, respectively. The mean deformity correction (ΔHKA) was significantly greater in TKA (16.3° ± 4.5°) than UKA (10.6° ± 3.1°) ($p < 0.001$), with significant changes in mLDFA and MPTA ($p < 0.001$). In UKAs, 68% of patients demonstrated a postoperative HKA between 171.2° and 175.7°, whereas in TKAs, values clustered between 174.5° and 179.8°. Mann-Whitney U test demonstrated no statistically significant differences between the UKA and TKA cohorts across all assessed PROMs, including OKS, SF-12, and FJS ($p > 0.05$ in all). 98% of the UKA an […]

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Acetabular superior wall fractures: a distinct acetabular fracture entity.

Gänsslen A, Krappinger D, Liodakis E, Clausen JD, Omar-Pacha T, Lindtner RA, Sehmisch S, Osche DB

INTRODUCTION: A transitional "grey-zone" exists between typical locations of classical trapezoidal anterior wall and large-fragment posterior wall acetabular fractures. Part of this grey zone is the superior acetabular dome, which is commonly involved in acetabular fracture patterns, especially in geriatric patients. The aim of this study was to characterize isolated acetabular superior wall fractures and to assess their relevance as a distinct fracture pattern.

METHODS: A retrospective analysis of pelvic fracture databases from three tertiary trauma centers was performed. Acetabular fractures with superior dome involvement were screened, and cases with isolated superior articular involvement were included. Overall, seven patients were identified. Clinical and radiological data were analyzed regarding fracture morphology, associated fracture characteristics (including dislocation, comminution, marginal impaction, intra-articular fragments, and femoral head injury), concomitant pelvic ring injuries, treatment, and clinical outcomes.

RESULTS: Isolated superior wall acetabular fractures occurred following both high-energy and low-energy mechanisms, including simple falls. The mean age was 48 years and four of seven patients were male. Hip dislocation or subluxation was observed in four cases (anterior-superior or purely superior). Surgical treatment was performed in six cases, p […]

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A novel biomechanical model for reproducing and analysing the causal mechanisms of PFNA "cut-in" phenomenon.

Rasappan K, Joshua K, Chou SM, Shaw LKRM, Huang D, Yew A, Kwek EBK

INTRODUCTION: "Cut-in" is a rare but serious complication associated with cephalomedullary nail fixation for intertrochanteric hip fractures, particularly with Proximal Femoral Nail Antirotation (PFNA). This phenomenon involves paradoxical superomedial migration of the helical blade through the femoral head. Although increasingly recognised, the underlying mechanisms remain incompletely defined. This study aims to describe a new superolateral tensile loading set up in a bidirectional loading model, with the aim of incorporating possible mechanical deviations introduced by abductor weakness.

MATERIALS AND METHODS: This study was conducted in two stages. An initial stage with synthetic femurs (SYNBONE® 2420) was used to calibrate the set up. Within this stage, intramedullary canal diameter was compared (12 mm vs. 18 mm). In the final stage, 5 osteoporotic synthetic femurs (Sawbones® 3503) were used. In both stages, standardised AO/OTA 31-A1.1 intertrochanteric fracture were created in the synthetic femurs and fixed with PFNA-II implants. All femurs were subjected to cyclical bidirectional loading with a universal testing machine. Each cycle consisted of axial compression (720 N) and superolateral tensile loading (120 N) at 2 Hz. This configuration was designed to simulate altered hip biomechanics postulated with the trendelenburg gait. Testing was continued until mechanical fail […]

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Routine intraoperative infection testing in presumed aseptic hip and knee revision: a matched cohort retrospective study.

Loppini M, Rocchi C, Di Maio M, Chiappetta K, Bulgarelli A, Grappiolo G

INTRODUCTION: Periprosthetic joint infections (PJIs) complicate 1-2% of primary and up to 4% of revision arthroplasties, and their diagnosis remains challenging due to the lack of a diagnostic gold standard. Differentiating PJI from aseptic failure is particularly difficult in low-grade or biofilm-related infections. Routine intraoperative microbiological screening is not recommended in presumed aseptic revisions. This study aimed to evaluate the association between routine intraoperative microbiological screening and implant survival in presumed aseptic revision arthroplasty. A secondary objective was to identify preoperative predictors of unexpected PJIs.

METHODS: Retrospectively collected data of patients undergoing presumed aseptic THA and TKA revisions between 2013 and 2019 were included. Two matched cohorts were compared: a screened group (2016-2019) undergoing routine intraoperative microbiological sampling, and an unscreened one (2013-2015) without systematic screening. Kaplan-Meier survival analysis with log-rank testing assessed overall and infection-free survival. Univariate conditional logistic regression and receiver operating characteristic (ROC) analyses evaluated potential associations between preoperative serum markers and unexpected PJI.

RESULTS: A total of 559 patients were included, of whom 295 underwent screening and 264 did not. Unexpected infections occ […]

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Artificial intelligence advancements for orthopaedic clinical reasoning: longitudinal assessment of newer models (ChatGPT-5, Grok-3, Gemini 2.5 Flash) compared to clinicians.

Agharia S, Soroush S, Ameen D, Zhou Y

INTRODUCTION: This descriptive study aimed to longitudinally evaluate the performance of contemporary large language models - ChatGPT-5, Gemini 2.5 Flash, and Grok-3 - on orthopaedic clinical multiple-choice tasks, benchmarked against pooled clinician consensus. A secondary aim was to assess whether recent advances in generative AI translated into improved alignment with clinician consensus compared with previous AI models.

MATERIALS AND METHODS: A total of 97 multiple-choice clinical cases spanning eight orthopaedic subspecialties were sourced from OrthoBullets and previously benchmarked against aggregated responses from thousands of practising clinicians. Using identical methodology to our 2023 study of ChatGPT-3.5, ChatGPT-4, and Bard, each model was prompted with standardised case stems and response options. The primary outcome was the proportion of AI responses matching the most popular clinician response; secondary analyses assessed agreement within 10% and 20% of clinician consensus, performance on 'controversial' (< 25% margin) questions, and inter-model concordance using Cohen's kappa coefficients.

RESULTS: Gemini 2.5 Flash achieved the highest alignment with clinician consensus (69.1%), followed by Grok-3 (66.0%) and ChatGPT-5 (58.8%). None of the LLMs refused to respond to any prompts, representing a reduction from 7.2% from our 2023 study. Subspecialty analysis de […]

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Trajectory of urinary incontinence symptoms following total hip arthroplasty for hip osteoarthritis.

Kadowaki M, Fukutani S, Ushio K, Uchio Y

INTRODUCTION: Urinary incontinence (UI), a prevalent quality-of-life-limiting condition, frequently coexists with hip osteoarthritis, potentially through pelvic floor dysfunction related to obturator internus muscle involvement. Several reports suggest improvement after total hip arthroplasty (THA), although postoperative time-course changes remain unclear. This study aimed to clarify the temporal trends in UI after THA and examine association of postoperative UI trajectories with functional recovery.

MATERIALS AND METHODS: This retrospective study included 118 female patients who underwent THA for hip osteoarthritis between October 2021 and March 2024. UI, assessed using the International Consultation on Incontinence Questionnaire-Short Form (ICIQ-SF), was defined as ICIQ-SF ≥ 1 preoperatively and at 3, 6, 9, and 12 months postoperatively, and classified as stress (SUI), urge (UUI), or mixed (MUI). Hip-related patient-reported outcomes were evaluated using the Japanese Orthopaedic Association Hip Disease Evaluation Questionnaire (JHEQ). Associations with BMI, age, surgical approach (direct lateral/Bauer vs. modified Watson-Jones/OCM), and UI type were tested using Mann-Whitney U, chi-squared, and repeated-measures ANOVA tests.

RESULTS: Of 118 patients (mean age at surgery, 67 years; mean BMI, 24.3 kg/m2) preoperatively, 73 (61.9%) reported UI (mean ICIQ-SF 7.88), including S […]

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The grand piano ratio: an adjunctive intraoperative screening tool for detecting excessive femoral component external rotation in total knee arthroplasty.

Avci Ö, Öztürk A, Güler BO, Bayrak HÇ, Kaya AÖ, Do■an Y

BACKGROUND: Accurate femoral component rotation is essential for optimal outcomes in total knee arthroplasty (TKA), yet reliable intraoperative assessment remains challenging. The "grand piano sign" has been described as a qualitative visual cue, but its quantitative clinical value has not been clearly established. This study aimed to evaluate the diagnostic performance of the grand piano ratio as an intraoperative tool for detecting excessive femoral component external rotation using advanced postoperative imaging as a reference standard.

METHODS: A retrospective analysis was conducted on 170 knees undergoing primary TKA. The intraoperative grand piano ratio was measured after completion of all femoral resections and before trial component implantation using the anterior femoral resection surface. Postoperative femoral component rotation was assessed using MAVRIC-sequence magnetic resonance imaging referenced to the surgical transepicondylar axis (sTEA). Excessive femoral component external rotation was defined as postoperative external rotation greater than 3° relative to the sTEA reference. Receiver operating characteristic (ROC) analysis was performed to evaluate diagnostic performance of the grand piano ratio and determine the optimal cutoff value, while multivariable logistic regression analysis was used to identify independent predictors of excessive external rotation. […]

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The shoulder stiffness scale: a validated tool for monitoring stiffness in arthroscopic rotator cuff reconstructions.

Bouaicha S, Biner M, Selman F, Wieser K, Müller AM, Audigé L

BACKGROUND: Postoperative shoulder stiffness after arthroscopic rotator cuff repair is common and clinically heterogeneous. Established shoulder scores assess pain and function, but do not specifically isolate stiffness during follow-up. The aim of this study was to validate the newly developed Shoulder Stiffness Scale (SSS) as a pragmatic monitoring tool for postoperative shoulder stiffness after arthroscopic rotator cuff repair (ARCR).

METHODS: From a cohort of 973 primary arthroscopic rotator cuff reconstructions, patients without concomitant joint pathology of the same or contralateral arm were included. The new SSS was based on three items related to shoulder pain, subjective and objective ROM ranging from 0 to 10 points. It was collected before surgery, as well as 6 and 12 months postoperatively, along with other pain level parameters, Constant Score (CS), Subjective Shoulder Value (SSV), Oxford Shoulder Score (OSS). Changes of the SSS and its subscales over time, Cronbach's alpha for internal consistency, bottom and ceiling effects were reported. Validity was investigated through SSS associations with other parameters of similar dimension, and functional scores.

RESULTS: The selected cohort with 740 patients showed postoperative shoulder stiffness in 12.7 %. The SSS averaged 5.1 points (SD 2.5) preoperatively, and decreased significantly to mean 2.9 (SD 2.4) and 1.6 (S […]

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Isolated screw fixation of posterior wall fractures.

Clausen JD, Özbek H, Omar-Pacha T, Sehmisch S, Gänsslen A

INTRODUCTION: Posterior wall (PW) fractures are common acetabular fractures and are most often treated by open reduction and internal fixation. The gold-standard of surgical stabilization is screw fixation of large wall fragments with additional buttress plating. Some reports focussed on screw fixation alone presenting adequate results.

MATERIAL AND METHODS: From a total of 208 PW-fracture treated between 1972 and 2008, 57 patients were identified with open reduction and internal fixation (ORIF) using isolated screw fixation. These patients were analyzed regarding demographical data (patient age, sex), type of accident, injury mechanism, Injury Severity Score (ISS), concomitant pelvic ring injuries, associated injuries, perioperative parameters and long-term results.

RESULTS: 44 patients were male and 13 were female with a mean age of 36.8 years. More than 90% sustained high-energy trauma with 34 sustaining multiple injuries or polytrauma. The mean Injury Severity Score was 11.5 points. All patients had fracture dislocations, which were reduced within 24 h after admission except one patient with secondary transfer. 11 patients presented with an initial sciatic nerve deficit. 29 patients had a single PW-fragment, 16 presented with two PW-fragments and in 12 patients marginal impactions were present. Surgery was performed in average after 7 days using the Kocher-Langenbeck appr […]

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Research progress on debridement, antibiotics, and implant retention (DAIR) for the treatment of periprosthetic joint infection after artificial joint replacement.

Feng W, Zhang Z, Pu R, Wang H, Liu R, Sun Y, Zhang G

Periprosthetic joint infection (PJI) is one of the most serious complications following artificial joint replacement, posing significant treatment challenges and a heavy clinical burden. Debridement, antibiotics, and implant retention (DAIR) has become the preferred treatment strategy for acute PJI due to its advantages of minimal invasiveness, functional preservation, and cost-effectiveness. This article reviews the latest advances in optimizing the indications, timing of intervention, key technical details, and multimodal anti-infection strategies for DAIR. It focuses on discussing patient selection models (e.g., KLIC score), the impact of modular component exchange on biofilm eradication, the synergistic role of chemical debridement, and the central position of biofilm-active therapy in penetrating biofilms. Furthermore, this article proposes that future efforts should integrate molecular diagnostics, intelligent antimicrobial materials, and artificial intelligence prediction models to achieve individualized precision therapy for PJI, providing a theoretical basis for clinical decision-making.

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Tranexamic acid in high-risk shoulder arthroplasty patients: safety across thromboembolic, cardiac, renal, and neurologic risk profiles.

Parmar T, Adio A, Handy A, Daher M, Kellish A, Khan A, Abboud J

INTRODUCTION: Tranexamic acid (TXA) effectively reduces blood loss and transfusion requirements in shoulder arthroplasty. However, concerns regarding thromboembolic, neurologic, cardiac, and renal complications have limited its use in medically high-risk patients. This study evaluated temporal trends and the safety of TXA in high-risk patients undergoing total shoulder arthroplasty (TSA).

METHODS: A retrospective cohort study was conducted using TriNetX, identifying patients who underwent shoulder arthroplasty (2012-2025). Patients were stratified by preexisting high-risk conditions, including prior thromboembolism, renal failure, atrial fibrillation, seizure disorders, and visual disturbances. Within each subgroup, patients receiving TXA were propensity score-matched to those who did not. Perioperative TXA utilization trends were assessed. Ninety-day postoperative outcomes were compared, including transfusion requirements, thromboembolic events, cardiac complications, renal failure, neurologic events, infections, readmissions, emergency department visits, and mortality. Outcomes were reported as odds ratios (ORs) with 95% confidence intervals (CIs).

RESULTS: Reported perioperative TXA use began in 2012 and increased through 2025, with over 70% of patients in both standard- and high-risk cohorts receiving TXA. TXA use was associated with significantly lower transfusion rates […]

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Osteoporosis treatment gap prior to femoral fracture and prevalence of pharmacological risk factors: a prospective observational study.

Böck L, Kerschan-Schindl K, Frossard M, Hajdu S, Crevenna R, Anditsch M, Stemer G, Antoni A

INTRODUCTION: Hip fractures are a major complication of osteoporosis and are associated with high morbidity, mortality, and costs. Despite clear guideline recommendations, pharmacological osteoporosis treatment remains underutilized. In addition, medications that increase fall and fracture risk further contribute to fracture occurrence. The aim of this study was to assess guideline concordance of osteoporosis therapy and to quantify the prevalence of medications associated with increased fall and fracture risk prior to the fracture in patients admitted with femoral fractures.

MATERIALS AND METHODS: In this prospective observational study, 145 patients aged ≥ 55 years admitted with femoral neck, per- and subtrochanteric, femoral shaft and distal femur fractures to a tertiary academic trauma center between April and June 2025 were included. Pre-admission medication was assessed using structured medication reconciliation. Guideline concordance of pre-fracture osteoporosis therapy was evaluated according to the current Austrian osteoporosis guideline (Dimai et al., 2024). Fall-risk-increasing drugs (FRIDs) were identified using the STOPPFall criteria, and fracture-associated medications were recorded based on Austrian osteoporosis guideline-defined drug classes.

RESULTS: Among 145 included patients (mean age 80.4 years; 71% female), 117 (81%) met criteria for pharmacological oste […]

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Gait speed recovery after iliofemoral ligament-preserving hip arthroscopy for femoroacetabular impingement: associations with early postoperative gait kinematics.

Machida S, Tsutsumi M, Utsunomiya H, Kudo S

INTRODUCTION: Whether iliofemoral ligament-preserving hip arthroscopy facilitates recovery of gait speed-an indicator of functional recovery following hip arthroscopy in patients with femoroacetabular impingement-remains unclear. We investigated whether gait speed improved after iliofemoral ligament-preserving hip arthroscopy and examined whether pelvic jerk, defined as the time derivative of acceleration and assessed using inertial sensors at 1 month postoperatively (1M), was associated with gait speed at 6 months (6M).

MATERIALS AND METHODS: Eighteen patients who underwent primary hip arthroscopy for femoroacetabular impingement were included. Gait speed and pelvic jerk were assessed during a 10-m walk at a self-selected speed, using an inertial sensor positioned at the third lumbar level. Peak pelvic jerks during the 1st and 2nd halves of the stance phase (1st - and 2nd -peak pelvic jerks) were calculated. Measurements were obtained preoperatively and at 1M and 6M. Gait parameters were compared across time points using a linear mixed model. Bayesian regression was used to compare the observed change in gait speed from preoperatively to 6M with previously reported values. Associations between gait variables at 1M and gait speed at 6M were also examined.

RESULTS: Gait speed was significantly higher at 6M than that pre surgery (preoperative, 1.15 [95% confidence interval, 1.0 […]

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Computed tomography findings in 11,504 adult patients with traumatic brain injury: a large real-world cohort study with a S100B subgroup analysis.

Clar C, Puchwein P, Moshammer M, Sadoghi P, Kramer D, Leithner A, Reinbacher P

OBJECTIVE: The aim of this study was to characterize acute cranial CT findings in a large cohort of adult patients with traumatic brain injury and to explore the diagnostic performance of S100B in a selected subgroup. The hypothesis was that most cranial CT scans in adult patients with traumatic brain injury would not demonstrate acute traumatic abnormalities, and that, within a selected subgroup, negative S100B levels would be associated with the absence of such findings on CT.

METHODS: This retrospective cohort study included 11,504 adult patients presenting with traumatic brain injury to a level I trauma center between 04/2016 and 05/2024. All patients underwent cranial CT, while serum S100B measurements were available in a selected subset. CT findings were classified as normal or showing acute traumatic abnormalities. In the S100B subgroup, biomarker levels were analyzed in relation to CT findings to assess their diagnostic performance for excluding acute pathology.

RESULTS: A total of 11,504 adult patients with traumatic brain injury were included; 27.4% were inpatients and 72.6% outpatients. Overall, 81.9% of CT scans showed no acute pathology, while 10.7% demonstrated abnormalities. 7.4% of all cases were excluded. Acute TBI-related interventions were rare (3.7% of inpatients vs. 0.3% of outpatients). In patients with dementia and those receiving anticoagulation, CT sc […]

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Early post-operative CRP is a better predictor of DAIR failure than pre-operative CRP in total knee PJI.

Beadel H, Chaffey R, Kim K, Zhu M, Coleman B

INTRODUCTION: Debridement, antibiotics and implant retention (DAIR) surgery is often performed as a first-line treatment for periprosthetic joint infection (PJI), however, success rates are variable. This study aimed to determine whether post-operative C-reactive protein (CRP) levels predict DAIR failure.

MATERIALS AND METHODS: A multi-centre retrospective cohort study was performed over a 15-year period. All total knee arthroplasty (TKA) patients undergoing DAIR for first episode PJI were included. CRP was measured at admission and for 6 weeks post-operatively. Treatment success was defined as implant retention without the need for revision surgery or long-term suppressive antibiotics. Trends in CRP were compared between two groups: failed and successful DAIR. Receiver operating characteristic (ROC) curves were analysed.

RESULTS: 189 DAIR procedures were included, of which 49% (92) failed and 51% (97) were successful. Mean follow-up was 7.5 years. Overall, CRP trended down following surgery. Mean CRP was significantly higher in the failed DAIR group at all time points. The greatest difference was at week one post-operatively (mean 76 vs. 101, $P < 0.01$). This remained significant when patients experiencing DAIR failure within 90 days were excluded (mean 76 vs. 107, $P < 0.01$). CRP was most accurate as a predictor of failure at week one post-operatively, with an area under the […]

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Tuberosity refixation in reverse proximal humerus fracture arthroplasty: a prospective cohort study with two-year follow-up.

Oberrauter CH, Hess F, Welter J, Pape HC, Mayer S, Dullenkopf A, Mazzucchelli R, Marintschev I

INTRODUCTION: To evaluate the outcomes of a simplified tuberosity refixation technique using a two-suture construct in patients with complex proximal humerus fractures treated with reverse shoulder arthroplasty (RSA). The technique reduces the number of fixation sutures rather than the traditional four to five, aiming to decrease operative complexity while promoting greater tuberosity (GT) healing and improving the healing rate. A secondary exploratory analysis compared clinical and radiological outcomes between cemented and uncemented humeral stem fixation.

MATERIALS AND METHODS: This prospective study included 73 patients who underwent RSA (Global Unite Fracture Reverse System, DePuy Synthes, Warsaw, Indiana, USA). Clinical and radiological outcomes were assessed at two-year follow-up. Bone quality was evaluated using the Deltoid Tuberosity Index (DTI). Functional outcomes and GT healing were analyzed, and stem fixation type was assessed in a secondary subgroup analysis.

RESULTS: The mean patient age was 79 ± 8 years (range 61-95), and 65 (89%) were female. Osteoporotic bone quality (DTI < 1.4) was observed in 51 (70%) of cases. At two years, the median outcome scores were: Subjective Shoulder Value, 90 (IQR 80-95); Constant-Murley score, 76 (IQR 73-81); active forward flexion, 140° (IQR 120-160); abduction, 140° (IQR 120-160); external rotation, 8 (IQR 8-8); and internal r […]

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Is it time for a systematic and objective assessment of knee laxity? The impact of objective knee laxity after TKA on patient-reported outcomes: a comprehensive review.

Vermue H, Klincke V, Stroobant L, Tampere T, Victor J, Arnout N

PURPOSE: Over the past decade, robot assistance has emerged as an innovative and rapidly advancing tool in total knee arthroplasty (TKA). Despite technological advancements, the evaluation of the soft-tissue envelope during TKA often relies on subjective and rudimentary assessments by the surgeon. This literature review explores the relationship between objective intraoperative and postoperative laxity measurements in the coronal and sagittal planes and patient-reported outcome measures (PROMs) following TKA.

METHODS: A literature review was conducted using PubMed/MEDLINE and Embase to identify studies on intra- or postoperative laxity measurements in TKA and their correlation with PROMs. Inclusion criteria required studies on TKA patients with quantifiable coronal and/or sagittal laxity measurements and postoperative PROMs linked to these measurements.

RESULTS: Thirty-four studies met inclusion criteria: ten on sagittal laxity, twenty-seven on coronal laxity, and three on both. Follow-up ranged from 1 month to over 8 years. Sagittal laxity thresholds of 5-10 mm under applied loads of 89-133 N were associated with better functional outcomes, whereas excessive (> 10 mm) or overly tight (< 5 mm) laxity correlated with reduced satisfaction. Medial laxity in the coronal plane was linked to lower PROMs, while the impact of lateral laxity on PROMs was less conclusive.

CONCLUSIONS: […]

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Silver-impregnated occlusive dressing is associated with a lower rate of acute surgical site infection after direct anterior total hip arthroplasty.

Tsai YY, Shen PH, Wu CC, Pan RY, Lin LC, Wang SH

INTRODUCTION: The direct anterior approach (DAA) for total hip arthroplasty (THA) has gained increasing adoption in recent years; however, it has been associated with a higher risk of wound-related complications compared with other surgical approaches. Silver-impregnated occlusive dressings have demonstrated antimicrobial potential, though their role in DAA THA remains underexplored.

MATERIALS AND METHODS: We conducted a retrospective cohort study with a historical control group, reviewing 321 consecutive primary DAA THAs performed by a single surgeon (September 2020-December 2025). Standard sterile gauze was used before August 2021 (n = 76) and a silver-impregnated occlusive dressing (AQUACEL® Ag SURGICAL; ConvaTec, Deeside, UK) thereafter (n = 245). Despite the chronological difference between cohorts, baseline demographic and clinical characteristics did not differ significantly between groups (Table 1). The primary endpoint was acute surgical site infection(SSI) within 30 days (CDC criteria); all patients had ≥ 3 months follow-up to capture early wound-related reoperations as secondary outcomes.

RESULTS: Four acute SSI events occurred (1.25%). SSI incidence was 3.94% (3/76) with gauze versus 0.41% (1/245) with the silver-impregnated dressing (Fisher's exact p = 0.043). This corresponds to an absolute risk reduction (ARR) of 3.54% (95% CI, 0.34-10.57) and a number needed t […]

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Minimally invasive stabilisation of anterior pelvic ring injuries through anterior infix versus percutaneous anterior retrograde pubic screw: a prospective comparative cohort study.

Goda El-Hamalawy A, Abdelmoneim M, Al-Sayed M, Youness M, Sadek F, Kassem E

BACKGROUND: Management of unstable anterior pelvic ring fractures has shifted toward minimally invasive fixation techniques. This prospective randomized cohort study compares anterior subcutaneous internal fixator with percutaneous retrograde pubic ramus screw fixation, evaluating radiological outcomes, functional scores, complications, and peri-operative parameters.

METHODS: Fifty adult patients with unstable anterior pelvic ring injuries (Tile B & C; Young-Burgess lateral compression and vertical shear mechanisms) were enrolled between 2023 and 2025 and randomized into two equal groups: group 1 (internal fixator) and group 2 (retrograde pubic screw). Both groups underwent individualized posterior ring fixation as required. Outcomes assessed included operative time, blood loss, fluoroscopy time, union rate, Matta radiological score at 6 months, Majeed and Pelvic Outcome Scores at 12 months, and post-operative complications.

RESULTS: Management of anterior pelvic injury showed that retrograde screw fixation demonstrated significantly shorter operative time (32.0 ± 9.2 vs. 50.3 ± 8.4 min, $p < 0.001$) and less intraoperative blood loss (68.3 ± 16.2 vs. 122.3 ± 31.8 ml, $p < 0.001$). Radiological outcomes were comparable: excellent Matta score in 68% vs. 64% ($p = 0.834$). Union time showed no significant difference (14.1 ± 2.4 vs. 15.1 ± 1.5 weeks, $p = 0.076$). Functional outcomes at […]

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Prioritisation of assessments, diagnostic classifications, and outcome measures in Perthes disease: a Delphi survey of international health professionals.

Ball S, Davies LM, Gray K, Pacey V

INTRODUCTION: The aim of this study is to reach consensus and prioritise the most clinically important assessments, diagnostic classifications, and outcome measures by an international panel of experts for Legg-Calve-Perthes disease.

MATERIALS AND METHODS: A three-round modified e-Delphi survey was conducted between April and May 2025. Participants rated the clinical importance of 14 clinical assessments, 26 radiological assessments, eight diagnostic classifications, and 26 outcome measures identified by a scoping review. Consensus was defined as ≥ 75% agreement. In the final round, items in each category were prioritised for overall condition and by specific age groups (< 6 years, 6-8 years, > 8 years).

RESULTS: Thirty-eight health professionals, from nine countries participated in round 1. Thirty-four completed all three rounds, a retention rate of 89.5%. Consensus and prioritisation for clinical use were achieved for 10 clinical assessments, three radiological assessments, three diagnostic classifications, and three outcome measures.

CONCLUSION: Paediatric specialists have reached consensus and prioritised methods to evaluate and diagnose Legg-Calve-Perthes disease in clinical practice. These results represent an important step towards improving continuity of care and the foundation for future development of clinical practice guidelines.

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Compensatory activation of the deltoid and remaining rotator cuff muscles in patients with rotator cuff tears using real-time tissue elastography.

Hoshikawa K, Oishi R, Yuri T, Uno T, Nagai J, Giambini H, Mura N

INTRODUCTION: Rotator cuff (RC) tears are a prevalent condition leading to dysfunction and shoulder pain. Spatial changes in the behavior of the deltoid and remaining RC muscles during arm elevation in patients with RC tears have not been investigated. The purpose of this study was twofold: (1) to investigate factors related to the changes in the deltoid muscle behavior during scapular plane abduction (scaption) in patients with RC tears, and (2) to examine the functional relationship between the deltoid and remaining RC muscles, particularly remaining infraspinatus (ISP) and teres minor (TMin) muscles, in these patients using ultrasound elastography.

METHODS: Twenty-four patients (mean age, 66 ± 6 years) with RC tears (mean antero-posterior diameter, 17.4 ± 8.3 mm) were evaluated. Antero-posterior and medio-lateral diameters of the posterosuperior RC and Subscapularis (SSC) tears were measured intra-operatively. Muscle elasticity was evaluated at passive and active states, as a proxy for muscle activation, at 30°, 60°, and 90° during scaption for the deltoid, ISP, and TMin muscles using real-time tissue elastography. Based on the deltoid muscle activity ratio at 30° and 90° scaption, patients were divided into normal and abnormal activation groups.

RESULTS: Activity of the deltoid muscle in the abnormal activation group exhibited consistently high activity across various ele […]

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Pin orthosis extension block pinning versus conservative treatment for doyle type 4B mallet fractures.

Col KA, Demirsu O, Ayd■n M, Surucu S, Yilmaz M, Atlihan D

INTRODUCTION: Although numerous treatment options have been reported for mallet fractures, a universally accepted gold-standard approach has not yet been established. The purpose of this study was to compare the clinical and radiographic outcomes of the pin-orthosis extension-block pinning technique (PO-EBPT) with those of conventional conservative treatment in patients with Doyle type 4B mallet fractures.

MATERIALS AND METHODS: This study included 62 patients with Doyle type 4B mallet fractures involving 20-50% of the distal interphalangeal (DIP) joint articular surface, treated between March 2022 and April 2024. Patients were randomized into two groups: Group 1 (n = 33) underwent PO-EBPT, whereas Group 2 (n = 29) received conservative treatment with splint immobilization. Outcome measures included DIP joint extension lag, range of motion, fracture union, complication rates and functional outcomes according to the Crawford criteria. Follow-up evaluations were performed at 2, 4 and 6 weeks and at 3, 6 and 12 months.

RESULTS: A total of 62 patients were analyzed (33 PO-EBPT; 29 conservative). No statistically significant differences were observed between the groups with respect to sex, affected side, injured finger, or complication rates (p = 0.461, p = 0.658, p = 0.763 and p = 0.165, respectively). However, the PO-EBPT group demonstrated significantly improved DIP joint exten […]

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Impact of cannabis use and tobacco smoking on outcomes of open carpal tunnel release surgery: a nationwide study in the United States.

Hoveidaei AH, Esmaeili S, Gilmor R, Chen Z, Conway JD, Ingari JV

PURPOSE: To evaluate the impact of cannabis use, tobacco smoking, and their concurrent use on postoperative complications following open carpal tunnel release (CTR) using a nationwide database.

METHODS: A retrospective cohort study was conducted using the PearlDiver database (2010-2022) to identify adult patients who underwent open CTR. Patients were categorized into four groups: cannabis-only users (n = 800), tobacco-only users (n = 167,803), concurrent users (n = 3529), and non-users (n = 167,803 matched controls). Postoperative complications, surgical site infection, wound disruption, joint stiffness, and complex regional pain syndrome, were assessed at 6 weeks and 3 months postoperatively.

RESULTS: At 6 weeks and 3 months, surgical site infection was more common among concurrent cannabis and tobacco users, cannabis-only users, and tobacco-only users. Wound disruption occurred more frequently in concurrent users and tobacco-only users, but not in cannabis-only users. Complex regional pain syndrome was associated only with tobacco use. No association was found between any substance use and joint stiffness.

CONCLUSION: Cannabis use, tobacco smoking, and especially concurrent use increase the risk of surgical site infection and wound disruption following open CTR. Tobacco use is independently associated with complex regional pain syndrome, while cannabis may have a protectiv […]

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Timing and microbiological profile influence long-term outcomes after debridement, antibiotics, and implant retention (DAIR) in acute hip periprosthetic joint infection.

Muñoz-Mahamud E, Perdomo-Lizarraga JC, Combalia A, Alías A, Serra A, Fernández-Valencia JÁ, Verdejo MÁ, Soriano Á

INTRODUCTION: The strategy of debridement, antibiotics, and implant retention (DAIR) represents one of the main therapeutic modalities for acute periprosthetic joint infection (PJI). However, reported outcomes remain highly variable, with success rates ranging from 16 to 92%. This study aimed to evaluate long-term treatment failure-free survival and identify risk factors for treatment failure in patients with acute hip PJI treated with DAIR using time-to-event analytical methods.

MATERIAL AND METHODS: This retrospective study evaluated the treatment failure rate in 115 patients treated with the DAIR strategy for acute PJI following primary or aseptic revision hip arthroplasty between 1999 and 2018. Potential predictors of treatment failure-free survival, including patient characteristics, infection microbiology, and surgical timing, were analyzed using univariate tests, Cox proportional hazards regression (hazard ratio [HR]), and Kaplan-Meier survival estimates.

RESULTS: Among the 115 patients included, 56 were women and 59 were men; 48 were aged 75 years or younger and 67 were older than 75 years. In 88 cases, the infection occurred after primary arthroplasty, whereas

27 followed aseptic revision surgery. The mean follow-up was 7.1 years (SD 4.4). At five years, treatment failure occurred in 30.4% of cases. Procedures performed within seven days of diagnosis showed lower fai […]

PubMed · DOI

Surgical treatment of varus unicompartmental knee osteoarthritis: indications, trends, and outcomes-a narrative review.

Burgio C, Bosco F, Cobisi C, Zepeda K, Giustra F, Capella M, Mayman D, Debbi E et al.

INTRODUCTION: Varus unicompartmental knee osteoarthritis (UC-KOA) is a frequent and heterogeneous clinical condition characterized by compartment-specific cartilage degeneration and frequently associated with lower-limb malalignment. The progressive understanding of knee biomechanics and alignment has led to the development of surgical strategies specifically tailored to address isolated compartment disease. The purpose of this narrative review is to evaluate indications, current trends, and clinical outcomes of the main surgical options for varus UC-KOA, with particular focus on realignment osteotomies and unicompartmental knee arthroplasty (UKA).

MATERIALS AND METHODS: A narrative review of the literature was conducted using major biomedical databases. Evidence related to biomechanics, alignment correction, osteotomy techniques, UKA indications, implant evolution, complication profiles, and comparative outcomes was analyzed and synthesized thematically.

RESULTS: Biomechanical and clinical evidence consistently identifies malalignment as a key driver of isolated compartment degeneration. Corrective osteotomies effectively restore physiological load distribution and provide durable outcomes in selected patients with extra-articular deformity. Advances in implant design and surgical techniques have substantially improved UKA survivorship, allowing broader patient selection wit […]

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Preoperative NarxCare overdose risk scores greater than 100 are associated with worse PROMs improvements and dissatisfaction after primary THA.

Khan ST, Emara AK, Patel SV, Elmenawi KA, Ibaseta A, Pasqualini I, Zhang C, Cleveland Clinic Adult Reconstruction Research Group et al.

INTRODUCTION: The NarxCare Overdose Risk Score (ORS) is a weighted measure of patient-specific prescription drug use. In patients undergoing primary total hip arthroplasty (THA), this study aimed to evaluate (1) the trend of the ORS from preoperative to 1-year postoperatively, and (2) the association of preoperative ORS with achievement of clinically meaningful improvements in patient reported outcome measures (PROMs), and satisfaction at 1-year.

METHODS: All patients who underwent primary unilateral elective THA at a USA tertiary healthcare system from January 2016-December 2022 were eligible. Patients with incomplete PROMs or missing baseline NarxCare ORS were excluded, leading to the inclusion of 5,424 patients. Multivariable regression models were used to assess the association between ORS and 1-year PROMs, including achievement of the minimum clinically important difference (MCID) and Patient Acceptable Symptom State (PASS) for the Hip Disability and Osteoarthritis Outcome Score (HOOS) subdomain scores, as well as satisfaction.

RESULTS: Preoperatively, 46.1% (n = 2501) had a NarxCare ORS of 0, 24.6% (n = 1336) had scores between 100 and 199, and 16.2% (n = 876) in the 200-299 range. After adjusting for confounding variables, a preoperative ORS of 100-199 was associated with failure to achieve MCID in HOOS Pain (OR 1.91;CI 1.32-2.77;p = 0.001), and HOOS JR (OR 1.54;1.10-2 […]

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Only a partial shield: local vancomycin postpones but does not prevent hip and knee prosthetic infections.

Darup A, Ettinger M, Savov P, Brand S, Seeber GH, Stauss R

INTRODUCTION: Periprosthetic joint infection (PJI) represents a severe complication following total hip arthroplasty (THA) and total knee arthroplasty (TKA). The intraarticular application of vancomycin powder is increasingly integrated into perioperative standards as an innovative strategy for PJI prevention. Yet the efficacy of this method remains a topic of debate. The aim of this retrospective cohort study was to evaluate whether topical vancomycin application reduces the incidence of PJI after primary total joint arthroplasty (TJA) and whether it is associated with an increased risk of wound complications.

MATERIALS AND METHODS: In this retrospective monocentric cohort study, all primary THA and TKA procedures performed between January 1, 2022, and December 31, 2023, were analysed. From January 1, 2023, intraarticular vancomycin powder was implemented as part of the standard perioperative protocol for PJI prevention. The PJI rates, postoperative surgery-related complications, and time to infection were compared between the vancomycin and control groups.

RESULTS: A total of 1,499 patients were included in the study. Patients were divided into two groups: a vancomycin group (VG, n = 818) and a control group (CG, n = 681). No statistically significant group differences were observed for wound healing disorders (p = 0.775) or PJI rates (1.0% VG vs. 0.9% CG, p = 0.846). Howev […]

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Validity and prognostic value of a novel intraoperative assessment tool for reduction in trimalleolar fractures: the SGSC checklist approach.

Gao H, Xu G, Ma J, Wang Y, Wang J, Zhou R, Liu Z, Wang B

BACKGROUND: Trimalleolar fractures present substantial surgical challenges, and the quality of fracture reduction significantly influences patient outcomes. Current intraoperative assessment methods, including fluoroscopy and arthroscopy, each possess inherent limitations when used independently. This study aimed to validate the effectiveness of a novel intraoperative evaluation tool, the SGSC (Step, Gap, Syndesmosis, Congruity) "traffic light" checklist, that integrates arthroscopic and fluoroscopic assessment, and to evaluate its predictive value for clinical prognosis.

METHODS: This single-center retrospective cohort study enrolled 36 patients who underwent surgical treatment for trimalleolar fractures. Validation proceeded in two stages: first, we calculated inter-rater reliability (Kappa statistic) and concordance with postoperative CT as the reference standard to establish the reliability and validity of the SGSC checklist (accuracy validation); second, patients were stratified into "All-Green" and "Non-All-Green" groups based on intraoperative SGSC assessment, and we compared AOFAS scores and early radiographic degenerative changes at final follow-up (prognostic validation).

RESULTS: For accuracy validation, inter-rater agreement between the two assessors demonstrated excellent reliability (Kappa = 0.827; 95% CI: 0.644-1.000). The overall concordance between SGSC check […]

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Radiographic assessment of femoral, acetabular and global offset following hip spacer implantation in staged total hip arthroplasty".

Cavagnaro L, Ferrari E, Providenti V, Marciante V, Carrega G, Formica M

BACKGROUND: The number of total hip arthroplasties (THAs) performed worldwide is steadily increasing, and, consequently, the demand for revision procedures is expected to rise as well. two-stage revision procedure continues to be considered the gold standard in many clinical scenarios. During the first step, failure to restore femoral offset has been implicated as a contributor to instability and mechanical failure, although a clear consensus is still lacking.

OBJECTIVES: The primary aim of this study is to evaluate the restoration of biomechanical parameters of the hip-specifically leg length discrepancy (LLD), femoral offset (FO), acetabular offset (AO), and global offset (GO)-following implantation of a specific type of articulating hip spacer during the first stage of a two-stage revision for PJI. A secondary objective is to assess the correlation between variations in offset parameters and the rate of interstage mechanical complications.

MATERIALS AND METHODS: We retrospectively reviewed all patients undergoing staged revision with a specific articulating spacer between 2020 and 2022 at a single institution. Harris Hip Scores, Oxford Hip Scores, and Visual Analogue Scales for pain were obtained at different time points. Radiographic analysis included LLD, FO, AO and GO measurements on the affected (spacer and post-reimplantation) and the contralateral side. Complications […]

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Simulated tears of antero-posterior rotator cuff force-couple and reduced glenoid concavity decrease anterior glenohumeral stability: a robot-assisted biomechanical analysis of a load

and shift sequence.

Karlsfeld MJ, Wermers J, Tänzler S, Sußiek J, Herbst E, Katthagen JC

BACKROUND: The glenoid concavity and the compression applied by the rotator cuff (RC) are essential for glenohumeral stability (GHS). This study aimed to determine how different simulated rotator cuff tears (RCT) and the glenoid depth influence GHS.

METHODS: A Load and Shift sequence was performed with eight fresh-frozen cadaveric shoulders in a robotic-assisted setup. Differently configured static loading of the reinforced RC and deltoid muscle (DLT) simulated intact RC and anterior, superior, anterosuperior, posterosuperior, mass, and complete RC plus DLT tears. Anterior dislocation forces and their changes were determined as indicators of stability. The glenoid depth and Bony shoulder stability ratio (BSSR) were defined as indicators of concavity. To assess GHS, the maximal force (Favg), the maximal force increase (dFavg), and their mean deviations (ΔFmax, ΔdFmax) to the intact configuration during the sequence were evaluated.

RESULTS: Simulated tears of the subscapularis tendon (SCP) (ΔFmax = 7.90 N, ΔdFmax = 1.54 N/mm) and simulated tears of the infraspinatus (ISP) + teres minor (TM) tendons (ΔFmax = 7.19 N, ΔdFmax = 1.48 N/mm) resulted in greater differences to the intact shoulder than simulated tears of the supraspinatus tendon (ΔFmax = 3.82 N, ΔdFmax = 1.16 N/mm). High correlations were observed between maximal force concerning glenoid depth (r = 0.81) and BSSR (r = 0 […]

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Non-tobacco nicotine dependence is associated with increased complications following clavicle open reduction internal fixation.

Adio AA, Goyal RK, Skiena A, Daher M, Horneff JG, Kassam HF, Hill BW, Abboud JA

BACKGROUND: Tobacco dependence is well established to impair bone healing and increase postoperative complications. However, the independent effects of nicotine, separate from combustion-related factors, remain unclear. This study aimed to evaluate the impact of non-tobacco nicotine dependence (NTND) following open reduction and internal fixation (ORIF) of clavicle fractures, including midshaft and lateral fracture subtypes.

METHODS: A retrospective cohort study was performed using a large national database. Patients undergoing ORIF of clavicle fractures were identified using procedural and diagnostic codes. Patients with non-tobacco nicotine dependence (e.g., vaping, patches, or gum) were identified and propensity score matched to (1) patients without nicotine exposure and (2) patients with tobacco nicotine dependence. In a secondary analysis, patients were stratified by fracture type (midshaft and lateral), and outcomes were compared between nicotine exposure and no nicotine exposure within each subgroup. Outcomes were assessed at 90 days and 1 year and included medical complications, number of opioid prescriptions, and implant-related outcomes. Risk ratios (RRs) with 95% confidence intervals (CIs) were calculated, with $p < 0.05$ considered significant.

RESULTS: After matching, 936 non-tobacco nicotine users were identified. Non-tobacco nicotine dependence was associated wit […]

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Association between timing of surgery and postoperative outcomes in older adults with distal femur fractures: a systematic review and meta-analysis.

Hoshika S, Yamamoto N, Saka N, Tsuge T, Sato T, Isaji Y, Nakao S

INTRODUCTION: The association between surgical timing and clinical outcomes in older adults with distal femur fractures remains unclear. We conducted a systematic review and meta-analysis to evaluate whether earlier surgery is associated with improved outcomes in this population.

MATERIALS AND METHODS: In accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines, two independent reviewers searched MEDLINE, Embase, the Cochrane Central Register of Controlled Trials, ClinicalTrials.gov, and the World Health Organization International Clinical Trials Registry Platform from inception to October 8, 2024. The primary outcome was 30-day mortality, and secondary outcomes included longer-term mortality and postoperative complications. Risk of bias was assessed using the Quality in Prognosis Studies tool, and the certainty of evidence was evaluated using the Grading of Recommendations Assessment, Development and Evaluation approach.

RESULTS: Of 6,101 records screened, 16 retrospective cohort studies comprising 31,213 patients met the inclusion criteria. Early surgery was not associated with reduced 30-day mortality (adjusted odds ratio [OR]: 0.77, 95% confidence interval [CI]: 0.56-1.05; low-certainty evidence) or with longer-term mortality at 90 days (crude OR: 0.91, 95% CI: 0.51-1.60), 180 days (crude OR: 0.46, 95% CI: 0.12-1.77), or 1 […]

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Is a central cavity necessary for bioactive glass-ceramic spacers in plated ACDF? A retrospective comparison of solid versus cavity designs.

Kang TH, Lee G, Kang B, Han J, Cho M, Lee JH

PURPOSE: Traditionally, interbody cages feature a central cavity for graft packing. However, for bioactive glass-ceramic (BGS-7) spacers achieving fusion via surface-mediated osseointegration, this cavity may be biologically redundant and structurally disadvantageous. We compared radiological and clinical outcomes of plated anterior cervical discectomy and fusion (ACDF) using BGS-7 spacers with or without a central cavity.

METHODS: We retrospectively reviewed patients undergoing plated ACDF using solid non-cavity (17 patients, 24 levels) or central-cavity (17 patients, 33 levels) BGS-7 spacers. Radiographic fusion was evaluated using dynamic interspinous distance (primary, ≤ 2 mm) and segmental angular motion (secondary, ≤ 4°), alongside CT-based osseointegration (direct contact ≥ 50%). Subsidence (≥ 2 mm) and clinical outcomes were evaluated at 12 months.

RESULTS: Both groups demonstrated comparable 12-month clinical improvements ($p > 0.05$). Radiographic fusion rates were equivalent between non-cavity and central-cavity groups based on the primary distance (91.7% vs. 84.8%, $p = 0.687$) and secondary angular criteria (79.2% vs. 63.6%, $p = 0.331$). The non-cavity group showed a trend toward superior segmental stability, with a smaller dynamic gap distance (0.82 ± 0.64 mm vs. 1.53 ± 1.99 mm, $p = 0.064$). CT-based fusion perfectly matched (90.5% vs. 90.9%, $p = 1.000$). Subsidence ra […]

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Clinical outcomes of internal iliosciatic fixation using a conventional reconstruction plate via an anterior approach for acetabular fractures involving the posterior column.

Xiong H, Chen H, Cao S, Chen X, Wang F

BACKGROUND: Acetabular fractures involving the posterior column remain surgically challenging because of the complex three-dimensional anatomy and restricted operative corridor. As an alternative to traditional posterior approaches, internal iliosciatic fixation through an anterior approach has gained increasing acceptance. Reconstruction plates, which offer the advantages of low cost and convenient intraoperative bending, are extensively used in clinical settings.

METHODS: A retrospective case series was conducted on 36 consecutive patients with acetabular fractures involving the posterior column who underwent internal iliosciatic fixation using an intraoperatively contoured reconstruction plate between July 2019 and November 2022. The modified Stoppa approach ($n = 26$) or the pararectus approach ($n = 10$) was employed. Fracture reduction was assessed using the modified Matta criteria, and hip function was evaluated using the modified Merle d'Aubigné-Postel score at one-year follow-up. Operative time, blood loss, fracture union, and complications were recorded.

RESULTS: The mean age of the patients was 42.6 ± 14.5 years (range: 21-71 years), with 25 males and 11 females. Mechanisms of injury included traffic accidents ($n = 21$), falls from height ($n = 10$), skiing injuries ($n = 3$), and others ($n = 2$). According to the Letournel-Judet classification, there were 13 both-column fra […]

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Novel UHMWPE sleeve for stabilizing loose suture anchors in rotator cuff repair.

Lin CK, Jaw FS, Sam SS, Siow TF, Yii SC, Chang YJ, Young TH

Suture anchors are frequently used for repairing torn rotator cuffs. However, post-operative anchor loosening is a common complication often caused by low bone density. The purpose of this study was to evaluate if a novel UHMWPE cortex locking sleeve could improve the anchor pullout strength after revision in synthetic bone. The initial failed anchor fixation was simulated by inserting a 4.5 mm anchor and pulling it out of the bone, which created an enlarged insertion hole for the revision anchor. After revision, two loosening conditions were simulated; (i) intra-operative anchor loosening where the revision anchor was directly pulled out from the block after insertion, and (ii) post-operative anchor loosening where the revision anchor was first subjected to cyclic loading and then pulled out. For both loosening conditions, the pullout strength and energy absorbed was recorded for the following configurations: (i) Group A, same-diameter anchor (SDA; 4.5 mm × 10 mm); (ii) Group B, larger-diameter anchor (LDA; 5.5 mm × 10 mm;); and (iii) Group C, same-diameter anchor combined with the novel sleeve (SDA + NS). For intra-operative anchor loosening, the highest pullout strength was recorded in the SDA + NS group, with a mean value of 370.077 ± 32.699 N, which was significantly greater than both SDA and LDA groups ($p < 0.05$). Similarly with post-operative anchor loosening, the SDAP […]

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Impact of rheumatoid arthritis on complications and hospital outcomes following reverse shoulder arthroplasty: evidence from 389,135 hospitalizations in the national inpatient sample.

Mahamid A, Elgabsi M, Khatib M, Murad H, Qawasmi F, Lavon E, Zahalka S, Yassin A et al.

OBJECTIVE: To evaluate the impact of rheumatoid arthritis (RA) on perioperative complications and hospital outcomes following reverse total shoulder arthroplasty (RTSA) using the National Inpatient Sample (2016-2021).

METHODS: Adult patients undergoing RTSA were identified using ICD-10 procedure codes. Outcomes included prolonged length of stay (LOS), high hospital charges, inpatient complications, and mortality. Multivariable logistic regression adjusted for demographic, clinical, and hospital characteristics. Survey weights were applied to generate nationally representative estimates.

RESULTS: A total of 389,135 patients underwent RTSA, including 18,140 (4.66%) with RA. In unadjusted analyses, RA patients had higher rates of prolonged LOS (17.8% vs. 14.8%) and acute blood loss anemia (10.7% vs. 8.6%). After multivariable adjustment, RA remained an independent predictor of prolonged LOS (adjusted OR 1.18, 95% CI 1.14-1.23, $p < 0.001$) and acute blood loss anemia (adjusted OR 1.25, 95% CI 1.18-1.33, $p < 0.001$), but not urinary tract infection. In-hospital mortality was rare and could not be modeled due to zero events in the RA group. RA patients also had slightly longer hospitalizations and higher hospital charges.

CONCLUSION: RA is independently associated with prolonged hospitalization and increased risk of perioperative blood loss anemia following RTSA. These findings high […]

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Long-term results of thumb carpometacarpal joint arthrodesis.

Vogler P, Benedikt S, Bode S, Keiler A, Stock K, Arora R

INTRODUCTION: Thumb carpometacarpal (CMC 1) arthrodesis remains an established treatment option for patients with high functional demands and strength requirements. Long-term data on mobility, complication rates and clinical outcomes remain limited. This study aimed to analyze long-term clinical and radiological results after CMC 1 arthrodesis.

MATERIALS AND METHODS: A single-center, retrospective study included patients who underwent CMC 1 arthrodesis between 2004 and 2024. Following medical record analysis, eligible patients were invited to a standardized clinical and radiological follow-up examination. Outcomes included DASH, PRWHE and MHQ scores, pain intensity, mobility, strength and patient satisfaction. Radiological assessment covered fusion signs, implant position and adjacent joint arthrosis. The median follow-up was 9 years.

RESULTS: 34 patients were included, 16 participated in follow-up. The median DASH score was 14 points, PRWHE 4 points and MHQ 80 (operated) vs. 84 (contralateral). Grip strength reached 85% of the contralateral side, with preserved pinch strength. Compensatory adaptation of adjacent joints was observed regarding range of motion. Patients reported no relevant resting pain and only minimal load-related discomfort. 15 of 16 patients were satisfied with the outcome. Non-union occurred in 4 of 34 cases (11.8%), associated with absent cancellous bone […]

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Does the tendon retract in flexor hallucis longus muscle laceration? A cadaveric study.

Ayd■n M, ■ahin Y, Ya■ar H, Çiçek F, Cerano■lu FG, Mert A, Ç■naro■lu S

INTRODUCTION: Flexor hallucis longus (FHL) tendon injuries are common, particularly in open injuries caused by sharp objects. However, data regarding FHL tendon retraction following hallux-level FHL tendon lacerations remain limited. This study investigated the retraction and shortening of the FHL tendon using cadaveric models.

MATERIALS AND METHODS: Six fresh-frozen below-knee amputated cadaveric feet (mean age: 65 years, range: 62-88) were utilized. FHL tendons were dissected, and lacerations were simulated at the hallux level. Tendons were subjected to various tensile forces (20-120 N) using a mechanical device. Force was measured in Newtons via a loadcell, and retraction distance was recorded in millimeters using an encoder. Data were summarized as mean ± standard deviation (SD), median, and quartiles.

RESULTS: The FHL tendon retracted gradually as applied force increased. An average retraction of 7 mm was measured at 20 N, reaching 21 mm at 60 N. Beyond 60 N (> 21 mm retraction), the other four toes also flexed. This occurs due to the anatomical connection between the FHL and flexor digitorum longus (FDL) tendons.

CONCLUSIONS: Only slight retraction was observed in hallux-level FHL tendon lacerations. At most, the findings suggest that distal FHL lacerations may exhibit limited proximal retraction under the tested conditions, which may have implications for surgical pla […]

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Early-onset rice body synovitis of the knee in children under three years: a case series and review of the literature.

Zhu L, Liu C, Liu Y, Mao B, Huang Y, Wen J, Liu Y, Qian T

OBJECTIVE: Rice body synovitis reports have been rarely reported in children under three years of age. We aimed to explore the clinical characteristics and treatment experience of rice body-like synovitis in children's knees.

METHODS: We retrospectively analyzed the clinical and imaging data of 4 cases (5 knees) of rice body-like synovitis in children treated at our hospital from January 2014 to December 2021. Among the patients, 1 was male and 3 were female, with an average age of 22.5 months. The average length of symptoms before admission was 1.86 months. Clinical presentations included swelling and limping, with no fever or rash.

RESULTS: Three patients with unilateral onset and one patient with bilateral onset underwent unilateral arthroscopic excision of synovial lesions. All 4 patients had pathological findings consistent with chronic inflammation, leading to a postoperative diagnosis of juvenile idiopathic arthritis (JIA). Postoperatively, all patients received standardized pharmacological treatment and were followed for 24 to 48 months in collaboration with the rheumatology department. None of the 4 knees undergoing surgery experienced recurrence, and the symptoms of the knee treated conservatively resolved spontaneously. In the last follow-up session, MRI and ultrasound showed only mild synovial thickening, with effusion and loose bodies disappearing.

CONCLUSION: R […]

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Extravertebral temperature exposure during spinal radiofrequency ablation: an experimental surrogate assessment of neural injury risk.

Weigert A, Hoffmann RT, Büttner A, Zaccaria R, Wegener V, Holzapfel BM, Wegener B

BACKGROUND: Radiofrequency ablation (RFA) has become an important minimally invasive option for the treatment of painful spinal tumors and benign bone lesions. While its effectiveness within the vertebral body is well established, there is ongoing concern about unintended heat exposure of nearby neural structures. Experimental data describing how heat propagates beyond the vertebra during spinal RFA are still limited. This experimental in vitro study aims to assess the extravertebral temperature exposure during spinal radiofrequency ablation as a surrogate parameter for potential neural injury risk under controlled conditions.

METHODS: This experimental in vitro study was performed on four fresh-frozen human lumbar spines. Radiofrequency ablation was carried out using a commercially available system and standard clinical protocols. Two anatomical configurations were examined: ablation within the vertebral body and ablation within the pedicle. Temperatures were measured at predefined locations outside the vertebra, including the spinal canal and cortical boundaries, using multiple thermocouples. To aid interpretation of radiofrequency-related measurement artifacts, an additional control setup using an electrothermal heating system without radiofrequency energy was included. Temperature data were evaluated descriptively.

RESULTS: In the control experiments, temperature profiles […]

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Giant cell tumor of bone: exploring HtrA1 as a potential predictor of recurrence.

Mirioglu A, Dalkir KA, Gokmen MY, Bagir M, Ates KE, Deveci MA, Gönlü■en G, Ozbarlas HS

INTRODUCTION: Giant cell tumor of bone (GCTB) is a locally aggressive tumor with a considerable risk of recurrence, but reliable predictive markers are limited. High-temperature requirement A serine peptidase 1 (HtrA1) has been implicated in the progression of several cancers and may also play a role in GCTB recurrence.

MATERIALS AND METHODS: This retrospective study included 34 patients with GCTB treated with curettage between 2002 and 2016. HtrA1 expression was evaluated separately in giant cells (GC) and mononuclear cells (MNC) using immunohistochemistry. Based on high or low expression in each cell type, patients were categorized into four expression patterns: High GC / High MNC, High GC / Low MNC, Low GC / High MNC, and Low GC / Low MNC. Clinical data, including recurrence status and time to recurrence, were analyzed.

RESULTS: The High GC / Low MNC group showed the highest recurrence rate at 71.4%, compared with 28.6% in the High GC / High MNC group and 31.6% in the Low GC / Low MNC group. No recurrence was observed in the Low GC / High MNC group, which included only one patient. The High GC / Low MNC group showed a non-significant trend toward higher recurrence compared with all other groups combined, with a similar trend toward shorter recurrence-free survival.

CONCLUSION: HtrA1 expression patterns in GCTB may provide preliminary insight into recurrence risk. Although […]

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Coronal hamate body fractures with dorsal fourth and fifth carpometacarpal joint instability: fracture patterns, injury mechanisms, surgical strategies, and outcomes.

Lee SH, Chung HJ, Cha SM

INTRODUCTION: Coronal hamate body fractures are rare injuries, and studies regarding their morphology and management are limited. This study analyzed fracture patterns, injury mechanisms, associated injuries, surgical strategy, and clinical outcomes in patients with coronal hamate body fractures involving more than one-third of the articular surface and concomitant carpometacarpal (CMC) joint instability.

METHODS: Fifty-eight male patients were retrospectively reviewed. Fractures were classified as dorsal oblique or coronal splitting types based on computed tomography findings. Surgical treatment involved open reduction and multiple screw fixation of the hamate using a K-wire-guided technique, with supplementary Kirschner wire stabilization for associated metacarpal base fractures or CMC joint instability. Radiographic and functional outcomes were evaluated after a minimum follow-up of 6 months.

RESULTS: Dorsal oblique fractures (63.8%) occurred more frequently than coronal splitting fractures (36.2%) and were most commonly associated with punching (OR, 4.9; $p = 0.01$), whereas falls from height were more common in coronal splitting types. Concomitant fractures occurred in 82.8% of cases, most frequently involving the fourth metacarpal base. Among 35 patients with a minimum 6-month follow-up, 34 achieved successful healing. In the 33 patients with functional outcome data, the […]

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Effect of spinal fusion length on hip and knee osteoarthritis: a comparative cohort study of no fusion, short-segment fusion, and long-segment fusion.

Sülek Y, Demirkale ■, Erto■rul R, Özdemir HM

PURPOSE: Lumbar fusion alters spinal-pelvic biomechanics and may influence degenerative changes in the lower extremities. This study investigated whether fusion length is associated with radiographic progression of hip and knee osteoarthritis.

METHODS: In this retrospective, single-center cohort study, patients aged 40-65 years who underwent lumbar surgery between 2010 and 2021 were grouped as no fusion (N), short fusion (S; ≤3 levels), or long fusion (L; ≥4 levels). Hip joint space width was assessed using minimum joint space (MJS) and superointermediate joint space

(SJS), including femoral head-standardized measures (sMJS and sSJS). Knee medial joint space width (KMJS) was measured when available. Spinopelvic parameters (LL, PI, PT, SS, PI-LL) and functional outcomes (ODI, HHS, WOMAC) were recorded. Multivariable linear and logistic regression analyses were performed to identify independent predictors of joint space narrowing and osteoarthritis risk.

RESULTS: Hip joint space narrowing differed significantly among groups, with the highest annual narrowing rates in the long-fusion group, followed by the short-fusion group, and the lowest in the no-fusion group ($p < 0.001$). In contrast, knee KL grade change and annual KMJS narrowing did not differ significantly between groups ($p > 0.05$). In logistic regression, each additional fused level increased the odds of postoperative MJ […]

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Biomechanical comparison of inverted triangle and L-shaped screw configurations with medial buttress and anteromedial support plate in Pauwels type III femoral neck fractures.

Erem M, Demirta■ U, Y■ld■r■m S, Selçuk E, Çopuro■lu C

INTRODUCTION: Surgical stabilization of Pauwels Type III femoral neck fractures remains a significant challenge due to high vertical shear forces. While the medial buttress plate is a recognized solution, it requires extensive deep dissection. This study aims to compare the biomechanical performance of various screw configurations combined with either a medial buttress or anteromedial support plate.

MATERIALS AND METHODS: Twenty-five third-generation synthetic femurs were used to create a standardized 70-degree (Pauwels III) fracture model. Specimens were divided into five groups ($n = 5$): (A) Inverted triangle (IT) screws with a Pauwels screw, (B) Inverted triangle (IT) with medial buttress plate (MBP), (C) Inverted triangle with anteromedial support plate (ASP), (D) L-configuration with medial buttress plate (MBP), and (E) L-configuration with anteromedial support plate (ASP). Axial loading was applied at 2 mm/min until construct failure, defined objectively by the real-time force-distance curve.

RESULTS: Although no statistically significant difference was found between groups ($p = 0.102$), a large effect size was observed ($\eta^2 = 0.309$). Group C (IT + ASP) demonstrated the highest mean failure load (1695 ± 494.6 N). Conversely, Group E (L-configuration + ASP) exhibited the lowest stability (977.2 ± 195.4 N) with a remarkably narrow standard deviation. The majority of failures […]

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Reverse total shoulder arthroplasty is a surgical option for patients over 60 years of age with locked shoulder dislocation.

Sebastia-Forcada E, Miralles-Muñoz FA, Martinez-Mendez D, Albero-Catala L, Climent-Peris V, Lizaur-Utrilla A

BACKGROUND: in locked dislocation of the shoulder, instead of reducing back to the glenoid, the humeral head remains incarcerated on the glenoid in a locked fashion. This clinical situation is fairly uncommon. It is essential to conduct an individual evaluation of each patient to determine the appropriate treatment.

OBJECTIVE: the aim of this study was to evaluate the functional outcomes of reverse total shoulder arthroplasty (rTSA) in the treatment of locked shoulder dislocation.

METHODS: patients with locked shoulder dislocation who underwent reverse shoulder prosthesis surgery and were admitted to our center between 2007 and 2023 were reviewed. The primary outcome was the Constant score. Secondary outcomes included the adjusted Constant, UCLA and DASH scores. Additionally, any signs of radiologic loosening were also documented.

RESULTS: the series consisted of 10 patients, six men and four women, with a mean age of 68.0 years. The average time from the traumatic injury to surgery was 7.5 months. All patients showed improved Constant, Adapted Constant, UCLA, and DASH scores compared to their preoperative values. When comparing the outcomes of chronic posterior and anterior dislocations, no differences in functional outcomes or shoulder motion were observed after rTSA implantation. There were no complications during or after surgery.

CONCLUSION: The results of the present […]

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Does the order matter? Comparing the order of stem placement and fracture reduction in revision total hip arthroplasty for Vancouver B2 and B3 periprosthetic femur fractures.

Antonioli SS, Khury F, Kennedy MF, Haider MA, Duke A, Schwarzkopf R, Aggarwal VK

BACKGROUND: Vancouver B2 and B3 periprosthetic femur fractures (PPFF) have posed significant treatment challenges due to stem instability and lack of adequate femoral bone stock. This study investigated subsidence, survivorship, and outcomes of Vancouver B2 and B3 fractures, based on the order in which revision stem placement and fracture reduction occurred during revision total hip arthroplasty (rTHA).

METHODS: This retrospective, cohort study included 46 rTHAs between June 2011 and April 2023. Included patients underwent rTHA for Vancouver B2 or B3 PPFF with minimum one-year radiographic and two-year clinical follow-up. All patients were treated with diaphyseal-engaging tapered fluted titanium stems and stem modularity decisions were based on surgeon preference. Cohorts were separated based on if stem placement (SF, n = 19), or fracture reduction (RF, n = 27) occurred first. Patient demographics, intraoperative information, and clinical and radiographic outcomes were collected.

RESULTS: The SF and RF cohort showed no statistically significant differences in rate of subsidence ≥5 mm [26.3%[SF], 22.2%[RF], P = 0.749), rate of subsidence ≥ 10 mm (15.8%[SF], 14.8%[RF], P = 0.928), nor average subsidence (4.1 mm[SF], 4.4 mm[RF], P = 0.861). We found no statistically significant differences in surgery-related clinical outcomes or all-cause revision rates within a two-year follow- […]

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Offset reconstruction in stemless and stemmed shoulder hemiarthroplasty: influence on long-term outcomes.

Trefzer R, Gurda A, Deisenhofer J, Weishorn J, Hariri M, Koch KA, Wolf M, Bülhoff M

BACKGROUND: Shoulder hemiarthroplasty (HA) is a well-established treatment for various humeral-side pathologies. Overstuffing should be avoided because it is associated with glenoid wear; however, specific radiological reconstruction targets remain unclear. This study assessed the association between radiological joint restoration in stemmed and stemless HA and long-term functional outcomes.

METHODS: Patients who underwent stemmed or stemless HA between 2001 and 2011 were included in a long-term follow-up. Cases with incomplete or non-calibratable digital radiographs or concomitant glenoid procedures were excluded. Clinical outcomes were evaluated using the Constant-Murley Score (CMS) and satisfaction questionnaires. Radiographic parameters-lateral glenohumeral offset (LGHO), lateral humeral offset (LHO), and center of rotation (COR)-were measured on standardized anteroposterior radiographs pre- and postoperatively by two independent observers. Differences from baseline to final follow-up (ΔLHO, ΔLGHO, ΔCOR) were calculated. Interobserver reliability was determined using the intraclass correlation coefficient (ICC). Associations between radiological measures and clinical outcomes were analyzed using linear regression; intergroup comparisons used unpaired t-tests ($p < 0.05$).

RESULTS: Forty-eight patients (12 stemmed, 36 stemless HA) were examined after a mean follow-up of 16.6 […]

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Emergency access to the subclavian vessels by non-thoracic surgeons: a cadaver-based learning model for orthopedic trauma surgery.

Grechenig P, Gänsslen A, Dauwe J, Wittig U, Sagmeister M, Koutp A, Puchwein P, Sadoghi P et al.

PURPOSE: The aim of this study was to evaluate the procedural time and safety of an infraclavicular approach to the subclavian vessels for damage control of major upper extremity vascular injuries. The study specifically focused on the performance of non-thoracic surgeons using a cadaveric model.

METHODS: A sample of 100 orthopedic trauma surgeons (residents, specialists, and attendings) was recruited from an AO cadaveric dissection course. Emergency infraclavicular access was performed on 50 cadavers over two consecutive days. The procedural time, successful vessel clamping, participant self-assessment, and the resulting learning curve were analyzed.

RESULTS: On day one, 27% (9/33) of participants who successfully achieved correct clamping of both subclavian vessels reported feeling confident. By day two, this proportion increased to 71% (55/77). Comparing day 1 to day 2 we found an improvement of 35% (subclavian artery) and 37% (subclavian vein) in the correct identification of subclavian vessels in our sample. The evaluations of this study show that there is no correlation between surgical experience and successful emergency access.

CONCLUSION: Anatomic dissection is of paramount importance for teaching rare and demanding surgical techniques. This study demonstrates that anatomical workshops significantly improve procedural safety and self-assessment when accessing subcla […]

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Operative timing and surgical complexity in hand trauma: a multicenter analysis of replantation and non-replantation centers in Germany.

Nicklas A, Will Marks P, Dragu A, Schmidt H, Rotärmel A, HandTraumaRegister DGH

BACKGROUND: Hand injuries are common traumatic conditions that often require specialized surgical care. Differences in hospital structures and levels of specialization may influence the timeliness and complexity of treatment. This study evaluates quality of care across different hospital types, focusing on time-to-skin-incision and surgical management of finger and hand injuries.

MATERIALS AND METHODS: This retrospective multicenter study is based on data from the HandTraumaRegister of the German Society for Hand Surgery (HTR DGH). A total of 16,726 surgically treated cases with documented finger or hand injuries recorded between 2018 and 2023 were analyzed. Hospitals were categorized as non-replantation centers, replantation centers, or FESSH-accredited replantation centers. Demographic characteristics, injury-patterns, treatment modalities, and time-related parameters were assessed. Statistical analysis included descriptive statistics, group comparisons, and multivariate regression models to evaluate factors associated with time-to-skin-incision.

RESULTS: The study population consisted predominantly of male patients (75.79%), and 91.6% were right-handed. Soft tissue injuries without fractures or amputations accounted for 46.57% of cases. Non-replantation centers demonstrated the longest time-to-skin-incision (median 13 h), whereas replantation centers (median 3.75 h) and FE […]

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Neglected acromioclavicular joint injury in coracoid process fractures: a retrospective cohort study.

Lee MS, Lee CH, Jo YH, Cho WT, Seo YW, Choi WS

INTRODUCTION: Coracoid process fractures are frequently associated with injuries to the superior suspensory shoulder complex, including the acromioclavicular (AC) joint. However, concomitant AC joint injuries may be overlooked in the acute phase, potentially limiting treatment options. This study aimed to investigate the association between coracoid process fractures and SSSC injuries and to identify imaging-based predictors of concomitant AC joint injury.

METHODS: A retrospective single-center cohort study including 60 patients with coracoid process fractures was conducted between February 2016 and April 2023. Coracoid fractures were classified based on anatomical location, and associated SSSC injuries were categorized according to the involved structures. These injuries were evaluated using 3D shoulder CT. Morphological deformity of the SSSC was assessed by measuring the glenoclavicular distance (GCD)-a longitudinal parameter-on final follow-up clavicle AP radiographs. In base fractures, the medial displacement gap was measured on sagittal CT. Statistical analyses included logistic regression and ROC curve analysis to identify predictors of AC joint injury.

RESULTS: Among the 60 patients, fractures were located anteriorly in 27 cases, in the middle region in 5 cases, and at the base in 28 cases. AC joint injury was identified in 15 patients, all of whom had base-type fractu […]

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Subclinical median nerve alterations after distal radius fractures: a comparative analysis of volar plate fixation and cast immobilization using dynamic ultrasonography and electrophysiology.

Karapinar SE, Arslan E, Oguzlar FC, Hatip AY, Akgun VO, Dincer R, Asci S, Kizilkaya V

BACKGROUND: Distal radius fractures may affect the median nerve through trauma-related and treatment-related mechanisms. While overt neurological deficits are uncommon, subclinical median nerve alterations may occur following both conservative and surgical treatment.

PURPOSE: To determine whether treatment modality influences subclinical median nerve behavior using combined dynamic ultrasonographic and electrophysiological assessment.

METHODS: This retrospective observational study included 42 patients with unilateral distal radius fractures. 18 patients were treated conservatively with circular casting, and 24 patients underwent volar plate fixation. Ultrasonographic assessment of the median nerve was performed at the distal radius level in neutral wrist position, as well as during wrist flexion and extension, using a dynamic ultrasonographic approach, and was combined with electrophysiological evaluation including sensory and motor nerve conduction studies. In the surgically treated group, only patients with Soong type 1 or type 2 plate positioning were included.

RESULTS: Clinical functional outcome scores were comparable between groups, despite a higher frequency of mild clinical signs such as positive Tinel test and night pain in the plate group. However, ultrasonographic evaluation demonstrated significantly greater position-dependent median nerve enlargement and flatte […]

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Impulsivity and inattention in hand tendon injuries: a case-control study revealing distinct profiles for work-related accidents.

A■ta■ Ertan E, Demir Karak■l■ç G, Ertan MB

INTRODUCTION: The impact of attention deficit, impulsivity, and anger on various types of injuries is a subject of ongoing research. Hand tendon injuries are frequently encountered clinical conditions; however, their potential relationships with the aforementioned factors have not been previously investigated. This study aimed to examine the mechanisms of hand tendon injuries from certain psychiatric perspectives.

MATERIALS AND METHODS: 32 patients presenting to the physical therapy and rehabilitation outpatient clinic with hand tendon injuries and 32 healthy controls were evaluated. The assessment included sociodemographic data, causes of injury, Quick Disabilities of the Arm, Shoulder, and Hand (Q-DASH) and Visual Analog Scale (VAS) scores, as well as the physical and functional consequences of the injury. Patients and controls were assessed for attention deficit, impulsivity, and anger using the Adult ADHD Self-Report Scale (ASRS), Barratt Impulsivity Scale-11 (BIS-11), and State-Trait Anger Expression Inventory self-report scales. Statistical analyses of the results were performed.

RESULTS: Patients with tendon injuries were found to have significantly higher scores in ASRS attention and BIS-11 Motor impulsivity compared to the control group ($p = 0.048$, $p = 0.040$). Individuals with hand tendon injuries resulting from work accidents demonstrated significantly lower ASRS to […]

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Comparative analysis of the improvement in shoulder rotation in complex shoulder fractures treated by reverse shoulder arthroplasty compared to open reduction and internal fixation: a systematic review and meta-analysis.

Albadran A, Alsugair R, AlHarthi A, Alshangiti H, Alqahtani H, Alsugair G, Ababtain R, Alhajress D et al.

PURPOSE: To compare the clinical and functional outcomes of open reduction and internal fixation (ORIF) and reverse shoulder arthroplasty (RSA) in patients aged ≥ 65 years with complex proximal humeral fractures (PHFs).

METHODS: A systematic review and meta-analysis was conducted according to the PRISMA guidelines. PubMed, Web of Science, ScienceDirect, EBSCO, and the Cochrane Library were searched for randomized controlled trials (RCTs) and cohort studies published in English without date restrictions. Eligible studies compared RSA and ORIF in elderly patients with PHFs and reported functional, radiographic, or complication outcomes. Pooled data were analyzed using a random-effects model.

RESULTS: Twenty-two studies involving > 34,000 patients were included. RSA was associated with greater forward flexion and a trend toward improved abduction, whereas internal rotation favored ORIF without reaching significance. No significant differences were observed in external rotation. The functional scores (Constant-Murley, Oxford Shoulder Score) were similar, and the complication and reoperation rates did not differ significantly between the groups.

CONCLUSION: RSA offers modest advantages in forward flexion compared with ORIF but does not consistently improve overall functional scores or reduce complications. RSA may be preferred in elderly patients with poor bone stock or rotator c […]

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Changes in kinematic behavior and collateral ligament strain after medially stabilized TKA using a novel intraoperative navigation platform: a cadaveric study.

Van de Vyver A, Taylan O, Peersman G, Demeulenaere M, Bialy M, Lauwagie T, Scheys L

OBJECTIVES: This study aimed to analyze to what extent a novel intraoperative navigation platform (Next-AR, Medacta) for total knee arthroplasty (TKA) allows to restore the native knee joint kinematics and strains in the medial collateral ligament (MCL) and lateral collateral ligament (LCL) throughout a squatting motion.

MATERIALS AND METHODS: Computed tomography (CT) scans of 6 native cadaver legs were used to design patient-specific guides. Bony landmarks and virtual single-line collateral ligaments were identified to acquire real-time intraoperative feedback on bone resection, implant alignment, tibiofemoral kinematics, and collateral ligament elongations using the Next-AR system. The specimens were subjected to squatting (35°-100°) motion using a physiological ex vivo knee simulator while maintaining a constant vertical ankle load of 110 N through active quadriceps and bilateral hamstring controls. Subsequently, each knee underwent a medially-stabilized TKA (Medacta) with mechanical alignment technique and was retested under the same conditions as in their native state. The tibiofemoral and patellofemoral kinematics, along with collateral ligament strains, were computed from 3D marker trajectories using a six-camera optical system (Vicon). MCL and LCL insertions-ant, mid, and post bundles-were identified in relation to bone-pin markers using a wand.

RESULTS: Both native a […]

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Regional differences in the management of patients with mild traumatic brain injury and antithrombotic therapy-an Austrian survey.

Sebek B, Dammerer D, Putzer D, Moser L

BACKGROUND/PURPOSE: The number of patients with mild traumatic brain injury (TBI) receiving oral anticoagulation is increasing. Up-to-date diagnostic and treatment algorithms are missing in Austria. The aim of this survey is to collect and analyse Austria-wide data on the care of this patient group.

METHODS: The survey was carried out using "Jotform" between 27 November 2023 and 31 January 2024, asking questions relating to the diagnosis and treatment of patients with mild TBI receiving antithrombotic therapy. Patient groups comprising individuals with alcohol addiction and hemophilia were incorporated as well with a view to the paucity of available literature on these groups. The survey was sent to all orthopedic-traumatological and neurosurgical departments in Austria.

RESULTS: 39 of 67 (58.2%) orthopedics & traumatology departments and 4 of 10 (40%) neurosurgery departments participated in the survey. The share of departments reporting routine submission of different patient groups to cranial computed tomography (CCT) was 38 (95%) for patients taking antiplatelet agents, 39 (97.5%) departments for patients taking vitamin K antagonists (VKAs), 39 (97.5%) for patients taking direct oral anticoagulants (DOACs), 11 (27.5%) for patients with alcohol addiction, and 28 (70%) for patients with known hemophilia. Considering the separate groups of patients not living in a nursing fa […]

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Extracorporeal shockwave therapy in bone fractures: a systematic review.

Santos TIR, Souza E Silva LC, Bonifacio M, Campos Melo CM, Sena BG, Simões MC, Risso Parisi J, da Silva Fernandes AC et al.

INTRODUCTION: The increasing incidence of fractures and cases of delayed union or nonunion has encouraged the search for non-invasive therapies. Extracorporeal shockwave therapy (ESWT) has been proposed as a strategy to stimulate bone regeneration and improve fracture healing.

MATERIALS AND METHODS: A systematic review was conducted following PRISMA guidelines, with searches performed in PubMed, Scopus, Web of Science, and Embase (2005-2025). Clinical studies evaluating ESWT for fracture treatment with radiographic assessment of bone healing and the presence of a control group were included.

RESULTS: A total of 834 studies were initially identified, of which 7 were included in the final analysis. Most studies demonstrated positive effects of ESWT, including improved bone healing, pain reduction, and functional recovery. However, there was considerable heterogeneity in treatment protocols and relatively small sample sizes.

CONCLUSION: ESWT shows potential as a non-invasive therapy to stimulate fracture healing, particularly in cases of delayed union or nonunion. However, further controlled clinical trials and standardized protocols are needed to confirm its clinical effectiveness.

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Clinical outcomes and complication rates of minimally invasive plate osteosynthesis (MIPO) versus open reduction and internal fixation (ORIF) in midshaft clavicular fractures: a comparative study.

Lin G, Zheng S, Zhang Y, Liu Q, Liu H

BACKGROUND: Conventional open reduction and internal fixation (ORIF) remains the standard approach for midshaft clavicle fractures, yet its traumatic nature has driven the need for minimally invasive techniques. Locking minimally invasive plate osteosynthesis (MIPO) demonstrates clinical potential through biomechanical superiority and tissue-preserving characteristics.

OBJECTIVE: To compare the clinical efficacy and safety of MIPO versus ORIF for midshaft clavicular fractures.

METHODS: A retrospective cohort study included 203 patients (MIPO group:101 cases; ORIF group:102 cases) treated between 2022 and 2024. Baseline characteristics, functional outcomes (DASH and Constant-Murley scores), and complications were analyzed using Mann-Whitney U and chi-square tests.

RESULTS: No baseline differences were observed ($P > 0.05$). Functional outcomes were equivalent between groups (DASH score: $P = 0.906$; Constant-murley score: $P = 0.646$). The overall complication rate was significantly lower in the MIPO group than in the ORIF group (7.92% vs. 36.27%, $P < 0.001$), particularly in nonunion (0% vs. 5.88%, $P = 0.039$) and sensory disturbance (6.93% vs. 27.45%, $P < 0.001$). This advantage is also evident in the subgroup of patients with AO type 2 C fractures, where MIPO similarly demonstrates a significantly lower incidence of these two complications.

CONCLUSION: MIPO achieves comparable fun […]

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Isolated scaphoidectomy for type II SLAC and SNAC wrists: retrospective case-series at long-term follow-up.

Altissimi M, Pataia E, Saracco M, Braghiroli L

BACKGROUNDS: Stage II scapholunate advanced collapse (SLAC) and Scaphoid non-union advanced collapse (SNAC) wrists are currently treated either by scaphoidectomy and Four Corner Fusion or by Proximal Row Carpectomy. In our retrospective study, isolated scaphoidectomy demonstrated clinical non-inferiority when compared to standard procedures and could be supplemented by a Four Corner Fusion only at a later time if needed.

METHODS: Between 2006 and 2011, 13 patients / 14 wrists (mostly males, manual workers) underwent the surgical procedure. 8 patients were affected by SNAC and 5 by SLAC. The average age was 56,5. All patients were manual workers. The mean follow-up was 15 years. These patients were reviewed a first time in 2012 and once again in 2024. Subjective outcome, DASH score, return to work, ROM and grip strength were evaluated. Radiographic assessment was done in all 14 wrists at the first follow-up and in 6 at the last long-time review. Statistical analysis was conducted using the SPSS software.

RESULTS: 12 patients had resumed the same work at a mean of 3 months after surgery. Only one patient required a Four Corner Fusion after 2 years. 11 of the 13 enrolled patients declared their satisfaction for the procedure at the longest follow-ups. Clinical results remained stable over time. Radiological deterioration was seen in 6/13 wrists at the first follow-up and in all […]

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Loop versus anchor tenodesis of the long head of the biceps tendon: a prospective randomized controlled pilot study.

Bischofreiter M, Kölblinger C, Gattringer M, Azesberger A, Gruber MS, Rittenschober F, Kindermann H, Ortmaier R

BACKGROUND: The long head of the biceps tendon is a frequent source of anterior shoulder pain, often treated with tenodesis. While anchor tenodesis is well established, it carries risks of implant-related complications. This study evaluates an implant-free "loop tenodesis" technique as a potentially safer and cost-effective alternative.

METHODS: In this prospective, double-blinded, randomized exploratory study, 48 patients undergoing arthroscopic biceps tenodesis were randomized to either loop (n = 24) or anchor (n = 24) tenodesis. Clinical outcomes were assessed using ASES, Constant, and DASH scores, as well as range of motion, supination strength, satisfaction, aesthetic appearance, and return-to-sport. Statistical analysis was performed using the Mann-Whitney U and Wilcoxon signed-rank tests to compare non-parametric data.

RESULTS: Both groups showed significant improvements with similar gains in functional scores. Supination strength recovery was greater in the loop group (103% vs. 93%). There were no significant differences in range of motion or complications. Notably, only the anchor group showed cases of Popeye deformity (2 patients). Return to sport was achieved in 95.3% of patients, typically within 3 months. No revision or implant-related issues occurred.

CONCLUSION: Loop tenodesis offers equivalent clinical outcomes to anchor-based techniques, with fewer aesthetic […]

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Split hamate fractures as part of complex ulnar carpometacarpal injuries: radiographic and functional outcomes.

Glaser J, Schmidt E, Aman M, Bickert B, Harhaus-Wähner L, Cordts T

BACKGROUND: Split hamate fractures, which most often are a complicating part of fracture dislocations of the 5th or 5th and 4th carpometacarpal joints, are rare and challenging injuries that can severely affect hand function. The injury usually requires surgical treatment. However, both short- and long-term data on the effectiveness of these interventions are limited. The objective of this study is to advance understanding of this rare hand surgical condition by investigating its epidemiology, etiopathogenesis, subsequent complications, and functional outcomes.

METHODS: Between 2010 and 2024, a total of 29 patients with a split hamate fracture were treated at a level-1 trauma center, of which 12 patients are included in this study. Using patient surveys (DASH, SF-36, PRWE), along with clinical and radiological examinations, the hand function of a total of 12 patients with split hamate fractures is evaluated.

RESULTS: Twelve male patients with a split hamate fracture were included, with a mean age of 28.1 years at the time of surgery. In 10 patients (83%) the dominant right hand was affected, most involving, the injuries resulted from direct trauma or a fall; surgery was performed on average of 12 days after injury. Functional outcomes were favourable, with a mean DASH score of 14.1 and PRWE score of 14.8, indicating minimal residual reduction of function in most cases. Radiol […]

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Outcomes and short-term survivorship of fixed-bearing all-poly unicompartmental knee arthroplasty. A retrospective single-center study of 42 prostheses with a main follow-up of 32.3 months.

Gaggiotti S, Gaggiotti G, Lustig S, Batailler C, Monllau JC, Gaggiotti S, Combalia A

PURPOSE: All Poly (AP) tibial unicompartmental knee arthroplasty (UKA) seems to have advantages in comparison with metal-based (MB) UKA, such as greater polyethylene thickness, wear resistance, bone stock preservation and lower implant costs. The objective of this retrospective study was to evaluate the short-term clinical-functional and radiographic results, complications and survivorship of AP UKA performed consecutively with a minimum follow-up of 2 years.

METHODS: Retrospective cohort analysis all consecutive knees operated with medial and lateral AP UKAs, including bilateral cases, with a minimum follow-up of 2 years. Demographic data were analyzed at the patient level and clinical-functional results were assessed per knee using the KSS, KOOS and VAS scales. Radiographically, mechanical axis (HKA) correction and the presence of radiolucency > 2mm were evaluated. Complications incidence and prosthetic survivorship was determined. Pre-postoperative comparisons were performed using linear mixed-effects models with random intercept per patient to account for the non-independence of bilateral cases. Subgroup analyses involving fewer than 10 observations were considered exploratory and no formal statistical inference is drawn. A value of $p < 0.05$ was considered statistically significant with a confidence interval (CI) 95%.

RESULTS: 42 AP UKAs were included, 5 lateral (11.9%) a […]

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Mix and match in reverse total shoulder arthroplasty: overview of current prostheses systems and potential combinations.

Seiß D, Schlichter A, Graf JM, Stange R, Ohlmeier M

INTRODUCTION: Mix and Match is considered a viable option for minimizing morbidity in total hip arthroplasty (THA). However in reverse total shoulder arthroplasty (rTSA) this strategy remains comparatively new. This study provides an overview of currently available implant systems and the potential combinations of glenoid and humeral components.

MATERIALS AND METHODS: A review of commercially available shoulder arthroplasty systems was performed. Nineteen manufacturers were contacted and asked to provide detailed information on glenoid and humeral component sizes for their rTSA systems. The reported component sizes were compiled and analyzed for possible combinations.

RESULTS: Thirteen manufacturers (68%) consented to the publication of their component size data. Two manufacturers (11%) do not produce rTSA systems, and four (21%) did not respond to repeated inquiries. A total of 21 implant systems were reported, comprising 10 distinct implant sizes. Glenosphere and humeral inlay diameters ranged from 32 mm to 48 mm, with 36 mm being the most commonly available size (16 of 21 systems, 76%).

CONCLUSION: Mix-and-match implantation in rTSA is likely employed more frequently than documented, yet remains poorly investigated. This study demonstrates that numerous systems can theoretically be combined during revision surgery, although the biomechanical implications are not yet under […]

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Radiolucent lines after robotic-assisted surface-matching kinematically aligned UKA: short-term incidence and lack of association with tibial alignment.

Itou J, Kuwasawa A, Nihei K, Okazaki K

PURPOSE: This study investigated the clinical outcomes of robotic-assisted unicompartmental knee arthroplasty (UKA) performed using a surface-matching kinematic technique that restores native joint morphology without alignment constraints. The primary aim was to determine the incidence and distribution of radiolucent lines (RLLs), and the secondary aim was to evaluate short-term revision rates and clinical outcomes.

METHODS: This retrospective cohort study reviewed 242 medial UKAs performed between November 2021 and August 2023 by the same high■volume surgeon using a surface■matching technique. Radiographic parameters and patient■reported outcome measures (PROMs) obtained before and 1 year after surgery were assessed, with radiographic evaluation performed independently of the clinical outcome assessment. A cemented JOURNEY II Uni implant was used in all cases, with intraoperative surface mapping used to guide kinematic placement of components without alignment constraints.

RESULTS: RLLs were seen in 21.9% of knees, exclusively on the tibial side and mainly in the most medial (zone 1: 14.8%, 95% confidence interval [CI] 10.9-19.9) and posterior regions (zone 9: 15.3%, 95% CI 11.3-20.3). A larger tibial component was the only independent predictor of RLLs (odds ratio 1.54, 95% CI 1.27-1.88; p = 0.04). All PROMs improved, with no differences between knees with and without RLLs. […]

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Radiographic parameters and mechanical complications of articulating vs. non-articulating hip spacers.

Lenz J, Baumann F, Baertl S, Straub J, Rupp M, Alt V, Freigang V

BACKGROUND: Periprosthetic joint infection (PJI) is a serious complication of total hip arthroplasty. Standard treatment for PJI is a two-stage revision surgery with antibiotic-loaded cement spacers to control infection and provide temporary joint stability. This study compares the radiological outcomes and complication rates between articulating and non-articulating hip spacers in patients undergoing treatment for PJI and native joint infections.

METHODS: We retrospectively reviewed 71 hip spacers (34 articulating, 37 non-articulating) in 38 patients treated between 2004 and 2022. Data on leg length discrepancy, femoral offset, infection control, and mechanical complications were obtained. For infection analysis, only patients treated exclusively with one spacer type were included. After excluding eight patients with mixed spacer types (29 spacers), 30 patients with 42 spacers were included in this subgroup.

RESULTS: Articulating spacers were significantly better at preserving leg length, with a mean discrepancy of -3.7 mm compared to -16.9 mm for non-articulating spacers ($p = 0.025$). However, non-articulating spacers maintained femoral offset (1.1 vs. 0.7, $p < 0.001$) closer to physiological offset. The rate of mechanical complications was higher in the articulating spacer group, with spacer dislocations occurring in 45% of cases compared to 10% in the non-articulating group […]

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The effect of Poller screw or monocortical plate augmentation on the stability of intramedullary nailing in proximal extra-articular tibial fractures: a biomechanical study.

■ahin MA, Durgut F, Yi■it ■, Akar MS, Özkul E, Atiç R, Çiftçi S

BACKGROUND: Proximal extra-articular tibial fractures present considerable biomechanical and technical challenges due to the short proximal fragment, the wide metaphyseal canal, and the deforming forces acting around the knee. These factors predispose isolated intramedullary (IM) nailing to malalignment, prompting the use of adjunct reduction aids such as Poller screws or supplemental plates. This study aimed to compare the mechanical behavior of IM nailing alone with that augmented by Poller screws or a monocortical plate in a standardized proximal tibia fracture model.

METHODS: Fifteen synthetic tibiae with an AO/OTA 41-A2.3 extra-articular proximal fracture model were randomized into three groups ($n = 5$ each): IM nailing alone, IM nailing plus two Poller screws, and IM nailing plus a four-hole medial monocortical plate. All specimens were instrumented with a 10 × 340-mm titanium nail and two proximal and two distal locking screws. Mechanical testing included axial compression to 800 N, axial loading to failure, and torsion at a rate of 25°/min. Displacement, stiffness, maximum load, and torque parameters were compared across groups.

RESULTS: Under 800 N axial load, the plate-augmented group showed the lowest displacement and highest stiffness (2.40 ± 0.22 mm; 332.4 ± 21.7 N/mm), followed by the Poller screw group (2.48 ± 0.25 mm; 321.6 ± 19.4 N/mm) and the IM-only group (3 […]

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Clinical cutoff value of lumbar bending range for predicting residual pelvic obliquity after total hip arthroplasty in dysplastic hip osteoarthritis.

Yokoi H, Osawa Y, Ozawa Y, Funahashi H, Takegami Y, Imagama S

INTRODUCTION: Residual pelvic obliquity (PO) after total hip arthroplasty (THA) for dysplastic hip osteoarthritis (DHOA) may adversely affect coronal balance and postoperative outcomes. Although lumbar bending range (LBR) is considered critical for PO improvement, the clinical cutoff value for predicting residual PO remains unclear. This study aimed to determine the cutoff value of LBR associated with residual PO after THA.

MATERIALS AND METHODS: Among 382 patients who underwent primary THA for unilateral DHOA between July 2019 and June 2024, 98 patients with preoperative downward PO of ≥ 2° on the affected side were included. Patients were classified into a residual group (postoperative PO ≥ 2° at 1 year) and an improvement group (postoperative PO < 2°). Demographic data and radiographic parameters of the hip, lower limbs, and spine were compared. Multivariate logistic regression analysis was performed to identify factors associated with residual PO, and receiver operating characteristic (ROC) analysis was used to determine the optimal LBR cutoff value.

RESULTS: Residual PO was observed in 28 patients (29%). Compared with the improvement group, the residual group demonstrated significantly greater Crowe index values, preoperative radiographic leg length discrepancy, and preoperative PO. Affected-side LBR was significantly smaller in the residual group (4.3° ± 3.0° vs. 7.7° ± […]

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Hidden blood loss after proximal femur fractures fixation: analysis of the first three postoperative days and influence of anticoagulants.

Mayor J, Winkelmann M, Schwinn LS, Özbek H, Aktas G, Omar-Pacha T, Oberthür S, Krammig RH et al.

PURPOSE: Hidden blood loss (HBL) is an underrecognized contributor to total perioperative blood loss after hip fracture surgery. This study aimed to quantify HBL during the first three postoperative days following proximal femur fracture fixation and to assess the impact of preoperative anticoagulant therapy on blood loss and transfusion requirements.

METHODS: A retrospective cohort study was conducted including patients aged ≥ 65 years who underwent surgical fixation of proximal femur fractures at a level I trauma center between January 2020 and December 2022. Total blood loss was estimated from perioperative hemoglobin changes and calculated blood volume. Patients were stratified according to anticoagulant use, and logistic regression analysis was performed to identify independent predictors of increased blood loss.

RESULTS: Among 260 patients analyzed, median total blood loss was 1184 ml, and 25% experienced losses exceeding 1675 ml. High blood loss correlated significantly with anticoagulant use, higher BMI, and longer operative time ($p < 0.001$). In multivariate analysis, anticoagulant therapy (particularly direct oral anticoagulants) remained an independent predictor of high blood loss (OR 3.06; 95% CI 1.47-6.37; $p = 0.003$). Patients in the highest quartile required significantly more transfusions, though short-term mortality did not differ.

CONCLUSION: Hidden blood los […]

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Visual processing and interference performance influences on knee angular impulse in ACLR individuals: a cognitive-biomechanical analysis of drop-jumps.

Santamaria-Guzman K, Holmes H, Kosek J, Peoples B, Harrison K, Campos-Vargas S, Weimar W, Neely K et al.

INTRODUCTION: Visual processing speed and cognitive interference control are critical for athletic performance and Anterior Cruciate Ligament (ACL) injury risk. However, how these cognitive functions relate to biomechanical performance after Anterior Cruciate Ligament Reconstruction (ACLR) remains unclear. This gap is clinically relevant given persistent re-injury rates of 20-25% in young athletes returning to sport, suggesting that rehabilitation may insufficiently address cognitive demands. This study examined cognitive differences between ACLR individuals and controls during drop-jump tasks, and their relationship with knee angular impulse.

MATERIALS AND METHODS: Thirty-two females (16 ACLR, 16 controls; 20 ± 1 years) completed cognitive assessments including the Stroop Color and Word Test, Trail Making Test, Digit Span Memory Test, and visual/auditory reaction time tests. Participants performed drop-jumps under four conditions: standard, choice, visual-cued, and audio-cued. Knee angular impulse was calculated for eccentric, concentric, and net phases. Binomial logistic regression identified cognitive predictors distinguishing groups, followed by factorial ANOVA to assess biomechanical differences. Spearman correlations examined relationships between cognition and knee angular impulse.

RESULTS: Three cognitive variables distinguished groups: cognitive interference score, v […]

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Perceived leg length discrepancy after total hip arthroplasty is associated with global spinopelvic coronal flexibility.

Minoji S, Ishikura H, Kudo M, Hatano M, Nishiwaki T, Tanaka S

BACKGROUND: Perceived leg length discrepancy (pLLD) is a common source of dissatisfaction after total hip arthroplasty (THA), even in the absence of significant radiographic discrepancies. While spinopelvic factors have been increasingly recognized, most previous studies have focused on sagittal alignment or lumbar mobility, and the impact of preoperative global spinopelvic coronal flexibility on pLLD has remained unclear.

METHODS: We retrospectively reviewed 114 patients who underwent primary unilateral THA for osteoarthritis between January and December 2023. pLLD was assessed using a four-point scale at 6 months postoperatively. Patients who reported no perception of discrepancy were classified as the non-pLLD group, while those reporting mild, clear, or strong perception were grouped as the pLLD cohort. Preoperative spinal flexibility was measured on coronal radiographs during maximal lateral bending, calculating changes in spinopelvic angle (ΔSPA), reflecting thoracic-to-pelvic coronal flexibility, and lumbosacral angle (ΔLSA), reflecting lumbar-to-pelvic coronal flexibility. Secondary parameters included radiographic leg length discrepancy, leg lengthening, and sagittal spinopelvic alignment. Multivariable logistic regression analysis was used to identify independent predictors of pLLD.

RESULTS: pLLD was reported in 47 patients (41.2%). The pLLD group exhibited signific […]

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Sex-related differences in final knee balance during robotic-assisted unicompartmental knee arthroplasty.

Stimolo D, Leggieri F, Secci G, Massenzi M, Civinini R, Innocenti M

INTRODUCTION: In robotic medial unicompartmental knee arthroplasty (mUKA), a preoperative alignment strategy can be defined and subsequently adjusted through stress testing to achieve optimal soft-tissue balance. However, the application of uniform planning principles does not necessarily yield equivalent balance in all patients. The aim of this study was to examine differences in intraoperative balance according to patient sex and the degree of preoperative varus deformity.

MATERIALS AND METHODS: A single-center retrospective cohort study including 254 MAKO® robotic-assisted medial UKAs (2018-2025) was performed using a uniform intraoperative planning protocol. Gap widths were measured in millimeters at full extension and 90° of flexion (F90) and classified as tight (< 0.5 mm and negative values), balanced (0.5-2 mm), or loose (> 2 mm). Patients were stratified by sex and preoperative varus severity, and gaps were analyzed quantitatively and categorically. The independent effects of sex, varus deformity, and operating surgeon on extension and flexion gaps has been assessed.

RESULTS: Mean preoperative alignment was 174.2 ± 2.8°, with no sex-related difference in baseline varus ($p = 0.058$). Flexion gaps were significantly tighter in female patients ($p = 0.0045$). A tight F90 gap was observed in 17.2% more female than male patients ($p = 0.023$), with an absolute mean difference o […]

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A novel pinless augmented reality-based navigation system for total hip arthroplasty in the supine position: a comparative study of acetabular cup angle accuracy.

Kurishima H, Ogawa H, Yamada N, Noro A, Tomata Y, Tsukada S

INTRODUCTION: We developed a novel pinless augmented reality (AR)-based portable navigation system for total hip arthroplasty (THA) in the supine position to eliminate pin-site complications and intraoperative navigation abandonment, and to reduce registration time. This study compared the measurement accuracy of this new pinless portable navigation system with that of a conventional AR-based portable navigation system requiring pin insertion.

MATERIALS AND METHODS: This retrospective cohort study compared primary unilateral THAs performed using the novel pinless AR-based navigation system and a conventional AR-based navigation system. The primary outcome was the absolute difference between the acetabular cup angles displayed on the navigation screen and those measured on postoperative computed tomography. Secondary outcomes included target achievement rates within 5° and 10°. Pin-site complications, intraoperative navigation abandonment, and dislocations were also recorded.

RESULTS: A total of 228 hips were analyzed (Pinless: n = 115; Conventional: n = 113). While there was no significant difference in the median absolute difference for inclination (2.5° vs 2.4°; $p = 0.366$), the median absolute difference for anteversion was significantly larger in the pinless group than in the conventional group (2.9° vs 1.9°; $p = 0.011$). There were no significant differences in the percent […]

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Impact of modified Henry approach with preservation of Flexor Carpi Radialis Tendon Sheath on wrist function in the treatment of distal radius fractures.

Ma D, Ji J, Zhu Y, Fang X, Wang Y, Fan J

OBJECTIVE: To compare the clinical efficacy of a modified Henry approach (preserving the FCR tendon sheath) versus the traditional approach (incising the sheath) for distal radius fractures (DRFs) treated with volar locking plates, and to evaluate the outcomes within 12 months postoperatively.

METHODS: A retrospective cohort study was conducted on 165 patients with unstable DRFs who underwent volar locking plate fixation from January 2023 to March 2025, who were retrospectively divided into the Preservation Group (n = 79, FCR tendon sheath preserved via the modified approach) and the Control Group (n = 86, FCR tendon sheath incised via the traditional approach). Perioperative indicators, radiological parameters, Visual Analog Scale (VAS) scores, Disabilities of the Arm, Shoulder and Hand (DASH) scores, grip strength, wrist range of motion (ROM) and complications were compared at the follow-up time points of 1 week, 1, 2, 3, 6 and 12 months postoperatively.

RESULTS: There were no significant differences in baseline data and postoperative radiological reduction quality between the two groups ($P > 0.05$). The Preservation Group had significantly lower VAS scores at all follow-up time points ($P < 0.001$), and lower DASH scores from 1 week to 6 months ($P < 0.001$), with no statistical difference at 12 months ($P = 0.550$). Grip strength of the Preservation Group was significantly super […]

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Clinical Outcomes, Return to Sport and Psychological Readiness After Arthroscopic Bankart Repair Using Knotless All-Suture Anchors at a Minimum 2Year Follow-up.

Rab P, Rupp MC, Pfarrmaier A, Vieider RP, Shirinskiy IJ, Scheiderer B, Siebenlist S, Lacheta L

PURPOSE: To report the clinical outcomes, return to sport (RTS) and psychological readiness of patients who underwent arthroscopic Bankart repair with knotless all-suture anchors with a minimum follow-up of 2 years.

METHODS: In this retrospective case series, consecutive patients who underwent primary arthroscopic Bankart repair using knotless all-suture anchors between 08/2019 and 07/2022 were included. Patient-reported outcomes were assessed using the Western Ontario Shoulder Instability Index (WOSI), American Shoulder and Elbow Surgeons (ASES) score, Disabilities of the Arm, Shoulder and Hand questionnaire (DASH), Shoulder Instability-Return to Sport after Injury (SI-RSI) scale, subjective shoulder value (SSV), and the visual analogue scale (VAS) for pain. Patient satisfaction, RTS, return to preinjury level of sport, instability recurrence and revisions were recorded. Receiver operating characteristic (ROC) curve was calculated to assess the discriminative performance of the SI-RSI scale, and the Youden's index was employed to determine the optimal cutoff for prediction of return to preoperative level of sports.

RESULTS: Of 57 patients eligible for inclusion, 46 patients (11.1% female, 28.7 ± 6.8 years at surgery) were available at a follow-up of 2.9 [2.3-3.4] years. Three patients (6.5%) reported a redislocation, one patient (2.2%) underwent a revision and was excluded f […]

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Impact of frailty on 12-month patient-reported outcomes after posterior lumbar fusion surgery.

Haffer H, Muellner M, Chiapparelli E, Dodo Y, Camino-Willhuber G, Zhu J, Pumberger M, Shue J et al.

INTRODUCTION: Frailty might increase the risk for postoperative complications and death. It is not well understood how frailty influences patient-reported outcomes (PROs) in spine surgery. The objective is to investigate on the impact of frailty on PROs 1-year after lumbar fusion.

MATERIALS AND METHODS: This was a prospective observational study of patients undergoing open posterior lumbar fusion for degenerative conditions with 12-month follow-up (12 M-FU). The modified 5-item frailty index (mFI-5) score was determined preoperatively. PROs included the Oswestry-Disability-Index (ODI), 12-item Short-Form-Healthy-Survey with Physical (PCS-12) and Mental-Component-Score (MCS-12) and Numerical-Rating-Scale (back/leg pain) before surgery and 6 weeks, 3, 6 and 12 months (6 W, 3 M, 6 M, 12 M) postoperatively. PROs were compared between groups stratified as frail (mFI-5 ≥ 2), pre-frail (mFI-5 = 1) and non-frail (mFI-5 = 0). PRO changes at 12 M-FU were evaluated with substantial clinical benefit (SCB) metrics. Regression models were used to determine associations between mFI-5 and PROs.

RESULTS: 135 patients (52.6% female, 62.1 years, BMI 29.1 kg/m2) were analyzed. 31.3% and 51.8% of the frail and non-frail patients reached ODI-SCB, respectively. Frailty patients reached a significantly smaller ODI change at 12 M-FU than pre- and non-frail patients (15.2% vs. 19.3% and 19.0%, p < 0.0 […]

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Rates of union and risk factors for continued nonunion following exchange nailing of tibial nonunion.

Mastracci JC, Averkamp B, Braswell M, Yu Z, Chen AT, Natoli RM, Farooq H, Mir H et al.

INTRODUCTION: The objective of this study was to evaluate the rate of nonunion repair success in tibia nonunions treated with exchange nailing. This retrospective cohort study was conducted across five academic Level 1 trauma centers and included 63 patients with tibia nonunions.

MATERIALS AND METHODS: Patients who sustained a tibia fracture (AO/OTA 42) treated with intramedullary fixation that developed a nonunion and were subsequently treated with exchange nailing were retrospectively reviewed. The primary outcome measure was nonunion repair success based on osseous union. Additional analyses included union rate by AO/OTA classification, nonunion type, implant(s) used, graft used, time from initial procedure, and infection status.

RESULTS: All patients sustained an AO/OTA type 42 fracture. Out of 63 patients, 47 tibias (75%) achieved osseous union after the index exchange nail procedure. The rates of nonunion revision success were similar across nonunion types and time from initial procedure until exchange nailing. There was no significant difference in union rate when infection was present. Complications included re-operation, readmission, infection, and implant failure.

CONCLUSIONS: This large, multicenter study with contemporary implants, instruments, and techniques for exchange nailing tibia nonunions demonstrates a nonunion repair success rate of 75%, consistent with […]

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Robotic-assisted unicompartmental knee arthroplasty is associated with lower odds of prolonged hospitalization and no higher odds of high-charge admission during the index hospitalization.

Sun Q, Xu Y, Hong X, Dai M, Wang J, Xie H, Fu Z

BACKGROUND: Robotic-assisted unicompartmental knee arthroplasty (RA-UKA) may improve implant positioning, but its impact on index-hospitalization length of stay (LOS), billed charges, and early inpatient outcomes in routine practice remains unclear.

METHODS: Using the National Inpatient Sample (2016-2022), we identified primary medial UKA for osteoarthritis. RA-UKA was defined by ICD-10-PCS robotic-assisted lower-extremity procedure codes. Primary outcomes were prolonged hospitalization (LOS ≥ the 75th percentile, ≥ 2 days), high-charge admission (total hospital charges [TOTCHG] ≥ the 75th percentile, ≥ $72,321), and in-hospital mortality. We performed univariable comparisons and multivariable logistic regression adjusting for demographics, payer, admission type, hospital teaching status, comorbidities, and calendar year fixed effects (YEAR 2016-2022) using the unweighted NIS discharge sample.

RESULTS: Among 7,154 UKAs, 1,297 (18.1%) were RA-UKA and 5,857 (81.9%) were C-UKA. Median LOS was 1 day (IQR 1-2) in both groups; however, prolonged hospitalization (LOS ≥ 2 days) occurred less frequently in RA-UKA (412/1,297 [30.7%] vs. 2,162/5,857 [37.2%]; P < 0.001). In adjusted analyses including YEAR fixed effects, RA-UKA was associated with lower odds of prolonged hospitalization (aOR 0.750, 95% CI 0.647-0.870; P < 0.001) and was not associated with high-charge admission (TOTCHG ≥ […]

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Is biceps augmentation effective for scope-assisted lower trapezius tendon transfer in posterosuperior irreparable rotator cuff tears? A retrospective short-term clinical

comparison.

Baek CH, Elhassan BT, Lim C, Kim JG, Kim BT, Kim SJ

INTRODUCTION: Although favorable clinical outcomes of scope-assisted lower trapezius tendon transfer (SALTT) have been reported in posterosuperior irreparable rotator cuff tears (PSIRCTs) patients, the limited restoring of static stability can be supported by biceps augmentation as a static stabilizer. This study aimed to compare the outcomes of scope-assisted lower trapezius tendon transfer (SALTT) with and without biceps augmentation in patients with posterosuperior irreparable rotator cuff tears (PSIRCTs).

MATERIAL AND METHODS: This retrospective clinical comparative study was performed with the inclusion criteria: patients who underwent SALTT for PSIRCT from January 2022 to April 2023; a follow-up period of more than 2 years; and availability for clinical assessment and MRI evaluation preoperatively and at 2 years postoperatively. The patients were grouped according to the management of biceps; Non-augmentation group (n = 26) and Augmentation group (n = 27). The clinical outcomes were evaluated using shoulder pain, patient-reported outcome measures (PROMs), and active range of motion (aROM). Radiological outcomes were performed to evaluate the progression of shoulder joint arthritis, biceps integrity, and graft integrity.

RESULTS: Both groups showed significant improvements in VAS score, PROMs, and aROM. Although postoperative PROMs of Non-augmentation group were signific […]

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Robotic-arm assisted total hip arthroplasty using a short metaphyseal filling collared femoral implant through a direct anterior approach: minimum two-year outcomes.

Marchand R, Kaczynski E, Taylor K, Smitterberg C, Mont MA

INTRODUCTION: Short, collared, metaphyseal-filling femoral stems preserve proximal bone and achieve stable fixation while reducing stress shielding compared to diaphyseal-engaging stems. Early six-month data for one such stem demonstrated improved patient-reported outcomes with no implant-related complications. This study extends follow-up of the same cohort to a minimum of two years to evaluate (a) functional outcomes, (b) implant-related complications, and (c) survivorship.

MATERIALS AND METHODS: The prospective cohort of 120 patients underwent robotic-arm assisted total hip arthroplasty (THA) through a direct anterior approach (DAA) using a short, collared, metaphyseal-filling stem. Patients had a mean age of 69.2 years (range 37-90) and a body mass index of 28.2 (range 18.1-49.1), and they were 71% women and were followed for a mean of 2.6 years (range 2.0-3.4). The Hip Disability and Osteoarthritis Outcome Score for Joint Replacement (HOOS-JR) was compared with baseline using Student's t-tests. Complications included all-cause revision, periprosthetic joint infection, periprosthetic fracture, and dislocation at final follow-up.

RESULTS: At mean follow-up, HOOS-JR scores improved from a mean of 54.5 ± 13.9 preoperatively to 94.4 ± 10.3 ($P < 0.001$). The overall survivorship was 98.1% following two septic revisions.

CONCLUSIONS: This short, collared, metaphyseal-filling st […]

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Second contralateral hip fractures reduce survival, mobility and daily activity : a matched pair analysis.

Blattner A, Sabath F, Röttinger T, Lisitano L, Mayr E, Fenwick A

BACKGROUND: Hip fractures in older adults are associated with substantial morbidity, functional decline, and mortality. Patients who experience a first fragility fracture are at high risk of subsequent fractures, with repeated events further exacerbating functional impairment and survival outcomes. Second contralateral hip fractures, while clinically important, remain under-characterized in terms of timing, long-term functional impact, and mortality.

OBJECTIVES: To investigate the incidence, timing, survival, mobility, and daily activity outcomes of second contralateral hip fractures using a matched pair analysis, and to situate these findings within the broader context of repeated fragility fractures.

METHODS: A retrospective cohort study was conducted at a single Level I trauma center, including all patients treated operatively for hip fractures (AO 31A1-A3, 31B) between January 2016 and June 2020. Patients with a second contralateral hip fracture were matched 1:1 with patients with a single fracture based on age, sex, fracture type, and Charlson Comorbidity Index. Demographic data, functional status (Barthel Index, Parker Mobility Score), walking aid use, living situation, and mortality were assessed at admission, discharge, and follow-up (minimum 3 years, maximum 7 years). Statistical analysis included paired tests and survival analysis.

RESULTS: Of 1,933 patients, 148 ([…]

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Short-term clinical outcomes of robotic-assisted total knee arthroplasty at 12-month follow-up: a prospective, multicenter, concomitant comparison to conventional total knee arthroplasty.

Nöth U, Rivkin G, Caldora P, Heller KD, Thienpont E, Hannouche D, Perets I, Cholewa J

INTRODUCTION: There is limited data available on short-term outcomes on a cut-block positioning robotic system. The purpose of this study was to compare 12-month clinical outcomes between robotic-assisted (raTKA) and conventional total knee arthroplasty (cTKA) with multiple outcomes and surgical centers.

METHODS: This was a non-randomized controlled trial of patients who received either raTKA (n = 120) or cTKA (n = 101) at 6 different surgical centers. Variables of interest included occurrence of soft tissue release, complications and revisions at minimum one-year follow-up. Satisfaction, pain (numeric rating scale [NRS]), 5-dimensional European Quality of Life (EQ-5D-5 L) questionnaire (index and visual analog scale [VAS]), Oxford Knee Score (OKS), and the Forgotten Joint Score (FJS-12) were collected pre-operatively, and at six weeks, three months, and 12 months post-operative.

RESULTS: There were significantly less soft tissue releases with raTKA (28/120, 23.3%) vs. cTKA (51/99, 51.5%), $p < 0.0001$). There were significantly fewer cases of medial/lateral instability in the raTKA group at six-weeks ($p = 0.038$) and three-months ($p = 0.007$) post-operative. At one-year follow-up, 96.3 and 92.5% of raTKA and cTKA patients were satisfied with the overall results of their surgery, respectively. Significantly more raTKA patients were very satisfied (32.1% vs. 14.6%) with their abil […]

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Clinical and radiological outcomes of using locked intramedullary nails in the treatment of severe frontal plane lower limb deformity in adolescents with hypophosphatemic rickets (mid-term results).

Galal S, Al-Abri AM, Al-Habsi G, Al Ghammari M, Elissawi A, Abou Senna WG, Arafa AS

INTRODUCTION: Lower limb deformities in patients with hypophosphatemic rickets are multi-apical and require multiple osteotomies for correction. Intramedullary nails (IMNs) are used to fix multiple osteotomies. Authors of this studyaimed to assess the clinical and radiographic outcomes of using IMNs for correcting lower limb deformity in adolescents with hypophosphatemic rickets.

METHODS: This prospective study included patients with hypophosphatemic rickets who underwent deformity correction between November 2020 and November 2023 using IMNs, with a minimum follow-up of 2 years. Clinical outcomes were assessed using the Lower Limb Deformity-Scoliosis Research Society (LD-SRS) score. Radiographic outcomes measured included the mechanical tibiofemoral angle (mTFA), mechanical axis deviation (MAD), mechanical lateral distal femoral angle (mLDFA), mechanical medial proximal tibial angle (mMPTA), and Stevens' knee joint zoning.

RESULTS: Twenty patients (25 limbs and 35 bones) were included (mean age: 16 years; range: 13-22 years). The LD-SRS score improved from 3.2 ± 0.4 preoperatively to 4.3 ± 0.7 postoperatively. Preoperative mechanical tibiofemoral angle was $27.8° \pm 12.1°$ and $17.33° \pm 7.9°$ in the varus and valgus groups, respectively, improving to $3.4° \pm 6.4°$ and $2° \pm 5.2°$, respectively. Preoperative mechanical axis deviation was 82.2 ± 35.6 mm and 41.8 ± 22.2 mm in the varus […]

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Risk prediction and surgical decision-making for pathological fractures in phalangeal enchondroma: validation of the 0.51 threshold and outcome comparison of prophylactic intervention in 65 patients.

Zhao J, Mo Z, Lao Y, Chen J, Li Q, Lin J

BACKGROUND: Enchondroma is the most common benign bone tumor of the hand, predominantly affecting the phalanges and metacarpals, and is frequently complicated by pathological fractures. Although previous studies have explored risk factors based on tumor size, a precise quantitative cut-off value remains undefined. Furthermore, quantitative research regarding the impact of “delayed treatment” on long-term functional outcomes is lacking. This study aims to establish a risk threshold for pathological fractures in phalangeal enchondroma and to provide an evidence-based rationale for optimizing surgical decision-making by comparing the clinical outcomes of prophylactic surgery versus post-fracture management. METHODS: We retrospectively analyzed the clinical data of 65 patients with phalangeal enchondroma treated surgically at our institution between 2017 and 2025. Patients were categorized into a Fracture Group (n = 35) and a Non-fracture Group (n = 30) based on the presence of a complete preoperative fracture. The tumor-to-phalanx length ratio was measured using the Picture Archiving and Communication System (PACS), and the degree of cortical thinning was assessed according to the Takedani classification. Multivariate logistic regression analysis was employed to identify independent risk factors, and Receiver Operating Characteristic (ROC) curve analysis was used to determine the […]

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Assessing PHILOS plate as an alternative fixation method for pediatric femoral neck fractures: a biomechanical comparison with cannulated screws.

Lim BS, Wang YP, Chung ST, Wang CH, Chang SW, Wang TM, Hong CK

BACKGROUND: Pediatric femoral neck fractures require stable fixation to avoid complications. It remains unclear whether fixation with the Proximal Humeral Internal Locking System (PHILOS) can serve as an alternative to cannulated screw fixation. The purpose of this study was to compare the biomechanical properties of PHILOS and cannulated screws for stabilizing unstable pediatric femoral neck fractures using a synthetic bone model.

MATERIALS AND METHODS: Twelve fourth-generation synthetic composite femurs were randomly assigned to screw fixation (Group S) or PHILOS fixation (Group P) (n = 6 each). A standardized vertically oriented Delbet type II osteotomy was created in all specimens. Group S was fixed with three 6.5-mm cannulated screws, whereas Group P received a PHILOS plate with 3.5-mm locking screws. Each specimen underwent a standardized loading protocol using a universal testing machine. Axial stiffness, cyclic displacement, ultimate failure load, and failure modes were recorded and statistically compared between groups.

RESULTS: No statistically significant difference was found in axial stiffness between Group P (746 ± 300 N/mm) and Group S (753 ± 256 N/mm) (p = 1.000). Displacement after cyclic loading was significantly greater in Group P (1.42 ± 0.3 mm) compared with Group S (0.57 ± 0.2 mm) (p = 0.004). The ultimate failure load was higher in Group S (2378 ± 513 N) […]

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Rerupture after flexor tendon repair of the hand and wrist: a retrospective risk factor analysis.

Marwieser SF, Vavricka S, Kaiser P, Schmidle G, Lindtner RA, Arora R, Konschake M

INTRODUCTION: Rerupture after primary flexor tendon repair of the hand and wrist is a serious complication that can substantially impair hand function and restrict activities of daily living, occupational performance, and recreation. Identifying factors associated with rerupture is essential to optimize surgical techniques and postoperative rehabilitation strategies and to improve patient outcomes. METHODS: A retrospective observational cohort study was conducted including patients who underwent surgical repair of flexor tendon injuries of the hand and wrist at a single institution between 2010 and 2022. The study included 292 patients with 429 complete tendon injuries. Collected data comprised demographic characteristics, injury mechanism and type, anatomical location, surgical timing and technique, and postoperative immobilization and rehabilitation protocols. Associations between potential risk factors and rerupture were explored using univariable analyses, followed by a multivariable logistic regression model to identify independent predictors. RESULTS: The overall rerupture rate was low but clinically relevant, occurring in a small proportion of patients and tendon lesions. Rerupture was more frequently observed in male patients, older individuals, and those with specific injury mechanisms. Surgical factors, particularly the need for pulley reconstruction, were associated […]

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Defining orthoplastic limb salvage centers: a systematic review.

Moussa O, Pacheco FJ, Raasveld FV, Gonzalez MR, Oflazoglu K, Ritt MJPF, Valerio IL, Rakhorst H et al.

INTRODUCTION: Limb salvage centers have increased in number over time, but lack standardized defining criteria. This systematic review aimed to assess organizational features of limb salvage centers and determine whether orthoplastic centers, in comparison to vascular limb salvage centers, represent a distinct care model that may benefit from standardization.

METHODS: We conducted a systematic review of publications related to limb salvage centers by searching MEDLINE, Embase, Web of Science, and Cochrane databases from their inception through 2024. We quantified binary data extraction as a reporting score of 26 organizational features across six structural care domains for limb salvage centers, based on a validated quality measurement framework. Organizational features differentiating distinct center types were identified to establish a quality framework for orthoplastic centers. Statistical comparisons between center types were performed using appropriate tests ($p < 0.05$).

RESULTS: Of 118 included studies, orthoplastic ($n = 43$) and vascular ($n = 48$) centers represented 77% of all studies. Recent increases in orthoplastic publications show substantial variability in organizational features. Orthoplastic center literature more frequently reported plastic surgery consultation criteria ($p < 0.001$), surgical outcomes ($p < 0.001$), and centralized network integration ($p \leq 0.006$), […]

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Male incidence rates outpacing females: results from a nationwide analysis on acetabular fractures in Germany.

Baumann F, Bärtl S, Szymski D, Hierl K, Alt V, Freigang V

INTRODUCTION: Acetabular fractures are uncommon, but serious injuries. Demographic changes may have a significant impact on planning healthcare structures to improve treatment outcomes. Aim of this nationwide, registry-based retrospective controlled study was to identify incidence trends, demographic characteristics, and care structures of patients with acetabular fractures in Germany.

MATERIALS AND METHODS: We analyzed inpatient data from the Institute for the Hospital Remuneration System (InEK). Based on 52 095 patients with primary diagnosis of an acetabular fracture between 2019 and 2024, we calculated incidence rates for different age-groups and put a spotlight on geriatric acetabular fractures (> 65 years of age).

RESULTS: Incidence rates in patients under 65 years remained stable, whereas patients over 65 years showed a significant age-dependent increase with an exponential rise in men aged 80 + with the highest incidence being 122.4/100 000 inhabitants annually. We recorded high levels of co-morbidity and nursing care dependency for elderly patients after acetabular fracture. Although 43% of patients were treated in hospitals > 500 beds, acetabular fractures were managed across all hospital sizes.

CONCLUSIONS: There is a rapidly increasing incidence of geriatric acetabular fractures, predominantly driven by elderly male patients over 80 years. Patients over 65 years […]

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Cemented total hip arthroplasty reduces early complications: a Japanese nationwide propensity-matched study.

Tanaka H, Tarasawa K, Mori Y, Baba K, Kurishima H, Fushimi K, Aizawa T, Fujimori K

INTRODUCTION: The optimal fixation method in total hip arthroplasty (THA) remains under debate. While cemented fixation has been associated with a lower risk of periprosthetic fracture, uncemented fixation predominates in Japan. This study aimed to compare early postoperative complications between cemented and uncemented fixation in elective THA using a nationwide inpatient database.

MATERIALS AND METHODS: We identified 198,102 patients aged ≥ 65 years who underwent primary THA for osteoarthritis, osteonecrosis, or rheumatoid arthritis between December 2011 and March 2023 from the Japanese Diagnosis Procedure Combination (DPC) database. After 1:1 propensity score matching for age, sex, body mass index (BMI), and Charlson Comorbidity Index, 36,859 patients were included in each fixation cohort. Surgical and medical complications, and in-hospital mortality were compared using multivariate logistic regression.

RESULTS: Cemented fixation was associated with a significantly lower risk of periprosthetic fracture (odds ratio [OR], 0.40; 95% confidence interval [CI], 0.30-0.53; p < 0.001), blood transfusion (OR, 0.76; 95% CI, 0.74-0.78; p < 0.001), and deep vein thrombosis (OR, 0.79; 95% CI, 0.74-0.84; p < 0.001). There were no statistically significant differences based on the predefined threshold (p < 0.001) in dislocation, infection, pulmonary embolism, cardiac or cerebrovascular […]

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Healthcare utilization following hip fractures based on social vulnerability status in the US: an analysis of 2016-2020 nationwide readmissions data.

Tilve R, Zhou G, Khan ST, Koroukian SM, Deren M, Piuzzi NS

INTRODUCTION: Social vulnerability (SV) influences rehabilitation and postoperative care for patients with hip fracture. However, most previous work relies on area-level measures that overlook interindividual variation. The recent adoption of ICD-10 Z-codes allows clinical identification of patient-level SV and may offer a better understanding of its impact. This study aimed to evaluate healthcare utilization, including readmissions, discharge disposition, and length of stay (LOS) in surgically treated hip fracture patients with and without clinically acknowledged SV.

METHODS: Adults surgically treated for hip fracture between 2016 and 2020 were included from the Nationwide Readmissions Database. SV was defined as having at least one documented relevant ICD-10 Z-code. Primary outcome measures included complications, LOS, discharge disposition, and 30- and 90-day readmissions, stratified by SV and evaluated using chi-square analyses. Multivariable logistic regression assessed long LOS (≥ 5 days) and discharge to home, adjusting for age, insurance/income status, and substance use.

RESULTS: Patients with SV were younger (35.6% with SV vs. 50.1% without SV were 81+), had a lower median household income (38.8% with SV vs. 25.7% without SV were in the lowest quartile), and were more often insured by Medicaid (19.3% vs. 3.8%). Alcohol/drug use disorders were significantly more preva […]

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Antibiotic prophylaxis for open lower extremity fractures: a comparative propensity-scored matched analysis of cefazolin vs. piperacillin-tazobactam.

Mody KS, Ponna AK, Santilli RT, Laychur A, Galloway J, Adams M, Reilly M, Lin S

INTRODUCTION: Optimal antibiotic prophylaxis for open fractures remains controversial, particularly regarding whether broader-spectrum regimens offer clinical advantages over cefazolin monotherapy. Despite guideline recommendations supporting first-generation cephalosporins for most open fractures, many centers routinely administer piperacillin-tazobactam. This study aims to evaluate differences in postoperative outcomes between cefazolin and piperacillin-tazobactam in lower extremity open fractures across all Gustilo-Anderson types.

MATERIALS AND METHODS: This retrospective cohort study was performed using the TriNetX database to identify adult patients with Gustilo-Anderson type I-III lower extremity open fractures who received either cefazolin or piperacillin-tazobactam. 1:1 propensity score matching controlled for age, sex, demographics, and relevant comorbidities. Outcomes were assessed at 90 days and 1 year using risk ratios (RR) with corresponding 95% confidence intervals.

RESULTS: A total of 47,692 patients met inclusion criteria prior to matching. After matching, 1,527 patients remained in each treatment group for the combined type I/II/III cohort and similar matched pairs were obtained for type I/II and type III subgroups. At 90 days, cefazolin was associated with significantly lower rates of surgical site infection (RR 0.569), osteomyelitis (RR 0.292), sepsis (RR 0 […]

PubMed · DOI

Venous thromboembolism prophylaxis in total shoulder arthroplasty: a matched cohort analysis.

Katakam A, Joshi T, Soussou T, Sirch F, Jones T, Yedikian T, Erickson J

BACKGROUND: Venous thromboembolism (VTE), including deep vein thrombosis (DVT) and pulmonary embolism (PE), is a potentially life-threatening complication following total shoulder arthroplasty (TSA). Despite the increased use of chemical prophylaxis, its effectiveness and safety in TSA populations remain unclear. This study evaluated the incidence of VTE, bleeding events, and related outcomes among TSA patients with and without postoperative chemical prophylaxis. MATERIALS AND METHODS: A retrospective cohort study was conducted using the TriNetX database, identifying patients who underwent TSA, stratifying into two cohorts: those who received chemical VTE prophylaxis and those who did not. Propensity score matching (1:1) was employed to balance demographics, comorbidities, and other confounding variables. Outcomes, including VTE, bleeding events, prosthetic joint infection (PJI), revision, and mortality, were assessed at 30 days, 90 days, and 6 months postoperatively. RESULTS: After matching, 9,859 patients were included in each cohort. There was no significant difference in the incidence of VTE, PE, or DVT at any time point between the groups. Patients who received prophylaxis showed a reduced risk of ischemic stroke at 30 days (HR 0.539; 95% CI 0.367–0.793; p = 0.001) and myocardial infarction (MI) at both 30 days (HR 0.469; 95% CI 0.306–0.718; p < 0.001) and 90 days (HR 0.72 […]

PubMed · DOI

Advancing pelvic and acetabular trauma education.

Pfeifer R, Gaensslen A, Lindahl J, Enocson A, Hildebrand F, Pape HC

Pelvic and acetabular injuries are uniquely demanding due to complex three-dimensional anatomy, the proximity of critical neurovascular structures, and the incidence of these fractures in elderly, a well as and polytrauma patients, is increasing. While existing courses focus heavily on acute surgical management and established techniques, they often omit critical subjects such as complication management, revision strategies, and interdisciplinary care. To establish an international benchmark for surgical excellence in pelvic surgery, the proposed program may utilize a competency-based framework divided into three parts: Part 1: Theoretical Foundation: Independent study of biomechanics, pathology, and operative planning, validated by a rigorous multiple-choice examination. Part 2: Surgical Skills Assessment: Evaluation of technical precision and intraoperative decision-making within a controlled wet lab environment. Part 3: Clinical Reasoning & Reflection: Expert-led discussions centered on the participant's own clinical cases, focusing on evidence-based reasoning and the management of complex revisions. By standardizing these advanced competencies, the program aims to ensure surgeons possess both the technical proficiency and the mature clinical judgment required to deliver high-quality care for complex pelvic and acetabular trauma cases.

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Mid- to long-term outcomes and survival of total hip arthroplasty using a Kerboull-type acetabular reinforcement plate: an analysis of associated factors.

Gosho S, Kaku N, Hosoyama T, Shibuta Y, Tanaka K

INTRODUCTION: Revision total hip arthroplasty (THA) for acetabular bone loss is challenging. The modified Kerboull-type (KT) plate has been adopted; nonetheless, mid- to long-term clinical outcomes remain poorly understood. We evaluated survival rates and radiographic outcomes of acetabular reconstruction using the KT plate and investigated risk factors for plate breakage or re-revision.

MATERIALS AND METHODS: We retrospectively included 120 patients (130 hips) who underwent acetabular reconstruction using the KT plate (1997-2024) and evaluated perioperative outcomes, Harris Hip Score (HHS), survival rates, and key radiographic parameters.

RESULTS: Mean age at surgery and follow-up duration were 69.0 ± 10.4 years and 107.4 ± 71.7 months, respectively. Mean blood loss and operative time were 630.2 mL and 284.3 min, respectively. HHS improved from 56.2 to 86.4. Fractures (5.3%) and dislocations (3.0%) were observed. Ten-year survival rates were 95.6% for re-revision and 93.0% for plate breakage. Failure and head migration occurred in 11 (8.5%) and 6 (4.6%) hips, respectively. Multivariate analysis identified younger age, use of morselized bone chips alone, and postoperative head migration as independent predictors of failure. In revision THA cases with ≥ 5-year follow-up and Paprosky classification type 3 A or 3B defects, age, Knight classification, and head migration differed […]

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Scaphoid waist fractures in the presence of carpal coalitions: a consistent biomechanical weak point?

Schmitt S, Winterholer D, Fritsche E

Carpal coalitions may alter intercarpal kinematics and load transfer. Reported coalition-associated scaphoid fractures show an exclusive predilection for the scaphoid waist, raising the possibility of a non-random biomechanical mechanism with focal stress concentration. These injuries have been associated with delayed union and nonunion. This pattern may justify CT-based assessment and consideration of a lower threshold for early surgical stabilization, including percutaneous screw fixation even in selected nondisplaced waist fractures. However, the currently available evidence is limited to case reports and does not allow firm treatment recommendations. Awareness of coalition anatomy and careful follow-up may be reasonable.

PubMed · DOI

Prophylactic percutaneous intra-aortic balloon occlusion reduces blood loss during anterior acetabular fracture surgery: a propensity score-matched study.

Kanezaki S, Miyazaki M, Hino A, Kawagishi M, Nishine J, Shigemi T, Kaku N

INTRODUCTION: Intra-aortic balloon occlusion (IABO) is used to control pelvic hemorrhage and has recently been adapted to elective surgery to limit intraoperative blood loss. Its role in acetabular fracture fixation, particularly via anterior approaches, remains unclear. This study evaluated whether prophylactic percutaneous IABO is associated with reduced blood loss during anterior open reduction and internal fixation (ORIF) of acetabular fractures, using propensity score matching (PSM) to minimize selection bias. MATERIALS AND METHODS: We conducted a retrospective cohort study including 80 consecutive adult patients who underwent anterior ORIF for acetabular fractures between 2013 and 2024. Twenty-four patients received prophylactic percutaneous IABO, and 56 served as controls. One-to-one PSM (caliper 0.2 SD of the logit) was performed on demographics, fracture characteristics, and surgical factors, yielding 20 matched pairs. Outcomes included intraoperative blood loss (IBL; primary), total blood loss (TBL; gross formula), operative time, transfusion, reduction quality, perioperative complications, and IABO-related parameters. RESULTS: IABO was associated with significantly lower blood loss both before and after matching. In matched pairs, median IBL was 525 g versus 1070 g in controls ($p = 0.004$), representing a 51% reduction, while TBL was 601 g versus 921 g ($p = 0.003$), co […]

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Comparison between the accuracy of conventional and portable computed tomography-based navigation systems in total hip arthroplasty.

Asai H, Osawa Y, Takegami Y, Funahashi H, Imagama S

BACKGROUND: Although conventional computed tomography (CT)-based navigation provides excellent placement accuracy and clinical outcomes, whether recently introduced portable systems can achieve comparable results remains unclear. This study aimed to evaluate the placement accuracy and surgical outcomes of portable CT-based navigation systems.

METHODS: This study assessed 56 hips of patients that underwent total hip arthroplasty (THA) using portable CT-based navigation. Using propensity score matching based on age, sex, and body mass index, we identified 51 hips treated with portable CT-based navigation (portable CTN group) and 51 hips with conventional CT-based navigation (CTN group). The evaluation parameters included cup orientation accuracy, cup positioning, operative time, blood loss, preoperative and postoperative Japanese Orthopaedic Association scores, and complications.

RESULTS: Regarding accuracy error, the portable CTN (radiographic inclination [RI]: 2.8 ± 2.8°, radiographic anteversion [RA]: 3.8 ± 3.2°) and CTN groups (RI: 2.7 ± 1.9°, RA: 3.0 ± 2.1°) did not significantly differ. For navigation error, the portable CTN group (RI: 3.2 ± 3.2°, RA: 3.5 ± 3.1°) had significantly inferior results to the CTN group (RI: 2.2 ± 1.7°, RA: 2.3 ± 1.7°) regarding anteversion. The portable CTN group demonstrated a significantly lower accuracy, as the proportion of hips with a nav […]

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Analysis of mental health outcomes in major versus minor upper extremity amputations: a retrospective national database study.

Nedder V, Wang J, Peek K

BACKGROUND: Upper extremity amputations are associated with developing psychiatric conditions. The aim of this study is to identify differences in mental health outcomes for patients undergoing major versus minor upper extremity amputations.

METHODS: Data were obtained from the PearlDiver database between 2010 and 2022 using Current Procedural Terminology and International Classification of Diseases codes. Patients aged 10 and above without prior mental health diagnoses or antidepressant prescription records who underwent upper extremity amputations were stratified by major (shoulder disarticulation, arm, forearm, wrist, transmetacarpal) versus minor (single metacarpal, digit, phalanx) amputations. Demographic characteristics and rates of mental health diagnoses, antidepressant prescriptions, and psychotherapy care were assessed for 90 days and one year postoperatively, using Welch's T-tests and Pearson's chi-squared tests.

RESULTS: Patients with major amputations had increased risk of mental health diagnoses at 90 days (OR: 3.29, $p < 0.001$) and at one year (OR: 2.01, $p < 0.001$). Both groups had higher rates of mental health diagnoses than the general population. Patients undergoing major amputations had higher odds of starting antidepressants at 90 days (OR 3.82, $p < 0.001$) and at one year (OR 2.38, $p < 0.001$). Psychotherapy care was significantly increased after major amput […]

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20 years of blood management and VTE prevention in orthopedic surgery in the Netherlands - a nationwide survey.

Kok RY, Gerritsen JTM, Verheyen CCPM, Németh B, Ettema HB

INTRODUCTION: Advances in arthroplasty care have progressively improved patient outcomes, including reduced bleeding and venous thromboembolism rates. Consequently, blood-saving measures applied pre-, peri- and postoperatively as well as thromboprophylaxis regimens have evolved over time. To evaluate these developments, we conducted a 2023 follow-up to the national surveys from 2002, 2007 and 2012, examining current blood-saving strategies and thromboprophylaxis use in The Netherlands.

MATERIALS AND METHODS: In 2023, a questionnaire designed to match the previous surveys was distributed to all orthopedic departments in the Netherlands. The collected data were summarized and compared with previous findings.

RESULTS: Responses were received from 61% of the Dutch orthopedic departments. Blood-saving practices have shifted significantly toward the use of tranexamic acid, accompanied by a decline in other measures. In particular, the use of drains, blood transfusions, and erythropoietin supplementation has decreased substantially. Thromboprophylaxis protocols showed fewer changes, although the average duration of prophylaxis has shortened, with low-molecular-weight heparin remaining the most commonly used anticoagulant.

CONCLUSIONS: Advances in surgical techniques and postoperative care around arthroplasty appear to have reduced concerns about perioperative blood loss to a level […]

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Optimal timing for hemodialysis in relation to hip fracture surgery for patients with end-stage renal disease.

House HE, Shimizu MR, Ma IH, Cohen JB, Summers HD, Levack AE

INTRODUCTION: End-stage renal disease (ESRD) is associated with a significantly increased risk of hip fractures and a higher rate of postoperative complications and mortality, particularly in patients on hemodialysis (HD). The timing of HD relative to surgery may influence outcomes by altering acid–base balance, electrolyte levels, and fluid status. This study aimed to evaluate the effect of perioperative HD timing on postoperative complication rates in ESRD patients undergoing hip fracture surgery. MATERIALS AND METHODS: This retrospective cohort study included patients with ESRD on HD who underwent operative treatment for hip fractures at a single Level I academic trauma center. Patient demographics, comorbidities, and perioperative variables were collected. Timing of HD was calculated as the interval in hours from the start of HD to the start of surgery. Disruptions in patients' routine HD schedules were also recorded. Univariable logistic regression was used to assess associations between HD timing and 90-day postoperative complications. RESULTS: Forty-one patients met inclusion criteria; 25 (61.0%) experienced a postoperative complication. The interval between HD and surgery did not significantly predict postoperative outcomes. Disruption of routine HD schedules, observed in 31 patients, was also not associated with increased risk of complications (OR = 0.94, 95% CI = [0.2 […]

PubMed · DOI

The use of communication applications in orthopaedics: a scoping review.

Olivelle A, Borra P, Sati R, Gupta MS, Oochit K, Patel A

BACKGROUND: Effective communication between clinicians is vital in orthopaedic practice, particularly in trauma and emergency care where optimal decision making is necessary. Although communication application use in orthopaedics is common within practice, there is little research to back up its ongoing use. This scoping review aims to synthesise the existing evidence on the use of clinician-to-clinician communication applications in orthopaedic practice, focusing on its prevalence, perceived utility, diagnostic reliability and associated challenges. MATERIALS AND METHODS: A scoping review was conducted in accordance with the PRISMA-ScR guidelines. This involved a literature search of MEDLINE, Embase, and the Cochrane Library from inception to September 2025. Studies involving clinician-to-clinician communication applications used within orthopaedic care were included. Data was extracted and synthesised using a methodological approach. RESULTS: Seven studies published between 2016 and 2023 were included. The evidence demonstrates widespread adoption of communication applications, particularly WhatsApp, for informal clinical communication, with consistent perceptions that these platforms improve communication efficiency and coordination of care. Four studies evaluating smartphone transmitted radiographic images reported good to excellent intraobserver diagnostic agreement when c […]

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Double-row wired anchors combined with "8" cross tension bands: an innovative procedure for the treatment of inferior pole fractures of the patella and its finite element and biomechanical evaluation.

Deng Z, Zhang W, Chen J, Yang Z, Gao W, Liu Y, Zhang X, Zeng W et al.

OBJECTIVE: To propose and verify the biomechanical properties of a new double-row suture anchor combined with figure-of-eight cross tension band technology for the treatment of inferior pole fractures of the patella, in order to overcome the shortcomings of traditional fixation methods in achieving stable fixation and reducing implant-related complications. METHODS: (1) Based on clinical CT data, a three-dimensional finite element model of patellar inferior pole fracture was reconstructed, and the new fixation technology (deep double-row anchors in the proximal bone fragment combined with figure-of-8 sutures) and Kirschner wire tension band wire fixation (TBW) were simulated and analyzed. Mechanical performance in the treatment of inferior pole fracture of the patella; (2) Divide the model into two groups according to surgical methods: suture anchor (SA) fixation group and TBW fixation group; (3) Carry out finite element analysis to compare the two groups at 500 Displacement and stress distribution of the knee joint at 45°and 135°of flexion under N load. RESULTS: Finite element analysis and biomechanical test results showed that there was no significant difference in displacement and stress between the two groups at different flexion angles. CONCLUSION: Double-row anchors combined with figure-of-eight suture technology exhibited excellent biomechanical properties, and is a feas […]

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Coordinated Return-to-Work model reduces sickness absences after hip or knee arthroplasty: a registry-based study.

Kangas P, Pamilo K, Soini S, Hirvonen M, Kervinen V, Kinnunen ML

INTRODUCTION: Working-age adults increasingly undergo total hip (THA) and knee (KA) arthroplasties. We evaluated how post-arthroplasty referral to occupational health services (OHS) using the Coordinated Return-to-Work (CRTW) model, affects return to work (RTW). MATERIALS AND METHODS: The CRTW model was evaluated through a benchmarking controlled trial. We used the electronic records of four hospitals in Finland to identify working-age THA and total/unicondylar KA patients before (control group, N = 668) and after (intervention group, N = 536) the CRTW model was implemented. We combined these data with sickness benefits registry data. The differences between the study groups' RTW were analyzed using a Cox regression model, adjusting for age, sex, body mass index, number of special reimbursement entitlements for medicines, and earnings as covariates. Subgroup analyses included intervention participants whose sick leaves were prescribed by a surgeon according to the CRTW protocol. RESULTS: After THA, the control group's mean RTW-duration was 87.8 days, compared to 74.9 days for the intervention group, with the mean difference being 12.9 days (95% CI 5.7–20.2). The intervention group was associated with earlier RTW; HR 1.35 (95% CI 1.13–1.61). After KA, the control group's mean RTW-duration was 107.8 days, while the intervention group's was 93.4 days, with the mean difference bein […]

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Minimum 10-year outcomes of cementless total hip arthroplasty using a 32-mm femoral head with CT-based navigation: a comparative study with manual technique.

Tone S, Naito Y, Kobayashi G, Sudo A, Hasegawa M

INTRODUCTION: Computed tomography (CT)-based navigation improves the accuracy of component placement in total hip arthroplasty (THA). To date, evidence of its long-term clinical impact is limited. This study aimed to compare minimum 10-year outcomes of cementless THA performed using CT-based navigation and a manual technique with a 32-mm femoral head. MATERIALS AND METHODS: This study included a minimum follow-up of 10 years. The CT-based group included 58 hips and the manual group included 53 hips. Postoperative cup orientation was assessed using CT. Clinical outcomes were evaluated using the Merle d'Aubigné and Postel score, Harris Hip Score (HHS), and Forgotten Joint Score-12 (FJS-12). Kaplan–Meier analysis was used to estimate the 10-year implant survival with revision for any reason as the endpoint. RESULTS: The CT-based group achieved a significantly higher cup placement accuracy (100% within Lewinnek's safe zone versus 49% in the manual group, p < 0.001). At the final follow-up, the CT-based group showed higher Merle d'Aubigné and Postel total score (16.8 vs. 16.1, p = 0.035) and HHS (90.3 vs. 85.0, p = 0.015), although the difference in HHS did not exceed the minimal clinically important difference. The FJS-12 scores did not differ significantly (71.1 vs. 62.2, p = 0.051). Ten-year survival rates were 98.2% in the CT-based group and 100% in the manual group (n.s.). CONC […]

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Long-term outcomes and revision-free survival following robotic-assisted unicompartmental and total knee arthroplasty.

Fleisher A, Kennedy M, Trudeau M, Piergrossi D, Madrid I, Meftah M

BACKGROUND: While robotic assistance improves precision in unicompartmental (R-UKA) and total knee arthroplasty (R-TKA), long-term comparative data are limited. This study evaluated perioperative outcomes and mid- to long-term survivorship for R-UKA versus R-TKA. METHODS: A retrospective review was performed of 163 patients who underwent primary, elective, R-UKA (n = 84) or R-TKA (n = 79) between 2011 and 2020 at a single, tertiary academic medical center, with a minimum follow-up of five years. Demographics, perioperative characteristics, and postoperative outcomes were compared between cohorts. Revision-free implant survivorship was evaluated using Kaplan–Meier analysis. RESULTS: R-TKA procedures had longer operative times than R-UKA (131 vs. 112.5 min; P < 0.001) and both were performed primarily for primary osteoarthritis. Cohorts had similar 90-day ED visit rates (3.8 vs. 4.8%; P = 1), for both surgical (2.5 vs. 3.6%) and medical causes (1.3 vs. 1.2%), while readmission rates were also similar (2.5 vs. 3.6%, P = 1). Median follow-up (9.55 vs. 6.07 years) and time since surgery (12.33 vs. 7.48 years) were significantly longer for R-UKA (both P < 0.001). At final follow-up, revision occurred in 3.8% of R-TKA and 4.8% of R-UKA cases (P = 1). Time to revision was shorter for R-TKA (1.36 vs. 5.22 years; P = 0.114). Kaplan–Meier analysis demonstrated 5-year revision-free survivo […]

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Comparison of 20-year results of total hip arthroplasty using first-generation annealed highly cross-linked polyethylene and zirconia heads in patients aged ≤ 50 and > 50 years.

Naito Y, Tone S, Kobayashi G, Hasegawa M

INTRODUCTION: This study investigated 20-year outcomes of total hip arthroplasty (THA) using zirconia heads on annealed highly cross-linked polyethylene (HXLPE) liners, comparing patients aged ≤ 50 and > 50. METHODS: We reviewed 117 hips (100 patients) who underwent cementless THA between 2001 and 2004. After excluding follow-up < 15 years, 22 hips (≤ 50) and 61 hips (> 50) were analyzed. Merle d'Aubigné and Postel score, University of California, Los Angeles activity score, Kaplan–Meier survivorship, femoral head penetration, and oxidative degradation in retrieved liners were evaluated. RESULTS: Patients ≤ 50 had higher activity. Three revisions for infection occurred in the > 50 group; none due to wear. Survivorship was 93% overall, 100% in ≤ 50, and 91% in > 50, with no difference. Steady-state wear rate was 0.01 mm/year in both groups. Retrieved liners showed oxidation. CONCLUSION: Zirconia-on-annealed HXLPE THA showed excellent 20-year wear resistance and survivorship, with comparable outcomes between ≤ 50 and > 50 groups.

PubMed · DOI

Effect of a one time guideline based educational video intervention on osteoporosis related knowledge.

Hauser AL, Noriega Urena JF, Saatashvili L, Esmail F, Lange T, Schulte TL, von Glinski A

INTRODUCTION: Osteoporosis remains underdiagnosed and undertreated, partly due to insufficient patient knowledge. This study aimed to assess osteoporosis-related knowledge and to evaluate the immediate and short-term effects of a one-time educational intervention through a guideline-based video in an adult online population. MATERIALS AND METHODS: In this prospective, randomized controlled trial, adults aged ≥ 18 years were recruited via an established online panel. Participants were randomized to an intervention group, which viewed a standardized educational video on osteoporosis, or a control group without intervention. Osteoporosis knowledge was assessed using the Facts on Osteoporosis Quiz (FOOQ) and the Osteoporosis Knowledge Assessment Tool (OKAT). Knowledge retention was evaluated in the intervention group after one week. Between-group comparisons were performed using Man-Whitney U tests, χ^2 tests, and odds ratios. RESULTS: A total of 513 participants were included (intervention group: n = 198; control group: n = 315), with no significant differences in baseline demographic or clinical characteristics. Immediately after the intervention, osteoporosis-related knowledge was significantly higher in the intervention group compared with controls, as measured by both OKAT (11.6 vs. 8.8 points, $p < 0.0001$) and FOOQ (13.2 vs. 11.1 points, $p < 0.0001$). The proportion of participa […]

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Modified rotational wedge distal metatarsal osteotomy versus chevron osteotomy for hallux valgus: long-term radiographic and clinical outcomes.

Aybar A, Gokkus K, Özden E, Çetin MH, Çetin MÜ

BACKGROUND: Distal Chevron osteotomy is commonly used for mild-to-moderate hallux valgus, but long-term loss of correction and radiographic recurrence remain concerns, particularly when distal articular alignment and sesamoid position are not adequately restored. We compared long-term radiographic and clinical outcomes of a modified rotational wedge distal metatarsal osteotomy versus standard distal Chevron osteotomy in adults with mild-to-moderate symptomatic hallux valgus. METHODS: In this single-center retrospective cohort study, 100 feet (100 patients) treated between 2010 and 2019 were analyzed (Modified, n = 46; Chevron, n = 54) at a mean follow-up of 101.2 ± 11.5 months. Soft-tissue balancing was standardized, with an intra-articular lateral release performed in both groups. Outcomes included radiographic measures (hallux valgus angle [HVA], intermetatarsal angle [IMA], distal metatarsal articular angle [DMAA], and medial sesamoid position), clinical scores (AOFAS, VAS), recurrence, and complications. Radiographic recurrence was defined as final HVA > 15°. RESULTS: Final AOFAS scores were similar between groups ($p = 0.621$), and the between-group difference in final VAS pain scores did not remain significant after Benjamini-Hochberg false discovery rate (BH-FDR) adjustment ($q = 0.057$). Compared with the Chevron group, the Modified group demonstrated superior final radiogr […]

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Tendon gel using the film model method promotes ligament healing in rabbits.

Yoshimizu R, Nakase J, Yoshioka K, Kuzumaki T, Torigoe K, Demura S

INTRODUCTION: Tendon gel is a translucent gel-like material secreted from the ends of a severed tendon. When mechanical stress is applied to a 3-day in vivo-preserved tendon gel, it matures into type I collagen–dominant tissue similar to normal tendon. This study aimed to evaluate the effects of transplanting a 3-day in vivo-preserved tendon gel into a knee medial collateral ligament (MCL) injury site in rabbits to promote intrinsic ligament regeneration. MATERIALS AND METHODS: Tendon gel was prepared from rabbit Achilles tendons using the film model method and harvested after 3 days of in vivo preservation. The 3-day tendon gel was transplanted into the knee MCL injury sites in another set of rabbits (n = 48). Additionally, the healing process was assessed at 1, 2, and 4 weeks postoperatively using mechanical and histological analyses. Ultimate load, peak stress, and elastic modulus were measured. Histological maturity was semi-quantitatively scored, and collagen type I and III expressions were examined by immunofluorescence staining. RESULTS: At 2 weeks, the tendon gel group demonstrated significantly higher ultimate load than the control group (12.25 ± 4.90 vs. 5.25 ± 2.40 N; $p = 0.02$). The tendon gel group had greater peak stress than the control group (3.54 ± 1.44 vs. 1.69 ± 0.78 MPa; $p = 0.02$).

Histological scores were higher in the tendon gel group than in the control gr […]

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Does the preoperative CPAK classification influence clinical outcomes after medial and lateral unicompartmental knee arthroplasty? Short-term outcomes and phenotype distribution in a retrospective series of consecutive cases.

Campi S, Esposito C, Manfreda A, Franceschetti E, Rizzello G, Vadalà G, Longo UG, Papalia R

INTRODUCTION: Recent evidence suggest that mechanically aligned TKAs may yield heterogeneous outcomes across CPAK categories, but whether this applies to unicompartmental knee arthroplasty (UKA) remains unclear. This retrospective study evaluated the influence of preoperative CPAK category on postoperative clinical outcomes after UKA, with additional analyses assessing CPAK distribution, residual varus alignment, and their relationship with patient-reported outcomes. MATERIALS AND METHODS: A total of 312 consecutive UKAs (280 patients) with ≥ 1-year follow-up were analyzed. Preoperative HKA and JLO were measured on weight bearing long-leg radiographs, and patients were classified according to the CPAK system. Outcomes were compared across the six most represented CPAK groups (I – VI) in 253 patients with complete PROM data. RESULTS: Mean postoperative OKS was 40.5 ± 5.3, with a mean ΔOKS of 21.3 ± 8.7 and 97% patients' satisfaction. No significant differences were observed among CPAK categories for OKS ($p = 0.80$), ΔOKS ($p = 0.90$), or satisfaction ($p = 0.70$), and alignment-based grouping (varus, neutral, valgus) showed similarly homogeneous outcomes. Residual varus ≥ 5° did not adversely affect results ($p > 0.10$), and no meaningful correlation was found between aHKA and postoperative OKS ($r = -0.003$, $p = 0.96$). CPAK distribution differed significantly by sex ($p < 0.001$). Five p […]

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Managing Vancouver B2 periprosthetic femoral fractures: is open reduction and internal fixation superior to stem revision? : A comparative analysis with a minimum 5-year follow-up.

Wang T, Xu H, Wu J, Wang X

PURPOSE: The aim of this study was to compare the long-term outcomes of stem revision (SR) and open reduction and internal fixation (ORIF) for the treatment of Vancouver B2 periprosthetic fractures of the femur. METHODS: From June 2013 to May 2023, 56 consecutive patients were studied at our institution. Four patients were lost to follow-up, four had incomplete data. Thus, 48 cases were included in the analysis. The patients were divided into a stem revision group (SR group with 25 patients) and an open reduction and internal fixation group (ORIF group with 23 patients). The surgical complications, perioperative parameters, and 1-year mortality rates were assessed, the functional outcomes were assessed with the Harris Hip Score (HHS), and the radiographic outcomes were assessed in accordance with the Beals and Tower criteria. RESULTS: In SR group, the mean follow-up time was 61.2 months, 36% of patients experienced complications, the mean HHS was 75.27, and 92% of the patients had "excellent–good" radiographic outcomes. In ORIF group, the mean follow-up time was 63.7 months, 21.7% of patients experienced complications, the mean HHS was 73.56, and 91.3% of the patients had "excellent–good" radiographic outcomes. The total number of postoperative complications, dislocation rate, blood loss volume, operation time and transfusion rate were lower in ORIF group, and two patients in S […]

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Outcomes and associated factors of revision procedures after failed total ankle arthroplasty: a comparative cohort analysis.

Pfahl K, Eder J, Simon D, Beckers G, Holzapfel BM, Walther M

INTRODUCTION: After failed total ankle arthroplasty (TAA), revision arthrodesis (RAA) and revision arthroplasty (RTAA) are treatment options, but comparative outcome data remain limited. The aim of this study was to compare mid-term survival of RAA and RTAA after failed primary TAA and to explore clinical and radiographic factors associated with failure.

MATERIALS AND METHODS: In this retrospective cohort study, 124 patients (RAA = 72, RTAA = 52) were reviewed after failed TAA between 2006 and 2020, with a minimum follow-up of 12 months. Kaplan-Meier analysis and Cox regression were used to assess survival, and decision tree models were used as exploratory tools to identify factors associated with failure.

RESULTS: Mean follow-up was 71.6 ± 42.7 months. Surgical failure occurred in 6.45% of patients, with no significant difference (RAA 6.9%, RTAA 5.8%). Five-year survival was slightly higher for RTAA (97% vs. 93%), although durability decreased beyond 87 months. Female sex, higher BMI, and younger age were associated with failure after RAA, whereas large periprosthetic cysts and elevated BMI were related to failure after RTAA.

CONCLUSIONS: RAA and RTAA demonstrate comparable mid-term survival after failed TAA. Revision strategy selection should be individualized, considering bone stock, patient characteristics, and failure patterns.

LEVELS OF EVIDENCE: III.

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Intramedullary headless compression screw fixation for metacarpal fractures: a retrospective clinical and radiological study.

Ku■cu B, K■na■ M

BACKGROUND: Intramedullary headless compression screw fixation has gained increasing popularity for the surgical treatment of metacarpal fractures owing to its minimally invasive nature and potential for early mobilisation without prolonged postoperative immobilisation. However, comprehensive data on clinical and radiological outcomes remain limited. METHODS: This retrospective study included patients with metacarpal shaft or neck fractures treated with retrograde intramedullary headless screw fixation. No postoperative immobilisation was applied, and early active mobilisation was initiated in all patients on the day of surgery. Demographic data, fracture characteristics, operative details, pre- and postoperative angulation, radiological union time, functional outcomes, QuickDASH score, and complications were recorded. Preoperative and postoperative angulation values were compared using a paired Student's t-test. RESULTS: A total of 37 patients were included, with a mean age of 37.5 ± 15.4 years. The majority of fractures involved the fifth metacarpal (70.3%). The most common mechanisms were direct impact/blunt trauma and falls. Mean operative time was 12.8 ± 3.9 min (range, 7–21 min). Mean preoperative angulation of 41.2° ± 12.2 improved significantly to 0.2° ± 0.71 postoperatively ($p < 0.001$). Mean radiological union time was 5.6 ± 0.8 weeks (range, 4–7 weeks). At final follo […]

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Persistent extensor strength deficit despite acceptable clinical outcomes after surgical treatment of patella fractures: a mid- to long-term follow-up study.

Dalaslan RE, Yücel MO, Sa■lam S, Ar■can M, Karaduman ZO, Sav ■

INTRODUCTION: To evaluate mid- to long-term clinical outcomes together with objective isokinetic measurements of knee flexor and extensor muscle strength in patients who underwent surgical treatment for AO/OTA type C patella fractures, and to investigate the relationship between fracture severity, clinical scores, and strength deficits. We hypothesized that significant quadriceps strength deficits would persist despite acceptable clinical outcomes and that these deficits would be associated with worse pain and functional scores.

MATERIALS AND METHODS: This retrospective cohort study included 58 patients treated surgically for patella fractures between 2015 and 2024, with a median follow-up duration of 60 months (IQR: 32-94; range: 13.1-132.0 months). The primary outcome was extensor muscle strength deficit measured by isokinetic dynamometry. Secondary outcomes included flexor strength deficit, Lysholm score, visual analog scale (VAS), and knee range of motion. Isokinetic dynamometry at 60°/s was used to measure concentric flexor and extensor peak torque values, and strength deficits were calculated by comparison with the contralateral limb. Outcomes were compared among fracture subtypes (C1, C2, C3), and correlations between strength deficits and clinical scores were analyzed. Non-parametric tests (Kruskal-Wallis) and partial correlation analysis (Pearson, adjusted for age) we […]

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Periacetabular osteotomy of the hip: an 8-year follow-up of 96 consecutive cases.

Enocson A, Wallensten R, Lundblad H

INTRODUCTION: A periacetabular osteotomy (PAO) is a joint-preserving surgical option for treatment of acetabular dysplasia. The procedure aims to prevent, or at least delay, the development of osteoarthritis, and subsequent need for total hip arthroplasty (THA). The conversion rate to THA differs widely in the literature, but most of the studies have few patients, and the follow-up time is often short for THA as an endpoint. The aim of this study was to evaluate the long-term outcome after PAO surgery with the rate of conversion to THA as the primary outcome. MATERIALS AND METHODS: Patients ≥18 years that underwent a PAO operation at the Karolinska University Hospital in Stockholm, Sweden from 2006 to 2022 were included. Radiological signs of hip osteoarthritis, and the lateral center-edge angle (LCEA) was calculated on pre- and postoperative radiographs or CT-scans. The national Swedish Arthroplasty Register was used to find cases who had a secondary operation with THA. RESULTS: The number of cases included was 96. Median age was 30 (18–46) years, and 84% (n = 81) were females. Median follow-up time was 99 (17–227) months (8 years). A total of 21 (22%) cases had a secondary THA. Cox regression analyses identified that age ≥ 30 years and smoking was associated with THA reoperation in both uni- (HR 2.8, CI 1.1–7.3, HR 3.7, CI 1.0–13) and multivariable (HR 4.0, CI 1.1–15, HR 7.8, […]

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Time-zero stability of uncemented standard versus intermediate revision stems after extended trochanteric osteotomy (ETO): a biomechanical study.

Hax J, Zderic I, Gueorguiev B, Kalbas Y, Leunig M, Pape HC, Rüdiger HA

INTRODUCTION: This time-zero biomechanical study investigates femoral stem stability after extended trochanteric osteotomy (ETO) without fragment refixation, representing a worst-case scenario. ETO is used in complex femoral revisions to improve exposure while preserving soft tissue and neurovascular structures. In the setting of a short ETO, surgeons may choose between standard (STD) and intermediate revision (REV) stems, although their primary stability remains unclear. Clinically, STD are still used after short ETO with low rates of aseptic loosening, aiming for proximal bone loading to reduce stress shielding. We therefore hypothesized that STD would demonstrate non-inferior axial and torsional stability compared with REV in a short open ETO model.

MATERIALS AND METHODS: A biomechanical model using SYNBONE® femurs and uncemented, collared, fully coated triple-tapered STD and REV 135° CCD was established. A standardized 60 mm ETO (vastus ridge to distal end of ETO) was performed and left unrepaired to simulate a worst-case scenario. Four groups (n = 6 each) were tested: (1) STD and (2) REV under axial loading (500 cycles at 20° adduction up to 500 N at 1 Hz, then static loading to failure), and (3) STD and (4) REV under torsional loading (500 cycles up to 10Nm internal rotation at 1 Hz, then failure testing). Axial stiffness (N/mm), torsional stiffness (Nm/deg), and failure […]

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Functional and quality-of-life outcomes after metal-on-polyethylene articulating spacer implantation for periprosthetic knee infection: a retrospective evaluation of a prospectively collected cohort.

Balato G, Festa E, Ascione T, De Mauro D, Di Gennaro D, Mariconda M

INTRODUCTION: This study aims to evaluate functional and quality-of-life (QoL) outcomes in patients with periprosthetic joint infection (PJI) undergoing implantation of a metal-on-polyethylene articulating knee spacer as part of a planned two-stage revision, and to explore preoperative predictors of meaningful postoperative improvement. METHODS: This was a retrospective observational study including 108 consecutive patients with chronic PJI treated with a metal-on-polyethylene articulating spacer between 2016 and 2020 and followed prospectively for clinical outcomes. QoL and joint-specific function were assessed preoperatively and before planned reimplantation using EQ-5D-5 L, Western Ontario and McMaster University (WOMAC), and Knee Society Score (KSS). Minimum clinically important difference (MCID) thresholds were derived using a distribution-based method. Receiver operating characteristic (ROC) analysis identified baseline predictors of achieving MCID. RESULTS: All outcome measures improved significantly after spacer implantation ($p < 0.001$), exceeding MCID thresholds for pain and function. The largest EQ-5D-5 L domain gain was for pain/discomfort. Predictive thresholds for achieving MCID were an EQ-5D-5 L index ≤ 0.44 and WOMAC index ≥ 42.5. Female sex was associated with better preoperative and postoperative WOMAC and KSS scores. CONCLUSION: Metal-on-polyethylene articulat […]

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Different sagittal cutting plane orientations do not affect posterior tibial slope in open wedge high tibial osteotomy: a Sawbone model study.

Gurbanov E, Cochard B, Bauer D, Miozzari H, Tscholl P

PURPOSE: Control of the posterior tibial slope (PTS) is crucial in high tibial osteotomy (HTO), since unintended changes can alter the normal knee kinematics and ligament loading. This study investigated the influence of sagittal cutting plane orientations on PTS in bi-planar medial open wedge HTO (MOWHTO) on a Sawbone model. METHODS: Sixty Sawbone® left-tibia models (A Pacific Research Company, USA) were used to perform biplanar MOWHTO with three sagittal orientations: (1) parallel to the medial tibial plateau (PO); (2) 10° anterior-inclined osteotomy (AIO); (3) 10° posterior-inclined osteotomy (PIO). The biplanar cut was performed distal to the tibial tuberosity in half the specimens and proximal in the remainder. Customized 3D-printed cutting guides ensured reproducibility. PTS and valgus correction were measured pre- and post-osteotomy using a navigation system. Group- and subgroup-specific ∆PTS were analyzed using a Bayesian multilevel model. Minimal clinically important difference (MCID) was set at 2.5°. Bayesian “power” analysis (assurance) was used to assess the sensitivity of ∆PTS and pairwise differences relative to the MCID. RESULTS: Across the three sagittal orientation and their distal/proximal subgroups, ∆PTS and pairwise ∆PTS did not exceed the predefined MCID of 2.5°. Bayesian assurance of equivalence exceeded 86% in every unit and pairwise comparison, indicatin […]

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Comparative assessment of the reaming characteristics of two different technologies for intramedullary bone graft harvesting.

Guttau S, Lange M, Finze R, Beimel C, Saifzadeh S, Gospos J, Wille ML, Kobbe P et al.

INTRODUCTION: The study aimed to compare the biomechanical impact and reaming performance of two intramedullary bone graft harvesting techniques: the established Reamer-Irrigator-Aspirator 2 (RIA 2) system and a novel aspirator+reaming-aspiration (ARA) concept. MATERIALS AND METHODS: In a preclinical in vivo sheep model, sixteen femora were assigned to either the RIA 2 or ARA group (n = 8 each). Biomechanical testing, computed tomography (CT)-based 3D bone geometry analysis, and fracture line assessment were performed postoperatively. Bending stiffness of the reamer shafts, cortical wall thickness pre- and post-reaming, and torsional stability of reamed femora were evaluated. Statistical analysis included TOST, Mann-Whitney U, and Shapiro-Wilk tests, considering values of $p < 0.05$ as statistically significant. RESULTS: Both reaming systems produced the greatest cortical thinning medially in the midsections of the analyzed region. No statistically significant differences were observed in postoperative cortical wall thickness, bone removal rates, or torsional stability between groups (all $p > 0.05$). Bending stiffness was significantly higher for the RIA 2 system ($p = 0.004$). In both groups, fractures predominantly initiated medially or at sites of previous drill holes. The ARA concept demonstrated equivalent biomechanical and reaming performance compared to the RIA 2 system, with […]

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External fixation versus reverse shoulder arthroplasty for proximal humerus fractures in the elderly: a retrospective comparative study.

Vadalà A, Benelli C, Suraci F, Carta B, Baldassari G, Maffulli N

BACKGROUND: The optimal management of displaced proximal humerus fractures (PHFs) in the elderly remains a subject of ongoing debate. This study aims to compare the clinical and functional outcomes of biologically driven external fixation (EF) versus functional joint replacement via reverse shoulder arthroplasty (RSA) in patients aged 65-80 years. METHODS: A retrospective comparative study was conducted on patients with displaced Neer two- or three-part PHFs treated between 2015 and 2022. The final analysis included 67 patients: Group A (EF; n=34) and Group B (RSA; n=33). Clinical and functional outcomes were quantified using the Constant-Murley Score (CMS), Simple Shoulder Test (SST), American Shoulder and Elbow Surgeons (ASES) score, Disabilities of the Arm, Shoulder and Hand (Quick-DASH) score, and Range of Motion (ROM) assessment. Patient satisfaction was evaluated via a 5-point Likert scale. RESULTS: Both cohorts were homogeneous regarding mean age (72.2 ± 3.5 vs. 73.4 ± 3.8 years). Group A demonstrated significantly shorter mean operative times (48.2 ± 10.5 vs. 92.4 ± 15.2 min; $p < 0.001$) and a lower requirement for blood transfusions (0% vs. 12.1%; $p < 0.05$). At the final follow-up, both cohorts achieved comparable CMS (58.4 ± 8.2 vs. 55.2 ± 10.1; $p = 0.42$), SST (7.6 ± 1.5 vs. 7.0 ± 1.4; $p = 0.38$), and ASES scores (69 ± 7.8 vs. 65 ± 9.4; $p = 0.51$). Group A exhibited sign […]

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Advances in the management of chest wall injuries - Influence of new technical options.

Beck P, Pape HC

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Trauma nursing as frontline health diplomacy: A binational ATCN program for Palestinian and Israeli nurses during conflict.

Jreis T, Kruyer VN, Kaim A, Becker D, Hani W, Mohammed A, Young L, Bkeirat W et al.

BACKGROUND: Nurses are central to trauma care in both peacetime and conflict, where training and expertise directly affect patient outcomes. Cooperative training between healthcare professionals from opposing communities is rare in regions of ongoing violence, yet may be a powerful mechanism for strengthening regional trauma systems and advancing health diplomacy. Operating Together is a binational initiative that brings Palestinian and Israeli trauma providers together for joint training. This study describes the establishment of a regional Advanced Trauma Care for Nurses (ATCN) program serving both Israeli and Palestinian communities, evaluates course effectiveness, and health■diplomacy attitudes.

METHODS: Between January 1 and December 31, 2025, six ATCN providers and two ATCN instructor courses were conducted. Participants completed anonymous questionnaires assessing satisfaction, perceived clinical relevance, and attitudes toward Palestinian-Israeli health cooperation. Descriptive statistics summarized responses, and comparisons used independent■samples t tests and Fisher's exact tests (significance $p < 0.05$).

RESULTS: Of 52 participants, 36 completed the study (50% Israeli, 50% Palestinian). Participants reported high course satisfaction ($M = 8.44/10$) and perceived educational impact ($M = 4.39/5$). Most (94.4%) expressed interest in maintaining professional contact. The […]

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Corrigendum to "Neutralizing the odds: Biomechanical protection by adiposity offsets physiological burden to explain the trauma.'obesity-paradox`" [Injury 57 (2) (2026) 112913].

Erdle B, Höne M, Diallo TD, Kalbhenn J, Schmal H, Mühlenfeld N

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Lateral epicondylar fractures in the pediatric population: Presentation, management, and outcomes.

Kushare I, Ghanta R, Smadi Z, Wunderlich N, Phillips T

INTRODUCTION: Lateral epicondylar fractures in young patients have received limited attention in the literature. This study aims to describe their mechanism of injury, concomitant injuries, management, and long-term outcomes in pediatrics.

METHODS: This IRB-approved retrospective case series included patients younger than 18 years diagnosed with lateral epicondyle fractures between (2015-2021). Data collected included patient demographics, fracture characteristics at presentation, management details, and clinical outcomes. Patient-reported outcome measures (PROMs) were assessed using the QuickDASH score and associated functional variables.

RESULTS: Twenty-two patients (10 male, 12 female) with a mean age of 10.5 years and BMI of 19.2 were included. Sports-related injuries were the most common mechanism (9 patients, 40.9%), followed by other high-energy falls such as from bicycles, stairs, or hoverboards (13 patients, 59.1%). Sixteen patients (72.7%) had nondisplaced fractures, while six (27.3%) had displaced fractures. Concomitant injuries were present in 12 patients (54.5%), including elbow fracture-dislocations (6, 27.3%), coronoid process fractures (4, 18.2%), and ligamentous injuries (2, 9.1%). Among those with fracture-dislocations, three (50%) also had additional elbow fractures, most commonly involving the medial epicondyle. Three patients (13.6%) required operative in […]

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Variation in surgical indications across national distal radius fracture guidelines: A comparative review.

Oftebro I, Olsen B, Enger MM, Hassellund SS, Solberg LB

AIMS: Surgical indications for dorsally displaced distal radius fractures (DRFs) are guided by national guidelines, yet variation in clinical practice persists. This study aimed to compare contemporary national guidelines, examine differences in recommended intervention thresholds, and assess their potential impact on treatment decisions and resource utilization.

METHODS: A structured search was conducted across biomedical databases, clinical knowledge platforms, guideline repositories, and national health agency websites. National guidelines were included if they provided surgical recommendations for dorsally displaced DRFs in adults. Radiographic thresholds, patient-related criteria, and methodological characteristics were extracted and analysed comparatively.

RESULTS: Eight national guidelines were included in the comparative analysis (one from the United States and seven from Europe). Ulnar variance, dorsal tilt, and intra-articular step-off were consistently identified as key radiographic indicators for surgery, although threshold values and age-related modifications varied. Additional parameters, including volar cortical alignment, fracture comminution, osteoporosis, and intra-articular gap, were incorporated inconsistently. Some guidelines applied fixed age thresholds, whereas others emphasised functional demand. Despite methodological variation, recommendations were d […]

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Prosthetic leg socket design: New insight on different tibia length and distal bevelled angle using finite element analysis.

Baharuddin MH, Abdullah NNAA, Kori MI, Gan HS, Abdul Kadir MR, Ramlee MH

Socket is one of the components that can determine the comfort of the user. Poor socket fit can induce pain and discomfort to the residual leg. Besides, poor selection of amputation morphological parameters such as length of bone and tibial bevelling angle can hinder the control of prosthesis. Thus, the surgeon needs to make a wise decision, or else it can cause problems later in prescribing prostheses for the amputee. The aim of the study is to investigate the effect of morphological variability of residual limbs such as tibial length, distal tibia bevelling angle, and stiffness of soft tissue. The residual leg of the volunteer was scanned using a 3D scanner, and a socket was designed based on the contour of the residual leg. The bone 3D model was derived from CT-scanned data and modelled as amputated bone, with the length of the tibia cut to 12.5 cm, 15 cm, and 17.5 cm and the distal is bevelled to 30°, 45°, and 60°. Stiffness of the soft tissue is defined as the D and C hyper-elastic values. From the result, tibial length of 12.5 cm, distal tibial bevelling angle of 45°, and stiffness of 55 kPa are found as the optimal parameters for the volunteer by selecting the lowest maximum von Mises stress of soft tissue for each category. No pain is predicted on the soft tissue during standing since the normal stress is not exceeding the pain threshold at fibular head region (0.89 MPa […]

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Untreated ulnar styloid fractures do not compromise wrist function in patients with distal radius fractures: A systematic review and meta-analysis.

Li C, Gao W, Yang C, Jia Z, Li J

BACKGROUND: Ulnar styloid fractures (USF) often occur in conjunction with distal radius fractures (DRF). This study aims to compare wrist prognosis between DRF patients with untreated USF and those with isolated DRF, and to assess the impact of different USF fracture types.

METHODS: A systematic search was conducted in databases such as PubMed, Embase, and the Cochrane Library, up to January 17, 2026. Randomized controlled trials (RCTs) and cohort studies comparing the wrist joint prognosis of patients with DRF combined with USF and those with isolated DRF were included. The primary outcome measure was the patient-reported wrist joint function score, and the secondary outcome measures were grip strength and wrist joint range of motion (ROM).

RESULT: 11 studies involving 1383 patients were included. In the short-term (≤6 months) follow-up, the wrist function score of patients with DRF combined with USF was statistically inferior to that of patients with DRF alone (SMD = 0.20, 95% CI: 0.05-0.34, P = 0.01), but the effect size was small and the clinical significance was limited. In the medium- to long-term (≥6 months) follow-up, there was no significant statistical difference in wrist function scores between the two groups (SMD = 0.11, 95% CI: -0.02-0.24, P = 0.11). Further analysis indicated that there was no significant statistical difference in prognosis between patients with […]

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The management of pink pulseless hands in paediatric supracondylar humerus fractures: A systematic review and meta-analysis.

Itte VN, Cowie J, Ali Yusuf Y, Zafar A, Wilson M, Jacob R, Asma F, Bourke G et al.

INTRODUCTION: The management of the pink pulseless hand (PPH) following paediatric supracondylar humerus fractures remains controversial, with no clear consensus between orthopaedic management and early neurovascular surgical exploration. This study aimed to compare outcomes of these strategies through systematic review and meta-analysis.

METHODS: A systematic review was conducted in accordance with PRISMA guidelines. Multiple electronic databases were searched up to December 2024. Retrospective cohort studies evaluating management of PPH in paediatric supracondylar fractures were included. Primary outcomes were return of radial pulse within 24 h following (1) closed reduction and percutaneous pinning (CRPP) without neurovascular exploration and (2) surgical exploration of the brachial artery. Secondary outcomes included comparative pulse return, conversion to neurovascular exploration, and long-term functional outcomes. Data were analysed using a random-effects model.

RESULTS: Twenty-five studies comprising 591 patients were included. Following CRPP alone, the pooled proportion of pulse return within 24 h was 0.61 (95% CI 0.48-0.74), with substantial heterogeneity (I^2 = 82.7%). In contrast, neurovascular exploration demonstrated a higher pooled pulse return rate of 0.94 (95% CI 0.89-0.99) with minimal heterogeneity (I^2 = 0%). Comparative analysis showed lower odds of immedia […]

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Acute-phase hyperbaric oxygen therapy induces differential regulation of NF-κB/COX-2 and NLRP3 inflammasome pathways in traumatic brain injury.

Puspadewi M, Dewi S, Hardiany NS, Wiwoho YY, Jusuf AA, Satriotomo I, Sadikin M

BACKGROUND: The initial injury from TBI damages brain tissue, disrupting blood flow and causing the brain to experience hypoxia. Hypoxia contributes significantly to the persistence of neuroinflammation and the progression of secondary brain injury, ultimately leading to long-term neurological deficits. Hyperbaric oxygen (HBO) therapy is a potential intervention to mitigate hypoxia. In our earlier work, we demonstrated that HBO treatment effectively reduced brain hypoxia in rat models, as indicated by decreased levels of hypoxia-inducible factor (HIF)-1α. Building on these findings, this study aims to investigate the impact of HBO therapy on neuroinflammatory responses following TBI.

METHODS: A total of 44 rats were randomly assigned to 11 groups: one control group, one TBI group, three normobaric groups, three 2 ATA HBO treatment groups, and three 3 ATA HBO treatment groups. Following TBI induction, the treatment groups received six sessions of 1-hour HBO exposure, administered every 12 h. Brain tissue was harvested on days 3, 7, and 14 post-TBI for analysis. Neuroinflammation was evaluated by measuring NF-κB, NLRP3 and caspase-1 expressions, as well as COX-2 activity. The percentage of intact neurons was assessed by hematoxylin-eosin (HE) staining.

RESULTS: On days 3, 7, and 14, both the 2 ATA and 3 ATA HBO groups exhibited a trend toward reduced NF-κB protein levels and CO […]

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The use of blocking "screws" in retrograde intramedullary nailing for distal femur fractures: A systematic review.

Hallak A, Amzallag N, Zahalka S, Ben-Tov T, Khoury A

INTRODUCTION: Distal femur fractures present significant challenges when achieving and satisfying alignment due to the wide metaphyseal canal and deforming muscular forces. Retrograde intramedullary nailing (RIMN) is a well-established fixation strategy for extraarticular distal femur fractures; however, malalignment remains a recognized limitation, particularly with metaphyseal comminution. Blocking screws have been introduced to guide nail trajectory and reduce diaphyseal-metaphyseal mismatch, as well as augment mechanical stability. This systematic review examines the indications, technical considerations, postoperative outcomes, and complication profile associated with blocking screw use during RIMN for distal femur fractures.

MATERIALS AND METHODS: A systematic search of PubMed and the Cochrane Library was conducted for studies published between January 1999 and December 2024. The search followed PRISMA guidelines and included randomized trials, comparative cohort studies, retrospective series, and biomechanical analyses evaluating blocking screws used in conjunction with RIMN. Two reviewers independently screened titles, abstracts, and full texts. Eligible studies reported radiographic and/or clinical outcomes related to malalignment, union, or construct stability. Studies without postoperative alignment metrics, those evaluating antegrade nails, open or periprosthetic f […]

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Comparison of three surgical methods in the treatment of displaced intra-articular calcaneal fractures: A retrospective study.

Zhao B, Ma C, Liu Q, Shen A, Du H, Li J, Gong M, Jiang X et al.

OBJECTIVE: To compare the efficacy of three surgical methods-Robot-Assisted Percutaneous Reduction and Screw Fixation (RA-PRSF), Sinus Tarsi Approach for Open Reduction and Plate Fixation (STA-ORPF), and Extended Lateral Approach for Open Reduction and Plate Fixation (ELA-ORPF)-in the treatment of displaced intra-articular calcaneal fractures (DIACFs).

METHODS: A retrospective cohort study was conducted involving 190 patients (202 fractures) with Sanders type II or III DIACFs. Patients were divided into three groups based on the surgical method. Perioperative outcomes, radiological parameters, functional scores, and complication rates were compared.

RESULTS: All three groups achieved comparable radiological outcomes and final AOFAS and VAS scores. However, the RA-PRSF group demonstrated significant advantages in intraoperative blood loss, weight-bearing time, and early functional recovery (AOFAS at 3 months, $P < 0.001$). The ELA-ORPF group had the highest wound complication rate (12.2%), significantly greater than the RA-PRSF (0%) and STA-ORPF (1.8%) groups ($P < 0.05$). No significant differences were observed in overall complication rates.

CONCLUSIONS: While all three methods are effective for DIACF management, RA-PRSF offers superior perioperative outcomes and early recovery with minimal soft tissue complications. STA-ORPF provides a balanced minimally invasive alternative, […]

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The impact of duration of infection on outcome of debridement, antimicrobial treatment and implant retention in fracture-related infections of the lower leg: A multicentre retrospective cohort study.

Sliepen J, Buijs MAS, Wouthuyzen-Bakker M, Depypere M, Rentenaar RJ, De Vries JPPM, Onsea J, Morgenstern M et al.

AIM: This study aimed to assess the association between the interval from initial fracture fixation to debridement, antibiotics and implant retention (DAIR) for fracture-related infection (FRI) (referred to as the FF-FRI interval) and treatment outcome in patients with FRI of the lower leg.

METHODS: This multinational retrospective cohort study included patients over 18-years of age who were diagnosed with an FRI of the lower leg, who underwent FRI surgery within 12 weeks after initial fracture fixation, between January 1st 2015 and July 1st 2020. All patients had a minimum follow-up of 12 months after cessation of therapy. Treatment success was defined as a combination of implant retention, eradication of infection, fracture healing and preservation of the affected limb. Both univariate and multivariable models were employed to ascertain factors associated with treatment outcome following DAIR.

RESULTS: A total of 100 patients were treated with DAIR. The median FF-FRI interval was 17 days (range: 1-63). A total of 57/100 patients (57%) had a successful treatment outcome after a median follow-up of 33.6 (P25-P75: 18.4-50.7) months. In 11 out of 22 patients (50%) identical pathogens were cultured at time of recurrent infection. The multivariable analysis demonstrated no statistically significant effect of the FF-FRI interval (HR: 1.01; 95%CI: [1.00-1.03]; p: 0.059).

CONCLUSIO […]

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Comparing the analgesic utility & safety of erector spinae plane block versus thoracic epidural for multiple rib fracture trauma: a retrospective cohort analysis.

Merchant N, MacLean DJ, Vincze SR, Chhabra J, Walker A, Finkel K

PURPOSE: Despite long-standing use of thoracic epidural block (TEB) for multiple rib fracture trauma, emerging research suggests that the erector spinae plane block (ESPB) offers similar analgesia and increased safety benefits. This study compared the analgesic effectiveness and safety of TEB versus ESPB for multiple rib fracture trauma.

METHODS: A historical cohort analysis was conducted, including 324 adult patients who received continuous TEB (n = 188) or ESPB (n = 136) plus multimodal analgesia following multiple rib fracture trauma. The primary objective compared average numeric pain scale (NPS) scores at rest and with activity over the first five days of local anesthetic infusion. Other outcomes included opioid consumption, length of stay data, venous thromboembolism (VTE) prophylaxis time measures, as well as vasoactive support, complication, and mortality rates.

RESULTS: Average pre- and post-block NPS scores at rest were greater in the TEB group [5.6 (2.1) vs. 4.8 (2.2), p = 0.002; 48-72 h: 3.7 (2.2) vs. 2.9 (2.2), p = 0.002, respectively). However, the daily changes in NPS scores throughout anesthetic infusion were similar between groups, without significant difference in opioid consumption. First VTE prophylaxis was given prior to block placement more frequently in the ESPB group (66.9 vs. 31.4%, p < 0.001), and median time between pre and post block VTE prophylact […]

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Geriatric tibial plateau fractures: A multicenter analysis of complications and treatment strategies.

Martin P, Springer F, Klaut M, Bormann M, Neidlein C, Fürmetz J, Böcker W, Keppler L

INTRODUCTION: Tibial plateau fractures are among the most complex periarticular injuries and pose a significant therapeutic challenge, particularly in geriatric patients, due to compromised bone quality, comorbidities, and limited mobility. Against the backdrop of demographic change, this multicenter study aims to analyze complications, risk factors for revision surgery, and treatment strategies for geriatric tibial plateau fractures.

METHODS: A retrospective cohort study was conducted at two supraregional trauma centers. Patients aged ≥ 70 years with surgically treated intra-articular tibial plateau fractures (AO/OTA 41-B/C) between 2010 and 2022 were included. Demographic parameters, fracture morphology, complications, risk factors, and surgical treatment were recorded. Statistical analyses were performed using chi-square tests and t-tests (significance p ≤ 0.05).

RESULTS: 148 patients were included (75% female; mean age 76.4 ± 5.3 years). Postoperative complications occurred in 36 patients (24.3%). Wound healing disorders (7.4%), deformities (6%), and infections (5.4%) were the most frequent. Classic risk factors showed no significant association with complications. Schatzker type VI fractures had the highest complication rate at 44.2%. Longer surgery times and the use of an external fixator were significantly associated with complications. N = 9 (6.1%) patients were treat […]

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Periprosthetic fracture following hip resurfacing arthroplasty: treatment guidance using the 'SAVE' classification system.

Al-Jabri T, Mirdad RS, Wong JM, Shankar S, Masterson S, Giannoudis PV

Periprosthetic fracture following hip resurfacing arthroplasty is an uncommon but important complication and represents a distinct failure mode compared with fractures around conventional stemmed total hip arthroplasty.

Existing periprosthetic fracture classifications are useful descriptively, but they do not fully address the biomechanical and biological considerations unique to hip resurfacing. Herein, the epidemiology, risk factors, mechanisms of failure, diagnostic workup, classification, and management of periprosthetic fractures following hip resurfacing arthroplasty is presented. Particular emphasis is placed on early femoral neck failure, avascular necrosis, femoral neck notching, varus component positioning, implant stability, adverse local tissue reaction, metal ion assessment, and the role of cross-sectional imaging. The article also proposes the SAVE classification, a treatment-oriented framework incorporating four key domains: Stability of the implant, Anatomical fracture location, Viability of the femoral head-neck segment, and local Environment. The SAVE classification provides a practical framework for determining whether the resurfacing construct can realistically be preserved or whether revision arthroplasty is more appropriate. By linking classification directly to treatment strategy, it may improve consistency in assessment and guide decision-making in these […]

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PRIoritisation of orthopaedic resources and interventions in trauma (PRIORI-T): A modified delphi consensus study.

Kotze JD, Amod M, Breedt C, Anley C, Ferreira N, Maimin D, Nortje M

BACKGROUND: Operative orthopaedic care in resource-constrained systems is frequently limited by theatre time, staffing, peri-operative support, implants, instruments and bed availability. In the absence of an explicit prioritisation framework, decisions about patients awaiting surgery may vary between clinicians and institutions. This modified Delphi study aimed to establish consensus on factors that should structure prioritisation of operative orthopaedic care in South African public hospitals.

METHODS: A three-round modified Delphi study was conducted among South African public-sector orthopaedic clinicians. Round 1 used open-ended responses to generate candidate prioritisation factors. Round 2 used a structured 1-9 importance scale to rate patient-specific factors, injury-specific factors, red-flag conditions and potential tie-breakers. Round 3 verified operational definitions, anchor levels and red-flag handling. The scope was confined a priori to admitted patients with stable, isolated orthopaedic injuries; polytrauma, spinal cord injuries and unstable vertebral fractures were excluded because they require individualised, time-critical prioritisation through established emergency pathways. Inclusion consensus was defined as a median score of at least 4 with at least 75% of respondents rating the factor 4-9. Strong inclusion consensus required a median score of at least 7 […]

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A retrospective cohort study on the clinical effectiveness of the modified masquelet technique with platelet-rich plasma for large infected bone defects.

Yang Z, Li X, Sun H, Guan H, Fang H, Zhao P, Wang Y, Qiao Z

OBJECTIVE: To evaluate the safety and efficacy of combining platelet-rich plasma (PRP) with a modified Masquelet technique for reconstructing segmental infected bone defects larger than 6 cm following trauma.

METHODS: A retrospective analysis was performed on 32 patients with large post-traumatic infected bone defects treated at our institution between January 2016 and September 2023. All patients underwent a two-stage procedure: Stage I involved thorough debridement with preservation of bone bridges, implantation of antibiotic-loaded PMMA bone cement, and stabilization using external fixation. Stage II, performed 6-8 weeks later, consisted of cement spacer removal, implantation of a PRP-autologous iliac bone graft mixture into the induced membrane cavity, and conversion to internal fixation. Outcome measures included bone healing time and rate, postoperative pain (VAS), reconstruction outcomes (Paley score), limb-specific function (DASH and LEFS), quality of life (SF-36), correlation between defect length and healing time, and subgroup comparisons of healing time (flap vs. non-flap, bone bridge preservation vs. absence). This study applied the Fracture-Related Infection (FRI) classification to systematically characterize patients' bone defect (F), host-related comorbidities (R), and soft tissue impairment (I) prior to treatment.

RESULTS: All 32 patients completed follow-up. […]

PubMed · DOI

Inclusion and acknowledgment of social determinants of health in trauma and injury research using national databases: A scoping review.

Kovacs MS, Paxton RJ, Wang JJ, Davis DR, Stoecker AK, Cucher DJ

Social determinants of health (SDOH) as defined by the WHO and CDC - socioeconomic status, one's physical environment, racism, discrimination, gender identity, etc. - influence the health of individuals and populations, and are understudied in trauma, injury, and surgical science. This paper quantifies the lack of awareness around excluding SDOH in retrospective injury research studies using large national databases. We reviewed 315 papers in total that use one of three national databases (ACS TQIP, NEISS, and H-CUP NEDS). For each database, we reviewed the four journals that most commonly publish papers using these databases. We conducted a scoping review. We asked two research questions of each article we reviewed: We found that 14.4% of TQIP authors in the examined journals include SDOH; 4.3% of NEISS authors include SDOH; and 83.3% of NEDS authors include SDOH. We also found that 13.6% of TQIP authors mention a lack of measuring SDOH relative to 25.5% of NEISS authors and 27.8% of NEDS authors. This scoping review of SDOH in large national injury databases finds varying levels of inclusion and acknowledgement of SDOH in trauma, emergency medicine, injury, and surgical publications. Health disparities and SDOH affect patients' baseline health status, recovery from illness and injury, and, potential injury recidivism.

PubMed · DOI

External fixators versus titanium elastic nailing for distal diaphyseal-metaphyseal junction tibial fractures: A finite element biomechanical analysis.

Wen Y, Xu C, Wang Q, Zhu D, Song B, Feng W

BACKGROUND: Computer modeling combined with finite element analysis was employed to assess the mechanical stability of external fixators (EF) and titanium elastic nails (TEN) in fractures involving the distal diaphyseal-metaphyseal junction (DMJ) of tibia.

METHODS: Transverse and oblique fracture models at the distal DMJ were constructed using computed tomography (CT) data from an 11-year-old girl's tibia■. Various fixation strategies were designed for comparison (EF and TEN with antegrade approach). Finite element analysis was utilized to evaluate the mechanical stability of various fixation techniques in the fracture models.

RESULTS: In both fracture models, EF exhibited superior rotational and bending stiffness compared to TEN, while the axial stiffness was lower than that of TEN. In the oblique fracture model, both TEN and EF demonstrated better rotational stiffness compared to the transverse fracture model, while their axial stiffness and bending stiffness were lower than those observed in the transverse fracture model. Although EF showed a decrease in bending stiffness in the oblique fracture model, it still outperformed TEN, even exceeding the bending stiffness of TEN in the transverse fracture model.

CONCLUSION: In clinical practice, postoperative weight-bearing restriction is a common treatment strategy, making rotational and bending stability of internal fixation p […]

PubMed · DOI

Interprofessional collaboration in trauma care in South Asia: A scoping review of barriers, interventions, and implications for patient and system outcomes.

Kunju SA, Rahman AA, Mohammed CA, Krishnan S V, Soy EM, Patil DS, Prakash H, Prabhu MM

BACKGROUND: Trauma is a leading global cause of preventable morbidity and mortality, with South Asia bearing a disproportionate burden due to fragmented trauma systems and resource limitations. Effective trauma care depends on coordinated personnel, infrastructure, clinical processes, and interprofessional collaboration (IPC).

OBJECTIVE: To map existing evidence on barriers to adult trauma care delivery in South Asia and to examine how IPC is implemented through educational, practice-based, and organisational interventions.

METHODS: A scoping review was conducted following the Arksey and O'Malley framework and reported in accordance with PRISMA-ScR guidelines. Four electronic databases (PubMed, Scopus, Embase, and Web of Science) and Google Scholar were searched for English-language studies published between 2005 and 2025. Eligible studies examined IPC, teamwork, or system-level challenges in adult trauma care across civilian healthcare settings. Data were charted and reported descriptively with multiple categories across system, provider, and patient levels.

RESULTS: Twenty-one studies were included from South Asia, with most evidence originating from India, highlighting important regional research disparities. Most were conducted in tertiary emergency departments or trauma centres and employed observational or quasi-experimental designs. Interventions were classified as IP […]

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Testosterone replacement therapy increases the risk of nonunion after surgical fixation of lower extremity fractures.

Smadi Z, Pereira DE, Mozny A, Irfan B, Boutros M, Hosseinzadeh P

INTRODUCTION: Testosterone replacement therapy (TRT) impacts bone mineral density and has been suggested to impact bone remodeling, yet its effects on fracture healing have not been completely characterized.

METHODS: A large-scale, multicenter retrospective cohort analysis was conducted utilizing a national database. Adult male patients who underwent surgical fixation for lower extremity fractures between 2005 and 2025 with a minimum of 2 years of follow-up were included. Patients were stratified into TRT and control cohorts based on documented use of testosterone replacement therapy, with 9067 patients in the TRT cohort and 88,510 in the control group. Following multivariate propensity score matching, 9027 patients were collated to each cohort. Statistical analyses included risk ratios (RR), confidence intervals (CI), and p-values, calculated using Student's t-tests and chi-square tests.

RESULTS: At two years, TRT patients demonstrated significantly elevated risks for nonunion (RR 1.172, $p = 0.017$) and revision (RR 1.329, $p = 0.003$). No significant differences were observed in deep vein thrombosis, pulmonary embolism or wound complications between the cohorts.

CONCLUSIONS: Testosterone replacement therapy (TRT) increases bone healing complications following surgical fixation of lower extremity fractures which was also associated with greater revision surgeries for nonunion […]

PubMed · DOI

Glucagon-like peptide-1 receptor agonist use for type 2 diabetes and risk of upper extremity fragility fracture.

Cathey JM, Fischer JP, Zhang J, Hammert WC, Klifto CS, Pidgeon TS

PURPOSE: Older adults with type 2 diabetes mellitus (T2DM) face an elevated risk of fragility fractures. Glucagon-like peptide-1 receptor agonists (GLP-1RAs) have grown in use for the management of T2DM over recent years, however, their potential skeletal effects remain incompletely understood. This study evaluated the association between sustained GLP-1RA use and the risk of upper extremity fragility fractures in this population.

METHODS: A retrospective cohort study was conducted using TriNetX, a global electronic health record platform. Adults of age ≥ 50 years with T2DM and active metformin use were included, excluding individuals with documented osteoprotective medication use. Patients were stratified by long-term GLP-1RA exposure (≥5 years of use). Cohorts were propensity-score matched at a 1:1 ratio, utilizing demographic (age, race, sex), clinical (comorbidities associated with fracture risk), and laboratory covariates (hemoglobin A1c). Pharmacotherapy initiation was defined as the initiation of metformin (control) or GLP-1RA therapy (exposure). Incidence of composite upper extremity fractures (distal radius, olecranon, proximal/distal humerus) was evaluated over the 5 years following pharmacotherapy initiation. Subgroup analyses were performed for high-risk populations and multivariate regression analyses evaluated the contribution of individual GLP-1RA agents to comp […]

PubMed · DOI

Factors influencing implant failure in cephalomedullary nails for proximal femoral fractures: A retrospective review.

Moreau J, Frackelton R, Tun SMT, Abed R, Brunton L

AIMS: The use of cephalomedullary nailing (CMN) to treat extracapsular hip fractures is a well-established fixation method. Nationally, the use of CMN has increased from 67.7% in 2012 to 90.9% in 2025 for subtrochanteric fractures (National Hip Fracture Database). Although rare, implant failure is a recognised complication which poses risk to patients who are otherwise already burdened with high morbidity and mortality. The purpose of this study is to review the outcomes of CMN fixations for proximal femoral fractures, with particular focus on implant failure. We analysed patient demographics, surgical factors, and fracture patterns to identify factors associated with rate of implant failure.

METHODS: This was a retrospective longitudinal cohort study of CMN fixations at a district general hospital between January 2020 and October 2024. All adult proximal femoral fractures treated with long CMN were included. Data collection was obtained from clinical records and relevant imaging. Measurement of radiological parameters was performed to evaluate quality of reduction. Multivariable logistic regression modelling was performed to analyse results.

RESULTS: Long CMN was used for 208 cases. Of these, the majority were female (74.5%) with a mean age 82 years. Open reduction and augmentation with cerclage wires was used in 27.4% of cases (n = 57). An overall implant failure rate of 4. […]

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Age-related macular degeneration and secondary fracture risk in patients with hip fractures.

Park M, Kim SH

Research on the relationship between age-related macular degeneration (AMD) and secondary fracture risk following hip fractures is scant. We aimed to examine whether AMD increases the risk of secondary fractures following hip fracture surgery. This retrospective cohort study was conducted using data from the Korean National Health Insurance Service Senior Cohort (2002-2019). Secondary fractures were defined as hip, wrist, humerus, spine, ankle, or pelvic fractures occurring within 6 months after hip fracture surgery. Patients diagnosed with AMD within 6 months postoperatively were identified. Covariates included age, sex, income, insurance type, Charlson Comorbidity Index, residential area, disability status, and hospital size and type. Associations between AMD and secondary fractures were analyzed using Cox proportional hazards models. Among 18 611 patients with hip fractures, 93 were diagnosed with AMD and 26 experienced secondary fractures. Patients with AMD had a significantly higher risk of secondary fractures than those without AMD (adjusted hazard ratio, 1.70; 95% confidence interval, 1.16-2.51). Overall, AMD was associated with an increased risk across all fracture types. Effective management of AMD may help reduce the risk of secondary fractures.

PubMed · DOI

The impact of urinary catheterization on urinary tract infection and renal function in elderly patients undergoing hip fracture surgery: A prospective randomized controlled trial.

Serttas MF, Uysal A, De■i■rmenci■ A, Gök Ü, Erten RA, Saglam F

INTRODUCTION: Urinary tract infection(UTI) is a common complication following hip fracture surgery in elderly patients. Urinary catheterization is frequently used to improve patient comfort during prolonged preoperative immobilization; however, its impact on UTI risk and renal function remains controversial. This prospective randomized study aimed to evaluate the effect of urinary catheter use on postoperative UTI incidence and renal outcomes in geriatric patients undergoing hip fracture surgery.

MATERIALS AND METHODS: This prospective randomized controlled trial included 150 patients aged > 65 years who underwent surgical treatment for proximal femur fractures between December 2022 and May 2025. Patients with UTI at admission, prior indwelling catheter use, high-energy trauma, multiple fractures, or a history of pelvic radiation were excluded. Patients were randomized into two groups: catheterized(n = 75) and non-catheterized(n = 75). Urinary tract infection was defined according to European Association of Urology (EAU) guideline criteria. Renal function was assessed using serum urea and creatinine levels measured preoperatively and on postoperative day 1. Demographic data, comorbidities, fluid intake, and perioperative variables were recorded and compared between groups.

RESULTS: The mean patient age was 79.8±7.2 years, and 61.3% were female. Postoperative UTI occurred in 2 […]

PubMed · DOI

Biomechanical evaluation of locked versus non-locked proximal subchondral raft screws in a Schatzker Type II tibial plateau fracture model.

Steinhoff AK, Beyer RS, Chalmers CE, Zamorano D, Bui CNH, McGarry MH, Lee TQ

OBJECTIVE: Schatzker Type II fractures involve a lateral split and intra-articular depression and are typically managed with a periarticular plate and subchondral raft screws. While locking plate technology is common in osteoporotic bone, its biomechanical advantage over non-locking screws for proximal subchondral rafting remains unclear. We aimed to compare construct stiffness and fragment displacement in a Schatzker Type II fracture model stabilized with a periarticular plate using proximal locked versus non-locked subchondral screws.

METHODS: Six matched pairs of fresh-frozen cadaveric tibias were osteotomized to create a reproducible lateral split-depression fracture. Fractures were reduced and stabilized with a 4-hole periarticular proximal tibia plate. One tibia per pair received proximal non-locking screws - the contralateral received proximal locking screws. Specimens underwent cyclic axial loading (10-100 N), followed by stepwise increases of 200 N (10 cycles each) until failure (>5 mm articular depression or hardware failure). Construct stiffness and displacement were measured using digital motion analysis and a Microscribe.

RESULTS: During cyclic loading, no significant differences were observed between locking and non-locking constructs in stiffness or fragment displacement. However, during load-to-failure testing, the non-locking construct demonstrated significan […]

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Volar marginal fragment escape in distal radius fractures. Results of salvage with a volar open wedge articular reorientation osteotomy.

Gavaskar AS, Srinivasan P, Ayyadurai P, Tummala NC

PURPOSE: To investigate the clinical and radiographic outcomes following volar open wedge osteotomy (VOWO) in patients with volar carpal instability resulting from volar marginal fragment (VMF) escape after distal radius fixation.

METHODS: This retrospective cohort study reviewed records of 21 consecutive adult patients treated with VOWO at two urban tertiary care centers in India. All patients had volar carpal translation following failed fixation of a VMF and a minimum follow-up of 1 year. Adjunctive procedures included short radiolunate ligament repair, refixation of large VMFs, and dorsal wrist-spanning fixation when indicated. The primary outcome was osteotomy healing with restoration of radiocarpal stability. Secondary outcomes included wrist range of motion, Mayo wrist score, and complications.

RESULTS: Osteotomy healing with restoration of radiocarpal stability was achieved in all patients. At a mean follow-up of 23 months (range, 14-48 months), significant improvements were observed in wrist motion and function. The mean wrist flexion-extension arc improved from 46° preoperatively (range, 25-70°) to 110°° postoperatively (range, 80-140°, $P < 0.001$), while the mean pronation-supination arc improved from 90° to 124° ($P = 0.023$). The mean Mayo wrist score improved from 23 preoperatively (range, 5-50) to 83 at final follow-up (range, 75-100, $P < 0.001$). Radiographic arth […]

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Analgesia for Kirschner wire removal in paediatric and adult populations: a narrative review.

Platt SR

BACKGROUND: Removal of Kirschner (K-) wires is a frequent office-based orthopaedic procedure in both paediatric and adult populations. The optimal analgesic strategy is poorly characterised and clinical practice varies substantially across institutions and specialties.

METHODS: A structured PubMed search was performed in April 2026 using terms encompassing K-wire and percutaneous pin removal, analgesia, anaesthesia, sedation, and distraction. The search was supplemented by a semantic search of the Semantic Scholar / OpenAlex corpus. Studies reporting clinical outcome data on procedural pain, anxiety or analgesic strategy during K-wire or related percutaneous hardware removal in the office or outpatient setting were eligible.

RESULTS: In paediatric populations, adequately powered randomised controlled trials demonstrate that oral paracetamol, oral ibuprofen and topical liposomal lidocaine each confer no measurable benefit over placebo. Mean procedural pain scores cluster between 3 and 4 on a ten-point scale, and a majority of caregivers do not consider pre-procedural analgesia necessary. Non-pharmacological distraction reduces both pain and anxiety; simple tablet-based distraction is non-inferior to immersive virtual reality on Level I evidence. In adults, prospective series demonstrate that awake clinic removal of percutaneous hardware without anaesthesia is well tolerated, w […]

PubMed · DOI

Risk Factors for Failure of Repair of Diaphyseal Tibial Nonunions: A Retrospective Cohort Study of 108 Patients.

Fan S, Wagner RK, Lehle CH, Musick AN, Gregg AT, Janssen SJ, Wijffels MME, Stenquist DS et al.

OBJECTIVES: To determine rates and risk factors for additional nonunion surgery and final radiographic healing following diaphyseal tibial nonunion repair.

METHODS: All consecutive patients aged ≥18 years who underwent diaphyseal (AO/OTA 42) tibial nonunion repair with (exchange) intramedullary nailing (IMN) or open reduction and internal fixation (ORIF) between 2001 and 2024 with ≥6 months follow-up were included. The primary outcome was the rate of additional nonunion surgery. The secondary outcome was radiographic healing (defined as Modified Radiographic Union Score for Tibia [mRUST] ≥11) at final follow-up. Multivariable logistic regression was used to determine the association of patient, nonunion, and treatment characteristics with additional nonunion surgery.

RESULTS: A total of 108 patients were included (70% male, median age 42 years [IQR 28-53]). Sixty patients (56%) were treated with IMN, of whom 54 (50%) underwent exchange IMN, and 48 (44%) were treated with ORIF. Autograft was used in 39 patients (36%) and bone graft substitutes in 21 (19%). Twenty-seven patients (25%) required additional nonunion surgery, of whom 5 (4.6%) ultimately underwent amputation. In multivariable analysis, current smoking (OR: 4.28, $p = 0.015$), initial open fracture (OR: 3.94, $p = 0.045$), and presence of a bone defect (OR: 9.20, $p < 0.001$) were associated with increased odds of addition […]

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Superior survival, hemodynamics, and metabolic recovery using PEG-20k-based low volume resuscitation in a polytrauma hemorrhagic shock model in rats.

Khoraki J, Payne C, Lust D, Mangino MJ, Liebrecht LK

BACKGROUND: Hemorrhagic shock is a major cause of preventable civilian and military trauma death. Although whole blood and blood products remain the standard of care, field■stable alternatives are needed for resource■limited environments and supply■chain disruptions. Blood■based resuscitation may also fail to optimally restore microvascular perfusion despite improved macrohemodynamics, contributing to ongoing ischemia-reperfusion injury, dysfunctional molecular cell dynamics, and capillary cell swelling. Prior controlled hemorrhage studies in rats and swine demonstrated superior survival, mean arterial pressure (MAP), and lactate clearance with polyethylene glycol 20,000 (PEG■20k) compared with crystalloid, colloid, and whole blood controls. These beneficial effects are attributed to the capacity of PEG-20k to augment water transfer from swollen ischemic cells into the vascular space, thereby decompressing capillary beds and restoring effective tissue perfusion. This study evaluated PEG■20k in a severe polytrauma rat model with combined controlled and uncontrolled hemorrhage.

METHODS: Adult male Sprague-Dawley rats (250-400-g) underwent a midline laparotomy, 75% tail amputation, and dual splenic transections. The spleen bled freely into the closed abdomen (uncontrolled, non-compressible hemorrhage), while the tail was intermittently clamped (controlled, compressible hemorrhage […]

PubMed · DOI

The forequarter lateral implosion injury: A diagnostic paradigm and Ch-Sh (ch■-sh■) classification.

George A, Paull TZ, Winter JD, Levine J, Cole PA

BACKGROUND: The lateral impaction force against the shoulder often causes multiple lesions to the forequarter including the chest wall. The purpose of this study is to describe the injury patterns and associated pathology in patients with combined chest wall and shoulder girdle injuries- forequarter lateral implosion injury and organize the findings into a classification framework to predict morbidity.

METHODS: Three hundred fifty-four consecutive patients with two or more rib fractures and ipsilateral shoulder girdle injury from a Level 1 trauma center, between 2018-2022, were retrospectively reviewed. Ch-Sh is pronounced ch■ sh■. Chest injuries were grouped into three categories: Ch1: 2-4 fractured ribs, Ch2:

5 + fractured ribs, or Ch3: 2 + fractured ribs with contralateral rib fractures and/or sternal fracture. Shoulder girdle injuries were grouped into: Sh1: clavicle or acromion fracture, and/or AC/SC joint dislocation, Sh2: scapula and/or proximal humerus fracture, or Sh3: combination of Sh1 and Sh2 injuries (clavicular chain and scapula/humerus). There were 9 total subgroups. Demographics, injury characteristics, associated injuries, and outcomes were recorded. Inter and intraobserver reliability of the classification system was studied.

RESULTS: As chest wall injuries scaled from Ch1 to Ch3, there were significantly higher rates of ICU admission, pneumothorax, pulmonar […]

PubMed · DOI

The fate of stage I Masquelet cement spacers in long bone reconstruction for acute fractures.

Dekle JK, Sisk CK, Litten RM, Alcaide DM, McIlwain RN, Wilson AL, Manson HC, Spitler CA et al.

INTRODUCTION: The Masquelet technique is widely used for reconstruction of segmental long bone defects following acute fractures; however, not all patients progress to the second-stage bone grafting procedure. The purpose of this study was to characterize Stage II completion following Stage I Masquelet cement spacer placement for acute long bone fractures, compare characteristics and outcomes between patients who underwent Stage I only versus completed Stage II reconstruction, and identify factors associated with progression to Stage II bone grafting.

METHODS: A retrospective review was conducted of adult patients who underwent Masquelet reconstruction for acute long bone fractures at a Level I trauma center between 2014 and 2024. Patients were stratified by completion of Stage II reconstruction. Demographic, injury-related, surgical, and clinical outcome variables were collected. Multivariable logistic regression was performed to identify factors independently associated with Stage II completion.

RESULTS: A total of 156 patients met inclusion criteria, of which 48 (30.8%) underwent Stage I only and 108 (69.2%) completed both stages. Patients who remained at Stage I underwent fewer orthopaedic operations (3.2 vs. 4.2, $p = 0.001$) and initial bony defect size did not differ between groups (84.1 vs. 78.8 mm, $p = 0.423$). Among patients with at least 6 months of follow-up, rates o […]

PubMed · DOI

Rethinking the Lisfranc injury: Rationale for and development of a new classification system for Lisfranc injuries.

Cawley WCF, Hwang P, Mittal R, Zochowski Y, Darwish I, O'Keefe R, Grogin J, Sivakumar B et al.

BACKGROUND: The Lisfranc joint comprises the articulations between the cuneiforms and bases of the metatarsals and is susceptible to a variety of complex injuries which are frequently missed or diagnosed late. This can result in morbidity with poor outcomes, and renders it challenging to standardise management and allow prognostication. Existing classification systems, including the Myerson-Hardcastle (MHC) and Nunley & Vertullo classifications, do not consider the propagation of force through the midfoot, which is key to identifying structures which may require stabilization. They also demonstrate poor clinical utility due to poor correlation with outcomes. This study aimed to develop a new classification system for Lisfranc injuries using data from a large single-surgeon case series from a tertiary level trauma hospital in Australia. The primary outcome measures for this study were inter- and intra-observer reliability.

METHODS: The Suthersan classification was developed to address shortfalls in current classification systems. Patient records were manually searched for Lisfranc injuries. Direct 'crush' Lisfranc injuries were excluded due to unpredictable injury patterns. Indirect Lisfranc injuries were included in analysis. Two orthopaedics foot and ankle surgeons and one hand surgeon reviewed patient radiographs and classified them using the MHC and Suthersan classification […]

PubMed · DOI

Injury and trauma care training in global health: A scoping review of costs and cost-effectiveness across resource settings.

Phan HT, Nguyen BT, Zhao T, Phung LT, Keakabetse TR, Dinh TTN, Nelson MR, Skinner M et al.

INTRODUCTION: Injury is a leading cause of death and disability worldwide, with a disproportionate burden in low- and middle-income countries, where health resources are constrained and funding decisions are highly cost-sensitive. Injury and trauma care training programs have demonstrated their effectiveness in enhancing the capacity of care providers to treat injured patients, but evidence on their costs and cost-effectiveness remains limited and fragmented. This scoping review aims to map and summarise current literature on costs and cost-effectiveness of injury and trauma care training across different resource settings.

METHODS: We conducted a systematic search on PubMed, MEDLINE Complete, Embase via Ovid, Web of Science, EconLit and Emcare. Papers published in English that reported implementation costs or cost-effectiveness of trauma care training programs were included. Cost data were converted and presented as USD 2024.

RESULTS: Twenty-eight studies were included, predominantly in low- and middle-income countries. Implementation costs varied considerably, starting from USD 3.70 (Lay-First-Responder) to USD 1433.5 (Advanced Trauma Life Support). Foreign instructor involvement was a particularly significant cost driver, accounting for 38-85% of total implementation costs in some programmes. Training-of-trainers and cascading models were effective in reducing costs, with […]

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Do reduced screws have a significant impact on fixation in medial open wedge high tibial osteotomy? A comparative fatigue and stability analysis.

Özkaya M

INTRODUCTION: Medial opening wedge high tibial osteotomy generally employs T-shaped plates with three or more proximal screws to ensure stabilization. This study aims to evaluate the biomechanical necessity of the standard three screw proximal configuration of the T-shaped plates and investigates whether fixations with a reduced number of proximal screws provide sufficient stability and fatigue life under diverse loading scenarios using finite element analysis.

MATERIALS AND METHODS: Three fixation models were analyzed, a standard T-shaped plate with three proximal screws in the first row and two modified models with two proximal screws in the first row with or without an empty hole. Models were subjected to physiological axial loading, centrically, anteriorly, or posteriorly applied in the sagittal plane, to measure total deformation, displacement of the tibial plateau, fixation stiffness, and von-Mises stress. Furthermore, a fatigue analysis was conducted to evaluate the endurance of fixations under these varying loading conditions.

RESULTS: The reduction in proximal screws resulted in a marginal stiffness decrease of less than 2%, with values remaining above 7700 N/mm under centric loading. Total deformation and tibial plateau displacement showed negligible differences between two screw and three screw configurations. In addition, all peak stresses remained well below the […]

PubMed · DOI

Moving beyond mortality measurements after road traffic injuries: Ten-year death risk after road traffic injuries.

Cherchiglia ML, Mougin V, Gray A, Wool EE, Malta DC, Melo APS, de Assis Acúrcio F, Junior AAG et al.

Road traffic injuries (RTIs) represent a significant global public health issue, particularly in low- and middle-income countries like Brazil. Despite the high mortality rates and associated costs, there is limited comprehensive literature addressing long-term mortality resulting from RTIs. Most research tends to focus on rates and hospitalization figures while neglecting the connection between hospitalizations and causes of death, partly due to the challenges of follow-up studies. This study employed record-linkage methodology to create a database that includes both hospitalization and mortality records. The aim was to estimate all-cause and cause-specific relative risks (RR) of death among inpatients due to road traffic accidents, compared to those hospitalized for other reasons, over a time span of three months to ten years. A retrospective cohort study examined hospitalization data for RTIs and mortality from 2000 to 2015 using Brazil's Unified Health System (Sistema Único de Saúde - SUS in Portuguese), excluding any obstetrics-related admissions and deaths, and calculated RR stratified by sex and age over 15 years old. The findings indicated a significantly higher mortality risk of 3.23 (95% Uncertainty Interval: 3.08- 3.39) for individuals previously hospitalized for RTIs compared to others during follow-up. Males aged 15- 29 and 30- 59, especially those with motorcycle, […]

PubMed · DOI

MAGPIE study author's reply.

Hartshorn S, Barrett MJ, Lyttle MD, Yee SA

PubMed · DOI

Can we conclude "Psychosocial intervention is not required"? Methodological reflections regarding a randomized controlled trial on post-traumatic limb amputation.

Zhao FY, Zhang WJ, Fu QQ, Zhu JY, Qian J

PubMed · DOI

Overutilization of helicopter emergency medical services compared to ground transport in moderate-severe penetrating abdominal trauma & associated outcomes.

Hus A, Brown A, Bundschu I, Prashar S, Kumar S, Indorewala Y, Elkbuli A

BACKGROUND: Helicopter emergency medical services (HEMS) play a crucial role in trauma transport, accounting for approximately 550,000 transportations annually. However, HEMS holds a disproportionate share of trauma causes, highlighting its potential overutilization.

METHODS: This retrospective cohort study utilized the ACS-TQIP-PUF dataset (2017-2020) to evaluate outcomes in adult (≥18 years) trauma patients in hemorrhagic shock with moderate-to-severe (AIS abdomen ≥ 2, ISS ≥ 15) penetrating abdominal injuries who activated massive transfusion protocol and were transported from scene to the hospital via either HEMS or GEMS. Patients were stratified by scene time and trauma center level. Primary outcomes included mortality, and secondary outcomes included transfusion requirements, intervention rate, and complications.

RESULTS: Patients transported via Ground Emergency Medical Services (GEMS) compared to HEMS were associated with significantly higher odds of 24-hour mortality (2.028, 95% CI: 1.1853.472, p = 0.010, SE: 0.274). However, no significant association with in-hospital mortality (1.355, 95% CI: 0.901-2.038, p = 0.144, SE: 0.208) was observed. Patients with a scene time of < 10 min demonstrated comparable outcomes, with no significant association with 24-hour or in-hospital mortality.

CONCLUSION: This national analysis demonstrates that outcomes of adult patients in h […]

PubMed · DOI

Early postoperative anxiety is associated with patient-reported recovery after hip fracture: A PROMIS-based longitudinal cohort study.

Achebe C, Cureton R, Rothberg D, Higgins T, Marchand L, Haller J

BACKGROUND: Hip fractures in older adults result in substantial functional decline. While physical factors influencing recovery are well established, the role of early postoperative psychological distress remains incompletely understood. We examined whether early postoperative anxiety was associated with achievement of minimal clinically important difference (MCID) in patient-reported recovery after hip fracture.

METHODS: We conducted a retrospective cohort study of patients aged > 55 years with native hip fracture treated operatively between 2015 and 2024 who completed PROMIS assessments. Baseline PROMIS assessments were defined as the first available domain-specific assessment during postoperative days 0-30. The primary outcome was MCID achievement in PROMIS Physical Function (PF) and Pain Interference (PI) using the final available assessment from postoperative days 31-365. A late-recovery sensitivity analysis restricted final follow-up to postoperative days 91-365. MCID was defined as a > =5-point improvement in PF or > =5-point reduction in PI. Multivariable logistic regression adjusted for baseline domain score, baseline anxiety, age, and Charlson Comorbidity Index (CCI).

RESULTS: The primary PF cohort included 304 patients, of whom 171 (56.3%) achieved MCID. The primary PI cohort included 285 patients, of whom 169 (59.3%) achieved MCID. Higher early postoperative anxie […]

PubMed · DOI

Association between neighborhood opportunity and incidence of pediatric trauma.

Monroy A, Tan JM, Arbogast H, Spurrier R, Nager A, Chaudhari PP

OBJECTIVE: To describe the epidemiology of pediatric trauma and its association with neighborhood opportunity in a large, urban metropolitan area.

STUDY DESIGN: This was a multicenter, retrospective, cross-sectional study of pediatric trauma activations in Los Angeles (LA) County between 2010 and 2021. Children under 18 years old with a home zip code in LA County were eligible. Our main exposure was the categorical Child Opportunity Index 2.0 (COI) score for a child's home zip code. The primary outcome was trauma activations per 100,000 children. Our secondary outcomes were variation in injury mechanism, severity and disposition by COI.

RESULTS: The final sample was comprised of 28,034 patients. The median age was 13.0 years (IQR 7.0, 16.0). Mortality was 1.8% (n = 493). Overall, geographic areas with lower COI overlapped with areas of higher trauma incidence. Falls and sports-related injuries were more common in higher COI areas (all $p < 0.001$). Conversely, being a pedestrian or bicyclist struck, firearm injuries and assaults/stabbings were more common in lower COI areas (all $p < 0.001$).

CONCLUSION: The incidence of pediatric trauma varies based on the COI of a child's home zip code. Our findings can help inform resource allocation for injury prevention programs targeted to specific neighborhoods at greatest risk.

PubMed · DOI

Mechanism of injury not associated with acute compartment syndrome in Tibia fractures.

Egbe S, Marks D, Christiano AV, Erdman MK, Wolf JM, Strelzow J

OBJECTIVES: To report the incidence of Acute Compartment Syndrome (ACS) after low-energy, ballistic tibial fractures and examine associations between injury mechanism, fracture location and morphology, and patient characteristics.

METHODS: Design: Retrospective Review Setting: Single-center, large, urban, level 1 trauma center Patient Selection Criteria: Consecutive skeletally mature patients presenting with a tibial fracture [OTA-41, OTA-42, OTA-43] were evaluated.

OUTCOME MEASURES AND COMPARISONS: Patients were stratified into blunt and ballistic fracture cohorts based on mechanism of injury. Patient sociodemographics, diagnosis of ACS, and established risk factors for the development of ACS (age, gender, fracture location, and morphology) were collected.

RESULTS: Over the study period, 281 patients were included. Study participants had a mean age of 37 years (range, 15-98) with the majority being male, (70%). Based on injury mechanism, 67% (n = 187) were blunt injuries, and 32% (n = 94) were ballistic injuries. Fractures were categorized according to the AO/OTA classification and included 69 [OTA-41], 163 [OTA-42], and 47 [OTA-43] fractures [1]. The overall incidence of ACS in this population was 2.1% (n = 6/281), with a incidence of 1% (n = 1/94) in the ballistic cohort and 2.7% (n = 5/187) in the blunt cohort. Mechanism of injury, sex, age, and open fractures did not sh […]

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Psychological outlook and recovery after orthopaedic trauma: Evidence and implications for Central India and other low-resource settings.

Salphale YS, Saodekar H, Kimmatkar N, Shinde G, Kothari P, Salphale AY

BACKGROUND: Orthopaedic trauma is frequently accompanied by psychological distress, including depression, anxiety, and post-traumatic stress, particularly in low- and middle-income countries. Beyond injury severity and surgical factors, psychological outlook-encompassing negative cognitions (e.g., catastrophising) and positive constructs (e.g., optimism, resilience, social support)-may substantially influence pain, disability, and functional recovery [3,4,6,17,18].

OBJECTIVES: To narratively review evidence on the impact of psychological outlook on clinical, functional, and psychosocial outcomes after orthopaedic trauma, with emphasis on India, Central India, and comparable low-resource settings [4,10,17].

METHODS: A structured narrative review of English-language studies (January 2000-October 2025) was conducted using PubMed, Scopus, and Google Scholar. Observational and interventional studies examining associations between psychological factors and pain, disability, function, quality of life, or rehabilitation outcomes in adult orthopaedic trauma populations were included [4,6,15].

RESULTS: Across diverse settings, approximately one-quarter to one-half of orthopaedic trauma patients experience clinically significant psychological distress, which consistently predicts higher pain, greater disability, and poorer quality of life independent of injury characteristics [2-4,14,1 […]

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Satisfaction with nursing care among Jordanian burn patients admitted to burn critical care units: A cross-sectional study.

Al-Ghabeesh S, Alnaeem M, Qaddoura N, Ali BA, Al-Masalha HA

BACKGROUND: Patient satisfaction with nursing care is a key indicator of healthcare quality in burn intensive care units (ICUs). However, limited evidence exists regarding predictors of satisfaction among burn patients in Jordan.

OBJECTIVE: The aim of this study was to assess satisfaction with nursing care among burn patients admitted to a burn intensive care unit in Jordan and to identify demographic and clinical predictors associated with satisfaction.

METHODS: A descriptive cross-sectional study was conducted between October 2024 and May 2025 in a specialized Burn ICU at a major military hospital in Jordan. Adult burn patients ($n = 146$) admitted for ≥ 48 h were recruited using convenience sampling. Satisfaction was measured using the Arabic version of the Patient Satisfaction with Nursing Care Quality Questionnaire (PSNCQQ). Independent t-tests, ANOVA, Pearson correlation, and multiple linear regression were used. Significance was set at $p < 0.05$.

RESULTS: The mean satisfaction score was 48.88 (SD = 13.83), indicating moderate satisfaction. Higher income was associated with greater satisfaction ($B = -6.529$, 95% CI: -11.65 to -1.40, $p = 0.014$). Facial deformity predicted lower satisfaction ($B = 7.743$, 95% CI: 1.30-14.18, $p = 0.019$). Age showed a weak but significant correlation ($r = -0.173$, $p = 0.037$).

CONCLUSION: Satisfaction among burn ICU patients was moderate and infl […]

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Flexible intramedullary nailing in the treatment of paediatric forearm fractures: 15-years' experience at a level one trauma centre, comparison of implant retention versus removal and cost-effectiveness analysis.

Ismayl G, Alabbas A, Howard A, Fishlock A, Sabouni MY

INTRODUCTION: Flexible intramedullary nailing (FIN) is a common treatment method for paediatric forearm fractures. They are routinely removed after fracture healing. This study aims to assess the 15-years' experience of managing paediatric forearm fractures with FIN, with comparison between the outcomes of implant retention versus removal and cost analysis. This is the first study on FIN retention in paediatric forearm fractures.

METHODS: This was retrospective cohort study for paediatric forearm fractures treated with FIN between 2009 and 2024 at Leeds General Infirmary, a level one trauma centre in the United Kingdom. We noted a significant decline in scheduled elective removal of forearm FIN following the Coronavirus pandemic (COVID-19). The cohort was therefore divided into two comparative groups: pre-COVID (routine implant "removal") and post-COVID (routine implant "retention"). Primary outcome measured was incidence of secondary (unplanned) surgery. Secondary outcomes measured were indications for surgery, time from index procedure to FIN removal and subjective patient/parents' satisfaction.

RESULTS: A total of 268 patients were included in the study. The pre-COVID group comprised of 212 patients, and the post-COVID group 56 patients. Elective implant removal was performed in 68.9% ($n = 146$) of patients in the earlier group and 3.6% ($n = 2$) in the latter. Secondary surg […]

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Association of Primary Trauma Care training with mortality in isolated traumatic brain injury: A sub-analysis of the PISTACHIO study in Vietnam.

Nguyen BT, Lam TT, Vu NHN, Pham TH, Do TD, Nguyen TTN, Nguyen HT, Skinner M et al.

BACKGROUND: The association of the Primary Trauma Care (PTC) program with Isolated Traumatic Brain Injury (TBI), where mortality is driven by secondary brain injury rather than haemorrhagic shock, remains underexplored in Low- and Middle-Income Countries.

METHODS: This prospective before-and-after study analysed 1388 patients with isolated TBI (head injury without significant extracranial trauma) at two provincial hospitals in Vietnam. Patients were stratified by admission Glasgow Coma Scale (GCS). The primary outcomes were 24-hour and 30-day mortality. Sensitivity analyses, including best- and worst-case scenarios, were performed to assess attrition bias.

RESULTS: Of 1388 patients (Pre: 712; Post: 676), baseline neurological severity was comparable ($p = 0.100$). Overall, 24-hour mortality decreased significantly from 6.7% to 1.8% (adjusted Odds Ratio-aOR 0.25, 95% CI 0.13-0.47, $p < 0.001$). This survival benefit was sustained at 30 days (10.8% vs. 4.1%; aOR 0.35, 95% CI 0.20-0.64, $p = 0.001$). The 30-day loss to follow-up rate was 38.7%, primarily due to pandemic disruptions. Although sensitivity analysis suggested no significant difference in Injury Severity Score ($p = 0.416$) or GCS ($p = 0.706$) between lost and retained patients, best- and worst-case scenario modelling (aOR 0.27 and 1.66, respectively) indicated that the potential for attrition bias remains a significant consi […]

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Peterson vs. Salter Harris physeal fracture classification systems: How consistent are they?

Charles-Lozoya S, Cobos-Aguilar H, Váldez-Cordova C, de la Parra-Márquez ML, Manilla-Muñoz E, Arriaga-Cazares HE, Guzmán-Delgado NE

INTRODUCTION: The Salter-Harris (S-H) classification establishes five types of physeal fractures and is the most widely used to describe them. However, the Peterson classification includes fracture patterns not covered by the S-H classification, such as fractures crossing the metaphysis and extending to the physis and open injuries with loss of physis. Nonetheless, it is necessary to validate and compare its reliability with that of the S-H to enhance diagnostic possibilities and clinical interpretation. This study aimed to compare the consistency of intra and interobserver agreement of the S-H and Peterson classifications.

METHODS: A cross-sectional study was conducted to evaluate the reliability of 100 radiographs in the anteroposterior and lateral views. Five pediatric orthopedic surgeons and five third- and fourth-year orthopedic residents were asked to classify both fracture types on two occasions, with an eight-week interval between the two occasions.

RESULTS: The intraobserver agreement for S-H in the expert and resident groups was $\kappa = 0.66$ and 0.61, respectively; for Peterson, it was $\kappa = 0.67$ and 0.61, respectively. The mean total agreement percentages for S-H and Peterson were 84.5% vs. 83.9%, respectively ($t = 0.254$; $P = 0.8$; Cohen's $d = 0.20$). The overall interobserver agreement for S-H in the first and second observations was $\kappa = 0.57$ and 0.63, respectively, and f […]

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Cemented versus uncemented femoral stem fixation in hip arthroplasty for displaced femoral neck fractures in elderly patients: A systematic review and meta-analysis of randomized controlled trials.

Malki AT, Babtain HW, Alharbi DM, Shahhatalsayed AA, Alshaban RM, Almaghrabi YT, Felemban RA, Almuffarh DS

BACKGROUND: The optimal method of femoral stem fixation in hip arthroplasty for displaced femoral neck fractures in elderly patients remains controversial. While cemented stems provide immediate fixation, concerns persist regarding perioperative risks. Uncemented stems avoid cement-related complications but may be associated with mechanical instability in osteoporotic bone.

METHODS: This systematic review and meta-analysis included randomized controlled trials comparing cemented and uncemented femoral stem fixation in elderly patients undergoing hemiarthroplasty or total hip arthroplasty for displaced femoral neck fractures. A comprehensive literature search of major databases was conducted from inception to July 2025. Outcomes of interest included mortality, periprosthetic fracture, dislocation, and wound infection. Data were pooled using a random-effects model, and subgroup analyses were performed based on follow-up duration and type of arthroplasty. To avoid duplication, only one dataset per study was included in the overall analysis.

RESULTS: Sixteen randomized controlled trials involving 3776 patients were included. Cemented fixation was associated with a statistically significant reduction in overall mortality compared with uncemented fixation (odds ratio 0.83, 95% confidence interval 0.70-0.97). The risk of periprosthetic fracture was substantially lower with cemented […]

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Impact of surgical timing on complications and mortality in periprosthetic hip fractures compared to native hip fractures.

Aerden LKP, Brouwers X, Sermon A, Ghijselings S, Vles G, Herteleer M, Metsemakers WJ, Hoekstra H

INTRODUCTION: Timely surgery within the next calendar day is recommended for native hip fractures (NHF) according to NICE guidelines. In contrast, no standardized fast-track protocols exist for periprosthetic hip fractures (PPFx), potentially resulting in surgical delays. This study aimed to evaluate time to surgery and reasons for delay in surgically treated PPFx patients, and to assess complications and mortality according to surgical timing. A previously published NHF cohort was used as a clinically relevant benchmark reflecting the established geriatric native hip fracture pathway.

MATERIALS AND METHODS: We retrospectively analyzed 217 surgically treated PPFx patients at a university level-one trauma center. Time to surgery, reasons for delay, complications, length of stay, and mortality were assessed. Outcomes were analyzed within the PPFx cohort according to surgical timing, defined as early surgery within the next calendar day versus delayed surgery. To improve comparability with the NHF benchmark cohort (all aged ≥75 years), additional analyses were performed in an age-restricted PPFx subgroup including patients aged ≥ 75 years.

RESULTS: Median time to surgery was significantly longer in the overall PPFx cohort than in the NHF benchmark cohort (43.6 h [IQR 21.1-85.7] vs. 16.8 h [IQR 9.7-23.1], $p < 0.001$). Similar findings were observed in the age-restricted PPFx subgr […]

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Antibiotic prophylaxis for preventing open fracture related infections - A scoping review.

Krakau RB, Hamilton ND, Dastrup AS, Klein K, Dreyer CH

PURPOSE: The primary aim was to report available literature on antibiotic prophylaxis against open fracture-related infections, in populations with antimicrobial resistance patterns similar to Denmark's as reported by the European Antimicrobial Resistance Surveillance Network. Countries included: Denmark, Finland, Germany, Great Britain, Iceland, the Netherlands, Norway, and Sweden. A secondary aim was to compare Danish regional guidelines with these findings.

METHODS: PubMed, Medline, and Embase were searched for studies published between January 1998 and March 2024. Danish guidelines were retrieved on May 7th, 2024, through a systematic search.

RESULTS: Out of 6492 studies, 120 were screened in full, and 19 were included for analysis. 9 studies recommended cephalosporins (primarily 1st and 2nd generations) for Gustilo-Anderson Grades I and II, while 10 studies supported further gram-negative coverage for Gustilo-Anderson grade IIIa-c fractures. One study found no significant benefit of antibiotic prophylaxis for reducing fracture-related infections. Danish guidelines varied: The Capital Region of Denmark recommends gram-negative coverage for all Gustilo-Anderson grades, whereas the Region of Southern Denmark and 10/19 studies only recommend it for Gustilo-Anderson grade IIIa-c fractures.

DISCUSSION: This review identifies a lack of data on pathogens causing FRI in regions […]

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The use of external fixation in the treatment of nonunion and malunion of the upper extremity.

Al Ashhab MGED, Abdelnaser A, Elashhab AM, Elmesalamy N, Hosny G

BACKGROUND: Nonunion and malunion of upper extremity fractures are uncommon but challenging complications, particularly when associated with infection, deformity, or multiple previous operations. Although internal fixation remains the standard treatment, external fixation based on Ilizarov principles offers advantages in complex situations including infection control, deformity correction, and gradual bone reconstruction. This study evaluates the indications, surgical technique, and outcomes of external fixation as a definitive treatment for nonunion and malunion of the upper extremity.

METHODS: A retrospective study was conducted including patients treated for nonunion or malunion of humeral and forearm fractures using external fixation as the definitive method of treatment. Hybrid or circular frames were applied according to fracture location. Corticotomy and gradual deformity correction were performed following the principles of Ilizarov and the law of tension-stress. Patients were evaluated clinically and radiographically for union, deformity correction, pain, function, and complications.

RESULTS: A total of 45 patients were included, consisting of 7 malunions, 8 aseptic nonunions, and 30 septic nonunions. Patients' ages ranged from 6 to 62 years. The mean time from injury to definitive treatment was 29.7 months. The average external fixation time was 7.9 months and mean […]

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Analysing geospatial variables in the context of trauma networks: A scoping review.

Harris M, Mosali T, Doyle F, Watts L

BACKGROUND: Trauma is a leading cause of preventable death and disability worldwide. Although nearly 90% of trauma-related mortality occurs in low- and middle-income countries, current reports on emergency medical care often come from high-income countries with fewer resource constraints. This scoping review investigates which geospatial variables are measured and reported in trauma networks, how they are defined, characteristics of relevant studies, analytical software used, and existing gaps in the literature.

METHODS: A search was conducted using PubMed, Embase, and Cochrane for studies published between 2004 and 2024. Data extracted included study characteristics, geospatial variables including prehospital time and distance, spatial access, transport modality, population measures, injury location, analytical methods, and patient outcomes such as mortality and injury severity. Descriptive analysis was performed, grouping studies according to key geospatial themes.

RESULTS: A total of 13,908 studies were identified, of which 41 were included, spanning seventeen countries. Prehospital time was reported by nineteen studies where longer prehospital intervals, including delays in dispatch and transport, were consistently associated with worse outcomes. Fifteen studies evaluated patients beyond golden hour access to care, demonstrating increased prehospital mortality with longer […]

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Validation of a novel osteoarticular fracture fragment decontamination-preservation system for delayed articular fracture reconstruction for limb preservation in a preclinical canine model pilot study.

Cook JL, Della Rocca GJ, Bozynski CC, Stoker AM, Cook CR, Crist BD, Nuelle JAV

INTRODUCTION: This study tests the hypothesis that a novel decontamination-preservation system (DPS) could effectively eradicate relevant pathogens in contaminated, devascularized, extruded osteoarticular fracture fragments while preserving essential viable chondrocyte density (VCD) for joint reconstruction resulting in functional limb preservation.

METHODS: With animal care and use committee approval, hounds were anesthetized for captive bolt-induced open bicondylar femoral fractures. One femoral condyle osteoarticular fragment was retrieved and randomly assigned to: Primary Bacterial Contamination (PBC) (n = 3 immediately processed using the DPS) or PBC+Secondary Methicillin-resistantStaphylococcus aureus(MRSA) Contamination (PBC+MRSA) (n = 3 directly inoculated with MRSA, incubated (24 h), then processed using the DPS). The injured limbs were stabilized with external skeletal fixators. One week following fracture, delayed joint reconstruction surgery was performed. Dogs were managed for pain and wound care, and assessed for safety (complete eradication of relevant pathogens) and efficacy (maintenance of VCD to allow for fracture healing and restoration of function, pain relief, and knee range-of-motion (ROM) to >80% of pre-operative levels).

RESULTS: All retrieved osteoarticular fragments produced polymicrobial growth with all in the PBC+MRSA cohort including MRSA. The DPS […]

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Orthoplastic approach to infected type 3B open tibial fractures: Management and complications.

Erol K, Er E, Vahabi A, Sügün TS

OBJECTIVE: This study aimed to describe the surgical strategies, the patterns of complications, and their subsequent management in patients with infected Gustilo-Anderson type 3B open tibial fractures.

MATERIALS AND METHODS: A retrospective chart review was conducted for the period between January 2010 and January 2024 at a single institution. A total of 27 patients with infected Type 3B tibial fractures were eligible for inclusion in the final analysis. The diagnosis of infection was established by incorporating clinical parameters, including wound and discharge characteristics, odor, microbiological cultures, and inflammatory markers. Demographic and clinical data, including age, gender, mechanism of injury, and total follow-up duration, were recorded. Furthermore, the interval between interventions, the presence and dimensions of bone defects, and the specific fixation methods were analyzed. Time to osseous union was determined based on serial follow-up radiographs. Pathogen identification was performed through a review of culture results, and the specific tissue transfer techniques utilized for soft tissue coverage were documented. Additionally, all observed complications and the frequency and extent of secondary interventions were recorded.

RESULTS: Mean age was 34.3 (± 14.6) years. Median follow-up period was 27 (12-140) months. For soft tissue coverage, an anterolatera […]

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Surgical treatment of fracture-related infections of the calcaneus with relevant soft-tissue defects.

Zhou F, Liu H, Zhang Y, Wang J, Chen H, Cai L

BACKGROUND: Fracture-related infection (FRI) of the calcaneus with associated soft tissue defects after open reduction and internal fixation (ORIF) presents significant therapeutic challenges. This study evaluates the efficacy of two filling techniques-bone cement with local flap transfer (BC+LFT) and free myocutaneous flap (FMF)-in managing these complex cases, with consideration of the FRI classification system.

METHODS: A retrospective analysis on patients treated between June 2017 and June 2024 was done. Based on post-debridement defect size, patients were allocated to BC + LFT (defects ≤3 cm) or FMF (defects >3 cm). All cases were classified according to the FRI criteria. Surgical management included radical debridement, dead space obliteration, and soft tissue reconstruction. Outcomes were assessed via AOFAS hindfoot scores, VAS pain scores, complication rates, and radiographic evaluation.

RESULTS: A total of 43 patients were included (25 BC+LFT, 18 FMF). The FMF group had longer operative times and greater blood loss ($p < 0.05$), but no significant differences were observed in debridement frequency, hospitalization, or complication rates. Both groups showed significant improvement in functional and pain scores at final follow-up. Fracture union was achieved in all cases without recurrence of infection.

CONCLUSION: Both BC + LFT and FMF are effective for managing calcan […]

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Corrigendum to "Post-orthotic brace upright radiographs in thoracolumbar compression fractures do not change initial management in the emergency department setting" [Injury 57 (2026) 113170].

Randall MM, Uchimura RN, Steinmann JC, Thomson BD, Labha J, Minahan T, Mesisca MK

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Retrieval of entrapped intramedullary fragments using laparoscopic instruments for fixation of femur and tibia fractures.

Gray MT, DiCosmo M, Schottel PC

Cortical fragments within the intramedullary canal can pose a challenge during intramedullary nail treatment of tibial and femoral shaft fractures. Complications including reamer entrapment, malunion, and inability to pass final implants have all been reported. To limit soft tissue injury and avoid an open approach, laparoscopic instrumentation can be a useful adjunct to manipulate or retrieve cortical fragments at distant sites. Our objective is to outline a method to remove cortical fragments during treatment of tibial and femoral shaft fracture and characterize the standard shaft lengths diameters, tip configurations and jaw opening widths for common laparoscopic instruments.

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Understanding the key correlates of post-crash clinical indicators and outcomes in road traffic crashes.

Ahmad S, Ahmad N, Dong Z

This study investigates the association of pre-crash factors (i.e., driver/rider behaviors and roadway conditions), and emergency response attributes (pre-hospital time (PHT) and mode of evacuation) with clinical outcomes directly and through the three post-crash clinical indicators, harnessing an extensive hospital-based dataset (N = 273,362) collected in Karachi (Pakistan). The three jointly estimated ordered probit models for post-crash clinical indicators including Glasgow Coma Scale (GCS), Systolic Blood Pressure (SBP), and Respiratory Rate (RR) outperformed their independent counterparts, while a multinomial logit model estimated the association of the three post-crash clinical indicators (along with other explanatory variables) with the final clinical outcomes (e.g., mortality, admitted, discharge etc.) in RTCs. The findings reveal that underage driving/riding significantly increases the chance of severe GCS (3-8), RR (<10 bpm), and SBP (<76 mmHg) by 1.959%, 0.957%, and 0.842%, respectively. Delay in pre-hospital time significantly worsens the three post-crash clinical indicators. On the other hand, the chance of severe GCS, SBP, and RR reduced significantly by 6.657%, 5.728%, and 5.528%, respectively, if a private vehicle was used for evacuation. Based on the clinical outcome model, severe GCS, SBP, and RR (compared to their normal values) increase the chance of mortali […]

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Excellent functional outcomes and low complication rate following open reduction for severe pediatric radial neck fractures: A retrospective analysis.

Puglia F, Tanel L, Priano D, Uboldi FM, Trezza P, Puce M, Memeo A

BACKGROUND: Pediatric radial neck fractures, though rare, pose significant management challenges with several treatment options. Open reduction is typically used for more complex cases, where greater fracture severity may contribute to higher complication rates and poorer outcomes. This study aims to evaluate functional outcomes, complications, and risk factors in a large cohort of pediatric patients treated with open reduction, performed after unsuccessful closed reduction.

METHODS: A retrospective cohort study was conducted on 53 pediatric patients (mean age 8 years) with radial neck fractures (Judet 4) treated surgically with open reduction, performed after unsuccessful closed reduction, at a single institution between March 2014 and October 2024. Data on surgical delay, complications, and functional outcomes (Oxford Elbow Score) were collected. Statistical analyses included correlation tests, t-tests, and multivariate regression models to assess predictors of outcomes and complications.

RESULTS: The mean Oxford Elbow Score was 96, indicating excellent functional outcomes. Complications occurred in 13.5% of patients, including heterotopic ossification (9.6%), posterior interosseous nerve injury (1.9%), and avascular necrosis (1.9%). Surgical delay was significantly longer in patients with complications ($p = 0.038$). Multivariate analysis revealed that complications were the […]

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Association between financial distress and levels of discomfort and incapability after musculoskeletal injury.

Parser N, Melendez JE, Shields C, Lewis A, Ring D, Jaarsma E

OBJECTIVES: To determine whether self-reported financial stress is associated with levels of capability during recovery from musculoskeletal injury.

METHODS: Design: Cross-sectional survey SETTING: Multiple metropolitan musculoskeletal outpatient specialty offices PATIENT SELECTION CRITERIA: Adult outpatients seeking specialty care within 6 months of a musculoskeletal injury (fracture, tendon rupture, sprain, or dislocation).

OUTCOME MEASURES AND COMPARISONS: Capability was measured using the PROMIS Physical Function Computer Adaptive Test. Financial stress was measured using the FACIT-COST Version 2 questionnaire. Explanatory variables included levels of distress and unhelpful thoughts regarding sensations, social health, and sociodemographic characteristics. Associations with capability were examined using bivariate analyses and multivariable linear regression.

RESULTS: Among 131 participants (56% men, mean age 42 ± 17 years), and an average of 8 ± 6 weeks had elapsed since injury. In multivariable analysis, higher capability was associated with longer time since injury (regression coefficient [RC] = 0.51, $p < 0.01$) and uninsured status (relative to Medicare coverage; RC = 13, $p < 0.01$). Lower capability was observed among unemployed participants (RC = -4.6, p = 0.033) and those with lower extremity injuries (RC = -4.5, $p < 0.01$). Financial stress demonstrated a modest inv […]

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Development and internal validation of a prediction model for in-hospital mortality in trauma patients with high-energy pelvic fractures admitted to a single major trauma centre.

Gavino SO, Read DJ, Russell M

BACKGROUND: Pelvic fractures, although relatively uncommon, are associated with high economic burden, morbidity, and mortality. Mortality is largely driven by severe associated injuries and high-energy mechanisms. While predictors of mortality are well established, their local applicability for the study institution has not been updated using site-specific data since an earlier regional study (2001-2008). As trauma management protocols have evolved, this study utilizes prediction model development and internal validation-adhering to the TRIPOD guidelines-to update the prognostic value of these established predictors within our institutional context by using site-specific data.

METHODS: Data from July 2010 - December 2022 were sourced from a Level I adult Major Trauma Centre's registry. The cohort included patients > =15 years old with Injury Severity Score (ISS)> 12 and pelvic fractures. In-hospital mortality was the outcome of interest. Model development utilized backward elimination for predictor selection into a multivariable logistic regression model, with internal validation via bootstrap methods. Model performance was assessed using the Brier scaled score, and discrimination (c-statistic), and calibration (calibration plot).

RESULTS: Out of 1564 included patients, 118 were non-survivors (mortality rate 7.5%). The optimism-adjusted prediction model identified ISS≥ 50 (OR […]

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Thermal advantages of zirconia drills in bone drilling.

Addepalli P, Sawangsri W, Wattanasinbumrung P, Ghani SAC, Thitiyanaporn C

Heat production during surgical bone drilling is a serious risk factor for osteonecrosis, and cortical bone damage is observed at temperatures exceeding approximately 47°C for 1 min. Traditional stainless steel (SS316L) drills have high thermal conductivity, which can cause rapid heat transfer, whereas zirconia (ZrO_2) has lower thermal conductivity and may result in lower thermal injury. This research comparatively studied SS316L and ZrO2 drills under dry-drilling conditions in terms of diameter (2.5, 3.0, 3.5 mm), drill speed (900, 1100, 1300 rpm), and feed rate (30 & 40 mm/min) using standardised bone blocks with thermocouple temperature monitoring. Findings proved that the two dominant factors were drill diameter and spindle speed. In the case of SS316L, a high temperature of 61 °C was observed at 3 mm dia, 1100 rpm, and 40 mm/min, which is above the necrosis temperature, whereas at the same conditions ZrO2 remained at low temperatures below 46 °C. An increase in feed rate (30-40 mm/min) decreased the Tmax of 2.5 mm SS316L drills (48.0 -39.3 °C) and raised it at 3.5 mm (48.5 -53.5 C). In all the parameters, ∆Tmax (SS316L - ZrO2) was positive (15 - 23 °C) with increased diameter and with increased speed, which demonstrates the consistent thermal merit of zirconia. In clinical practice, zirconia drills are better in mid-diameter osteotomies and high-speed operations, whereas S […]

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Correlation analysis between PFNA insertion point and neck-shaft angle in intertrochanteric fractures, and selection of PFNA (240 mm) under varying radius of femoral curvature.

Han HS, Li X, Liu HJ, Yan J, Han LR

BACKGROUND: The Intramedullary fixation of unstable intertrochanteric fractures is usually performed using the proximal femoral nail anti-rotation (PFNA), but the problems of optimal insertion point of intramedullary nail and anterior cortical impingement have not been solved.Meanwhile, the variation of the femoral neck-shaft angle (NSA) may also result in the deviation of the PFNA's insertion point. The purpose of this study to determine the ideal intramedullary nail placement position in the treatment of intertrochanteric fractures through the three-dimensional (3D) models, and to investigate the relationship between the differences in the RFC and the selection of intramedullary nail length when the ideal insertion position is established.

METHODS: We retrospectively collected the femur computed tomography (CT) scans of 200 adults, and the DICOM-format CT scan images of each patient were imported into Mimics software to creat virtual femoral models. By inserting the PFNA into the femoral model, we recorded the length of each femur model and the neck-shaft angle (NSA); the vertical distance between each bony landmark and the optimal insertion point were measured in the projected plane;the distance of the insertion point to the apex of the greater trochanter (L2); the distance of the insertion point to the vastus lateralis muscle ridge (L3); the distance of the insertion point […]

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Antibiotic prophylaxis protocol for gunshot wound-related open fractures: A single-center retrospective analysis.

Gubara K, Gordy D, Echeverria N, Rehman S

BACKGROUND: Gunshot-induced open fractures (ballistic OFs) present significant challenges in trauma management due to high risk of infection and lack of standardized antibiotic prophylaxis protocols. This study evaluates the implementation of an institutional antibiotic prophylaxis protocol for ballistic OFs at a Level I trauma center, assessing its impact on infection rates and protocol adherence.

METHODS: A retrospective analysis of patient records was conducted, including all cases of ballistic OFs treated between June 7, 2021, and March 31, 2024. Patients were stratified into pre-implementation and post-implementation groups, with the latter further divided into adherent and non-adherent subgroups. Infection rates, time to antibiotic administration, and demographic and clinical variables were analyzed using chi-squared tests and logistic regression.

RESULTS: Among 289 patients included, post-implementation protocol adherence was observed in 36% of cases. Infection rates were 7.1% (12/168, 95% CI 4.1-12.1%) before implementation, and 11.6% (14/121, 95% CI 7.0-18.5%) after with no significant difference (Fisher's exact test, p = 0.276). On multivariable analysis, infection was associated with vancomycin use (adjusted OR 5.24, 95% CI 1.95-13.77, p = 0.001), use of irrigation and debridement (adjusted OR 4.60, 95% CI 1.85-12.24, p = 0.001), and use of flap closure or split-th […]

PubMed · DOI

Linking hospital-based violence intervention program implementation with patient-reported outcomes.

Kapa HM, Liu Z, Berardi J, Glavin CM, Myers RK

BACKGROUND: Understanding patient perspectives on the short-term impact of hospital-based violence intervention programs (HVIPs) is crucial to providing patient-centered care. However, HVIP evaluations often fail to integrate patient-reported outcomes (PROs), relying predominately on distal outcomes (e.g., reinjury, fatality) captured by administrative data sources to determine impact. We aimed to examine how observable program implementation metrics may be related to PROs, including satisfaction with and short-term impacts of a pediatric HVIP.

METHODS: We conducted a retrospective cohort study of 357 patients (ages 8-18 years) who participated in our HVIP located within an urban pediatric trauma center. We linked HVIP implementation metrics from case management records and PROs to proxy caregiver-reported satisfaction and outcome data from the HVIP-Client

Satisfaction Questionnaire (HVIP-CSQ). We summarized patient demographic and injury characteristics, as well as program implementation metrics and HVIP-CSQ responses. We conducted multivariate linear regression analyses to examine associations between implementation metrics and caregiver-reported satisfaction and outcomes.

RESULTS: Patients with (n = 199) and without PROs data (n = 158) had similar demographic, clinical, and HVIP implementation characteristics, though those with PROs were more likely to have been retained i […]

PubMed · DOI

Autologous bone marrow aspirate as a standalone treatment in patients with impaired fracture healing: A scoping review.

Singh T, Rouleau DM, Laflamme GY, Leduc S, Therrien E, Chapleau J

INTRODUCTION: Fracture nonunion develops in 5-10% of all fractures and the industry offers biological grafting solutions to address this. Autologous bone marrow aspirate (BMA) may represent an inexpensive alternative, minimally invasive, biologically active adjunct to surgery.

OBJECTIVE: This scoping review aimed to analyze the current evidence on the effectiveness and safety of autologous bone marrow aspirate as a standalone treatment in patients with impaired fracture healing (minimum 3 months post-fracture).

METHODS: The data was extracted from Embase, Medline, Cochrane Library, Web of Science, and ClinicalTrials.gov, up to October 30, 2024, and conducted according to the Joanna Briggs Institute Scoping Review Manual and Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews. Randomized controlled trials, non-randomized trials, prospective or retrospective cohort studies, and case series with a minimum of five participants were included. Studies had to report on BMA as a standalone treatment for nonunion and assess at least one outcome related to effectiveness (ex: time to union, radiographic healing, etc.). Two members of the research team independently carried out the screening, selection, and data extraction processes, with a third member resolving any discrepancies. Descriptive data were presented in tabular form, while other f […]

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Factors associated with older adult fracture visits to United States emergency departments, 2013-2022.

Matter AL, Pierce MF, Beal JR, Sahmoun AE

INTRODUCTION: Fractures in older adults are increasingly burdensome on U.S. emergency departments (EDs) and the greater healthcare system. As the older adult population continues to grow, the number, cost, and complexity of fall-related injuries, particularly fractures are rising. This study aimed to identify national trends and factors associated with fracture-related ED visits among older adults.

METHODS: We conducted a retrospective analysis using 2013-2022 National Hospital Ambulatory Medical Care Survey-Emergency Department (NHAMCS-ED) data. Adults aged ≥65 with a primary fracture diagnosis were included. Fractures were then categorized by location and mechanism. Demographic and clinical parameters were compared across age groups: 65-74, 75-84, and ≥85 years. Analyses used SPSS Complex Samples v30.0, accounting for survey design, with $p < .05$ considered significant.

RESULTS: An estimated 5.5% or 12.5 million older adult ED visits were fracture-related from 2013 to 2022. Adults aged 65-74 accounted for 39.1% of visits, followed by 75-84 (33.2%) and ≥85 (27.7%). Falls were the most frequent cause of fracture (74.5%), ($p = .002$). The proportion of femur fractures rose steadily with age, accounting for 10.9% of fractures in adults 65-74, 21.2% in those 75-84%, and 30.1% in those ≥85 years ($p < .001$). History of osteoporosis was associated with increasing age along with ambul […]

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Perioperative multidose tranexamic acid combined with erythropoietin and iron isomaltoside supplementation reduces transfusion requirements of elderly patients with intertrochanteric fractures.

Ma H, Li M, Liu X, Liu X

INTRODUCTION: Perioperative anaemia and blood transfusion are common complications in elderly patients with intertrochanteric fractures, impacting recovery. This study aimed to evaluate whether a combined approach of early, continuous antifibrinolytic therapy with tranexamic acid (TXA) plus erythropoietin (EPO) and intravenous iron isomaltoside (IM) reduces transfusion requirements compared to TXA alone or standard care.

METHODS: This retrospective analysis included 134 patients aged > 70 years with intertrochanteric fractures treated with proximal femoral nail antirotation. Three perioperative blood management protocols were compared: Control group receiving a single 1 g dose of intravenous TXA; TXA group receiving sequential 1 g TXA every 12 h for 48 h; and TXA + EPO/IM group receiving the sequential TXA regimen plus a single 1000 mg dose of IM and daily 40,000 U EPO. The primary outcome was total perioperative transfusion rate. Secondary outcomes included hidden blood loss (HBL), haemoglobin (Hb) levels, and 90-day complications.

RESULTS: From post-traumatic day 2 (POD 2) to POD 3, HBL was significantly lower in the TXA and TXA + EPO/IM groups compared to the control group ($P < 0.05$). On PODs 2 and 3, Hb levels in the TXA + EPO/IM group were significantly higher than in the TXA group ($P < 0.05$). The total transfusion rate was significantly lower in the TXA + EPO/IM group ([…]

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Risk factors for postoperative infection after non-elective femur fracture fixation in pediatric patients.

Travis LM, Blanc NK, Hinchberger V, Hellinger R, Pavlis W, Grottkau BE

BACKGROUND: Pediatric femoral shaft fractures are common and often require surgical fixation. Postoperative infections remain a serious complication, yet risk factors specific to this population are not well defined.

METHODS: We analyzed 14,037 pediatric patients undergoing non-elective femur fracture fixation in the 2019 and 2022 Kids' Inpatient Database (KID). Multivariable logistic regression identified independent predictors of postoperative soft tissue, device-related, and overall infection. Secondary outcomes included length of stay (LOS) and hospital charges.

RESULTS: Infections occurred in 0.6% of patients (0.4% soft tissue, 0.3% device-related). Nondisplaced fractures had sharply increased odds of device infection (OR 35.2, 95% CI 4.7-189.7, $p < 0.001$), while open fractures were protective (OR 0.12, 95% CI 0.01-0.58, $p = 0.039$). External fixation conferred the highest infection risk versus intramedullary nailing (OR 4.65, 95% CI 2.25-9.13, $p < 0.001$). Blood transfusion consistently predicted infection across categories (OR 2.6-3.5, all $p < 0.01$). Soft tissue infection prolonged LOS (20.5 vs 4.2 days, $p < 0.001$) and raised hospital charges ($421,798 vs $111,845, $p < 0.001$).

CONCLUSION: This study identifies novel and actionable risk factors for postoperative infection following pediatric femur fracture fixation. Recognition of high-risk groups, including those receiv […]

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The DRoPH score: development of a tool for predicting reduction difficulty in proximal humerus fracture osteosynthesis.

Barthel C, Rudolph F, Kraus M, Pape HC, Allemann F

BACKGROUND: Proximal humerus fractures (PHFs) are among the most common and challenging fractures due to complex anatomy and variable fracture patterns. While anatomical reduction is critical for successful outcomes, no tool currently predicts the intraoperative difficulty of achieving it. Despite advances in surgical techniques, complications persist. Existing classification systems, such as Codman's, Neer's, AO/OTA, and Mayo-FJD, focus on fracture morphology and displacement but do not address intraoperative complexity. This study aims to develop a new score for the early prediction of intraoperative reduction difficulty in proximal humerus fracture surgery, the DRoPH score (Difficulty of Reduction of Proximal Humerus Fractures).

METHODS: An online survey was conducted among experienced upper extremity surgeons in Switzerland to identify factors influencing the difficulty of reducing and retaining proximal humerus fractures. Inclusion criteria for surgeon grading comprised experience with more than 100 surgically treated proximal humerus fractures. Eligible surgeons rated the relevance of factors influencing reduction difficulty using a five-point Likert scale. A rank-based weighting system derived from expert ratings was applied, assigning two points to the three highest-ranked factors. The resulting DRoPH-Score ranged from 0 to 10, with higher values indicating greater sur […]

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Interface stability of intramedullary nail locking configurations under combined axial and torsional loading.

Marianne H, Martin W, Claus G, Anja B, Peter A

The biomechanical performance of intramedullary nails in metaphyseal regions strongly depends on distal locking configuration. This study aimed to evaluate screw-bone interface stability under combined axial and torsional cyclic loading for different distal locking configurations of intramedullary nails, focusing on the effects of screw number, spatial distribution, interscrew distance, and multiplanar versus coplanar fixation using a standardized synthetic bone model. Six distal locking configurations (n = 5 per group) were tested under stepwise cyclic axial compression, superimposed with a constant torsional moment (± 2 Nm, 2 Hz). Primary endpoints were cycles to failure, cycles to 1 mm axial displacement, and cycles to 0.5° rotation. All constructs failed by progressive screw cut-through within the synthetic bone. The three-screw multiplanar configuration exhibited the highest endurance (≈ 36,000 cycles to failure) and lowest micromotion (≈ 34,000 cycles to 1 mm displacement), significantly outperforming the two-screw configurations overall (p = 0.002). Increasing interscrew distance from 10 mm to 34 mm improved migration resistance by approximately 25%, whereas the short 10 mm spacing led to premature failure. Blocked and free-hand configurations showed comparable behavior to the standard setup, indicating no added benefit from toggle elimination alone. Locking configuratio […]

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Three-dimensional finite element analysis for internal fixation of mandibular condylar fractures: a scoping review.

Kakei Y, Sader R, Ebener P, Tanaka A, Akashi M, Kuroda R, Thoenissen P

PURPOSE: Three-dimensional finite element analysis (3D-FEA) has emerged as an essential computational tool for evaluating the biomechanical behaviors of internal fixation systems for mandibular condylar fractures. This scoping review maps and synthesizes the existing literature on 3D-FEA studies investigating the internal fixation of mandibular condylar fractures and explores, in a descriptive manner, differences in methodological focus between European and non-European research centers.

METHODS: PubMed and Cochrane Library databases were systematically searched following PRISMA-ScR guidelines. Original English-language research articles using 3D-FEA to evaluate internal fixation of mandibular condylar fractures were included. Data were extracted using a standardized charting form.

RESULTS: Thirty-six studies met the inclusion criteria. European studies (n = 12) predominantly focused on development of novel plate designs and mechanical testing protocols, whereas Asian and other nationality studies (n = 24) emphasized material comparisons between titanium and bioabsorbable systems, patient-specific modeling, and loading condition analysis. Consensus findings include: (1) double-plate fixation provides superior stability; (2) among 3D plate designs, trapezoid plates demonstrate the best performance for subcondylar and condylar base fractures, whereas alpha and lambda plates may […]

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Initial surgical management of injuries to the upper extremities for patients with multiple and/or severe injuries - a systematic review and clinical practice guideline update.

Berk T, Könsgen N, Lechler P, Imach S, Schär M, Bieler D, Horst K, Hildebrand F et al.

PURPOSE: Our objective was to update the evidence-based and consensus-based recommendations for the initial surgical management of upper extremity injuries in patients with suspected multiple and/or severe injuries based on current evidence. This guideline topic is part of the 2025 update of the German Guideline on the Treatment of Patients with Multiple and/or Severe Injuries.

METHODS: MEDLINE and Embase were systematically searched to September 2024. Further literature reports were obtained from clinical experts. Randomised controlled trials (RCTs) or observational studies reporting risk-adjusted outcomes were included if they compared early versus delayed surgical treatment for fractures, vascular injuries, or nerve injuries affecting the upper extremities in patients with multiple and/or severe injuries. Studies comparing limb salvage versus amputation or comparing amputation criteria for the upper extremities were also included. We considered patient-relevant outcomes such as mortality and limb salvage, as well as the sensitivity and specificity of scores for predicting upper extremity amputation. Risk of bias was assessed at the outcome level using ROBINS-I for observational studies and AMSTAR-2 for systematic reviews. We used available meta-analyses if possible; alternatively, we synthesised the evidence narratively. We used GRADE to rate the certainty of evidence. Expe […]

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Transosseous sutures versus vertical cerclage wire for distal pole patellar fractures: a retrospective comparative cohort study.

Laso JI, Muñoz JT, Bianchi S, Martínez D, Rojas T, Franulic N, Olivieri R

PURPOSE: To compare transosseous sutures (TOS) and vertical wire cerclage (VCW) for treating distal pole patellar fractures, evaluating clinical outcomes, patient-reported outcome measures (PROMs), radiographic outcomes, and complications.

METHODS: We conducted a retrospective comparative cohort study of patients surgically treated for patellar distal pole fractures (AO/OTA 34A1b) with either TOS or VCW between 2015 and 2023. Patient characteristics, surgical technique, postoperative data, patellar height, and complications were recorded. Functional outcomes were evaluated at a minimum of 12 months postoperatively using PROMs (Kujala, Lysholm, and Bostman scores). The primary outcome of this study was functional recovery, assessed via the Bostman score at 12 months.

RESULTS: A total of 46 patellar distal pole fractures were included (30 TOS and 16 VCW). Baseline characteristics showed no statistically significant differences between groups. No significant differences were found regarding clinical or functional outcomes. Regarding radiographic outcomes, patellar height remained stable; however, both groups showed significant patellar length variation, with elongation being significantly more pronounced in the TOS group (mean 3.77 mm) compared to the VCW group (mean 0.44 mm; p = 0.001). While overall complication rates showed no statistically significant difference, all five ca […]

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Re■fracture risk following implant removal in consolidated hip fractures: a multicenter retrospective cohort study of 575 patients.

Mohammed J, Mili-Schmidt V, Wadsten M, Juto H, Wolf O, Crnalic S, Sköldenberg O, Mukka S et al.

PURPOSE: The risk of re-fracture after implant removal in healed hip fractures remains a clinical concern. This study aimed to determine the incidence of re-fracture following implant removal after osteosynthesis of a hip fracture.

METHODS: We conducted a retrospective multicenter cohort study including patients aged ≥ 50 years who underwent implant removal between 2003 and 2023 after radiologically confirmed consolidation of a hip fracture. Patients were identified using procedural codes. Baseline variables included age, sex, ASA classification, fracture type (femoral neck or trochanteric), and implant type (pins/screws, sliding hip device [SHD], or short/long cephalomedullary nail [CMN]). Patients were followed from implant removal until re-fracture, conversion to arthroplasty, death, or end of follow-up, with a minimum follow-up of 1 year. Cox proportional hazards regression was used to assess associations between implant type and re-fracture risk, adjusting for age and sex. Because the proportional hazards assumption was violated, a time-stratified Cox regression and restricted mean survival time analyses were applied.

RESULTS: A total of 575 patients (median age 73 years, IQR 65-81) were included, with a median follow-up of 53 months (IQR 18-100). Lateral hip pain was the most common indication for implant removal (72.5%). The overall re-fracture incidence was 10.4%. Ris […]

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Physical functionality two years after major trauma: predictors from a single center cohort study.

Fetz K, Koetsenruijter J, Rodrigues Recchia D, Rutetzki J, Lefering R, Kaske S

PURPOSE: Traumatic injuries are a major global public health concern, often leading to impaired physical function and disability. Long-term physical function impairments can greatly affect a patient's health related quality of life, influencing their emotional state and social interactions. This study examines patient-reported physical functionality before and after major trauma, identifying key predictors of poor functional recovery.

METHODS: This retrospective single-center cohort study investigates physical functionality 23 months after a traumatic injury using standardized questionnaires combined with clinical data. Inclusion criteria were Injury Severity Score (ISS) ≥ 9, requiring ICU treatment, and age ≥ 18 resulting in an eligible sample of 515 patients. Health outcomes were assessed using the Trauma Outcome Profile (TOP), the Abbreviated Injury Scale (AIS), and the ISS.

RESULTS: Physical functionality significantly decreased from 94.8 (SD 11.5) to 72.9 (SD 26.7) 23 months post-trauma with almost half of participants still scoring below the critical cutoff of 80 points. The strongest predictor for functional physical impairment was pre-trauma physical functionality (OR 8.21, $p < 0.001$). Other determinants included age (age 50-59 versus 18-29 OR 2.75, $p = 0.001$), as well as an impacted lower extremity including pelvic (OR 1.60, $p = 0.018$).

CONCLUSION: The most importan […]

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Association between fresh frozen plasma transfusion and mortality stratified by Glasgow Coma Scale scores in isolated traumatic brain injury: a nationwide cohort study in Japan.

Nagamura T, Aoki M, Yamada K, Terayama T, Seno S, Tomura S, Kiyozumi T

PURPOSE: Fresh frozen plasma (FFP) is widely used to manage trauma-induced coagulopathy. However, its effectiveness in isolated traumatic brain injury (TBI) management remains controversial, particularly when stratified by severity based on the Glasgow Coma Scale (GCS) scores. This study aimed to examine the association between FFP administration within 24 h and mortality in patients with isolated TBI, overall and stratified by GCS on arrival.

METHODS: We conducted a multicenter retrospective cohort study using the Japan Trauma Data Bank (2019-2023). Adults with isolated TBI (intracranial AIS ≥ 3 and extracranial AIS < 3) were included. Patients were classified into FFP and non-FFP groups based on transfusion within 24 h of admission. Subgroup analyses were performed by TBI severity (severe: GCS ≤ 8; mild-to-moderate: GCS > 8), craniotomy status, and FFP dose. The primary outcome was 28-day mortality. Propensity score matching was applied to adjust for confounding.

RESULTS: Among 12,480 patients (median age 72 years [IQR: 55, 82]; 66.6% male), 513 received FFP. After matching, FFP was not significantly associated with 28-day mortality (OR 0.80; 95% CI 0.59-1.08). In subgroup analyses, FFP was associated with higher mortality in mild-to-moderate TBI (OR 2.38; 95% CI 1.07-5.31) and lower mortality in severe TBI (OR 0.61; 95% CI 0.42-0.88). No significant differences were observ […]

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Percutaneous fixation of posterior pelvic ring lesions in the elderly: current evidence and recent advances.

Liodakis E, Schreiber S, Giannoudis VP, Fritz T, Giannoudis PV

Transsacral percutaneous fixation is increasingly seen as a safe, effective minimally invasive treatment for geriatric posterior ring fractures The use of Navigation appears a safe and efficient method for percutaneous screw placement Multiple new implants have been described for management of these injuries. Level I evidence is still lacking for treatment options in the management of posterior pelvic ring lesions with percutaneous fixation

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Infrared thermography and dual-layer spectral CT for the detection of femoral vascular compromise: an experimental large animal study.

Hübner CT, Ricklin J, Müller AL, Klingebiel FK, Stoeck CT, Weisskopf M, Carneiro Gomes A, Cinelli P et al.

BACKGROUND: Rapid identification of vascular compromise remains a clinical challenge in the acute trauma setting. Conventional diagnostic tools such as CT angiography (CTA) and Doppler-ultrasound might provide decent anatomical and functional assessment. However, they are limited by ionizing radiation, contrast exposure or time requirements. Although not validated, infrared thermography (IRT) might offer a non-invasive alternative capable of detecting surface temperature asymmetries that may reflect underlying perfusion deficits. This study aimed to evaluate the feasibility of thermographic imaging in detecting vascular lesions.

METHODS: As part of a porcine polytrauma model ($n = 28$), vascular lesion of the right femoral artery was defined as a ≥ 50% diameter reduction on CTA. Temperature differences between both extremities were measured using infrared thermography. SDCT based iodine concentrations distal to the stenosis were quantified to assess perfusion changes. Correlation and regression analyses were performed between stenosis degree, temperature difference, and tissue iodine uptake.

RESULTS: Six animals fulfilled the criteria for relevant vascular compromise. This resulted in a mean vessel diameter reduction behind the lesion of 68% ±18%. The vascular lesion group demonstrated a mean inter-limb temperature difference of 6.93 °C ± 1.78 °C compared to 0.71 °C ± 1.27 °C i […]

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Fixation strategies for periprosthetic femur fractures around knee replacements: a comprehensive single-center analysis of 102 consecutive cases.

Nomdedéu J, González-Morgado D, Minguell-Monyart J, Joshi-Jubert N, Teixidor-Serra J, Tomàs-Hernández J, Andrés-Peiró JV

PURPOSE: To evaluate 1 year mortality, complication rates, and functional outcomes after internal fixation of femoral periprosthetic knee fractures following total knee arthroplasty, and to identify predictors of adverse outcomes and increased healthcare resource utilization.

METHODS: This retrospective cohort study included 102 consecutive patients with femoral periprosthetic knee fractures classified as UCPF VB3 and VC3 treated with internal fixation at a single tertiary center between 2010 and 2023. Interprosthetic fractures and combined fixation techniques were excluded. Primary outcomes were 1 year mortality and postoperative complications. Secondary outcomes included length of stay, discharge destination, and ambulatory status at follow up.

RESULTS: Patients were markedly frail, with a mean age of 82.4 ± 10.8 years and mean Charlson Comorbidity Index of 5.6. One year mortality was 16.7% and was associated with older age, higher ASA class, higher Charlson index, impaired pre fracture ambulation, and fracture related infection. Overall complication rate was 29.4%, including 15.7% infections and 13.7% mechanical failures. Open reduction was more frequent in plate fixation and resulted in longer operative times, without increasing complication rates. Median hospital stay was 18 days, 66.3%

required discharge to nursing facilities, and only 7.8% regained independent ambulati […]

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Epidemiology and management of pediatric femoral fractures at a level I trauma center: a 10-year retrospective monocentric study.

Dietz SO, Traub F, Schmidt E, Schippers P, Wegner E, Schierjott L, Gercek E, Nowak TE

PURPOSE: Pediatric femoral fractures are relatively uncommon but represent severe injuries frequently requiring inpatient trauma care. Epidemiological characteristics, fracture patterns, and treatment strategies vary between institutions and regions. This study aimed to analyze the epidemiology, fracture distribution, treatment modalities, complications, and fracture-predisposing comorbidities of pediatric femoral fractures treated at a German level I pediatric trauma center over a 10-year period.

METHODS: All patients aged 0-14 years treated for femoral fractures between January 2012 and December 2021 were retrospectively reviewed. Fractures were categorized using the AO Pediatric Comprehensive Classification of Long-Bone Fractures (PCCF). Demographic variables, fracture characteristics, management approaches, time to consolidation, complications, and relevant comorbidities predisposing to fracture were analysed descriptively in accordance with STROBE recommendations.

RESULTS: A total of 183 patients met the inclusion criteria (67.2% male; mean age 5.2 years). Children aged 2-5 years constituted the largest group (42.1%). Femoral shaft fractures were the most frequent femoral fracture type (70%). Surgical management, most frequently elastic stable intramedullary nailing (ESIN), was increasingly utilized with advancing age. The overall complication rate was 14.2%, with the ma […]

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Blood-based biomarkers GFAP/UCH-L1 for the diagnosis of mild traumatic brain injury (mTBI): a single-center implementation experience.

Maegele M, Breidenbach L, Kaufmann J, Keles Y, Uhl E, Schroeer P

INTRODUCTION: Mild traumatic brain injuries (mTBI) affect millions of people worldwide every year as one of the most common clinical presentations in the emergency department. Diagnosis is mainly based on clinical criteria and computed tomography scans. The use of computed tomography causes high costs, long waiting times in daily clinical practice and radiation exposure. GFAP (glial fibrillary acidic protein) and UCH-L1 (ubiquitin carboxyl-terminal hydrolase-L1) turned out to be potential biomarkers for the diagnosis of mTBI. This study retrospectively evaluates the possible use of these biomarkers combined as negative predictors for excluding brain injuries in patients with suspected mTBI in the emergency department.

METHODS: Adult patients (n = 320) registered in the emergency department at a level 1 trauma emergency center in Germany (Cologne Merheim Medical Center/CMMC) between 11/2023 and 04/2024, with suspected mTBI, Glasgow Coma Scale (GCS) score 13-15 and within 12 h after trauma were considered. All evaluable patients underwent cranial CT (cCT) scans and blood tests for GFAP and UCH-L1 serum concentrations.

RESULTS: Biomarkers GFAP and UCH-L1 were tested positive in 261 patients (82%) while CT detected intracranial injuries in only 29 patients (9%). Biomarkers combined had a sensitivity of 97% and a negative predictive value (NPV) of 98% in mTBI diagnosis with a nega […]

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Resuscitative endovascular occlusion of the aorta restores cerebral metabolic markers of ischaemia induced by haemorrhagic shock.

Bader SE, Magnuson A, Brorsson C, Wallin G, Löfgren N, Löfgren F, Blind PJ, Öman M et al.

BACKGROUND: Resuscitative endovascular balloon occlusion of the aorta (REBOA) is used as an adjunct in haemorrhagic shock (HS) to restore proximal perfusion. Its cerebral metabolic effects during haemorrhagic shock, particularly in the presence of elevated intracranial pressure (ICP), remain incompletely characterised.

METHODS: In a controlled porcine model, HS was induced by controlled bleeding to a mean arterial pressure of approximately 40 mmHg. Animals were allocated to a normal ICP group or an experimentally elevated ICP group. Total REBOA (zone 1) was applied for 90 min. Cerebral metabolism was assessed using intracerebral microdialysis with measurements of lactate, pyruvate, and lactate-pyruvate ratio (LPR). Cerebral haemodynamics and ICP were continuously monitored.

RESULTS: HS was associated with a statistically significant increase in LPR in both groups, indicating cerebral metabolic disturbance. Following aortic occlusion, LPR gradually decreased toward baseline in both groups. Animals with elevated ICP demonstrated a transient delay in metabolic normalisation during the early post-occlusion phase. Statistically significant differences between groups were limited to the first 10 min following occlusion. Overall metabolic trajectories were similar thereafter.

CONCLUSIONS: Total REBOA restored cerebral metabolic markers of ischaemia during haemorrhagic shock, as ref […]

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Delayed single-stage surgical management using a peroneus brevis muscle flap for fracture-related infection after lateral malleolus fixation: a retrospective case series.

Dyreborg K, Jørgensen AI, Peterlin AAN, Gottlieb H

BACKGROUND: Fracture-related infection (FRI) following lateral malleolus fracture fixation complicated by soft tissue breakdown represents a challenging clinical problem with no established standard of care. This study evaluates a treatment strategy consisting of prolonged antibiotic suppression until fracture union, followed by delayed single-stage surgical management.

METHODS: This retrospective consecutive case series included 11 patients treated between 2020 and 2023 at a multidisciplinary orthopaedic infection unit. All patients were managed according to a standardized protocol involving oral antibiotic suppression until radiologically confirmed fracture consolidation. This was followed by a single-stage surgical procedure including implant removal, radical debridement, deep tissue sampling, local antibiotic therapy, and soft tissue reconstruction using a distally based peroneus brevis muscle flap with split-thickness skin grafting.

RESULTS: Eleven consecutive patients (median age 58 years) were treated according to the protocol. Six patients underwent antibiotic suppression for a median duration of 147 days prior to surgery. Among patients available for outcome assessment, infection control was achieved in 10 of 10 surviving patients. One patient died of unrelated causes within 14 days postoperatively and was not classified as treatment failure. Soft tissue complication […]

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Mixed reality improves agreement on surgical approach selection and patient positioning in tibial plateau fracture planning compared to CT, 3DCT and 3D printing.

Dust T, Henneberg JE, Hartel M, Reiter A, Kylies J, Korthaus A, Streckenbach A, Keller J et al.

PURPOSE: Tibial plateau fractures require precise preoperative planning, particularly regarding treatment concept, patient positioning, and surgical approach. Mixed reality (MR) may enhance three-dimensional understanding of fracture morphology beyond conventional imaging and physical 3D models. This study investigated whether MR improves agreement on key preoperative planning decisions compared to computed tomography (CT), 3D computed tomography (3DCT), and 3D-printed fracture models.

METHODS: Twelve orthopaedic trauma surgeons (6 junior, 6 senior) evaluated 22 surgically treated tibial plateau fractures (AO/OTA type B or C). Each case was assessed in four steps using (1) CT, (2) 3DCT reconstructions, (3) physical 3D-printed models, and (4) MR visualization using Microsoft Hololens 2. Raters selected (i) treatment concept, (ii) patient positioning, and (iii) surgical approach. Interobserver agreement was analysed using Fleiss' kappa (κ) and percentage match (PM).

RESULTS: Mixed Reality (MR) yielded the highest interobserver agreement for both surgical approach (PM 32%, $\kappa = 0.30$) and patient positioning (PM 57%, $\kappa = 0.35$), particularly among junior surgeons (approach PM 39%, $\kappa = 0.28$; positioning PM 57%, $\kappa = 0.39$). Compared to CT, 3DCT, and 3D printing, MR demonstrated consistent improvements, while agreement on treatment concepts remained high across all modalities (PM > 82% […]

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Timing of fracture fixation for femur and pelvis fractures in patients with severe traumatic brain injury - an analysis of the TraumaRegister DGU®.

Bayer TP, Teuben MPJ, Halvachizadeh S, Shehu A, Lefering R, Pfeifer R, Pape HC, Jensen KO

PURPOSE: Fracture fixation timing and strategy in polytrauma patients with traumatic brain injury (TBI) remain controversial. This study investigates treatment patterns and outcomes for femoral and/or pelvic fractures stratified by TBI severity.

METHODS: Patients in the TraumaRegister DGU® (2016-2022) with pelvic and/or femoral fractures (AIS ≥3) and TBI (head AIS ≥3) were included. Strategies were non-operative management (NOM), early total care (ETC), and damage-control orthopedics (DCO). Outcomes included treatment allocation, fixation timing, and in-hospital mortality.

RESULTS: 985 patients were included (mean age 52.5, SD 26.3 years; ISS 27.8, SD 8.1). Allocation was NOM in 320 (32.5%), ETC in 336 (34.1%), and DCO in 329 (33.4%) patients. Head AIS was 3 in 48.5%, 4 in 31.1%, and 5 in 20.3%. NOM patients were older, had the highest ISS and estimated mortality, and showed the largest proportion of critical TBI (AIS 5: NOM 30.9%, ETC 14.3%, DCO 16.1%). Femoral ETC was mainly performed within the first day (median 0, IQR 0-1 days), whereas pelvic ETC was delayed with increasing TBI severity (median 3, IQR 0-5 days for head AIS 3; 5, IQR 0-7 days for AIS 4). Observed mortality was 37.2% after NOM, 9.2% after ETC, and 10.3% after DCO.

CONCLUSION: ETC in patients with moderate TBI (AIS 3) was associated with reduced observed mortality relative to NOM and matching DCO. Increasi […]

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From single to dual platelet inhibition FIBTEM assay: fibrinogen replacement thresholds for critical bleeding after trauma.

Latona A, Winearls J, Ho A, Mitra B

PURPOSE: The FIBTEM assay in Rotational Thromboelastometry (ROTEM®) measures fibrin-based clot formation. The reagent transitioned from single to dual-platelet inhibition with addition of tirofiban to cytochalasin D, offering more complete platelet suppression. The aim of this study was to assess the impact of this change on transfusion thresholds in trauma.

METHODS: In this retrospective cohort study, we included adult patients with major trauma across Queensland who presented between 2022 and 2025 and had results available for paired ROTEM and conventional coagulation tests on arrival. Patients were sub-grouped into single and dual-platelet inhibition assay groups. Correlations between FIBTEM-A5 and Clauss fibrinogen were analysed, comparing correlation strength and diagnostic accuracy for fibrinogen levels of < 2.0 g/L.

RESULTS: The relationship between FIBTEM-A5 and Clauss fibrinogen was similar using single platelet inhibition assays (R^2 0.61;95%CI: 0.51-0.70) and dual-platelet inhibition assays (R^2 0.70;95%CI: 0.62-0.77; p = 0.94). Predicted FIBTEM-A5 amplitudes at 2 g/L fibrinogen were 7.83 mm (95%CI: 7.42-8.23) and 7.54 mm (95%CI: 7.22-7.87). Diagnostic performance for detecting fibrinogen < 2.0 g/L was comparable between assays, for both FIBTEM-A5 ≤ 10 mm and FIBTEM-A5 ≤ 8 mm (all p > 0.05). For the dual-platelet inhibition assay, sensitivity and specificity were 97. […]

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Pattern of blast injuries. A systematic review: Part 2 - Landmines, unexploded ordnance and terrorism.

Neubert A, Gaeth C, Seidelmann M, Bieler D

PURPOSE: This systematic review analyzes the patterns and distribution of blast injuries from landmine explosions, UXOs and terror attacks including suicide bombing, in civilians and military personnel.

METHODS: A search was conducted in two databases on 26. March 2025 for observational studies reporting blast injuries in civilian and military populations of all ages. Studies were clustered by explosion typ. A narrative data synthesis was performed. The JBI tool for case series was used.

PROSPERO: CRD42023391736.

RESULTS: 126 studies were included in the overall systematic review. 45 studies on terror & suicide bombings and 12 studies on landmines & UXOs are analyzed in the present paper. Studies on landmines from 1980 to 2006 included 13,270 individuals (age: 24.0 years (Range 8.2 years - 32.2 years); mainly male (88.9%)) in post-conflict low- and middle-income countries. Lower extremity injuries including amputations were most frequent, followed by eye injuries. Studies on terrorist attacks with a total of 12,518 mainly civilian victims (average age: 33,1 years; 61.6% males) reported on incidents in post-conflict regions and high-income cities mainly from the early 2000s. The injury pattern was multidimensional, with extremity injuries most frequent (28%-30%), including fractures, lacerations, and traumatic amputations (18%-20%).

CONCLUSION: This Systematic Review Part 2 […]

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Is nail plate fixation the way forward for distal femur fractures with long segmental metaphyseal comminution? - A study of 56 patients.

Mahesan H, Sundaram VP, Devendra A, Ramesh P, Dheenadhayalan J, Rajasekaran S

BACKGROUND: Mechanical instability due to long metaphyseal comminution and medial bone gap after lateral locking plate results in varus collapse and implant failure. We believe that fixation in the form of Nail and plate combination provides better stability and prevents complications.

MATERIALS: We performed a retrospective analysis of 56 distal femur fractures with long metaphyseal comminution treated in our institution with nail and plate construct from 2017 to 2023. The mean age was 43.1 years and the mean follow up was 24 Months (4-36 months). Of these 56 patients, 16 were closed injuries underwent primary fixation, and 40 were open injuries requiring staged reconstruction. At the final follow up, patients were assessed for shortening, knee range of movements, Sander et al. functional score, radiological union and loss in alignment.

RESULTS: The mean length of metaphyseal comminution was 10.15 cm(5-18 cm). Primary Iliac crest bone grafting was done in 25 patients. Of the 56 patients, 51 achieved primary union and the mean time to union was 6 months. Of the remaining, 3 patients united after secondary bone grafting and 2 patients had nonunion. The mean loss in aLDFA at the final follow up was 1.75° in 7 patients. Shortening of the limb was noted in 16 patients, and 4 patients had significant shortening of 2 cm. Three patients developed deep SSI, of which one patient requi […]

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Impact of modified ERAS protocol implementation on length of stay and complications in elderly hip fracture patients: a before-and-after study.

Liu Y, Ou G, Xie M, Wu F, Wang X, Yang B

AIM: Hip fractures in elderly patients pose significant healthcare challenges with high complication rates and prolonged hospital stays. Enhanced Recovery After Surgery (ERAS) protocols have shown promise in various surgical specialties, but their application in elderly hip fracture patients, particularly in Traditional Chinese Medicine (TCM) hospital settings, remains limited.

METHODS: This retrospective before-and-after study was conducted at a TCM hospital from July 2022 to July 2025. A total of 300 elderly hip fracture patients (≥ 65 years) were included: 146 in the control group (July 2022 - October 2023) receiving conventional care and 154 in the ERAS group (February 2024 - July 2025) receiving modified ERAS protocol incorporating TCM elements. Primary outcomes were length of hospital stay and perioperative complications within 30 days. Secondary outcomes included functional recovery [Harris Hip Score (HHS), Modified Barthel Index (MBI)], time to first mobilization, pain scores, and healthcare costs.

RESULTS: The ERAS group demonstrated significantly shorter length of stay compared with controls (12.38 ± 3.75 vs. 16.24 ± 5.10 days; mean difference 3.86 days, 95% CI: 2.84-4.88; $P < 0.001$). Total perioperative complications were reduced (7.79% vs. 15.07%; risk difference: -7.28%, 95% CI: -14.5% to - 0.1%; $P = 0.047$). The ERAS group showed superior functional outcomes with […]

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Prehospital endotracheal intubation and 30-day survival in severe trauma: a matched cohort study from Navarra, Spain.

Zulet Murillo D, Ferraz Torres M, Fortún Moral M, Belzunegui Otano T, Tamayo Rodríguez I

PURPOSE: Prehospital endotracheal intubation (ETI) is frequently used in major trauma, but its association with 30-day survival remains debated. We evaluated the association between prehospital ETI and 30-day survival and assessed whether sex modifies the age-related increase in mortality risk.

METHODS: We conducted a retrospective cohort study using the Navarra Major Trauma Registry (2010-2019). Patients aged ≥15 years with New Injury Severity Score (NISS) ≥15 were included; patients who died before hospital admission or had incomplete records were excluded. To mitigate confounding by indication, intubated and non-intubated patients were matched 1:1 using prehospital Glasgow Coma Scale (GCS), respiratory distress, and prehospital cardiac arrest as matching variables. Because the proportional hazards assumption for Cox regression was not met, parametric Weibull survival models were fitted, including a sex-by-age interaction term.

RESULTS: Overall, 1,909 patients were included; 212 (11.1%) underwent prehospital ETI. Matching yielded 190 pairs (n=380). In the matched cohort, prehospital ETI was not independently associated with 30-day survival (adjusted HR 0.91, 95% CI 0.65-1.27). Mortality risk was independently associated with lower GCS (per 1-point increase: HR 0.86, 95% CI 0.82-0.91) and respiratory distress (HR 7.63, 95% CI 4.34-13.4). A significant sex-by-age interaction […]

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Mechanical and manual cardiopulmonary resuscitation in traumatic cardiac arrest: a retrospective observational study.

El-Menyar A, Ahmed K, Howland IR, Mollazehi M, Mekkodathil A, Rizoli S, Cameron P, de Oliveira Manoel AL et al.

BACKGROUND: Prehospital Traumatic cardiac arrest has a high fatality rate despite advances in trauma systems and resuscitation strategies. Mechanical chest compression devices have been increasingly adopted to optimize cardiopulmonary resuscitation (CPR) during Emergency Medical Services (EMS) transportation. We aimed to evaluate the impact of CPR (mechanical during transportation versus manual CPR only at the scene) on survival among trauma patients.

METHODS: A retrospective analysis was conducted for patients who received CPR at the scene and during transportation to the hospital between 2016 and 2024.

RESULTS: A total of 610 patients sustaining blunt traumatic cardiac arrest were included. The mean age was 34 ± 12.6 years; 94.6% were male, and 5.4% were female patients. Two hundred twenty (36.1%) received mechanical CPR during transportation to the hospital, and 390 (63.9%) received manual CPR only at the scene. Compared with the manual CPR group, the mechanical CPR group had higher rates of primary chest injury (30.5% vs. 20.5%; p = 0.006), bystander CPR (15.9% vs. 6.7%; p = 0.001), adrenaline administrations (95% vs. 26.4%; p = 0.001), and initial non-shockable rhythm (80% vs. 27.2%; p = 0.001). The two groups were comparable in median Injury Severity Score (ISS), head Abbreviated Injury Score (AIS), and total transport time. Mechanical CPR was associated with a markedly […]

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Cervical spine clearance in pediatric trauma patients - Consensus algorithm of the Pediatric Spinal Trauma Group of the Spine Section of the German Society for Orthopedic and Trauma Surgery (DGOU).

Bolte J, Rüther H, Brecht P, Wiersbicki DW, Youssef Y, Jarvers JS, Disch AC

PURPOSE: Although cervical spine injuries in pediatric patients are rare, clinical and radiological examinations are clinical routine. Currently, no standardized decision algorithm does exist for detection of these injuries in the European region. The proposed algorithm is aiming to safely identify significant injuries through consistent diagnostics while minimizing immobilization and radiation exposure at the same time.

METHODS: The Pediatric Spinal Trauma Group of the Spine Section of the German Society for Orthopedic and Trauma Surgery (DGOU) developed an algorithm based on own scientific investigations and considering primary and secondary literature. The algorithm was developed through a formal consensus process and thereby aligns with international recommendations.

RESULTS: The algorithm uses the initial pediatric Glasgow Coma Scale (pGCS) to group patients to three treatment pathways. Resulting recommendations for imaging, clinical re-assessment, and further management include the child's age, trauma mechanism, and clinical evaluations.

CONCLUSION: Despite their rarity, potential cervical spine injury in children requires accurate diagnosis. Clinical examinations can be challenging, especially in children with compromised consciousness. Increased radiation exposure from X-rays and computer tomography (CT) scans must be considered, particularly as magnetic resonance im […]

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Development of a low-cost external fixation system: an off-label solution for resource-limited settings.

Seiser M, Deininger C, Fierlbeck J, Hollensteiner M, Coles P, Reuter C, Tempfer H, Traweger A et al.

PURPOSE: External fixators are essential for the surgical management of long-bone fractures, particularly in resource-limited settings where access to commercial systems is often restricted by cost and logistics. This study aimed to develop a low-cost external fixation concept using widely available construction materials and evaluate its biomechanical performance under axial loading.

METHODS: An external fixator composed of M10 threaded rods, washers, and nuts was developed. Five configurations were tested: SP (Steinmann pins), KT (threaded K-wires), KS (smooth K-wires), and BKS (bilaterally applied, smooth K-wires). A commercial external fixator served as control (CG). Juvenile bovine ulnae with a standardized fracture gap were subjected to cyclic axial loading. Construct stiffness, elastic deformation, and plastic deformation were assessed.

RESULTS: CG showed the highest stiffness (107.80 ± 14.12 N/mm), followed by SP (87.73 ± 17.92 N/mm), BKS (83.81 ± 10.01 N/mm), KS (49.84 ± 3.80 N/mm), and KT (46.66 ± 2.39 N/mm). Post-hoc analyses showed no significant differences between CG vs. SP and CG vs. BKS, whereas KT and KS were significantly less stiff. Elastic deformation followed a similar pattern. Plastic deformation was greatest in CG, while SP and BKS showed significantly lower values.

CONCLUSIONS: The threaded-rod fixator provides relevant mechanical stability at extreme […]

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Supplementary fixation in multifragmentary midshaft clavicle fractures: do interfragmentary screws or suture cerclage improve outcomes compared with plate-only fixation?

Barthel C, Hinkelmann MA, Schibler D, Frei F, Pape HC, Allemann F

BACKGROUND: Clavicle fractures are common injuries. As displaced or multifragmentary midshaft patterns carry a substantial risk of malunion and nonunion with nonoperative treatment, they have contributed to the increasing use of surgical fixation. Modern operative techniques include plate fixation, intramedullary devices, and hybrid strategies to address comminution and improve fragment stability. Supplementary interfragmentary screws and cerclage constructs have been proposed to enhance fragment control and optimize load transfer in multifragmentary fracture patterns. This study aimed to compare three fixation strategies for operatively treated midshaft clavicle fractures managed with plate osteosynthesis: Group IS (interfragmentary screw), Group CA (cerclage augmentation), and Group POF (plate-only fixation), with respect to operative time, complication rates, and potential biological and mechanical differences.

METHODS: This retrospective study included patients aged ≥ 16 years who underwent plate fixation for multifragmentary clavicle fractures between 2016 and 2024. Patients were categorized into three groups: Group IS (supplementary interfragmentary screw), Group CA (suture cerclage augmentation), and Group POF (plate-only fixation). Demographic, surgical, and radiographic data were collected. Callus formation was assessed by three independent observers. Statistical anal […]

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Cerebral glycerol during haemorrhagic shock in normal and raised intracranial pressure resuscitated with total REBOA: an experimental porcine study.

Bader S, Magnuson A, Brorsson C, Wallin G, Löfgren N, Löfgren F, Blind PJ, Öman M et al.

BACKGROUND: Haemorrhagic shock (HS) is frequently associated with secondary cerebral injury due to compromised cerebral perfusion. Resuscitative endovascular balloon occlusion of the aorta (REBOA) is an effective haemorrhage control strategy; however, concerns persist regarding its cerebral effects, particularly in the presence of elevated intracranial pressure (ICP). Cerebral microdialysis (CMD) derived glycerol (CGly) is a recognised marker of cellular membrane stress and injury.

OBJECTIVE: To investigate CGly dynamics during prolonged total REBOA (tREBOA) in HS and to compare cerebral cellular responses between animals with normal and elevated ICP.

METHODS: In this experimental porcine study, eighteen pigs were subjected to controlled HS and resuscitated with tREBOA for 90 min. Animals were allocated to either a normal ICP group (NICPG, n = 9) or an elevated ICP group (EICPG, n = 9). Continuous monitoring of proximal arterial pressure, ICP, and cerebral perfusion pressure was performed. CGly concentrations were measured using CMD throughout the experimental protocol.

RESULTS: tREBOA effectively restored proximal arterial pressure and cerebral perfusion pressure in both groups. CGly concentrations remained relatively stable during the haemorrhage phase and increased progressively during prolonged aortic occlusion. Importantly, CGly responses did not differ significantly be […]

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Transport and destination hospital for patients with suspected multiple and/or severe injuries - a systematic review and clinical practice guideline update.

Pavlu F, Könsgen N, Bieler D, Goossen K, Gliwitzky B, Häske D, Faul P, Wölfl C et al.

PURPOSE: Our aim was to update the evidence-based and consensus-based recommendations on criteria for transport and destination hospital for patients with suspected multiple and/or severe injuries based on available evidence. This guideline topic is part of the 2025 update of the German Guideline on the Treatment of Patients with Multiple and/or Severe Injuries.

METHODS: MEDLINE and Embase were systematically searched to January/February 2024. Further literature reports were obtained from clinical experts. Randomised controlled trials (RCTs) or observational studies reporting risk-adjusted outcomes were included if they compared different transport strategies or destination hospitals for patients with (suspected) multiple and/or severe injuries. We considered patient-relevant outcomes such as mortality and neurological outcomes. Risk of bias was assessed at outcome level using ROBINS-I for observational studies and RoB 2 for RCTs. We conducted meta-analyses if possible; alternatively, we synthesised the evidence narratively. We used GRADE to rate the certainty of evidence. Expert consensus was used to develop recommendations and determine their strength.

RESULTS: A total of 127 studies were identified. They addressed helicopter vs. ground emergency medical services (EMS) transport, ground EMS with vs. without a physician, on-scene time, and handover on arrival at the hospital […]

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Does the addition of peripheral nerve blocks improve analgesia during distal radius fracture reduction? A prospective, randomized, multicenter clinical trial.

Guerrero-Serrano JA, Santamaría-López A, García-Lamas L, Jiménez-Díaz V, González-Fernández Á, Cecilia D

PURPOSE: This study aimed to evaluate whether adding a median nerve and a superficial branch of the radial nerve block to hematoma block improves analgesia during distal radius fracture reduction and to identify factors associated with increased pain perception.

METHODS: A prospective, randomized, controlled study was conducted in two tertiary hospitals, including 180 adult patients with displaced distal radius fractures requiring closed reduction. Patients were allocated either to an isolated hematoma block group or to a combined block group receiving hematoma block with median nerve and superficial branch of the radial nerve blocks. Within the combined group, half of the median nerve blocks were performed under ultrasound guidance and half using anatomical landmarks. The primary endpoint was digital pain during traction and fracture reduction, assessed using a visual analog scale (VAS, 0-10). Secondary outcomes included wrist pain at different procedural stages and the influence of demographic and clinical variables including age, sex, laterality, participating hospital, body mass index (BMI), fracture type (AO/OTA), and baseline psychiatric or chronic analgesic medication use.

RESULTS: Digital pain was significantly lower in combined group compared with isolated hematoma block during traction (4.54 ± 2.53 vs. 7.39 ± 1.76; $p < 0.001$) and reduction (5.36 ± 2.56 vs. 8.33 ± 1. […]

PubMed · DOI

Under pressure - a sensor-based analysis of simulated prehospital pressure dressing application for hemorrhage management.

Brauckmann V, Menzel J, Welke B, Decker S, Macke C

BACKGROUND: The concept of "x-ABCDE", with "X" representing exsanguinating hemorrhage, highlights the priority of immediate bleeding control in trauma care. Despite being a fundamental intervention, pressure dressing techniques lack standardization. Studies show high variability in achieved pressures, often below therapeutic targets or exceeding harmful levels, with no correlation between subjective estimates and objective measurements.

OBJECTIVES: To objectively assess current practices of pressure bandage application and identify influencing factors including material type, technique, and provider experience.

METHODS: This prospective, single-center, simulation-based observational study was conducted from August 2024 to January 2025 at a Level I Trauma Center. Emergency medical providers applied pressure dressings according to personal preference using available materials. Pressure distribution was measured via two calibrated capacitive pressure sensors (61 measurement fields). Analysis included correlation tests, group comparisons, and multiple regression.

RESULTS: A total of 124 emergency medical providers (75% male; 36% paramedics, 44% emergency medical technicians, 13% trainees, 7% emergency physicians) completed pressure dressing applications, with 116 datasets included in final analysis. Mean maximum pressure was 169.35 ± 84.33 mmHg (range 50-412 mmHg) with applicati […]

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Clinical impact of early postoperative routine radiology in isolated pelvic injuries: a retrospective study.

Bolierakis E, Ioannou P, Schubert SL, Groven RVM, Bläsius FM, Bode M, Alabdulrahman H, Hildebrand F et al.

PURPOSE: The clinical value of routine immediate postoperative conventional radiography following surgical fixation of pelvic fractures remains debated. This study assessed the impact of early postoperative X-rays on management decisions and revision surgery in patients with isolated pelvic ring or acetabular fractures.

METHODS: We conducted a retrospective cohort study at a Level 1 trauma center, including adults undergoing surgical treatment for isolated pelvic ring or acetabular fractures between 2020 and 2023. All patients received high-quality intraoperative fluoroscopy and postoperative conventional radiographs. The primary outcome was any change in standard postoperative care based on radiographic findings. The secondary outcome was revision surgery prompted by conventional postoperative imaging.

RESULTS: A total of 216 patients were included (median age 63 years), with 156 pelvic ring injuries and 60 acetabular fractures. Postoperative radiographs were obtained at a median of 3 days post-surgery. Changes in postoperative care based on radiographs occurred in 2 patients (3.3%) within the acetabular fracture group. Revision surgery triggered by conventional imaging was required in 1 patient (0.6%) among pelvic ring injuries. In all cases, additional CT imaging preceded definitive management changes.

CONCLUSION: Routine immediate postoperative conventional radiography a […]

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Older age and post-traumatic organ failure a TraumaRegister DGU® analysis of 34,469 elderly major trauma patients.

Bewersdorf TN, Lefering R, Stein S, Streblow J, Findeisen S, Schmidmaier G, Grossner T

PURPOSE: As the number of elderly major trauma patients (EMTP) increases continuously, this study sought to determine the incidence of organ failure (OF) and multiple organ failure (MOF) and their impact on mortality and recovery status in this cohort.

METHODS: Based on the German TraumaRegister DGU® database this study included EMTP aged between 65 and 99 years with an Injury Severity Score (ISS) of ≥ 16 between 2014 and 2023. OF was defined as Sequential

Organ Failure Assessment (SOFA) score of ≥ 3 and MOF as failure of ≥ 2 organs.

RESULTS: The database revealed 34,469 EMTP, here 37% suffered from OF (16.0%) or MOF (21.0%). Mortality was significantly higher in OF and MOF patients compared to non-OF patients (non-OF: 7.4%; OF: 45.0%; MOF: 59.0%; $p < 0.001$). 80.9% of all deaths were associated with failure of at least one organ and 51.1% of non-survivors suffered from MOF. Highest odds ratios (OR) for mortality were found for liver (OR: 5.13, $p < 0.001$), central nervous system (OR: 3.63, $p < 0.001$) and renal failure (OR: 3.25, $p < 0.001$), followed by cardiovascular (OR: 1.95, $p < 0.001$) and lung failure (OR: 1.16, $p = 0.010$). Haemostatic failure had no impact on mortality.

CONCLUSION: OR and mortality differ significantly between distinct OFs due to varying degrees of success in treatment options and systemic impact of these OFs. Clinicians should be aware that OF and MOF o […]

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Denis pattern-dependent biomechanical behavior of posterior trans-iliac plate fixation compared with bilateral triangular osteosynthesis in unilateral sacral fractures: a finite element study.

Yang J, Böker KO, Li C, Schilling AF, Ritter Z, Acharya M, Lehmann W

OBJECTIVE: Posterior trans-iliac plate osteosynthesis (TPO) is commonly used for the stabilization of unilateral vertically unstable sacral fractures. However, its biomechanical performance across different Denis fracture patterns remains unclear. This study aimed to investigate the fracture pattern-dependent biomechanical behavior of posterior trans-iliac plate osteosynthesis (TPO) in unilateral vertically unstable sacral fractures, using bilateral triangular osteosynthesis (BTO) as a high-stability reference construct.

METHODS: A validated three-dimensional finite element model of the lumbopelvic complex was developed to simulate unilateral vertically unstable sacral fractures corresponding to Denis type I, II, and III patterns. Two posterior fixation strategies, TPO and BTO, were constructed for each fracture type. Seven physiological loading conditions were applied, including standing, flexion, extension, axial rotation, and lateral bending. Von Mises stress distribution and fracture displacement were evaluated. Fracture micromotion was quantified by calculating relative displacement changes between paired points along the fracture gap.

RESULTS: When interpreted relative to the high-stability BTO reference construct, TPO showed fracture pattern-dependent variation in fracture displacement. The difference was most pronounced in Denis type I fractures and progressively decr […]

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Factors associated with in-hospital mortality in necrotising soft tissue infections. a multicentre retrospective cohort study.

Podda M, Ceresoli M, Virdis F, Rosa F, Pilia T, Murzi V, Dessì A, Tedesco S et al.

PURPOSE: Necrotising soft tissue infections (NSTIs) are rare but life-threatening conditions associated with high mortality rates. This multicentre study aimed to identify admission variables associated with in-hospital mortality.

METHODS: This retrospective study included adult patients with surgically confirmed NSTIs treated at four high-volume academic referral centres in Italy between 2010 and 2024. Demographic, clinical, physiological, and laboratory variables available at hospital admission were analysed. Categorical variables were compared between survivors and non-survivors using the chi-square test or Fisher's exact test, while quantitative variables were compared using the Student's t test or Mann-Whitney U test. Multivariable logistic regression analyses were performed to identify independent predictors of in-hospital mortality. Results were reported as ORs with 95%CI. A prognostic nomogram was developed from the final multivariable model, including variables independently associated with mortality.

RESULTS: A total of 379 patients were included. In-hospital mortality was 16.7%. In subgroup comparisons, mortality was 17.0% in necrotising fasciitis and 22.4% in Fournier's gangrene ($P = 0.275$). In the overall NSTI population, age (aOR 1.061, 95%CI 1.037-1.087) and serum creatinine at admission (aOR 1.301, 95%CI 1.040-1.629) were independently associated with mortalit […]

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Influence of frailty on short-term mortality in older patients with multiple trauma in the emergency department.

Çelik ■, Toker A, Biçer S, Cebecio■lu ■K

PURPOSE: To investigate the association between frailty and 28-day mortality in older patients presenting to the emergency department with multiple trauma and to assess the discriminative performance of PRISMA-7 and the Trauma-Specific Frailty Index (TSFI) in frailty assessment in relation to commonly used trauma severity scores.

METHODS: This prospective observational study included patients aged ≥ 65 years with multiple trauma admitted to the emergency department between July 1 and August 31, 2024. Frailty was assessed using PRISMA-7 and TSFI. Patients were stratified into frail and non-frail groups and further categorized according to age groups (65-74, 75-84, and ≥ 85 years). Clinical characteristics, trauma severity scores, and outcomes, including 28-day mortality, were compared. Receiver operating characteristic (ROC) analysis was used to evaluate discriminative performance, and logistic regression analysis was performed to assess the association between scores and mortality.

RESULTS: A total of 144 patients were included. Frailty was significantly associated with increased 28-day mortality, higher ICU admission rates, and longer hospital stays ($p < 0.001$). Mortality increased with age, reaching 30.0% in patients aged ≥ 85 years ($p = 0.006$). Frailty scores (PRISMA-7 and TSFI) increased significantly with age, whereas ISS did not differ significantly across age groups. R […]

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Mortality prediction in older adults with earthquake-related trauma: a model combining MEWS and laboratory biomarkers.

Ozen A, Acehan S, Satar S, Bulut EA, Gulen M, N■g■z K, Sevd■mbas S, Sonmez GO et al.

PURPOSE: This study aimed to identify laboratory parameters and clinical scores that can predict in-hospital mortality among patients aged 65 years and older who presented to the emergency department (ED) with earthquake-related trauma following the February 6 and 20, 2023 earthquakes.

METHODS: In this retrospective observational study, 362 patients with earthquake-related trauma between February 6-21, 2023, were included. Demographic characteristics, laboratory results, and clinical scores such as the Modified Early Warning Score (MEWS), Injury Severity Score (ISS), and Shock Index (SI) were analyzed.

RESULTS: Of the included patients, 53.9% ($n = 195$) were female, and the median age was 73 years (IQR: 69-80). The overall mortality rate was 9.4%. Multivariable logistic regression analysis identified MEWS (OR: 3.535; 95% CI: 2.397-5.212; $p < 0.001$), urea (OR: 1.009; 95% CI: 1.002-1.015; $p = 0.008$), and hs-Troponin I (OR: 1.001; 95% CI: 1.000-1.002; $p = 0.016$) as independent predictors of mortality. Receiver operating characteristic (ROC) analysis demonstrated an AUC of 0.936 for the combined model including MEWS, urea, and hs-Troponin I (95% CI: 0.875-0.996; $p < 0.001$). Among the variables, MEWS showed the highest AUC (0.903; 95% CI: 0.829-0.976). The optimal cutoff value for MEWS was determined as 1.5, with a sensitivity of 70.6% and a specificity of 97.6%.

CONCLUSIONS: In o […]

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The economic burden in terms of cost of illness and generic health-related quality of life of posttraumatic long bone non-unions among the adult population of the Netherlands from a societal perspective.

van der Broeck LCA, Shchurov DG, Gabrio A, Lodewijks AJL, Poeze M, Blokhuis TJ, Evers S

PURPOSE: This study assessed the societal economic burden in terms of cost of illness and health-related quality of life (HRQoL) of posttraumatic long bone non-unions in the Netherlands.

METHODS: An incidence-based bottom-up approach was used, focusing on adult patients with posttraumatic long bone non-unions. The analysis included healthcare costs, patient and family costs, and productivity losses, measured using the iMCQ and iPCQ questionnaires. Cost evaluations followed Dutch costing guidelines, and productivity losses were calculated using the friction cost method. HRQoL was assessed with the EQ-5D-5L. A deterministic one-way sensitivity analysis varied baseline characteristics by ± 10%. Scenario analyses were conducted from healthcare and patient perspectives, as well as for patient subgroups.

RESULTS: Average costs per patient during the three months before the initial visit at a non-union clinic were €8,928 (healthcare, N = 78), €1,360 (patient and family, N = 58), and €4,313 (productivity losses, N = 59), the average of the observed data calculates to 10,831 (N = 44). The mean EQ-5D-5L utility score was 0.390 (± 0.29 SD). Subgroup analysis showed no significant cost increase for patients with infections or open fractures. However, a higher Non-Union Severity Score and a lower HRQoL were significantly associated with higher total costs.

CONCLUSION: Posttraumatic long […]

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Nonoperative treatment of Maisonneuve fractures.

Frühauf J, Bartoní■ek J, Fojtík P, Hor■áková A, Tu■ek M

PURPOSE: The aim of this study is to describe the basic pathoanatomical characteristics of a stable Maisonneuve fracture and mid-term results of its nonoperative treatment.

METHODS: The study included 17 prospectively collected patients with a mean age of 59 years. The postinjury ankle CT had to meet the following criteria: nondisplaced or minimally displaced (up to 1 mm) fracture of medial malleolus, medial clear space less than 3 mm, nondisplaced or minimally displaced (up to 2 mm) fracture of posterior malleolus, anatomical position or minimal malposition of the distal fibula in the fibular notch (widening of the tibiofibular space up to 2 mm or external rotation of the distal fibula up to 10°). The average follow-up period was 34 months, the final follow-up included CT examination and functional evaluation based on AOFAS and FAOS scores.

RESULTS: A medial malleolus fracture was recorded in 12% cases, a posterior malleolus fracture in 29% patients and a Tillaux-Chaput tubercle fracture in 18% cases. All fractures of the proximal fibula, medial and posterior malleolus healed within 3 months. The position of the distal fibula in the fibular notch did not change in 11 cases compared to the post-injury CT scan, improved slightly in 5 cases, and worsened slightly in 1 case. The average final AOFAS hindfoot score was 96.9 points and the average final FAOS score was 98.7%.

CONCL […]

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Age-related differences in prehospital triage and critical interventions: a multicenter TraumaRegister DGU® analysis.

Koch DA, Hagebusch P, Hackenberg L, Pavlu F, Jakobi T, Münzberg M, Lefering R, Schweigkofler U et al.

PURPOSE: Older trauma patients are known to be undertriaged in the prehospital setting. However, it remains unclear whether these disparities translate into differences in the delivery of critical prehospital interventions. This study aimed to evaluate age-related differences in prehospital triage, allocation, and treatment within a physician-staffed emergency medical service system.

METHODS: A retrospective multicenter analysis of the TraumaRegister DGU® was conducted, including trauma patients aged ≥ 20 years between 2020 and 2023 with trauma team activation and subsequent intensive care unit admission. Patients were stratified into two age groups, 20-59 vs. ≥60 years, with additional subgroups up to 90-99 years. Prehospital interventions, transport modality, and trauma center allocation were analyzed. Furthermore, predefined high-risk scenarios (severe motor vehicle trauma, traumatic brain injury with GCS ≤ 8, and pneumothorax) were evaluated.

RESULTS: A total of 67,069 patients were included, with comparable injury severity across age groups (mean ISS 20.2). Older patients were less frequently transported by air and less often allocated to supra-regional trauma centers. While overall intervention rates were similar, differences emerged in high-risk scenarios. Older patients received endotracheal intubation and tranexamic acid less frequently despite comparable injury seve […]

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Prediction of risk of death in severely injured patients: the revised injury severity classification score, version 3 (RISC III).

Lefering R, Imach S, Bieler D

PURPOSE: Trauma Registries with a focus on severely injured patients use survival as their primary outcome. In order to compare hospitals, interventions, and changes over time, a precise risk of death prediction is mandatory. The German TraumaRegister DGU® uses the RISC II model (Revised Injury Severity Classification, version II) since 2013. The aging trauma population and an improved handling of missing data required the present revision.

METHODS: A total of 53,738 seriously injured trauma patients documented in 2022-2023 served as basis for development and validation (3:1 ratio). Missing values should now be considered to be within normal physiological range except in patients with specific findings indicative for an altered physiology. These findings were suggested by clinical experts and validated by registry data. The increasing age of trauma patients was addressed by additional categories and higher point weights. A logistic regression analysis provided new point weights for all predictors. Precision (observed versus predicted mortality) and discrimination (area under the receiver operating characteristic curve, AUROC) were calculated in the development and validation dataset.

RESULTS: Patients in both datasets were well comparable, with a mean age of 55 years, 69% males, and an average Injury Severity Score (ISS) of 18 points. The rate of missing values ranged from 0% […]

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Dislodgement forces and cost effectiveness of prehospital securement of peripheral intravenous catheters on moist skin: a randomized controlled clinical trial in healthy volunteers.

Vogeler J, Schumann S, Heinrich S, Hell J, Schmutz A

OBJECTIVES: Peripheral intravenous catheters (PIVC) are widely used clinical devices for administering fluids or essential drugs to patients. The failure rate of PIVC remains high and Emergency Medicine has been shown to be a risk factor for dislodgement. When the skin is moist from sweat or fluids, standard intra-hospital dressings and securements fail. In emergency situations, a failed catheter can then critically delay intravenous therapies. The most effective dressing to prevent accidental removal of a prehospital PIVC remains unclear. It was the aim of this study to compare the force required to dislodge a PIVC with four different methods of securing PIVCs used in emergency medicine. In addition, the costs were calculated.

METHODS: Artificial sweat was applied to the skin of 180 volunteers. PIVCs were attached onto the forearm using four different securements (elastic gauze, cohesive gauze, clingfilm and a velcro securing device). Continuously increasing traction force was applied until dislodgement of the respective securement. For statistical tests, either Friedman's test or repeated measures ANOVA was used.

RESULTS: Clingfilm showed the greatest resistance to increasing pulling force with cohesive bandages as close second. The velcro securing device was strongest at resisting low level of forces but fell off sharply at higher force. Elastic bandages were the weakest i […]

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Injury to the thorax is a dominant contributor to post-traumatic neutrophil mediated inflammatory response.

van Eerten FJC, Duebel LV, de Fraiture EJ, Hanlo MF, Breuking EA, Brands SR, van den Bosch TD, van Wessem KJP et al.

PURPOSE: Traumatic injury induces a neutrophil associated inflammatory response mediated by the release of damage-associated molecular patterns (DAMPs). This response can be maladaptive, increasing risk of infection. Since blunt thoracic injuries are associated with high infection rates, we hypothesized that these injuries elicit a stronger neutrophil associated inflammatory response than injuries to other body regions.

METHODS: A cohort study was conducted in adult patients presenting at the trauma bay of a level 1 trauma center between March 2020 and November 2022. These patients underwent routine analysis of their neutrophil associated inflammatory response by phenotyping blood neutrophils using automated flow cytometry. Injury per body region was counted when the Abbreviated Injury Severity (AIS) scores were ≥3 to focus on substantial injuries. The primary outcome was an extensive post-traumatic inflammatory neutrophil response.

RESULTS: 861 patients were included. Blunt internal thoracic injuries showed the strongest association with severe neutrophil associated inflammation (OR 5.7), followed by abdominal (OR 2.6), pelvis (OR 2.5) and lower extremities (OR 2.3), even when corrected for ISS, age and hemodynamic instability. In multivariable analysis, only thoracic internal injuries remained a statistically significant contributor to excessive neutrophil associated inflam […]

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Major improvement using peroperative 3D compared to 2D imaging for SI screw fixation in traumatic pelvic fractures.

van Eerten FJC, Benders KEM, van Baal MCPM, Hietbrink F

INTRODUCTION: Pelvic fractures, though accounting for only 3% of skeletal injuries, carry a high mortality rate, primarily due to the associated risk of hemodynamic instability. Sacroiliac (SI) screw fixation is a common surgical intervention, which can be technically challenging due to the complex pelvic anatomy. Imaging techniques have evolved from 2D to 3D imaging for its improved accuracy in screw placement. The impact of this transition for SI screw placement was evaluated.

METHOD: A single-center observational retrospective study was performed between 2013 and 2023. Patients ≥ 18 years who underwent SI screw placement following pelvic injuries were analyzed. 2D imaging was used until 2021, after which 3D imaging was implemented. Outcomes included the need for revision surgery during initial admission and < 18 months after trauma.

RESULTS: 2D imaging was used in 42 patients and 3D imaging in 44 patients. Baseline characteristics and injury severity of both groups were comparable. In the 2D group, 18 patients (42.9%) had suboptimal placement of the SI screw. Of these, 3 (7.1%) required revision surgery due to malpositioned screws. In contrast, no patients (0.0%) in the 3D group had suboptimally positioned screws or underwent revision surgery, ($p < 0.001$, $p = 0.22$).

DISCUSSION: Intraoperative 3D imaging enhanced the accuracy of SI screw placement, leading to an encouragin […]

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Minimally invasive versus open fixation for flail chest in elderly patients: a prospective cohort study on perioperative safety and long-term outcomes.

Tian Y, Zheng Z, Ji Z, Lu W, Song N, Wang J

OBJECTIVE: To systematically evaluate the perioperative safety and short- and long-term efficacy of minimally Invasive fixation using a reversed shape-memory alloy plate versus conventional open fixation for the treatment of flail chest in elderly patients.

METHODS: This prospective cohort study enrolled 130 elderly patients (≥ 60 years) with flail chest between January 2022 and January 2026. After 1:1 propensity score matching, 100 patients were included (minimally invasive group, n = 50; open group, n = 50). Perioperative parameters (operative time, blood loss, C-reactive protein(CRP) at 24 h/48 h/72 h, complications), short-term outcomes (resting/cough-evoked Numerical Rating Scale(NRS) pain scores, non-steroidal analgesic drug duration, rescue analgesic frequency, Patient-Controlled Analgesia Pump(PCA) consumption, time to ambulation, hospital stay), and long-term outcomes (implant dislodgement, 6-month pulmonary function, Douleur Neuropathique 4 Questions Naire(DN4), Medical Outcomes Study 36- Item Short Form Health Survey(SF-36)) were compared.

RESULTS: The minimally invasive group had significantly fewer incision-related complications ($P = 0.046$), lower resting NRS scores on postoperative days 1 and 3 ($P < 0.01$; $P = 0.041$), lower cough-evoked NRS score on day 1 ($P = 0.011$), shorter non-steroidal analgesic duration ($P < 0.01$), and lower PCA consumption ($P = 0.027$). At 6 […]

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Epidemiology and outcomes of traumatic sternal fractures and associated blunt cardiac injury: a nationwide cohort study in the Netherlands.

Klei DS, Benders KEM, Leenen LPH, van Wessem KJP

PURPOSE: Comprehensive data on epidemiology, trauma mechanisms, associated injuries, and outcomes of traumatic sternal fractures are scarce. This study analysed nationwide data to improve diagnosis and management within the Dutch healthcare system.

METHODS: This nationwide retrospective cohort study using the Dutch National Trauma Registry included adult patients admitted with traumatic sternal fractures between 2015 and 2023. Patients with prehospital cardiopulmonary resuscitation or penetrating trauma were excluded. Incidence, patient characteristics, trauma mechanisms, associated injuries, and in-hospital outcomes were analysed. Subgroup analyses evaluated patients with concomitant blunt cardiac injury (BCI).

RESULTS: Of 568,399 adult trauma admissions, 4,765 patients (0.84%) sustained traumatic sternal fractures. Median age was 62 years; 60% were male. Motor vehicle accidents (48%) and falls (28%) were the leading mechanisms. 35% were severely injured (ISS ≥ 16). Associated injuries included rib fractures (51%), spinal fractures (36%), and lung contusions (18%). Critical care unit admission was 40%, with median mechanical ventilation duration of 4 days; median hospital stay was 5 days. In-hospital mortality was 5.7%, and 30-day mortality 6.0%. BCI occurred in 9.5% of patients and was associated with a higher number of injuries and increased injury severity, emergency inte […]

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Trauma patients ≥ 65 years old constitute 40% of severely injured adult trauma patients in Sweden - a 10-year analysis of geriatric trauma prevalence.

Bredberg S, Modalen E, Jöneby C, Nilsson D, Lapidus O

BACKGROUND: The elderly population in Sweden is steadily increasing. Age is an independent predictor of mortality in trauma patients. There is growing evidence that geriatric patients constitute an increasing proportion of the Swedish trauma population. No studies have analyzed temporal trends in the proportion of geriatric trauma patients in the context of the national trauma population in Sweden.

AIM: To determine the proportion of geriatric patients in the Swedish trauma population and analyze temporal trends in geriatric trauma prevalence.

METHODS: The study cohort was 11,969 severely injured trauma patients (NISS > 15) treated in Sweden during 2013-2022, obtained from the Swedish Trauma Registry. The proportion of geriatric patients was compiled annually. Temporal trends were analyzed using weighted linear regression models. Two subgroups were defined based on injury severity (NISS ≥ 25 and NISS ≥ 40).

RESULTS: The proportion of patients ≥ 65 years in the adult trauma population increased from 30% to 40% during 2013-2022 ($p = 0.003$), while the proportion of patient ≥ 80 years also increased, from 8.6% to 17% ($p = 0.009$). Similar increases were also seen in the NISS ≥ 25 and NISS ≥ 40 subgroups.

CONCLUSION: There is likely an increase in geriatric trauma prevalence in Sweden, but the full extent of this temporal trend remains uncertain. Trauma patients ≥ 65 years old co […]

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A totally laparoscopic strategy combining microwave ablation and autologous blood transfusion for patients with initial WSES grade IV traumatic splenic rupture in selected transient responders: a proof-of-concept study.

Feng Z, Xie D, Hong W, Zheng C

PURPOSE: Splenic artery embolization (SAE) is the standard intervention for transient responders with high-grade splenic trauma. When SAE is unavailable, contraindicated, or declined, alternative minimally invasive approaches may be warranted. This study evaluates a totally laparoscopic strategy combining microwave ablation (MWA) for hemostasis with intraoperative autologous blood transfusion (ABT) in selected transient responders with initial WSES Grade IV splenic rupture.

METHODS: This retrospective study included 10 consecutive transient responders (October 2021-June 2024) with initial WSES Grade IV splenic rupture who underwent totally laparoscopic MWA combined with intraoperative ABT. All patients exhibited transient hemodynamic response to limited resuscitation (≤ 1500 mL fluids) and preoperative shock index > 0.9. Intraoperative injury was graded by the AAST system. Outcomes included technical success, transfusion requirements, complications, and follow-up.

RESULTS: All 10 patients underwent successful spleen-preserving laparoscopy without conversion. AAST grades: II (n = 2), III (n = 4), IV (n = 4). Mean operative time was 191.3 ± 101.7 min. Mean blood salvage was 1587.5 ±

663.9 mL, yielding 490.8 ± 80.2 mL autologous red cells. No patient required postoperative allogeneic transfusion. Hemoglobin, platelets, and lactate improved significantly by postoperative day 3 (P […]

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Pediatric pancreatic trauma: a 10-year retrospective analysis from two high-complexity centers in Medellín, Colombia.

Barrientos ML, Calderon SP, Zapata CAL, Lopez CAD

PURPOSE: Pediatric pancreatic trauma is uncommon, but it is associated with significant morbidity and an ongoing debate regarding its management. Hemodynamic stability and pancreatic duct involvement are key factors in treatment decisions.

METHODS: A retrospective study analyzed 25 pediatric patients with pancreatic trauma treated between 2010 and 2020 at two high-complexity hospitals in Medellin. Clinical variables, trauma mechanism, imaging findings, management, and complications were analyzed.

RESULTS: Blunt trauma was the predominant mechanism (64%). Preoperative computed tomography was performed in 12 patients (48%); radiological AAST grading was not retrievable in 13 (52%). Immediate surgery was performed in 13 patients (52%), predominantly driven by associated intra-abdominal injuries rather than the pancreatic injury itself. Non-operative management was initially attempted in 9 patients (36%), with a 56% success rate; failures (44%) were related to clinical deterioration and progression of peripancreatic collections. Clinically relevant postoperative pancreatic fistula (ISGPS Grade B) occurred in 3 of 13 operated patients (23%), with a mean closure time of 55.6 days (range 18-110); no Grade A or Grade C fistulas were recorded. Other complications included post-traumatic pancreatitis (24%), pancreatic pseudocyst (8%), and organ/space surgical site infection (8%); most […]

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Influence of Antithrombotics on Severely Injured Patients Over 50 With Blunt Abdominal Trauma.

Eibinger N, Limberger B, Hörlesberger N, Puchwein P, Lefering R, Bayer TP, Zumsteg JS, Pape HC et al.

INTRODUCTION: Severe abdominal trauma represents a critical subset of injuries, particularly in the aging European population, where the prevalence of preexisting anticoagulant or antiplatelet therapy is increasing. While the impact of such medications on outcomes in traumatic brain injury is well studied, limited data exist regarding their influence in patients with abdominal trauma.

MATERIALS AND METHODS: This retrospective cohort study used data from the TraumaRegister DGU® between January 2015 and December 2023. Inclusion criteria were patients aged ≥ 50 years with severe blunt abdominal trauma (AIS Abdomen ≥ 3) without relevant head injury (AIS Head ≤ 3) from Austria, Germany, or Switzerland. Patients were grouped according to pre-injury antithrombotic medication (no medication [NM] vs. antithrombotic medication [AM], with subgroups). Statistical analysis included descriptive comparisons, standardized mortality ratios (SMR) using RISC III, and multivariate logistic regression to identify independent predictors of hospital mortality.

RESULTS: Of 328,281 patients, 4,069 met inclusion criteria (2,831 NM; 1,238 ATM). Patients in the AM group were older (74.1 vs. 62.3 years) and had higher mortality (25.0% vs. 10.4%), particularly in the DOAC subgroup (30.3%). SMR analysis demonstrated increased mortality in the AM group compared to expected rates. In multivariate logistic re […]

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Predicting limb amputation in trauma-related necrotizing fasciitis of the extremities: insights from a 10-year cohort.

Yang S, Ren Z, Li Y, Zhao Z, Sun C, Liu L, Jin L, Hou Z

BACKGROUND: Necrotizing fasciitis (NF) is a rapidly progressive, life-threatening soft tissue infection that in most cases requires aggressive surgical management. Major limb amputation is a devastating limb-related outcome. We retrospectively analyzed NF cases to quantify the rate of amputation and identify associated risk factors.

METHODS: We retrospectively reviewed the medical records of all patients with trauma-related extremity NF treated at Hebei Medical University Third Hospital from Jan 2014 to Dec 2024. Demographics, comorbidities, admission laboratories and key time intervals were compared between patients who did and did not undergo major limb amputation. Multivariable logistic regression identified independent predictors, and receiver operating characteristic (ROC) analysis assessed the discriminative power of continuous variables.

RESULTS: Of 134 patients, 22 (16.4%) required amputation. Compared with the non-amputation group, amputees had a markedly longer pre-hospital delay (median 242 h vs. 65 h), more multiple injuries (63.6% vs. 13.4%) and a higher prevalence of diabetes (59.1% vs. 21.4%). Admission lactate dehydrogenase (LDH) was substantially higher (median 589 vs. 182 U/L). In multivariable analysis, elevated LDH, prolonged time from traumatic injury to hospital admission, multiple injuries and diabetes were independently associated with amputation. LDH […]

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Is 18 F-FDG PET-CT-guided geographic debridement superior to conventional debridement in appendicular chronic osteomyelitis? a retrospective comparative cohort study.

Elsheikh A, Elseidy H, Saad H

BACKGROUND: Determining the surgical extent of chronic osteomyelitis and fracture-related infection (FRI) remains challenging. Adequate resection margins prevent recurrence; however, no standardised preoperative tool objectively defines resection borders.

OBJECTIVE: To compare PET■CT-guided surgical planning versus the local standard■of■care imaging pathway (predominantly plain radiographs, with selective CT or MRI in a minority of patients) for chronic limb osteomyelitis and FRI outcomes.

METHODS: Retrospective comparative cohort study of 126 patients treated 2019-2024. Group 1: 84 patients underwent 18 F-FDG PET-CT imaging and map-based geographic debridement. Group 2: 42 patients received surgery after local standard■of■care imaging pathway (predominantly plain radiographs, with selective CT or MRI).

PRIMARY OUTCOME: infection recurrence at 12 months. Multivariable logistic regression controlled for confounders.

RESULTS: PET-CT group achieved 3.6% recurrence versus 16.7% in the conventional group ($p = 0.011$), representing 78.5% relative risk reduction and a number-needed-to-treat of 8. Critically, the PET-CT cohort had significantly worse baseline complexity: 3.3-fold longer infection duration (43.1 vs. 13.1 months, $p < 0.001$) and 3.6-fold more prior surgeries (2.48 vs. 0.69, $p < 0.001$). Multivariable analysis confirmed PET-CT-guided surgery as an independent protective […]

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Digital coordination in mass casualty incidents: a retrospective analysis of prehospital distribution times before and after IVENA-MANV implementation.

Brauckmann V, Ringe B, Caviglia M

BACKGROUND: Digital coordination tools have been proposed to improve management during mass casualty incidents (MCIs). In 2021, the IVENA-MANV system was introduced in the Hannover region as an extension of the established IVENA eHealth platform to enable digital patient tracking and hospital capacity management. However, no real-world evaluations have yet quantified their operational impact.

OBJECTIVES: This study aimed to provide the first real-world evaluation of the IVENA-MANV digital coordination system and to determine how its implementation affected prehospital distribution processes and time intervals during mass casualty incidents.

METHODS: A retrospective observational study to assess the impact of the IVENA-MANV digital coordination platform on prehospital times during MCIs in Germany was conducted using dispatch records and IVENA-MANV logs from all MCIs between January 2018 and July 2025 in the Hanover-Region in Germany. Primary outcomes were the triage-to-evacuation (TtE) and total prehospital time (TPHT) per patient. Mann-Whitney U tests compared pre- and post-implementation groups, and multiple linear regression models examined associations between IVENA-MANV use, MCI-level, and time intervals.

RESULTS: Seventy-three MCIs met inclusion criteria, including 188 documented individual casualty characteristics. 41.1% of incidents used IVENA-MANV. Most MCIs were tra […]

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The real-world management of acute mesenteric ischemia in Spain: results from a multicenter national survey.

González-Castillo AM, Calsina Juscafresa L, Gelabert A, Velescu A, Marrano E, Vikatmaa P, Tolonen M

BACKGROUND: Acute mesenteric ischemia (AMI) remains one of the most lethal vascular emergencies, with in-hospital mortality rates frequently exceeding 50%. Although early diagnosis and timely revascularization are critical, clinical practice remains highly variable. This study aimed to evaluate the real-world management of AMI in Spain, identifying gaps in resources, protocols, and interhospital coordination.

METHODS: A national cross-sectional survey was conducted between March and August 2024 using the Survio® platform. The questionnaire was distributed to general surgeons through national surgical societies and included 27 items covering hospital infrastructure, clinical protocols, diagnostic and therapeutic availability, personal experience, and perceived system-level barriers.

RESULTS: A total of 291 surgeons responded. The median age was 40 years (IQR: 17). Most were consultants (76.3%) working in tertiary (42.6%), secondary (33%), or community hospitals (24.4%). While 97.6% reported access to 24/7 multiphasic CT and 90.4% to round-the-clock radiology, only 51.9% had 24/7 interventional radiology and 60.1% vascular surgery. Just 26.8% had institutional AMI protocols. The median distance to a referral center was 25 km (range: 2-250 km), and 68.4% reported difficulty in patient transfers. While 96.9% felt competent managing AMI, only 36.4% were familiar with the term "int […]

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Changes in operative and in-hospital outcome of proximal femoral fractures before and after the implementation of work hour restrictions in Switzerland.

Tewes A, Canal C, Schöb O, Neuhaus V

PURPOSE: Surgeon fatigue is a recognized risk factor for patient safety; however, reduced working hours may limit surgical training exposure. In 2005, Switzerland introduced a maximum 50-hour workweek for resident surgeons. In view of current discussions on further reductions (42 + 4 h), concerns persist regarding training quality and workforce capacity. This study evaluates the impact of the 2005 Swiss labor law on patient outcomes, operative characteristics, and intraoperative teaching utilizing a standardized trauma procedure.

METHODS: A retrospective analysis of anonymized data from a national database was performed. All patients undergoing operative fixation of trochanteric femur fractures (ICD-10 S72.1) were included. Two four-year periods were compared: pre-regulation (2001-2004) and post-regulation (2016-2019). Primary endpoints were in-hospital complications, mortality, and length of stay. Secondary endpoints included patient characteristics, surgeon seniority, teaching status, operative duration, and time to surgery.

RESULTS: Post-regulation patients were older and exhibited higher American Society of Anesthesiologists classifications despite fewer comorbidities. Complications increased from 9.1% to 17.7%, and mortality from 1.6% to 3.5%. Length of stay decreased from 14 to 9 days. Operative duration decreased by 10 min across all surgeon levels. Resident surgeons i […]

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Early discharges in severely injured patients by ISS score: exploring injury patterns and coding practices.

Parouei F, Krüsemann EJZ, Poeze M, de Jongh MAC, Hietbrink F, Landelijke Beraadsgroep Traumachirurgen (LBTC)

PURPOSE: This study examines the prevalence of early discharge (ED) among patients classified as severely injured in the Dutch National Trauma Registry (DNTR) and evaluates whether identifiable patient-, injury-, and system-related factors are associated with the occurrence of ED in this population.

METHODS: This retrospective cohort study analyzed DNTR data from 2015 to 2022, focusing on severely injured patients (Injury Severity Score [ISS] ≥ 16). Patients were grouped as early discharge (ED; discharged within 48 h without in-hospital mortality) or non-ED (NED). Patients' characteristics were compared using descriptive statistics, and a mixed-effect model identified factors associated with ED.

RESULTS: From 2015 to 2022, a total of 37,626 severely injured patients were registered including 1,454 (3.9%) ED patients. This proportion increased from 3.2% in 2015 to 4.8% in 2022. ED was more common in younger patients, males, general practitioner (GP) or self-referred cases, those with severe head injury (Abbreviated Injury Scale [AIS] ≥ 3) and fewer registered injury codes. In the mixed-effects model, younger age (OR up to 2.0), increasing head injury severity (OR: 1.16 per AIS point), less than 3 registered AIS codes (OR: 0.60), referral by a general practitioner (OR 1.68) or self-referral (OR 1.92), and admission year (OR: 1.08 per year) independently predicted early discharg […]

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Trauma team activation for pediatric patients in Denmark: a multicenter study of criteria, organization, and injury severity.

Nordestgaard CH, Gude MF, Leth SV, Raaber N

BACKGROUND: Pediatric trauma team activation (TTA) is intended to ensure timely management of severely injured children, yet both prehospital visitation practices and in-hospital TTA criteria vary across trauma systems. The aim of this study was to describe and compare pediatric TTA criteria, organizational models, and injury severity across all Danish level I trauma centers.

METHODS: We conducted a retrospective multicenter cohort study including all pediatric trauma patients (< 18 years) admitted via TTA at the four Danish level I trauma centers between 2014 and 2024. Center-specific prehospital and in-hospital TTA protocols were obtained through structured inquiries. Injury severity was assessed using Injury Severity Score (ISS) and Abbreviated Injury Scale (AIS). Overtriage was defined as TTA in patients with ISS < 15.

RESULTS: A total of 3,452 pediatric trauma patients were included. Two distinct TTA models (single-criterion and point-based) and four organizational structures were identified. Overall, 14% of patients had ISS ≥ 15, corresponding to high overtriage across all centers. Overtriage rates ranged from 81% to 95% and were highest at centers using point-based TTA models. Mortality was low and did not differ significantly between centers. Among the most severely injured patients (ISS ≥ 25), head and thoracic injuries predominated.

CONCLUSION: Pediatric trauma tri […]

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Does the operative technique (MIPO vs. ORIF) influence surgical outcomes and complications in proximal humerus fractures? A matched-pair analysis.

Barthel C, Rudolph F, Kraus M, Kalbas Y, Russek R, Pape HC, Allemann F

INTRODUCTION: Proximal humerus fractures account for approximately 4-6% of all fractures and are among the most common fractures in the elderly. Surgical treatment is increasingly applied for displaced and unstable fracture patterns. The two most commonly used techniques are open reduction and internal fixation (ORIF) via a deltopectoral approach and minimally invasive plate osteosynthesis (MIPO) using a deltoid-split or anterolateral approach. ORIF allows direct fracture visualization and anatomical reduction but is associated with more extensive surgical exposure. MIPO aims to preserve soft tissues and vascular supply through indirect reduction techniques. This study compares ORIF and MIPO in proximal humerus fractures with regard to surgical parameters, radiographic outcomes, and complications.

METHODS: A retrospective single-center cohort of 392 proximal humerus fractures was analyzed. Seventy-nine patients treated with MIPO were matched to 79 patients treated with ORIF using exact matching for fracture type (Mayo FJD classification) and propensity score matching for age and sex. Primary outcomes included operative time, reduction quality, and complication rates.

RESULTS: The MIPO group showed significantly shorter operative times (97.3 ± 30.8 min vs. 112.2 ± 34.0 min, p = 0.004). No significant differences were observed in reduction quality or overall complication rates […]

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Risk factors and predictors of in-hospital mortality in necrotizing fasciitis: a five-year retrospective cohort study from a tertiary care center in Colombia.

Barrientos ML, Cordoba ASR, Valencia MDML, Valencia MMM, Toro DAM, Zapata CAL, Delgado CA

PURPOSE: Necrotizing fasciitis (NF) is a life-threatening soft tissue infection with limited epidemiological data from Latin America. This study aimed to characterize clinical outcomes, identify factors associated with in-hospital mortality, and develop a clinical prediction model in patients with NF at a tertiary care center in Medellín, Colombia.

METHODS: A retrospective cohort study was conducted including all adult patients (≥ 18 years) with surgically or histopathologically confirmed NF between January 2019 and December 2024. Demographic, clinical, microbiological, and therapeutic variables were collected. Bivariable analysis and multivariable logistic regression were performed to identify independent mortality predictors. A clinical risk score was developed and internally validated using bootstrap resampling and 10-fold cross-validation.

RESULTS: Of 112 identified patients, 111 met inclusion criteria. Overall in-hospital mortality was 23.4% (26/111). Deceased patients were significantly older (median 69.5 vs. 52 years; $p < 0.001$). Serum lactate demonstrated strong prognostic value (AUC 0.719; 95% CI 0.596-0.842; 2.9 vs. 1.8 mmol/L; $p = 0.003$). Septic shock was present in 34.2% and doubled mortality risk (34.2% vs. 17.8%; $p = 0.060$). Polymicrobial infections predominated (74.8%), with a predominantly gram-negative profile and high antimicrobial resistance rates (carbapen […]

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ESTES recommendations regarding distal radius fractures.

Lameijer CM, Teuben MPJ, van Delft EAK, Tomazevic M, Kastelec M, Nau C

Distal radius fractures (DRFs) account for 15% of all fractures in adults. Optimal treatment of DRFs depend on patient and fracture characteristics. Non- or minimally displaced DRFs can be treated nonoperatively with a cast. Although surgical treatment is becoming more popular, the majority of patients with a DRF are still treated nonoperatively. In addition, these considerations may be different for elderly patients. Conflicting evidence exists on the different aspects of treatment of DRFs. These recommendations regarding distal radius fractures describe considerations regarding diagnostic modalities, associated soft tissue injuries, different treatment options, rehabilitation protocol, considerations for treatment of elderly patients, follow-up decisions and possible complications. The aim is to aid the orthopedic traumasurgeon in decision making and treatment of distal radius fractures.

PubMed · DOI

Small traumatic intracranial hemorrhages identified in routine radiology reports are associated with a low risk of adverse events: a retrospective cohort study.

Gerdås K, Wigermo A, Offenbartl M, Vestlund S, Rezk F, Vedin T

PURPOSE: To determine whether routinely available radiology reports, together with basic clinical data, can identify patients with traumatic intracranial hemorrhage (TICH) who are at low risk of adverse events.

METHODS: This retrospective cohort study of adults with TICH in Region Jönköping, Sweden (2019-2021). Clinical data, findings from radiology reports and outcomes were extracted from medical records. Hemorrhage size was classified as small (≤ 4 mm or described as minimal/very small/discrete) or larger. Adverse events were defined as neurosurgical intervention or death directly attributable to the TICH. Risk difference (RD), relative risk (RR), and Firth's penalized logistic regression were used to assess associations with adverse events.

RESULTS: Among 527 included patients, 195 (37%) had small TICHs. None of these patients experienced adverse events, compared with 13.6% neurosurgical interventions and 13.0% trauma-related deaths in the group with larger TICHs (RD 24.5% points, 95% CI 18.4-29.6; RR 97.1, 95% CI 6.1-1557.1; $p < 0.001$). Small TICH size had the strongest association with absence of adverse events. Normal neurological status and GCS 14-15 were also associated with a low risk of adverse events. Anticoagulant or antiplatelet therapy showed no significant association with adverse events.

CONCLUSION: Routinely available radiology reports, combined with basic c […]

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From fracture to joint injury: association between CT-based fracture characteristics and surgically treated non-bony injuries in tibial plateau fractures.

Neidlein C, Colcuc S, Ullerich F, Schöllmann H, Steinhausen ES, Kröpil P, Brinkmann N, Dudda M et al.

PURPOSE: Tibial plateau fractures (TPF) are often associated with non-bony injuries that may impair knee stability and long-term outcomes. However, it remains unclear in which patients preoperative MRI is indicated to detect these injuries, and clear recommendations for surgical decision making are lacking. This study aimed to develop a CT-based predictor of surgically relevant non-bony injuries.

METHODS: This retrospective single-center study included all intra-articular TPF treated between January 2022 and May 2025 at a Level I trauma center. Fractures were classified according to the 10-segment and three-column classifications. Operative reports were reviewed to identify non-bony injuries requiring surgical treatment. Descriptive statistics and multivariate logistic regression were performed to determine predictors of surgically relevant non-bony injuries.

RESULTS: Among 243 patients (mean age 49,3 ± 14,9 years), 34,6% had at least one surgically treated non-bony injury. Posterior column involvement was significantly associated with a higher injury rate (41,4%), particularly affecting meniscus (20,7%), Anterior cruciate ligament (ACL) (17,2%), and medial collateral ligament (MCL) (12,3%). The posterolateral-central (PLC) segment showed the highest segment-specific injury rate (37,6%) and was an independent predictor of ACL injury. Increasing fracture complexity was associa […]

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Transferrals and clinical pathways of unstable pelvic fractures over the last 10 years - a retrospective analysis of the Trauma Register DGU®.

Weuster M, Pfeifer R, Seekamp A, Lippross S, Lefering R, Dgu T, Menzdorf L

PURPOSE: Pelvic fractures are classified as stable or unstable. They correlate with a severity of trauma and the initial medical treatment is decisive. This study evaluated the transferrals of such fractures and described the initial treatment as well as the clinical course.

METHODS: We analysed retrospective data from a large cohort of the TraumaRegister DGU® (TR-DGU), covering the period from 2014 to 2023, comprising a total of n = 397,910 patients. All patients aged ≥ 16 years were included. Injury patterns were described according to the Abbreviated Injury Scale (AIS), the mechanically unstable fractures were classified with an AIS ≥ 3. We considered all participating hospitals within Germany. The patients were subdivided in three groups: Group 1 = primary admitted patients with outcome, group 2 = pre-treated patients transferred in from other hospitals, and group 3 = primary admitted and early (< 48 h) transferred out.

RESULTS: The majority of the patients was male and about 53 years old. Blunt trauma was the leading trauma mechanism. Concomitant injuries (AIS 2+) affected thorax (56%), spinal cord (41%), lower extremities (38%), head (31%) and abdomen (24%). Among primary admitted cases with pelvic fractures (n = 36,398), 21,091 cases (57.9%) had an unstable pelvic fracture (AIS pelvis 3-5). Level 1 trauma centers not only treated 12,836 primary admitted cases with unst […]

PubMed · DOI

Surgical treatment of coronal shear fractures: short- to mid-term results and risk factor analysis.

Bauer A, Cornel L, Sauter M, Jakobi T, Münzberg M, Klug A

BACKGROUND: Coronal shear fractures of the distal humerus are rare but severe injuries. Reconstruction is often challenging, especially in comminuted cases, which is why many surgeons opt for an elbow arthroplasty in those cases. However, total elbow arthroplasty is associated with a variety of potential problems itself. Therefore, the aim of this study was to present the functional and clinical outcome of coronal shear fractures treated by osteosynthetic reconstruction in a short- to mid-term follow-up, and to identify possible risk factors for an inferior outcome.

METHODS: We performed a retrospective follow-up assessment of 51 consecutive patients (30 women; median age 56 years, (IQR 39-62)) who underwent osteosynthetic reconstruction for coronal shear fractures between 2012 and 2022 after a minimum follow-up period of two years. The Mayo Elbow Performance Score, Oxford Elbow Score, and Disabilities of the Arm, Shoulder and Hand score were evaluated, and all available radiographs were analyzed. All complications and revision procedures were assessed. Univariable and multivariate regression analyses were performed to identify potential risk factors for a poor outcome following osteosynthetic reconstruction.

RESULTS: After a median follow-up period of 43 (IQR 28-78) months, the median Mayo Elbow Performance Score was 100 (IQR 85-100), the median Oxford Elbow Score was 42 (IQ […]

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Publisher Correction: Experience with patient-specific guides, instead of general surgical experience, improves the accuracy of 3D-guided corrective osteotomies.

Buist ML, Meesters AML, Colaris JW, Doornberg JN, Ten Duis K, Tabernée Heijtmeijer SJC, Jutte PC, Pijpker PAJ et al.

PubMed · DOI

Assessment of a new technique for biological augmentation in sternoclavicular joint dislocations: a cadaveric feasibility and biomechanical evaluation.

Somberg O, Förster E, Rosteius T, Bernstorff M, Schildhauer T, Königshausen M

AIMS: Biological augmentation is necessary in chronic sternoclavicular dislocations. However, the placement of drill holes for a graft can be difficult at that region. The aim of this study was to evaluate, as an early proof-of-concept, the feasibility of using a locally available clavicular periosteal flap (CPF) or sternal periosteal flap (SPF) for sternoclavicular joint (SCJ) stabilization and to biomechanically test the periosteal flaps.

METHODS: A cadaveric anatomic feasibility and biomechanical study (level of evidence V) was performed on nine SCJs. The CPF was mobilized from three sides by lifting the flap off the bone, preserving its attachment near the SCJ. Two 1.6 mm drill holes were created in the sternum, and the CPF was folded over the SCJ and fixed with a non-resorbable wire. Additionally, the non-resorbable wire was used in terms of a figure-of-8-bracing through a 1.6 mm clavicular drill hole to secure the joint position. Flap dimensions and sternal overlap were measured. Vice versa SPFs were prepared in selected specimens and fixed in a similar fashion. The CPFs were biomechanically tested using a uniaxial tensile testing machine to obtain values for stress at 50% strain (σ (50%)), maximum force (F_{max}), maximum stress (σ_{max}), strain at maximum force ($\varepsilon(F_{max})$), and strain at failure (ε at failure).

RESULTS: CPF mobilization and fixation were successfully achieve […]

PubMed · DOI

Differentiating injury patterns and outcomes in accidental, suicidal and occupational falls from heights.

Beyersdorf C, Hussmann B, Wergen N, Lefering R, Schiffner E, Maus U, Jaekel C, Dgu T

BACKGROUND: Falls from significant heights are a leading cause of severe injuries and can result from accidents, suicide attempts, or occupational incidents. While previous studies have explored general injury patterns associated with falls, little is known about the specific differences between accidental, suicidal, and occupational falls. This study aims to identify distinct injury patterns and outcomes among these fall types, thereby enhancing primary care and decision-making algorithms.

METHODS: We conducted a retrospective analysis using data from the German TraumaRegister DGU® (2014-2023), focusing on patients who experienced falls from heights. Only patients admitted to trauma centers with trauma team activation and a Maximum Abbreviated Injury Scale (MAIS) severity ≥ 3 were included. All falls were categorized in accidental, suicidal, or occupational and distinguished between high (≥ 3 m) and low (< 3 m) falls. Statistical analyses included chi-squared tests for categorical variables and Mann-Whitney U tests for continuous variables, with significance set at $p < 0.05$.

RESULTS: Among 188,708 patients, 29,991 (15.9%) experienced high falls, while 50,674 (26.9%) fell from low height. High falls were predominantly accidental (82.3%), with suicidal falls accounting for 17.7%. Accidental high falls mainly caused thoracic (58.9%) and head injuries (46.2%), whereas suicidal f […]

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Predictive performance of ISS and NISS for clinical outcomes in severely injured trauma patients: a retrospective registry study.

Säkkinen A, Partio N, Mattila V, Mäenpää H, Riuttanen A

BACKGROUND: The Injury Severity Score (ISS) and New Injury Severity Score (NISS) are widely used in evaluation of injury severity in trauma patients. The aim of this study is to assess the predictive accuracy of these scoring systems on clinical outcomes.

METHODS: We conducted a retrospective registry study of 1,112 severely injured trauma patients (NISS ≥ 16) treated at Tampere University Hospital (TAUH) between 2015 and 2024. Outcomes included in-hospital mortality, prolonged hospital and ICU stay (≥ 75th percentile), in-hospital intubation, and blood transfusions. Predictive performance was assessed using the area under the receiver operating characteristic curve (AUC) and the Hosmer-Lemeshow (H-L) test for calibration. Subgroup analyses were performed for patients with significant (AIS ≥ 3) head, thorax, and extremity injuries.

RESULTS: A total of 848 (76%) patients had a higher NISS than ISS with a median increase of 9 points. Both scoring systems generally demonstrated fair to considerable (0.7 < AUC < 0.9) discrimination for mortality and intubation, and poor to fair (0.6 < AUC < 0.8) discrimination for blood transfusions and prolonged hospital/ICU stay. ISS had superior discrimination for blood transfusions in the total cohort (AUC: 0.669 vs. 0.630, p = 0.019), head (AUC: 0.737 vs. 0.672, p = 0.004), and thorax (AUC: 0.656 vs. 0.613, p = 0.006) subgroups, and for intu […]

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Surgical and anesthetic activity in a french emergency field hospital responding to devastating floods.

Gaudray E, Brac L, Agopian P, Arnaud I, Ribelles P, Buland S, Couret A, Simonet J et al.

PURPOSE: Floods are the most frequent natural disasters, causing widespread mortality and devastation. In 2023, Storm Daniel triggered catastrophic flooding in Derna, Libya, leading to over 11,300 deaths and leaving more than 10,000 people missing. In response, the French Emergency Medical Team (EMT-FRA) ESCRIM, deployed via the European Union Civil Protection Mechanism, arrived to provide critical medical aid. METHODS: The team set up a field hospital equipped with adult and pediatric care, ICU, radiology, and laboratory services, adhering to strict aseptic conditions. Data collection was performed daily by designated physicians, with records transcribed for analysis. Over the course of the 22-day mission, the EMT-FRA provided care to over 1,600 patients, with 7.6% undergoing surgery. RESULTS: The anesthesia team performed 64 procedures, including 11 critical care assessments and 65 pre-anesthetic consultations. Surgical cases were primarily related to infections (49%), visceral issues (27%), and orthopaedic injuries (25%), including wound debridement, VAC therapy, amputations, and laparotomy. The surgical team carried out 64 surgeries and 126 consultations, with 83% of patients receiving emergency care for the first time. A significant number of procedures (78%) were dirty or infected, reflecting the local health challenges. CONCLUSIONS: The deployment of EMT-FRA demonstrated […]

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Hook plate versus locking plate fixation in lateral clavicle fractures: operative efficiency and cumulative surgical burden in a propensity score-matched cohort study.

Schibler D, Hinkelmann MA, Frei F, Pape HC, Allemann F, Barthel C

INTRODUCTION: Hook plates remain controversial in the management of lateral clavicle fractures despite their biomechanical advantages. Concerns include implant-related complications, subacromial irritation, and the need for routine implant removal. The aim of this study was to compare operative efficiency and surgical burden of hook plate fixation with locking plate fixation in lateral clavicle fractures. METHODS: A retrospective cohort analysis was conducted at a single tertiary trauma center between January 1, 2016, and December 31, 2024. Surgically treated lateral clavicle fractures were identified. Propensity score matching was performed based on age, sex, and AO fracture classification to reduce confounding, and matched pairs were compared between hook plate fixation (Group HP) and locking plate fixation (Group LCP). Outcome measures included operative time, complications, implant removal, radiographic callus formation, and refracture rates. RESULTS: A total of 101 lateral clavicle fractures were identified. After propensity score matching, 44 matched pairs (Group HP vs. Group LCP) were analyzed, with comparable baseline characteristics between groups. Operative time was significantly shorter in Group HP (77.8 ± 28.4 vs. 93.5 ± 30.2 min, p = 0.018). Complication rates were higher in Group HP but did not reach statistical significance (18.2% vs. 4.5%, p = 0.093). Implant re […]

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Diagnostic value of hypoxia-inducible factor 1 alpha and adrenomedullin in acute mesenteric ischemia.

Bayindir E, Küçükceran K, Özer MR, Ayranci MK, Kilinç ■, Kökbudak N, Bayindir GK

BACKGROUND: This study aims to evaluate the diagnostic significance of adrenomedullin and hypoxia-inducible factor 1 alpha (HIF1-alpha) levels in an animal model of acute mesenteric ischemia.

MATERIALS AND METHODS: In our study, a total of 21 male New Zealand rabbits were used, and the animals were divided into three groups for the study. Blood samples were taken for adrenomedullin and HIF1-alpha levels at 0, 1, 3, and 6 h from the control, sham, ischemia groups.

RESULTS: HIF levels of the ischemia group at the 3rd and 6th hours were statistically significantly higher than those in the sham (p = 0.002, p < 0.001, respectively) and control groups (p = 0.002, p < 0.001, respectively). Adrenomedullin levels of the ischemia group at the 1st, 3rd, and 6th hours were statistically significantly higher than those in the sham (p = 0.004, p < 0.001, p < 0.001, respectively) and control groups (p < 0.001, p < 0.001, p < 0.001 respectively).

CONCLUSION: In this experimental model of acute mesenteric ischemia, hypoxia-inducible factor-1 alpha and adrenomedullin levels were significantly higher in the ischemia group compared with the control and sham groups. Adrenomedullin demonstrated an earlier increase, while hypoxia-inducible factor-1 alpha increased significantly at later time points. These biomarkers may provide supportive diagnostic value when interpreted alongside clinical and im […]

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Videolaryngoscope-assisted fibreoptic intubation in anticipated difficult airways of maxillofacial trauma: a prospective randomised comparison with videolaryngoscope-guided nasotracheal intubation.

Jain K, Nair R, Goyal A, Makkar JK, Singla K, Arora S, Bhatia N

BACKGROUND: Maxillofacial trauma poses airway management challenges due to edema, distorted facial anatomy and associated bleeding. Further, its fixation requires unhindered access to facial fractures, often necessitating securing the airway naso-tracheally during the intraoperative period. Traditional technique of nasotracheal intubation is associated with 17%-77% incidence of epistaxis. Fibreoptic bronchoscope helps guide nasotracheal tube under vision, decreasing incidence and severity of epistaxis. Moreover, as patients with maxillofacial trauma have difficult airways, every effort should be made to secure these challenging airways with best possible technique that increases first-attempt success rate. Use of fibreoptic bronchoscope as a part of hybrid videolaryngoscope-assisted fibreoptic intubation(VAFI) technique may provide these benefits. Hence, we conceptualized this study comparing the VAFI technique with traditional videolaryngoscope-assisted technique for securing nasotracheal tube in patients with anticipated difficult airways following maxillofacial trauma. METHODS: Following a written, informed consent, we included adult patients with maxillofacial trauma, scheduled for fracture fixation under general anaesthesia, requiring nasotracheal intubation. All patients were randomised into two groups of 30 patients each, with patients in group-VLS undergoing videolaryng […]

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Malignancy risk and management outcomes in adult intussusception: a single-institution retrospective study.

Chang HC, Kang JC, Pu TW, Chen CY

PURPOSE: Adult intussusception is rare and frequently caused by a pathological lead point, including malignancy. We assessed malignancy rates by CT-based anatomical subtype/location, identified predictors of malignancy, and summarized management outcomes to support risk-stratified decision-making in the emergency setting. METHODS: We conducted a single-institution retrospective study at Tri-Service General Hospital (Taipei, Taiwan), including consecutive adults with CT-diagnosed intussusception from January 2015 to August 2024. We collected demographics, symptoms, hemoglobin, CT-based type/location, management, and histopathology. Malignancy was defined by index-admission pathology (malignant vs. benign). Anemia was defined as hemoglobin < 13 g/dL in men or < 12 g/dL in women. Predictors of malignancy were evaluated using univariable and multivariable logistic regression. Primary analyses included pathology-confirmed cases; sensitivity analyses additionally included presumed benign cases. RESULTS: Forty-seven patients were identified; 38 had pathology-confirmed outcomes. In the primary cohort (n = 38), 18 (47.4%) were malignant. Lower hemoglobin was associated with malignancy (OR per 1 g/dL increase, 0.709; 95% CI, 0.529–0.950; p = 0.021), and anemia strongly predicted malignancy (OR, 11.667; 95% CI, 2.438–55.830; p = 0.002). In multivariable analysis adjusting for age and colo […]

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Operative burden and resuscitation resource utilisation in patients with traumatic shock requiring urgent surgical or endovascular intervention.

Noonan M, O'Loan M, Hendel S, Fitzgerald M, Ban EJ, Huynh F, McKay S, Scorer A et al.

BACKGROUND: Patients with traumatic shock represent a high-risk subgroup within major trauma populations, yet the operative burden and resuscitation resource utilisation associated with urgent surgical and endovascular intervention in trauma systems managing predominantly blunt injury remain incompletely described. AIMS: To describe the epidemiology, resuscitation resource utilisation, and operative burden of patients presenting with traumatic shock who require urgent surgical or endovascular intervention. METHODS: We conducted a retrospective observational study at an Australian level 1 trauma centre using prospectively maintained trauma registries. Adult major trauma patients (≥ 16 years, Injury Severity Score ≥ 13) meeting institutional shocked trauma activation criteria between December 2022 and December 2024 were included. Resource utilisation, blood product transfusion, operative and endovascular interventions, procedural timing, and critical care outcomes were described. Comparisons were performed between shocked patients requiring urgent surgical or endovascular intervention and those managed without urgent procedures. RESULTS: Of 3667 major trauma patients, 324 (8.8%) met shocked trauma criteria, and 138 (42.6% of shocked patients) underwent urgent surgical or endovascular intervention. These patients demonstrated substantial operative complexity; 37.7% required combin […]

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Incidence of ROTEM®-defined coagulopathy at hospital arrival among polytrauma patients receiving fibrinogen concentrate in prehospital setting: a prospective observational study.

Vrbica K, Keller F, Stingl J, Vanickova K, Stepanova R, Dolecek M, Hrdy O, Gal R

PURPOSE: Trauma-induced coagulopathy affects approximately one-quarter of polytrauma patients and is associated with increased mortality. Given the central role of fibrinogen and its early depletion after injury, supplementation is recommended after hospital admission in the presence of hypofibrinogenaemia or suspected major haemorrhage. As coagulopathy develops within minutes of trauma, prehospital fibrinogen administration may help mitigate its progression before hospital arrival. This study evaluated the incidence of coagulopathy at hospital admission, assessed using ROTEM®, among polytrauma patients receiving fibrinogen concentrate in the prehospital setting. METHODS: In this single-centre, prospective observational study, adult patients with polytrauma (Injury Severity Score ≥ 16) were enrolled after prehospital administration of fibrinogen concentrate. The primary outcome was the presence of coagulopathy at hospital arrival, defined according to predefined ROTEM® Delta thresholds (FIBTEM MCF ≤ 8 mm and/or EXTEM CT ≥ 80 s and/or EXTEM MCF ≤ 49 mm). The secondary outcomes were laboratory coagulation status, 28-day all-cause mortality, and thromboembolic complications. RESULTS: Thirty patients were enrolled, of whom 28 were included in the final analysis. Twenty-seven patients received 4 g of fibrinogen concentrate and one received 2 g due to short transport time. ROTEM®-def […]

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Timing of laparoscopic cholecystectomy after percutaneous transhepatic gallbladder drainage and the risk of bailout surgery: a restricted cubic spline analysis.

Yoshitake H, Uchida K, Kataoka N, Mizobata Y

BACKGROUND: The optimal timing of laparoscopic cholecystectomy (LC) after percutaneous transhepatic gallbladder drainage (PTGBD) for acute cholecystitis remains controversial. Previous studies have often categorized surgical timing using arbitrary cutoffs and have not adequately evaluated potential non-linear associations. In particular, the interval from PTGBD to LC has rarely been analyzed as a continuous variable when assessing operative difficulty. This study aimed to assess the relationship between the interval from PTGBD to LC and intraoperative technical difficulty using restricted cubic spline (RCS) analysis. METHODS: This retrospective single-center study included patients with acute cholecystitis who underwent LC after PTGBD between November 2014 and November 2024. The primary outcome was bailout surgery (conversion to open surgery, subtotal cholecystectomy, or fundus-first approach). Secondary outcomes included operative time, intraoperative blood loss, postoperative length of stay, and major postoperative complications (Clavien–Dindo ≥ III). The interval from PTGBD to LC was analyzed as a continuous variable using multivariable regression models incorporating RCS. Models were adjusted for age, C-reactive protein (CRP) at diagnosis, Tokyo Guidelines 2018 severity grade, previous abdominal surgery, anticoagulant therapy, and Charlson Comorbidity Index. RESULTS: A tota […]

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Early orthoplastic coverage reduces fracture-related infections in Gustilo III B/C tibial fractures - a systematic review and meta-analysis.

Ruprecht N, Klingebiel FK, Hübner CT, Hax J, Teuber H, Jensen KO, Hierholzer C, Pfeifer R et al.

BACKGROUND: Gustilo-Anderson type III B/C open tibial fractures are among the most complex and high-risk injuries in trauma care. Their management demands a multidisciplinary approach, and early intervention is critical. While early debridement has been established as essential, the optimal timing for definitive orthoplastic coverage remains less clearly defined. OBJECTIVE: To evaluate whether early orthoplastic coverage reduces the risk of fracture-related infections (FRIs) and complication rates compared to late coverage in type III B/C open tibial fractures. METHODS: We conducted a PRISMA-compliant systematic review and meta-analysis of studies comparing early versus late orthoplastic treatment in patients with Gustilo-Anderson III B/C open tibial fractures. Databases included PubMed and EMBASE. Primary outcome was FRI; secondary outcomes included amputation, non-union, and revision surgery rates. Risk of bias was assessed using the Newcastle-Ottawa Scale. RESULTS: Of 3746 studies screened, five studies with a total of 454 patients met the inclusion criteria. Cutoff for early vs. late treatment was defined variably across studies (either within 72–168 h). Meta-analysis showed that early treatment significantly reduced the odds of infection (OR: 0.34; 95% CI: 0.23–0.52; $p < 0.001$) compared to late treatment. Subgroup analyses confirmed consistent benefits regardless of timing […]

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Long-term renal function after high-grade renal trauma: a nuclear imaging based cohort study.

Saxena M, Dheer Y, Roy S, Singh U, Deswal S, Kumar N, Singh A, Jaiswal V et al.

PURPOSE: Renal trauma constitutes approximately 1% of all trauma cases, yet data on long-term renal function following severe injury remain scarce. This study aimed to evaluate long-term functional and morphological outcomes in patients sustaining Grade 3 or higher renal trauma, and to identify predictors of residual renal function using nuclear imaging. METHODS: This retrospective study included patients with Grade ≥ 3 renal injuries who presented for follow-up ≥ 6 months after trauma at a tertiary care centre between January 2018 and June 2023. Patients with nephrectomy or comorbidities affecting renal function were excluded. Serum renal function tests, DTPA scans (differential GFR), and DMSA scans (cortical morphology and dye uptake) were performed. Multivariable linear regression was used to determine predictors of residual function based on injury grade, gender, and duration of follow-up. RESULTS: Thirty-six patients (mean age 20.50 ± 12.08 years; 75% male) were included with equal representation across Grades 3–5. Mean follow-up was 35.92 ± 18.09 months. The affected kidney demonstrated reduced function (mean GFR 24.96 ± 12.50 mL/min; split function 31.83 ± 12.69%; dye uptake 35.71 ± 13.40%). Grade 5 injuries were significantly associated with lower differential GFR and dye uptake ($p < 0.01$), while longer follow-up duration and female sex predicted better renal recovery ([…]

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Experience with patient-specific guides, insteatientad of general surgical experience, improves the accuracy of 3D-guided corrective osteotomies.

Buist ML, Meesters AML, Colaris JW, Doornberg JN, Ten Duis K, Tabernée Heijtmeijer SJC, Jutte PC, Pijpker PAJ et al.

PURPOSE: This study investigated whether the accuracy of 3D-guided osteotomies is influenced by surgeon experience. Two types of experience were assessed: (1) general surgical experience, defined as self-reported years in practice as an orthopaedic trauma surgeon, and (2) prior use of patient-specific guides (PSGs). Secondary aim was to evaluate whether accuracy varies by anatomical location. METHODS: 24 orthopaedic-trauma surgeons performed 75 corrective osteotomies on cadaveric long-bones (43 single-cut, 32 double-cut; 107 cuts in total). Each osteotomy was preoperatively planned on CT-derived 3D-reconstructions, and PSGs were manufactured to guide the cuts. Postoperative CT-scans were used to compare planned and executed osteotomy planes. Surgeons completed a questionnaire reporting years of practice and annual PSG usage. A Spearman's rank-order correlation was used to test differences in accuracy versus experience, a Kruskal-Wallis test was performed to explore differences across anatomical locations. RESULTS: General surgical experience ranged from 0 to 25 years (median 9), and PSG-usage from 0 to 20 cases annually (median 2). No association was found between general surgical experience and osteotomy accuracy (p = 0.406 and p = 0.548). In contrast, PSG-experience correlated with improved accuracy for angular and translational deviation of the osteotomy plane (p = 0.027 and […]

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Analysis of DRG remuneration of acute fracture procedures versus elective total arthroplasty for the German health care system - Is elective total arthroplasty really more cost-effective?

Hierl K, Kerschbaum M, Baumann F, Alt V

PURPOSE: Elective arthroplasty is considered a cost-effective procedure compared to trauma-related surgical interventions. This study aims to compare remuneration, certain ratios regarding implant costs, operation (OR) time and length of stay (LOS) as well as contribution margins including personnel costs for acute fracture interventions and elective arthroplasty procedures at the hip and shoulder joint in the aG-DRG system. METHODS: Based on billing data of 212 inpatient treatment cases from a University Orthopaedic Trauma Department reimbursements of nail fixation and hemi-arthroplasty (HA) for proximal femoral fractures versus elective total hip arthroplasty (THA) for osteoarthritis were analysed. Locking plate fixation and reverse total shoulder arthroplasty (rTSA) for proximal humeral fractures were compared to elective rTSA for osteoarthritis. Different ratios of the DRG remuneration in relation to LOS, OR time and implant costs but also personnel costs and contribution margins were calculated. RESULTS: DRG reimbursement was highest for fracture arthroplasty of the shoulder (9.050 EUR, rTSA) and the hip (6.930 EUR, HA). Regarding LOS, DRG reimbursement was highest for elective rTSA (1.641 EUR/day). Nail fixation of proximal femoral fractures showed the highest DRG reimbursement regarding OR time (111 EUR/min) and implant costs (15 EUR/EUR). The contribution margin was hig […]

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Percutaneous screw osteosynthesis for the treatment of intra-articular displaced calcaneus fractures.

Wilmsen L, Neubert A, Deichsel A, Schulz D, Hoelscher-Doht S, Windolf J, Icks A, Thelen S

PURPOSE: This systematic review investigates how percutaneous screw osteosynthesis for the treatment of intra-articular displaced calcaneus fractures differs from open reduction and internal fixation (ORIF). METHODS: Randomized controlled trials and controlled clinical trials investigating adults with closed intra-articular displaced calcaneus fracture (Sanders type II - IV) treated with percutaneous screw osteosynthesis or ORIF were included. Severe vascular and neurological diseases or polytrauma patients were excluded. On January 29, 2024, five databases and two trial registries were searched. The Risk of Bias Tool of Cochrane was used. The protocol was registered on PROSPERO (CRD42021244695). RESULTS: Five studies with 654 participants (682 calcaneus fractures) were included. The risk of bias was moderate to high. None of the single studies found a significant difference regarding severe complications. However, meta-analysis revealed a lower risk for the percutaneous screw osteosynthesis group compared to ORIF (relative risk (RR) 0.40 (95% Confidence Interval [0.17; 0.92]; I^2 = 0%; p = 0.03). One study showed less pain in the percutaneous screw osteosynthesis group up to 4 weeks postoperatively. No study assessed health-related quality of life. CONCLUSION: Percutaneous screw osteosynthesis in intra-articular displaced calcaneus fractures might be beneficial to ORIF in terms […]

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Conservative versus operative treatment of LC-1 pelvic ring fractures: radiographic and long-term functional outcomes.

Metzger F, Buschbeck S, Swartman B, Schweigkofler U

PURPOSE: To compare radiographic and long-term functional outcomes of conservatively and operatively treated traumatic lateral compression type 1 (LC-1) pelvic ring fractures, considering fracture morphology and injury severity. METHODS: This single-center observational cohort study included adult patients with traumatic LC-1 pelvic ring injuries treated between 2012 and 2020. Radiographic outcomes were assessed using the Matta–Tornetta system, and functional outcomes were evaluated at long-term follow-up using the normalized Majeed Pelvic Score and SF-12. Baseline characteristics and complications were analyzed descriptively. RESULTS: Seventy-seven patients were included (38 conservative, 39 operative). Operatively treated patients had higher injury severity and more complex fracture morphology. Despite worse initial alignment, radiographic outcomes at follow-up were similar between groups, with most patients achieving excellent or good results. Radiographic consolidation was observed in nearly all patients. Long-term functional outcomes did not differ significantly between groups. CONCLUSION: In this observational cohort with substantial baseline differences, no significant differences in long-term functional outcomes were observed between treatment strategies. These findings support an individualized, morphology-based treatment approach. However, results should be interprete […]

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Correction: Clinically relevant bleeding risk in low-energy fragility fractures of the pelvis in elderly patients.

de Herdt CL, Loggers SAI, van Embden D, Bijlsma T, Joosse P, Ponsen KJ

PubMed · DOI

A novel concept for splint fixation of midshaft clavicle fractures: a biomechanical feasibility study.

Ziegenhain F, Zderic I, Richards RG, Gueorguiev B, Groven RVM, Varga P, Hildebrand F, Pape HC et al.

BACKGROUND: Minimally invasive treatment of clavicle shaft fractures has become increasingly popular due to reduced soft tissue damage, smaller incisions, lower rates of infections, and minimal scarring. Most commonly described options are titanium elastic nailing and plate osteosynthesis. Clinical studies report promising results with good healing tendencies, particularly in children and adolescents. However, there are still some downsides to consider—titanium elastic nails have been associated with soft-tissue irritation, implant associated discomfort, and notable complications rates including lateral perforation of the cortex and medial protrusion of the implant. Implant removal after a few months is standardly carried out, leading to a second surgery with corresponding risks and costs. Furthermore, biomechanical studies report a low rotational stability for these implants. Therefore, the aim of this biomechanical feasibility study was to explore a new concept for minimally invasive splint fixation of clavicle shaft fractures in comparison to the titanium elastic nailing. METHODS: Twelve anatomical composite clavicles with standardised oblique midshaft fractures AO/OTA 15.2 A were randomized to two groups for fixation with either a 2.5 mm titanium elastic nail (Group 1) or a 5.0 mm rib splint (Group 2). Each specimen underwent (1) non-destructive quasi-static pure-bending te […]

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Development and external validation of a novel prediction model for the TraumaTriage App.

Gulickx M, Lokerman RD, van der Sluijs R, Tuinema RM, van Vliet R, Hietbrink F, Groenwold RHH, van Heijl M

PURPOSE: Accurate pre-hospital trauma triage is essential for optimizing survival and functional outcomes within inclusive trauma systems. A prediction model incorporated into the TraumaTriage App (TTApp) has been shown to improve patient allocation in the Netherlands, but was developed based on a relatively small cohort. This study aims to redevelop and improve the performance and applicability of the TTApp’s prediction model before nationwide implementation. METHODS: In this prospective multicenter cohort study, data from all trauma patients transported within two emergency medical service (EMS) regions (i.e., Brabant and Utrecht), between February 2015 and October 2019, were used to develop and externally validate a prediction model to identify severely injured adults (Injury Severity Score [ISS] ≥ 16) using routinely available pre-hospital data. A Gradient Boosting

Decision Tree (GBDT) algorithm was applied, and model performance was evaluated in terms of discrimination and calibration. RESULTS: The development cohort included 51,001 patients (median age 63.8 years, median ISS 9), and the external validation cohort included 29,737 patients (median age 62.1 years, median ISS 9). In external validation, the GBDT model showed excellent discrimination (c-statistic 0.850; 95% CI, 0.837–0.863) and good calibration (calibration-in-the-large 0.009; slope 0.952). Sensitivity was 91. […]

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Association between high ratio of platelet concentrate to red blood cell and survival among massively transfused patients.

Aoki M, Okada Y, Fujiwara G, Katsura M, Matsumoto S, Kiyozumi T, Tomura S, Matsushima K

PURPOSE: Although balanced transfusion protocols (platelet concentrate (PC): red blood cell (RBC) = 1:1) are standard for severe trauma, the impact of a higher PC: RBC ratio (> 1:1) on survival remains unclear. This study aimed to investigate the effectiveness of a high PC to RBC among severely injured patients requiring massive transfusion. METHODS: We performed a retrospective cohort study using the Japan Trauma Data Bank (2019–2023). Adult patients receiving massive transfusion (> = 10 units of RBC) within the first 24 h of injury were included. Patients were classified into high PC (> 1) and low PC groups (< = 1). The primary outcome was 24-hour mortality, analyzed using inverse probability of treatment weighting and modified Poisson regression. Additionally, we performed spline curve analysis to investigate the nonlinear relationship between the PC: RBC ratio and outcome. RESULTS: Among 3,067 patients (mean age 57.1 years, 67.2% male), 934 were in the high PC group and 2,133 in the low PC group. The high PC group had lower 24-hour mortality (8.9% vs. 15.7%). Adjusted analysis showed a significantly lower risk of mortality in the high PC group (risk ratio 0.54; 95% CI 0.43–0.69). A nonlinear analysis suggested that increasing PC: RBC ratio was associated with lower mortality. CONCLUSION: A high PC: RBC ratio was associated with lower 24-hour mortality in severely injured pa […]

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Treatment and outcomes of periprosthetic femoral fractures in patients aged 90 years or older - a retrospective bicentre cohort study with 107 patients.

Müller F, Wulbrand C, Probst A, Langenhan R

PURPOSE: The surgical treatment of periprosthetic femoral fractures (PFFs) is highly demanding, especially in very old patients. For patients aged 90 years or older, data regarding internal fixation (IF) and revision arthroplasty (RA) are lacking. METHODS: This retrospective case series was conducted in two different regional centres in one country. We enrolled patients with an age of 90 years or older, and who had PFFs with cemented or cementless standard stems. The treatment period was from 01/2010 to 12/2024. The exclusion criteria were interprosthetic femoral fractures or periprosthetic acetabular fractures. Treatment was performed with IF (locking plates) or RA (noncemented revision stems). The primary end point was any revision within the follow-up period. The secondary end points were death, serious adverse events, and function with Parker score. The minimum follow-up for living patients was one year after surgery. Living patients received a final contact by phone. RESULTS: A total of 107 patients were included in this study. The mean age of the cohort was 92.0 years (range 90 to 100), and most patients were female (n = 85). Following the Vancouver classification and its recommendations, 17 out of 22 patients with type B1 fractures were treated with IF, 44 out of 48 patients with type B2/3 fractures were treated with RA, and 34 out of 37 patients with type C fractures we […]

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The graft-reposition-on-flap technique for fingertip amputations: a patient-reported outcomes study of 111 patients.

Thushara KR, Sankaran A, Kolleri MN

PURPOSE: Graft-reposition-on-flap (GRF) is a reconstruction method described for non-replantable fingertip amputations. We hypothesised that this technique provides acceptable outcomes for such an injury, as measured by a patient-reported survey.

METHODS: This retrospective study evaluated 111 patients [118 digits], operated between January 2019 and March 2023 for Allen 3 or 4 fingertip amputations. All patients had crush avulsion amputations not fit for replantation and had undergone flap cover with grafting of the amputated nail bed and bone. 80 digits had undergone a flap with bone and nail bed graft, while 38 digits received only a nail bed graft with the flap. The flaps used were based on the defect, including local and pedicle flaps. Outcomes were assessed using an original patient-reported questionnaire administered telephonically at least 1 year after injury. The survey contained six items rated on a 1-10 Likert scale, with higher scores indicating better outcomes.

RESULTS: Nail growth was present in 105 of the 118 digits (89%). 107 patients of 111 (96%) were willing to undergo a similar procedure again if indicated. The median rated score for the Length of the finger was 9 [Interquartile range (IQR) 8-10], for Fingertip shape 8 [IQR 7-9], for Donor appearance 9 [IQR 8-10], for Nail appearance 8 [IQR 5-10], and for Sensibility 9 [IQR 7.5-10]. The median Overall Satisf […]

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Mechanical comparison of different plate-and-screw configurations for repairing forearm diaphysis fractures: a systematic review.

Zdero R, Brzozowski P, Schemitsch EH

PURPOSE: This is the first systematic review of all studies that adjusted plate and/or screw variables to determine the most mechanically stable implant configuration for forearm fracture repair. METHODS: PubMed and Scopus archives were searched using the phrase “biomechanics” AND “fracture” AND “plate” AND (“forearm” OR “radius” OR “ulna”). Eligibility criteria were applied: (i) studies that mechanically analyzed implant performance by modulating plate and/or screw variables, such as plate geometry, plate material, screw number, etc.; (ii) forearm diaphysis “shaft” fractures of the radius and/or ulna; (iii) studies published at any time; (iv) English language studies. RESULTS: There were 34 eligible papers. There was greater mechanical stability for these configurations: (i) plates used alone or double-stacked; (ii) plates that were longer, wider, thicker, straight or curved at the bottom, non-ribbed at the bottom, and/or non-grooved at the sides; (iii) plates made of material with higher elastic modulus; (iv) plates that were locking; (v) plates not elevated above bone; (vi) plates affixed to the anterior side of bone; (vii) screws that were longer, thicker, and/or numerous. There was a range of numerical outcomes: (i) interfragmentary motion; (ii) peak stress in cortical bone or plates; (iii) cortical bone stress under the plate to assess bone “stress shielding”; (iv) fatigu […]

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Postoperative complications of xenogenic dural substitutes in hemicraniectomy.

Tanrikulu L, Alhasan AA, Dömer P, Woitzik J

PURPOSE: Decompressive hemicraniectomy (DHC) is a treatment option for refractory intracranial pressure (ICP) elevation in cases of cerebral infarction, hemorrhage, and traumatic brain injury. Several surgical techniques are used for musculocutaneous, bony, and dural exposure. The method of dural closure, specifically whether to use expanding duraplasty with or without xenogenic dural substitutes, remains a topic of debate. This study focuses on the occurrence of postoperative surgical side effects associated with these approaches. METHODS: A total of 258 patients (mean age 52.8 ± 16.7 years; 140 male, 118 female, 79 infarctions, 57 traumatic brain injuries, 55 subarachnoid hemorrhages, 32 intracranial hemorrhages, 30 subdural hematomas, 5 other indications) who underwent DHC between 2015 and 2024 at our institution were reviewed. Complete clinical and radiological data were available for all patients. The patient population was divided into two groups: • Group A: 201 patients who received the xenogenic dural substitute. • Group B: 57 patients who did not receive a dural substitute. RESULTS: Baseline demographics (age, gender, underlying pathology) were comparable between both groups. No significant differences were found in the following parameters: • Duration of DHC surgery: 102.0 min (Group A) vs. 108.0 min (Group B); p = 0.24. • Postoperative CSF leakage: 3.5% (7/201) in Gr […]

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Rethinking routine chest radiography after central venous catheterisation: a rapid review of current evidence and alternatives.

Orso D, Guglielmo N, Pangallo R, Montanar V, Favaro S, Bove T

BACKGROUND: Despite growing evidence supporting ultrasound (US)- and ECG-guided protocols, chest radiography (CXR) remains the most widely used method for post-procedural confirmation of central venous catheter (CVC) position in clinical practice. METHODS: We conducted a rapid review of observational studies, randomised controlled trials, systematic reviews, and guidelines comparing CXR with US- and ECG-based confirmation modalities in adult patients across ICU, emergency, and perioperative settings. Systematic searches were performed in PubMed, Scopus, and Web of Science through December 2025. Outcomes of interest included diagnostic accuracy for tip position and complications, procedural time, and clinical and economic impact. FINDINGS: Ninety-four studies were included, comprising more than 20,000 CVC placements. Evidence from meta-analyses, multicentre cohorts, and comparative studies suggests that US-based protocols – particularly those incorporating agitated saline ("bubble test") and right-atrial visualisation – and intracavitary ECG (IC-ECG) can achieve high diagnostic performance for tip confirmation in appropriately selected patients, with several reports describing sensitivity and specificity of approximately 95% or higher. Across studies, bedside strategies enabled faster confirmation than CXR (often 15–40 min), avoided radiation exposure, and facilitated earlier de […]

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Comparison of CATCH, PECARN and CHALICE rules in children with minor head trauma.

Ferhatlar ME, Cumhur BA, Koca Y, Bozan O, Kalkan A

PURPOSE: The aim of this study is to prospectively compare the diagnostic performance of the PECARN, CATCH, and CHALICE clinical decision rules in pediatric patients presenting with minor head trauma, within a rural secondary-level emergency department characterized by limited resources, high patient volume, and restricted access to specialist physicians.

METHODS: The study was designed as a prospective observational study. The PECARN, CATCH, and CHALICE clinical decision rules were applied to the patients included in the study, and the data were recorded. The data included in the study were statistically analyzed using the SPSS (Statistical Package for Social Sciences; SPSS Inc., Chicago, IL) 27 program. A value of $p < 0.05$ was considered statistically significant.

RESULTS: A total of 1,032 patients were included in the study. The PECARN did not miss any of the 15 patients with pathological brain CT and had 100% sensitivity and 52.1% specificity. CATCH failed to detect pathological brain CT in 1 patient and had 93.3% sensitivity and 53.3% specificity. CHALICE failed to detect pathological brain CT in 2 patients, with 86.7% sensitivity and 87.8% specificity. Overall, PECARN was more successful in detecting pathological brain CT, but it also had a high false-positive brain CT rate.

CONCLUSION: PECARN criteria are more sensitive than other criteria in predicting brain CT patho […]

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Effect of head injury on blood transfusion requirements in severe trauma: a post-hoc analysis of the RESTRIC trial.

Nagasawa H, Omori K, Tanaka N, Maekawa C, Hamada M, Ota S, Ohsaka H, Yanagawa Y et al.

PURPOSE: To evaluate whether head-region injury, as defined by Abbreviated Injury Scale (AIS) coding, is associated with transfusion requirements in patients with severe trauma. METHODS: This study is a post-hoc analysis of the RESTRIC trial, a multicentre, cluster-randomised, crossover, non-inferiority trial comparing restrictive (7–9 g/dL) and liberal (10–12 g/dL) blood transfusion strategies for patients with severe haemorrhagic trauma. Adult patients with severe trauma anticipated to require transfusion were included. Head injury was defined as an AIS score ≥ 2 in the head region. The primary outcome was cumulative red blood cell (RBC) transfusion volume within 28 days. Key secondary outcomes included RBC transfusion volume within 24 and 48 h. Propensity score matching (PSM) was used to balance baseline injury severity using Injury Severity Score (ISS) and, separately, AIS scores for body regions excluding the head. Sensitivity analyses were conducted using generalised linear mixed models (GLMM) with random intercepts for study centre. RESULTS: Among the 411 patients, 157 (38.2%) had head AIS ≥ 2. In unadjusted analyses, 28-day RBC transfusion volumes were similar between patients with and without head injury (median 10 vs. 10 units). After adjustment using PSM and GLMM, no evidence of an independent association between head-region injury and RBC transfusion volume was obse […]

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Emergency laparotomies in trauma management: a six-year case series.

Ahrens G, Hoffmann M, Proksch N

INTRODUCTION: Emergency laparotomies, defined as procedures performed within 24 hours of hospital admission following trauma, are associated with considerable mortality. Notably, blunt abdominal injuries carry higher mortality rates compared to penetrating injuries. The primary aim of this study was to evaluate whether emergency trauma laparotomies performed at our Level I trauma center—serving a population with a comparatively higher proportion of penetrating injuries—achieve outcomes comparable to contemporary benchmark data. METHODS: We conducted a retrospective analysis of patients treated between 2019 and 2024 in the surgical trauma bay at Asklepios Klinik St. Georg, Hamburg, who underwent laparotomy within 24 hours of hospital admission. Mortality was defined as survival until hospital discharge. Functional outcomes were assessed using the Glasgow Outcome Scale (GOS). RESULTS: A total of 64 patients were included. Of these, 39.0% sustained penetrating trauma and 61.0% underwent laparotomy following blunt trauma. Among penetrating injuries, stab wounds predominated (92.0%), while 8.0% resulted from gunshot wounds. The most frequent mechanisms of blunt trauma were traffic accidents (30.8%) and falls from height (25.6%). Overall survival was 71.9% (mortality 20.0% for penetrating trauma; 33.3% for blunt trauma). Prehospital cardiopulmonary resuscitation (CPR) had been perfor […]

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Correction: Comparison of total intravenous anesthesia vs. inhalational anesthesia on brain relaxation, intracranial pressure, and hemodynamics in patients with acute subdural hematoma undergoing emergency craniotomy: a randomized control trial.

Preethi J, Bidkar PU, Cherian A, Dey A, Srinivasan S, Adinarayanan S, Ramesh AS

PubMed · DOI

Non-scaphoid carpal fractures: lessons learned from a series of 86 cases.

Herrero Alonso S, Fernández González-Cuevas J

PURPOSE: To describe the epidemiology, injury mechanisms, diagnostic approach, management, and clinical outcomes of non-scaphoid carpal fractures, which are frequently overlooked or diagnosed late. METHODS: A retrospective review was conducted of patients with non-scaphoid carpal fractures treated at our institution between 2013 and 2020. Demographic data, fracture characteristics, time to diagnosis, treatment strategy, rehabilitation requirements, and clinical outcomes, including complications, were analyzed. RESULTS: 76 patients with 86 fractures were included; 54 (71%) were male. The most commonly affected bones were the triquetrum (37%) and hamate (33%). Isolated fractures accounted for 89% of cases. Advanced imaging was frequently required, with CT performed in 53% and MRI in 7% of patients. The median diagnostic delay was 6 days (range: 1–40). Conservative treatment was used in 84% of cases. Surgical management was significantly more frequent in lunate (75%) and hamate fractures (25%) ($p = 0.008$). Surgically treated patients required rehabilitation more often than those managed conservatively (58% vs. 31%, $p = 0.012$). Functional range of motion (> 40° flexion/extension) was achieved in 97% of patients. However, 16% developed chronic wrist pain, 8% reported prolonged soreness, and one patient developed reflex sympathetic dystrophy. CONCLUSIONS: Although functional outcomes […]

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A machine-learning-based tool for stroke risk prediction in blunt cerebrovascular injury: development and preliminary evaluation.

Wagner V, Preston JD, De Leon Castro A, Mueller WF, Nguyen J, Garcia-Toca M, Benjamin ER, Todd SR et al.

PURPOSE: Stroke risk correlates with the Biffl grading system in blunt cerebrovascular injury (BCVI). Although anti-thrombotic therapy is the mainstay of stroke prevention, no point-of-care clinical decision-support tool exists to guide timing for therapy. We sought to develop an interactive online calculator that incorporates patient-specific demographic and injury characteristics to estimate stroke risk and risk reduction with anti-thrombotic (AT) administration.

METHODS: Data from BCVI patients ($n = 1,197$) at a Level I Trauma Center were retrospectively collected. Six machine learning methods were employed to predict stroke risk with and without AT therapy. Class imbalance was addressed using downsampling and/or class weighting. Model performance was assessed using 10-fold cross-validation. The model was implemented as an R-based Shiny online application.

RESULTS: Stroke rate among the population was 4%, and the strongest predictors for stroke were the greatest Biffl grade of carotid (aOR [95%CI] = 2.02 [1.62-2.53]) and vertebral injuries (1.44 [1.18-1.77]). The least absolute shrinkage and selection operator (LASSO) model outperformed all others, achieving 66% [33%-100%] sensitivity and 74% [62%-82%] specificity for stroke prediction, with an area under the receiver operating characteristic curve of 0.79 [0.57-0.95]. This model was integrated into an interactive online to […]

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Association between screw angulation and union after inverted-triangle cannulated screw fixation of femoral neck fractures.

Çatal T, Bayrak A, T■ng■r M, Sapmaz ■E, K■l■çkaya ME, Mercan N, Kural C, Duramaz A

PURPOSE: The aim of this study was to evaluate whether screw angulation relative to the inter-ischial(horizontal) axis and femoral anatomical axis was associated with union after inverted-triangle cannulated screw fixation of femoral neck fractures and to explore angle intervals associated with higher union rates. METHODS: Between January 2017 and December 2022, 83 patients operated on with closed reduction-percutaneous cannulated screws by the same surgical team were retrospectively analyzed. The angles of the calcar, anterosuperior, and posterosuperior screws were measured according to both the femoral anatomical axis and the horizontal axis. Multivariable logistic regression analyses adjusted for lateral angulation and Garden or Pauwels classification were additionally performed for the horizontal-axis screw-angle parameters. Exploratory angle intervals were assessed using quartile and sliding-window analyses. RESULTS: A total of 64 patients achieved union, with a mean union time of 8.69 ± 3.03 weeks. The nonunion group had higher lateral angulation and greater postoperative neck length($p = 0.006$, $p = 0.042$, respectively). In angle values relative to the horizontal axis, calcar and anterosuperior screws exhibited higher measurements in the union group($p = 0.027$, $p = 0.013$, respectively). In exploratory sliding-window analyses, high union intervals were identified as follows: […]

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Response to the letter to the editor to discuss "Factors to consider when selecting an injured control group in case-control studies on skiing injuries".

Rutsch N, Gewiess J, Albers CE, Tinner C

PubMed · DOI

A study on prolonged use of resuscitative endovascular balloon of the aorta (REBOA) with shunt flow in a porcine hemorrhagic shock model.

Kiriu N, Saitoh D, Yamamura K, Sasa R, Kiyozumi T, Tomura S

PURPOSE: To evaluate the future efficacy and prolonged availability of double-balloon resuscitative endovascular balloon occlusion of the aorta (DB-REBOA) to determine whether bypassing upstream blood at the occlusion site improves survival in a porcine hemorrhagic shock model. METHODS: Twenty-five pigs (approximately 40 kg, estimated circulating blood volume 25%) were subjected to hemorrhagic shock by controlled phlebotomy. Animals were treated with standard REBOA (control, $n = 12$), internal shunt DB-REBOA (internal shunt, $n = 4$; an intraluminal conduit was set parallel to the DBREBOA in the aorta), or external shunt DB-REBOA (external shunt, $n = 9$; upstream blood from the occlusion site was shunted by connecting the internal carotid and femoral arteries with a pressure-resistant tube). Zone 1 aortic occlusion was maintained for 90 min, followed by balloon deflation. Survival, systolic blood pressure, and serum lactate levels were analyzed. RESULTS: Ten of 12 control animals developed cardiac arrest shortly after balloon deflation. All animals in the internal shunt group survived, whereas 5 of 9 animals in the external shunt group experienced cardiac arrest. Internal shunt DB-REBOA maintained stable systolic BP during prolonged occlusion, resulting in significantly lower serum lactate levels at 90 min compared to controls. External shunt DB-REBOA resulted in unstable hemodynam […]

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Checklist- and algorithm-based simulation training produce comparable improvements in teamwork-related shared mental models: a cluster-randomized study in trauma resuscitation teams.

Beaugé Y, Kusch S, Thorsson L, Zentgraf M, Schauwienold J, Muenzberg M, Verboket R, Lefering R et al.

PURPOSE: Shared mental models (SMMs) reflect the extent to which team members hold a similar understanding of roles and team processes. In trauma resuscitation, higher SMM alignment is considered a prerequisite for coordinated interprofessional teamwork. This study compared the effects of checklist-based versus algorithm-based in situ simulation training on teamwork-related SMMs in trauma room teams. METHODS: In this multicenter, cluster-randomized pre–post study, 29 interprofessional trauma teams (186 participants) from six hospitals received either checklist-based training (n = 15 teams) incorporating the Trauma Room Manual checklists or algorithm-based training (n = 14 teams), both grounded in ATLS principles. Teams completed an SMM questionnaire adapted to the trauma setting, based on a previously published instrument, assessing similarity in (1) task responsibility and (2) team communication. Team-level similarity scores were calculated pre- and post-training. Training effects were analyzed using mixed repeated-measures ANOVA. RESULTS: Across both training formats, teamwork-related shared mental models increased significantly from pre- to post-training in total similarity, task responsibility, and communication. No significant between-group differences or time × group interactions were observed. The interaction for task responsibility approached significance (p =.06; $\eta^2 p$ = […]

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The accuracy of the injury severity score in patients below the age of 18 years, a trauma registry data study.

Edwards BHJ, van Heumen JJP, Nelen SD, Edwards MJR, Jaarsma RL, de Blaauw I, Hermans E

PURPOSE: The Injury Severity Score (ISS) is an anatomical score that estimates both severity of trauma and risk of mortality in adults. Although widely used, its applicability in pediatric trauma remains uncertain. This retrospective cohort study evaluated the predictive value of ISS for pediatric mortality and clinically relevant outcomes. METHODS: Patients (< 18 years) treated at a level-1 trauma centre with traumatic injury between 2015 and 2023 were included in the Dutch National Trauma Registry. Receiver operating characteristic (ROC) analyses were used to assess the discriminative performance of ISS for mortality, prehospital advanced care, emergency interventions, imaging, length of stay, surgical burden, and functional outcome. RESULTS: 1733 patients were included. ISS showed excellent discrimination for mortality (AUC 0.97) and good performance for emergency interventions, prehospital advanced care, functional outcome, and admission to higher levels of care, but limited value for predicting number of surgeries. Optimal thresholds for ISS were 16 or higher, except for CT imaging (ISS ≥ 10). CONCLUSION: ISS reliably predicts mortality and several clinically relevant outcomes in children. While ISS ≥ 16 remains appropriate for identifying severe injury, optimal thresholds vary by outcome. These findings emphasize the need for outcome-specific interpretation of ISS, for be […]

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Epidemiology of acute ice-hockey-related orthopedic fractures in high school- and college-aged players in the United States from 2006-2023.

Reiad T, Dinh P, Khan H, Bruni D, Chang E, Owens B, Marcaccio S

BACKGROUND: This investigation quantified the incidence and injury patterns of acute orthopedic fractures attributed to ice hockey among high-school and college-aged athletes in the United States. METHODS: We performed a retrospective analysis of the National Electronic Injury Surveillance System (NEISS) database focusing on ice hockey fractures in patients aged 14–23 years from 2006 to 2023. The cohort was stratified into high-school-aged (14–18) and college-aged (19–23) groups. Weighted national estimates were calculated to evaluate anatomical sites, fracture subtypes, and hospitalization rates. RESULTS: An estimated 24,350 ice hockey-related fractures occurred in the study population. Males accounted for 94.8% of cases. Upper extremity and trunk fractures comprised 81.6% of all fractures, with shoulder (27.0%), wrist (19.3%), and lower arm (11.3%) being the most common sites. In comparison to high-school-aged players, college-aged athletes were significantly more likely to sustain hand fractures (OR 2.82; p = 0.001) and lower leg fractures (OR 3.67; p = 0.042), while less likely to sustain lower arm fractures (OR 0.31; p = 0.039). Hospitalization was required for 4.2% of fractures overall, with the neck (82.6%) and upper leg (76.3%) having the highest admission rates, and several areas like the elbow, hand, finger, and knee having admission rates < 1%. Temporal analysis demo […]

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Incidence, mortality, and factors associated with fat embolism syndrome in patients with long bone fractures at a trauma referral center in Bogotá, Colombia: 2016-2019 and 2022-2023.

Bernal OYG, Lozano NRC, Cordero JFB, Tenjo EAA, Lazaro JS, Acosta DCB, Aguilera CSQ, Valero JDG et al.

PURPOSE: To estimate the incidence, mortality, and factors associated with the development of fat embolism syndrome (FES) in patients with long bone fractures treated at a trauma referral center in Bogotá, D.C., Colombia. METHODS: Retrospective cohort study including patients >18 years with diaphyseal fractures of humerus, femur, tibia, or fibula, treated between 2016-2019 and 2022-2023. COVID-19 pandemic years (2020-2021) were excluded. Statistical analyses included univariate, bivariate, and multivariate logistic regression. RESULTS: Among 3,475 patients with long bone fractures, FES incidence was 4.3% (n=148) with 5.4% in-hospital mortality. Multivariable analysis identified femoral fractures as the strongest risk factor (aOR=4.63; 95%CI:2.99-7.22; p<0.001), while early surgical fixation (<24h) demonstrated significant protection (aOR=0.42; 95%CI:0.25-0.71; p<0.001). Obesity independently increased FES risk (aOR=2.63; 95%CI:1.33-4.93; p=0.004). Most cases (97%) resulted from high-energy trauma, predominantly motorcycle accidents. Respiratory compromise was severe, with 93.9% of FES patients requiring advanced ventilatory support versus 16.8% in controls (p<0.001). CONCLUSION: This study demonstrates that early surgical fixation significantly reduces FES risk and identifies humeral fractures as an underrecognized risk factor. These findings support immediate implementation of […]

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Association between fracture location and one-year mortality in patients aged 65 years and older after lower extremity trauma: a retrospective cohort study.

Helen A, Michelle Z, Christoph H, Thomas L

PURPOSE: Fractures in older patients are associated with increased mortality. Our study evaluated 30-day and one-year mortality among older patients based on fracture location. METHODS: A descriptive retrospective cohort analysis was conducted including all patients aged 65 years or older hospitalized at a level I trauma center with a pelvic or lower extremity fracture between January 1, 2016 and December 31, 2019. Mortality rates were compared with age- and gender-matched data from the Swiss general population. In addition, an exploratory multivariable logistic regression analysis was performed to assess the association between fracture location and one-year mortality, adjusting for age and sex. RESULTS: A total of 1,739 patients were included. The overall 30-day mortality rate was 12%, and the one-year mortality rate was 34%. In comparison, the age- and gender-matched one-year mortality rate of the general Swiss population was 8% (p < 0.001). Unadjusted one-year mortality was highest after femur fractures (41%), followed by pelvic fractures (33%) and lower leg or foot fractures (11%). Among specific fracture types, pertrochanteric femur fractures showed the highest one-year mortality (47%), followed by Tile type C pelvic fractures (43%), distal femur fractures (40%), and femoral neck fractures (39%). In contrast, patients with fractures of the lower leg or foot had a one-year […]

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Proximal femoral replacement for oncologic and non-oncologic indications: a retrospective study over a 13-year period.

Ramón R, Gökmen A, Mayor J, Özbek H, Niemann A, Marín L, Omar Pacha T, Omar M et al.

BACKGROUND: Proximal femoral replacement (PFR) is a critical limb-salvage strategy for managing massive proximal femoral bone loss due to oncologic and non-oncologic conditions. Although oncologic outcomes are well described, evidence surrounding PFR for infection, periprosthetic fracture, and mechanical failure remains limited. This study evaluates short- to mid-term outcomes following PFR for multiple indications and compares primary with revision procedures. METHODS: We retrospectively reviewed all patients undergoing PFR at a Level I trauma center from 2009 to 2023. Cases were categorized by procedure type (primary vs. revision) and indication (tumor, infection, other causes). Outcomes included operative time, complications according to the Henderson classification, reoperation rates, and functional outcomes. RESULTS: A total of 81 patients (84 procedures) were included with a median follow-up of 7 months (IQR 0–31); 57.1% were lost to follow-up. The overall complication rate was 42.9%, significantly higher in revision procedures than in primary PFRs. Infection (Henderson Type IV) was the most frequent complication and occurred predominantly in revision surgeries. No cases of aseptic loosening (Type II) were identified. Reoperation was required in 29% of all procedures, with the highest rates observed in infection cases. CONCLUSIONS: PFR remains a reliable limb-salvage opti […]

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Interleukin-6 to identify mildly injured patients in the trauma resuscitation room - a clinical feasibility study.

Bewersdorf TN, Wimmer M, Popp E, Streblow J, Baumann L, Leowardi C, Stein S, Schmidmaier G et al.

PURPOSE: Overtriage in the trauma resuscitation room (TRR) is often a consequence of admission based solely on trauma mechanism criteria. As interleukin-6 (IL-6) can assess injury severity in critically injured patients, the hypothesis of this study was that IL-6 can differentiate upon admission between mild (Injury Severity Score (ISS) 0-8) and moderate (ISS 9-15) trauma and that the parameter correlates with injury severity in lower injury severity groups (ISS < 16) during the first 24 h. Subsequently, we evaluated the ability to define IL-6 cut-off values with an adequate sensitivity, specificity, and negative predictive value to distinguish between moderate and mild injuries.

METHODS: Fifty patients admitted to the TRR solely based on trauma mechanism criteria and with an ISS < 16 were included in the study. IL-6 was measured at admission, 1, 6, 12 and 24 h later.

RESULTS: Thirty-one patients were mildly injured (ISS 0-8), while nineteen patients sustained moderate injuries (ISS 9-15). IL-6 correlated significantly with ISS at all time points. Moderately injured patients showed significantly higher IL-6 levels than mildly injured patients during the first 6 h. IL-6 levels at admission predicted moderate injuries for the IL-6 cut-off value of 2.9pg/ml with a sensitivity of 0.933, specificity of 0.370 and a negative predictive value of 0.970 resulting in undertriage rates o […]

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Attenuating pulmonary injury and inflammation with C5/CD14 inhibition therapy: results from a porcine polytrauma model with blunt chest trauma.

Mert Ü, Groven RVM, Qin K, Greven J, Balmayor ER, Mollnes TE, Huber-Lang M, van Griensven M et al.

BACKGROUND: Thoracic injury is prevalent among polytrauma patients, affecting up to 45% of these patients and often leading to complications such as pulmonary dysfunction up to acute respiratory distress syndrome (ARDS). This study investigates the impact of surgical invasiveness and C5/CD14 inhibition therapy on pulmonary tissue damage and the local immune response of the lungs in a porcine polytrauma model. METHODS: A post hoc analysis focusing on the pulmonary consequences of blunt thoracic trauma and multiple trauma was performed in a previously published porcine trauma model, in presence or absence of immunomodulation. A control group (n = 6; CG) was used that received identical instrumentation, anesthesia, nutrition, and fluid intake, but no trauma. For the trauma model, male pigs underwent a standardized polytrauma, followed by allocation into three treatment groups: early total care (n = 8; ETC), damage control orthopedics (n = 8; DCO), and ETC with C5/CD14 inhibition (n = 4). Pulmonary histopathological changes were assessed at 72 h after trauma through lung injury scores and wet/dry ratios, while ex vivo cytokine expressions from isolated alveolar macrophages (AMs) were analyzed using enzyme-linked immunosorbent assays. RESULTS: Histological results revealed significant lung damage in both the ETC and DCO groups compared to the control as well as the C5/CD14 inhibitio […]

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Hemodynamic rescue after blast injury: intravenous administration outperforms intramuscular vasopressin in a swine model.

Renberg M, Gustavsson J, Rehn M, Günther M

BACKGROUND: Hemorrhage remains the leading preventable cause of trauma mortality. Blast trauma combines primary blast effects with complex secondary and tertiary injuries, precipitating profound hemodynamic instability and microcirculatory disruption. While intramuscular (IM) arginine vasopressin (AVP) stabilizes arterial pressure in isolated hemorrhagic shock, vasopressin's efficacy in blast-associated hemorrhagic shock is unknown. METHODS: In a randomized, blinded study, 22 swine underwent femoral blast injury and controlled class II hemorrhage. Animals received IM AVP 40 U (two doses 60 min apart due to short half-life and transient effects; n = 5), IM terlipressin 2 mg (n = 5), intravenous (IV) terlipressin 2 mg (n = 5), or saline control (n = 7). All subjects received a 500 mL autologous whole blood transfusion and were observed for 120 min. The primary outcome was systolic arterial pressure (SAP). Secondary outcomes included systemic vascular resistance index (SVRI), cardiopulmonary and metabolic variables, and serum AVP to assess IM uptake. RESULTS: IV terlipressin rapidly stabilized hemodynamics, increasing SAP (mean difference 24 mmHg, p = 0.01) and SVRI (mean difference 44593 dynes·s·cm■■·kg, p = 0.004) versus controls, and increased mixed venous oxygen saturation and urine output. Respiratory and metabolic variables were similar across groups. Neither IM AVP nor IM t […]

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Prevalence and outcomes of frailty in emergency laparotomy: a single-centre cohort study.

Rival PM, Bellgrove C, Ejaz MU, Van Weel JM, Stokes MAR, Pilgrim CHC

PURPOSE: Frailty is common among patients undergoing emergency laparotomy and is associated with adverse postoperative outcomes, yet routine frailty assessment remains inconsistently implemented despite international guideline recommendations. This study evaluates the prevalence of frailty using three rapid assessment tools and examines their associations with postoperative outcomes following emergency laparotomy. METHODS: We conducted a single-centre retrospective cohort study of adults undergoing open emergency laparotomy over a 12-month period. Frailty was assessed retrospectively using the Clinical Frailty Scale (CFS ≥ 5), Emergency Surgery Frailty Index (EmSFI ≥ 7), and five-item Modified Frailty Index (mFI-5 ≥ 2). Primary outcomes were 30- and 90-day mortality. Secondary outcomes included postoperative complications, ICU admission, hospital length of stay, and discharge destination. Unadjusted analyses were descriptive. Multivariable logistic regression models were constructed with frailty measures specified as the primary exposures and adjusted for age, sex, operative indication, and anaesthetist-assigned ASA grade. RESULTS: Among 102 patients (median age 67 years, IQR 54–79; 53% female), frailty prevalence was 26% by CFS, 24% by EmSFI, and 27% by mFI-5, with 37% meeting at least one frailty threshold. In unadjusted analyses, patients living with frailty experienced high […]

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Timing of surgery and predictors of canal reduction after short-segment fixation for thoracolumbar burst fractures.

Aono H, Takenaka S, Okuda A, Kikuchi T, Takeshita H, Nagata K, Ito Y

PURPOSE: In patients with thoracolumbar burst fractures, surgical delay is often unavoidable due to concomitant injuries. Concerns persist that delayed surgery may compromise indirect reduction of retropulsed fragments via ligamentotaxis. This study aimed to determine whether the timing of surgery affects the quality of intracanal fragment reduction and identify independent predictors of successful reduction following short-segment posterior fixation. METHODS: This multicenter study included 252 patients who sustained a single traumatic thoracolumbar burst fracture and underwent short-segment fixation. The canal compromise ratio (CCR) was measured on pre- and postoperative CT scans, and the reduction rate was calculated using this formula: Reduction rate = (Preoperative CCR – Postoperative CCR) / Preoperative CCR × 100. Multiple linear regression analysis identified predictors of reduction quality including age, sex, affected level, surgical timing, AO classification, load-sharing score (LSS), preoperative CCR, and concomitant vertebroplasty, RESULTS: The study included 141 males and 111 females. The mean time from injury to surgery was 3.8 days. The mean CCR improved from 45% to 24%, yielding a mean reduction rate of 44%. Multiple linear regression analysis revealed that T11 to L1 and AO type B fractures were associated with a significantly higher reduction rate. Notably, the […]

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Outcomes following implementation of a structured diagnostic and treatment approach for combat abdominal trauma in a hybrid war setting: a retrospective cohort study.

Oleh H, Cerkauskaite D, Kashtalian M, Yaroslav H, Kvasnevskyi I, Okolets A, Dulskas A

OBJECTIVE: This study evaluated whether implementation of an optimized diagnostic and treatment protocol improved early and intermediate outcomes among combat casualties.

BACKGROUND: High-energy ballistic and blast mechanisms during the war in Ukraine caused a surge of complex abdominal injuries requiring intensive resource utilization and rapid coordination of surgical, critical care, and evacuation capabilities.

METHODS: A retrospective cohort study was conducted including 496 male soldiers with combat-related abdominal injuries treated at military hospitals from 2014 to 2017. Patients were stratified into pre-implementation (n = 161) and post-implementation (n = 335) groups. The structured protocol included perfusion index-guided triage, expanded use of FAST, diagnostic laparoscopy, and standardized damage control surgery principles. Primary outcomes were postoperative complications, in-hospital mortality, and return-to-duty rates; secondary outcomes included missed intra-abdominal injuries and hospital length of stay.

RESULTS: After protocol implementation, FAST utilization increased (68.1% vs. 19.3%) and diagnostic laparoscopy increased (30.7% vs. 14.9%), while the rate of missed intra-abdominal injuries decreased from 7.5%

to 3.6% ($p = 0.04$). Median decision-to-surgery time shortened by 26 min ($p = 0.001$). Postoperative complications declined (39.1% vs. 27.8%, $p < 0.05$ […]

PubMed · DOI

Association between preoperative NT-proBNP levels and mortality among patients undergoing acute high-risk abdominal surgery: a prospective cohort study.

Fattori S, Kanstrup CTB, Serup CM, Kleif J, Lundstrøm LH, Bertelsen CA

PubMed · DOI

Systemic injury extension and mechanism of trauma determine ICU admission in CT-confirmed maxillofacial fractures.

Usta T, Bayramlar OF, Yüce S, Demirtakan T, Sert E, Fettaho■lu S, ■ncealt■n O

PURPOSE: To evaluate whether intensive care unit (ICU) admission in CT-confirmed maxillofacial trauma is primarily associated with facial fracture burden or with radiologically defined systemic injury extension. METHODS: This retrospective cohort study included 1,511 patients with CT-confirmed maxillofacial fractures treated at a tertiary trauma center. Injury mechanism, fracture multiplicity (mFISS-face), and systemic extension (cranial and thoracic involvement) were assessed. Multivariable Firth penalized logistic regression was performed to identify independent predictors of ICU admission. Model discrimination, calibration, and internal validation were evaluated using AUC, calibration metrics, Brier score, and bootstrap resampling. RESULTS: ICU admission occurred in 37 patients (2.45%). In multivariable analysis, age (aOR 1.04 per year, $p < 0.001$), traffic-related mechanism (aOR 2.74, $p = 0.038$), thoracic injury (aOR 25.08, $p < 0.001$), and cranial extension (aOR 98.24, $p < 0.001$) were independently associated with ICU admission. Fracture multiplicity demonstrated borderline significance (aOR 1.32, $p = 0.071$). The model showed good discrimination (optimism-corrected AUC 0.874), acceptable calibration (calibration slope 0.98), and low overall prediction error (Brier score 0.016). CONCLUSION: In CT-confirmed maxillofacial trauma, ICU admission is predominantly associated with s […]

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Better & systematic trauma care (BEST) in Norway: from a local quality improvement project to a national collaborative.

Brattebø G, Wisborg T

PURPOSE: Healthcare professionals in Norwegian local hospitals have focused on the need for improved trauma care since the late 1990s. Because of lack of national guidelines the hospital management of patients with serious trauma varied. The authors and colleague health care providers felt a need for improvement. Therefore, a project called Better & Systematic Trauma care (BEST) was established. The goal was to improve trauma care through training and preparation for infrequently occurring situations, with a focus on the team. METHODS: The project began as a bottom-up initiative in four local hospitals. Through cross-professional team training and local simulation, staff built competence in trauma care and practised non-technical skills such as communication, cooperation, leadership, and decision-making in their local setting. Creating a national meeting place for all hospitals was essential for fostering collaboration across institutions. RESULTS: A 1-day team course was developed and adopted nationwide. Network meetings have enabled the collaborative development of tools and training materials without national strategic leadership or external influence. Over time, the focus shifted from individual professions to the trauma team. A damage-control surgery course was created for all surgical teams. All materials remained freely available, and the project operated independently o […]

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Conservative management of 900 pediatric distal radius fractures using the schede cast: a multicenter study.

Scherer S, Schultz J, Rausch T, Schoof B, Altmeier L, Hertel J, Sahin M, Esser M et al.

BACKGROUND: Distal forearm fractures are among the most common pediatric injuries and are typically managed with cast immobilization. Volar-flexion ulnar-deviation splints (Schede casts) have been proposed to reduce displacement and prevent secondary dislocation. However, data on efficacy and safety in children remain limited. This study aimed to evaluate the clinical outcomes of Schede cast immobilization across four pediatric trauma centres over a 16-year period. METHODS: We conducted a retrospective analysis of patients under 18 years with distal forearm fractures treated with Schede casts. Demographic data, fracture characteristics, initial and post-cast displacement, incidence of secondary dislocation, and complications were recorded and analyzed. RESULTS: A total of 900 patients (mean age 9.47 years) were included. The most common mechanism of injury was sports-related trauma, and transverse metaphyseal fractures predominated. Mean initial displacement was 17.7° (± 9.4°), reduced to 5.9° (± 5.5°) after cast application and 6.9° (± 5.4°) at consolidation after a mean of 35 (± 16) days. Secondary dislocations were effectively prevented at flexion angles > 50° ($p < 0.001$). Complications were rare: tingling paresthesia occurred in 22 patients (2.4%), and prolonged movement restriction in 8 patients (0.9%). All adverse events resolved within days to weeks without long-term seq […]

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Publisher Correction: Clinical outcomes and quality of life of patients after surgical treatment of a tibia plateau fracture.

Höller LF, Höller S, Klinger E, Henkies D, Leha A, Lehmann W, Dobroniak CC, Hoffmann DB

PubMed · DOI

Does the Hertel criteria correlate with postoperative avascular necrosis and failure in proximal humerus fractures? A retrospective cohort study.

Ismailov S, Bacaksiz T, Maden M, Akan I, Onder Y, Kazimoglu C

PURPOSE: Avascular necrosis and failure are significant complications following surgical fixation of proximal humerus fractures. The Hertel criteria are a widely used morphological classification system intended to predict humeral head ischaemia. This study aimed to evaluate the correlation of Hertel criteria with the development of postoperative avascular necrosis and failure and to assess the prognostic value of other radiological parameters. MATERIALS AND METHODS: In this retrospective cohort study, 94 patients with proximal humerus fractures treated with locking plate fixation and a minimum 2-year follow-up were included. Preoperative radiographs and computed tomography scans were used to assess Hertel criteria, Neer classification, and proximal humeral cortical thickness. Postoperative radiographs were evaluated for avascular necrosis and failure, defined as a composite of nonunion, severe avascular necrosis, screw penetration, severe deformity, or significant arthrosis. Clinical outcomes were assessed using the Visual Analog Scale and Disability of Arm, Shoulder and Hand scores. RESULTS: The presence of an anatomical neck fracture and tubercle displacement greater than 10 mm were significantly associated with both avascular necrosis and failure ($p < 0.05$). In contrast, other factors in the classical Hertel criteria, such as metaphyseal head extension less than 8 mm, media […]

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Learning gain of an ATLS®-based interprofessional and multidisciplinary in-situ simulation training of trauma resuscitation.

Hammer S, Lock JF, König S, Happel O, Paul MM

PURPOSE: Team performance in polytrauma management determines patient outcome and is crucially shaped by Crew Resource Management (CRM). This study explored patterns of self-perceived learning following an interdisciplinary, interprofessional, in-situ, simulation- and ATLS®-based trauma team training with a focus on CRM principles. METHODS: We conducted a retrospective analysis of routine educational data collected immediately after 36 training sessions (03/2022–11/2023) including 238 participants at a single Level I trauma center. Participants completed a retrospective pre-test/post-test self-assessment questionnaire. Exploratory analysis, including EFA, were used to descriptively structure self-perceived learning patterns and subgroup differences. RESULTS: Participants came from anesthesiology, surgery and radiology in equal proportions and differed in working experience, professional role, and exposure to polytrauma management. Exploratory factor analysis identified three descriptive dimensions of self-assessed CRM-related concepts: (i) personal operational competence, (ii) team communication, and (iii) decision making. Patterns of self-perceived change varied by experience level, with larger reported gains in personal operational competence among less experienced participants. CONCLUSION: This exploratory analysis describes patterns of self-perceived learning following in-s […]

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Multifaceted intestinal defense following experimental blunt abdominal trauma.

Meisen S, Rayatdoost F, Oßwald K, Palmer A, Breunig M, Huber-Lang M, Halbgebauer R

PURPOSE: Blunt abdominal trauma (AT) can cause significant intestinal injury and act as a trigger for remote posttraumatic organ dysfunction, including the development of multi-organ dysfunction syndrome (MODS). A deeper understanding of its impact on intestinal barrier integrity and immune regulation is essential for developing therapeutic strategies that support posttraumatic recovery. METHODS: We assessed jejunal tissue 24 hours following a standardized murine blast-induced direct blunt AT. Histological, immunohistochemical, proteome profiling, Western Blot, and molecular analyses were applied to assess epithelial integrity, barrier-associated proteins, intestinal mucus composition, and immune responses. RESULTS: The jejunum showed no trauma-specific histomorphological changes. Consistently, the expression of key epithelial barrier proteins, such as E-cadherin and tight-junction protein 1, remained stable or were even upregulated. Local inflammatory responses were evident; AT induced activation of defense-related pathways, accompanied by a marked increase in S100A8 expression and enhanced infiltration of CD3+ T cells. Mucus-covered areas of the intestinal surface were reduced, although transcriptional levels of MUC2 and C1GALT1 were elevated. In addition, the antimicrobial peptide CRAMP was upregulated in jejunal tissue following injury. CONCLUSIONS: These findings suggest t […]

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From surgery to rehabilitation: a cross-sectional survey on the feasibility of live digital ward rounds between acute care and rehabilitation clinics in trauma patients.

Schreiber L, Halvachizadeh S, Schäfer FP, Tschui F, Sturzenegger C, Johannes S, Pape HC, Heining SM

INTRODUCTION: Telemedicine is increasingly embraced by both patients and healthcare professionals. By bridging the gap between acute care hospitals and rehabilitation clinics, it offers a promising approach to enhance communication and continuity of care. This study aimed to assess the feasibility of telemedicine in the rehabilitation of patients with musculoskeletal injuries or polytrauma. METHODS: A cross-sectional survey was conducted involving healthcare professionals engaged in the telemedical management of patients with musculoskeletal injuries or polytrauma during acute care and rehabilitation clinics. We established digital ward rounds between an acute care hospital and rehabilitation clinics with live patient presentation. A self-developed questionnaire assessed four domains: overall satisfaction, data security, technical setup, and the rehabilitation process. Each item was rated using a 5-point Likert scale. Participation was voluntary and anonymous. RESULTS: A total of 39 participants completed the survey, with 23 respondents (59%) from a rehabilitation institute and 14 (41%) from an acute care hospital. The majority were physicians (n = 32, 82.1%), with the most common age group being 50–60 years (n = 12, 30.8%). Overall satisfaction with the telemedicine process was high, with 31 participants rating their experience as either 4 or 5 (very satisfied). Only one respo […]

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Health-Related Social Needs Are Associated With Worse Physical Function, Pain, and Mobility in Hip and Knee Osteoarthritis Patients at Presentation.

Winter AD, Hayashi CL, Túa-Vargas LA, Chau N, Gibson DH, Rubin LE, Molloy IB, Wiznia DH

BACKGROUND: Health-related social needs (HRSNs) are mandatory reporting measures per the Centers for Medicare & Medicaid Services. We sought to assess the prevalence of HRSNs and associations with clinical presentation among preoperative hip and knee osteoarthritis patients.

METHODS: This prospective single-institution cross-sectional study assessed HRSNs (living situation, food, transportation, utilities, safety, financial strain, substance use, and mental health); Patient-Reported Outcomes Measurement Information System (PROMIS) Physical Function, Mobility, Pain Interference scores; and demographics (sex, race, education, income, household size, marital status) via in-person interviews in an academic arthroplasty practice. Kellgren-Lawrence Osteoarthritis Score, Charlson Comorbidity Index, and insurance were also determined. Odds of HRSNs by demographics were analyzed using logistic regression and relationship between HRSNs and PROMIS scores was determined by weighted least square linear regression.

RESULTS: The study included 253 patients. Prevalences of each HRSN were 15% for living situation, 13% for food insecurity, 11% for transportation, 4% for utilities, 0.4% for safety, 58% for mental health, 31% for substance use, and 27% for financial strain. Black patients and patients on Medicaid had higher HRSN prevalences. PROMIS scores were significantly worse in patients wit […]

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Striving for LGBTQI+ Health Equity in Arthroplasty.

Ross LA, Bellamy JL, Scott CE

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Social and Demographic Health Disparities in Knee Osteoarthritis and Total Knee Arthroplasty.

Dash AS, Ghanem OA, Tyler WK, Hung CT, Chahine NO, Sarpong NO

There has been increased awareness around the interplay between patients' social and demographic background and health outcomes. The influence of social determinants of health on patients outcomes has recently been recognized in orthopaedic literature. As knee osteoarthritis (OA) is one of the leading causes of disability in the world and the procedural volume of total knee arthroplasty (TKA) is rising with increased access and an aging population, better appreciation for these factors is imperative. This review seeks to examine the effect of sociodemographic factors on OA care and disease burden; and TKA outcomes. Our goal is to provide the arthroplasty surgeon community with a better understanding of how inequities in care and research impact our patients' OA and TKA outcomes. Previously published research has focused mainly on race and ethnicity, but the literature is limited, with what is available showing that there are disparate presentations and outcomes in both OA and TKA based on different social determinants of health. There is substantial work that will be required to better understand and address nonmodifiable risk factors and optimize surgery outcomes across all demographics.

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Barriers to Care for Total Joint Arthroplasty Patients With Medicaid Insurance: A Survey Study of the Surgeon Perspective.

Bloise C, Hall L, Sisco-Wise L, Randell T, Leonardi C, Cohen-Rosenblum A

BACKGROUND: The purpose of this study was to assess the demographic characteristics of orthopaedic surgeons in Louisiana taking care of patients with Medicaid insurance and to assess any perceived barriers to care, with a focus on total joint arthroplasty (TJA).

METHODS: An electronic survey was distributed to all practicing surgeon members of the Louisiana Orthopaedic Association with questions about practice type, subspecialty, demographics, estimated percentage of Medicaid patient volume, and perceived barriers to care. Responses were collected using a secure database, and statistical analysis was performed.

RESULTS: The 114 respondents were mostly male (95.6%), white (87.7%), and in private practice (61.4%). Most respondents (73.8%) did not accept Medicaid insurance (22.8%) or took care of <25% patients with Medicaid (50.9%). Fifty-eight respondents (50.9%) reported performing TJA. The majority of surgeons performing TJA (76.6% total hip arthroplasty [THA], 93.7% total knee arthroplasty [TKA]) either did not accept Medicaid insurance (36.2% THA, 47.3% TKA) or performed less than 25% of their TJA volume on patients with Medicaid insurance (40.4% THA, 36.4% TKA). The most commonly reported perceived barriers to care were lower physician and facility reimbursement, and difficulty referring to ancillary services. Surgeons in academic practice were significantly more likely to […]

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The Impact of Social Drivers of Health on Patient Portal Utilization for Total Joint Arthroplasty Patients: A Qualitative Study.

Antonioli SS, Alpert Z, Mishra S, Onakomaiya D, Vallurupalli N, Bosco JA, McLaurin T, Lajam C

BACKGROUND: Despite similar rates of osteoarthritis, minority populations undergo fewer hip and knee arthroplasties and have more complications. More than 90% of US hospitals have certified electronic health records, yet only 40% of patients utilize electronic patient portals (EPPs), with lower rates across some demographics. Adverse quality metrics and lower patient-reported outcome survey completion rates were noted for patients with inactive portals. Activation rates for EPPs can be lower among underrepresented groups, perpetuating existing disparities in access to care. Barriers to EPP activation must be identified to design interventions that improve portal utilization and, therefore, outcomes. We designed this study as there are no published reports analyzing factors which impede EPP utilization by total joint patients.

METHODS: In this IRB-exempt qualitative study, sixty-six arthroplasty patients were interviewed using a questionnaire designed to reveal reasons for EPP nonusage. Demographic factors including language, age, sex, insurance type, and zip code were collected from the electronic health records. Dedoose, a qualitative research tool, was used to analyze data and abstract trends from interview notes and transcripts.

RESULTS: We found arthroplasty patients' demographics and social drivers of health influenced utilization of EPPs. Older adults struggled with dig […]

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Association Between Regional Social Vulnerability and Race-Based Differences in Total Knee Arthroplasty Use Among Medicare Beneficiaries.

Cruse JJ, Schloemann DT, Danielson EC, Ricciardi BF, Balkissoon R, Franklin PD, Thirukumaran CP

BACKGROUND: There is inequity in total knee arthroplasty (TKA) use between racial groups. The social vulnerability index (SVI) may characterize mechanisms underlying disparity, which is essential for identifying targeted interventions. We examined the relationship of the overall SVI and its themes with TKA use rates for Black and White Medicare beneficiaries and the difference in TKA utilization between these groups.

METHODS: We used 2013-2017 Medicare data to calculate annual TKA use rates per 1000 beneficiaries in each hospital referral region (HRR). Multivariable mixed-effects linear regressions estimated the relationship between social vulnerability and TKA use.

RESULTS: The mean (standard deviation) TKA rates were 9.4 (1.7) and 5.9 (2.2) per 1000 White and Black beneficiaries, respectively. HRRs with highest social vulnerability (quartile 4) had lower TKA rates for White (estimate for quartile 4: -1.7, 95% confidence interval [CI]: -2.2 to -1.1, $P < .001$) and Black beneficiaries (quartile 4: -1.7, 95% CI: -2.3 to -1.0, $P < .001$) compared to HRRs with the least vulnerability. The SVI's minority status and language theme was linked with wider rate differences between White and Black beneficiaries (quartile 4: 0.6, 95% CI: 0.0-1.2, $P = .046$).

CONCLUSIONS: Worse overall SVI was related to lower TKA use rates for both White and Black Medicare beneficiaries. Areas with the hi […]

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Bridging the Empathy Gap: Why Health Equity Must Matter to All of Us.

Tucker KK

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Sociodemographic and Clinical Factors Associated With Response to 1-Year Postarthroplasty Patient-Reported Outcome Measures.

Sylla F, Stern BZ, Burla M, Suleiman LI, Franklin PD

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Rural vs Urban Outcomes in Total Knee and Hip Arthroplasty: A Study of Patient-Reported Outcomes and Web-Based Tool Efficacy.

Taylor JM, Dipane MV, Sassoon AA

BACKGROUND: Rural patients in the United States often face higher tobacco use, obesity rates, and lower health literacy, impacting arthroplasty outcomes. This study investigates whether rural residency is directly associated with poorer patient-reported outcomes (PROs) following total knee (TKA) and total hip arthroplasty (THA). It also evaluates the effectiveness of web-based PRO tools across urban and rural populations. Understanding these relationships is crucial for addressing disparities in care and improving surveillance for rural populations.

METHODS: From May 2021 to April 2023, TKA and THA patients at our institution were categorized based on their home zip codes, using the 2020 Census Bureau binary urban-rural classification. Web-based PROs were administered via text or email and assessed using the Knee Injury and Osteoarthritis Outcome Score, Joint Replacement; Hip Injury and Osteoarthritis Outcomes Score, Joint Replacement; University of California, Los Angeles Activity Score; Patient-Reported Outcomes Measurement Information System Global Physical and Mental Health Scores; and Forgotten Joint Scores. PROs were recorded preoperatively and at 6 weeks, 3 months, and 1 year postoperatively. Statistical analysis encompassed chi-square and independent samples t-tests.

RESULTS: Among 781 THA and 1094 TKA cases, survey response rates (71.9%-83.8%) were consistent across […]

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Health-Related Social Needs Associated With Worse Patient-Reported Outcomes and Increased Adverse Events Following Total Joint Arthroplasty.

Hayashi CL, Winter AD, Túa-Vargas LA, Chau N, Kalia A, Gibson DH, Rubin LE, Molloy IB et al.

BACKGROUND: Health-related social needs (HRSNs) are the individual-level adverse social conditions that negatively impact a person's health. This study characterizes the association of HRSNs with patient-reported outcomes and adverse events following total hip (THA) and knee arthroplasty (TKA).

METHODS: This single-institution cross-sectional study utilized the Centers for Medicare & Medicaid Services HRSN Screening Tool. In-person interviews captured HRSNs, patient-reported outcome measures, and demographic data from postoperative THA and TKA patients in an academic arthroplasty practice. Charlson Comorbidity Index, American Society of Anesthesiologists scores, discharge data, and 90-day complications were collected via chart review. Regression analysis was used to determine associations between HRSNs and outcomes.

RESULTS: Among 190 patients, food insecurity had a significant association with reoperation (odds ratio (OR) = 5.78, 95% confidence interval 1.44-23.2, P = .013). Patients with food and transportation HRSNs had significantly worse postoperative physical function, pain, and mobility (all P < .05). Black patients had significantly higher odds of visiting the emergency department (OR = 2.15, 95% CI 1.10-4.20, P = .025) or being readmitted (OR = 2.70, 95% confidence interval 1.11-6.58, P = .029) within 90 days postoperation compared to White patients.

CONCLUSIONS: Fo […]

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Impact of Rurality on Total Joint Arthroplasty Access and Outcomes: A Systematic Review.

Agrodnia SL, Call CM, Lachance AD, McDonald JC, Rana AJ, McGrory BJ

BACKGROUND: Total joint arthroplasty (TJA) is one of the most frequent orthopaedic surgeries in the United States; however, disparities in utilization and outcomes based on race, sex, and socioeconomic level have been well documented. The impact of rural geographic location, a characteristic that may impact access to TJA care and postoperative outcomes, remains understudied. This systematic review investigated associations between rural location and several metrics in total hip arthroplasty and total knee arthroplasty to investigate whether disparities are present based on rural geographic location.

METHODS: In November 2024, PubMed, Web of Science, Scopus, EMBASE, and Cochrane Review databases were queried. Ten studies investigating TJA utilization and outcomes that included patients in a rural location were included. Study quality was assessed using a modified Newcastle-Ottawa Scale.

RESULTS: Ten articles were included in the final analysis. Differences in complications, readmission rates, and length of stay due to rural location of patient and/or hospital were reported. Results indicate urban TJA patients more frequently discharge to rehab. Patient-reported outcome measures following TJA do not appear different between rural and urban patient groups.

CONCLUSIONS: Rural geographic location is associated with different TJA utilization and outcomes and may point to dispariti […]

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Patient Social Determinants of Health Are Associated With Hip Dysfunction and Osteoarthritis Outcome Scores and Outcomes After Total Hip Arthroplasty.

Yacoub G, Janney CA, Wang Z, Meremikwu R, Caird MS, Kheir MM

BACKGROUND: Disparities in access and outcomes persist for total hip arthroplasty (THA). The Hip Dysfunction and Osteoarthritis Outcome Score (HOOS) is a validated patient-reported outcomes and measure and a proxy for quality of care. We examined whether patient social determinants of health were associated with HOOS scores and outcomes.

METHODS: We retrospectively identified 2938 THAs at a single institution from 2015 to 2023. Multivariate analyses were performed to determine if associations existed between patient factors (race, sex, marital status, age, insurance carrier, among others) and HOOS scores (preoperative, 6-week, and 1-year postoperative) or outcomes (length of stay [LOS], discharge disposition, or perioperative complications).

RESULTS: Lower preoperative HOOS scores were associated with being younger, unmarried, Black, higher body mass index, higher Elixhauser score, government insurance, posterior approach, and female ($P \leq 0.01$). Lower 6-week and 1-year postoperative HOOS scores were associated with being Black and higher Elixhauser scores ($P \leq .05$). Decreased HOOS score improvement at 1 year was associated with older age ($P < .01$). Longer LOS and discharge to rehabilitation facility were associated with being older, unmarried, black, female, higher Elixhauser score, and posterior approach ($P < .01$).

CONCLUSIONS: Several social determinants of health were ass […]

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Arthroplasty Surgeons Taking Care of the Highest Social Risk Populations Are More Often Osteopathic Trained, Less Experienced, and More Likely to be Penalized by the Merit Based Incentive Payment System.

Holle AM, Gill VS, Lin E, Bingham JS, Spangehl MJ, Clarke HD

BACKGROUND: The purpose of this study was to examine how hip and knee arthroplasty surgeon Merit-based Incentive Payment System (MIPS) performance, surgeon demographics, practice characteristics, and patient population varied based on the social risk of their caseload in 2017 and 2021.

METHODS: Multiple databases published by the Centers for Medicare and Medicaid Services were combined and utilized to examine all US hip and knee arthroplasty surgeons. Surgeons were placed into quintiles of social risk based on the proportion of dual-eligible Medicare-Medicaid patients in their patient population. Demographics, Distressed Community Index scores, patient population information, and MIPS performance were compared between quintiles in years 2017 and 2021.

RESULTS: In 2017, arthroplasty surgeons with the highest social risk caseloads scored lower on MIPS (70.0 vs 73.5, P = .012) and were more likely to receive a negative payment adjustment (odds ratio: 1.64; 95% confidence interval: 1.01-2.68, P = .046) compared to surgeons with the lowest social risk caseload. In 2021, arthroplasty surgeons with the highest social risk caseloads scored higher on MIPS (86.2 vs 82.4, P < .001), but remained more likely to receive a negative payment adjustment (odds ratio: 3.98; 95% confidence interval: 1.63-10.53, P = .003) compared to those with low social risk caseloads.

CONCLUSIONS: These findi […]

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